ENT ATLAS

CLINICAL

Mouth & throat

18 articles · imaging key facts · November 2026

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CLINICAL · MOUTH & THROAT

Tonsillitis and peritonsillar abscess

The tonsils are the most operated-on lymphoid tissue in the body and among the most misunderstood. Most sore throats need no antibiotic and most tonsils need no surgery; but the patient who cannot swallow his own saliva, whose uvula is pushed across and whose jaw will not open has a quinsy that must be drained, and a child with a membrane on the tonsils and a bull neck may have diphtheria. Know the tonsil, the infections and the operation, and the dangerous throat stands out from the ordinary one.

AT A GLANCE

- Sore throat with trismus means quinsy.
- Drain and treat, watching the airway.
- Use Paradise criteria for tonsillectomy.
- An asymmetric adult tonsil needs biopsy.
- Intracapsular tonsillectomy means less pain and bleeding; codeine is contraindicated under 18 for OSA.

1 CHAPTER ONE Applied anatomy

The palatine tonsil.

1.1 Position and surfaces

Each tonsil is an ovoid mass of lymphoid tissue in the lateral wall of the oropharynx, in the *tonsillar fossa* between the anterior (palatoglossal) and posterior (palatopharyngeal) pillars. It is larger than it looks, extending up into the soft palate, down into the base of tongue and forwards under the anterior pillar.

- **Medial surface:** non-keratinising stratified squamous epithelium dipping into 12 to 15 *crypts*; the largest, near the upper pole, is the *crypta magna* (intratonsillar cleft), the remnant of the second pharyngeal pouch. Crypts fill with cheesy debris that can be expressed by pressing on the anterior pillar.
- **Lateral surface:** a well-defined fibrous *capsule*, separated from the bed by loose areolar tissue: the plane of dissection at tonsillectomy and the space where pus

collects in a quinsy.

- **Upper pole:** reaches into the soft palate under a semilunar fold enclosing the *supratonsillar fossa*.
- **Lower pole:** attached to the tongue, separated from it by the *tonsillolingual sulcus*, a hidden site for carcinoma.

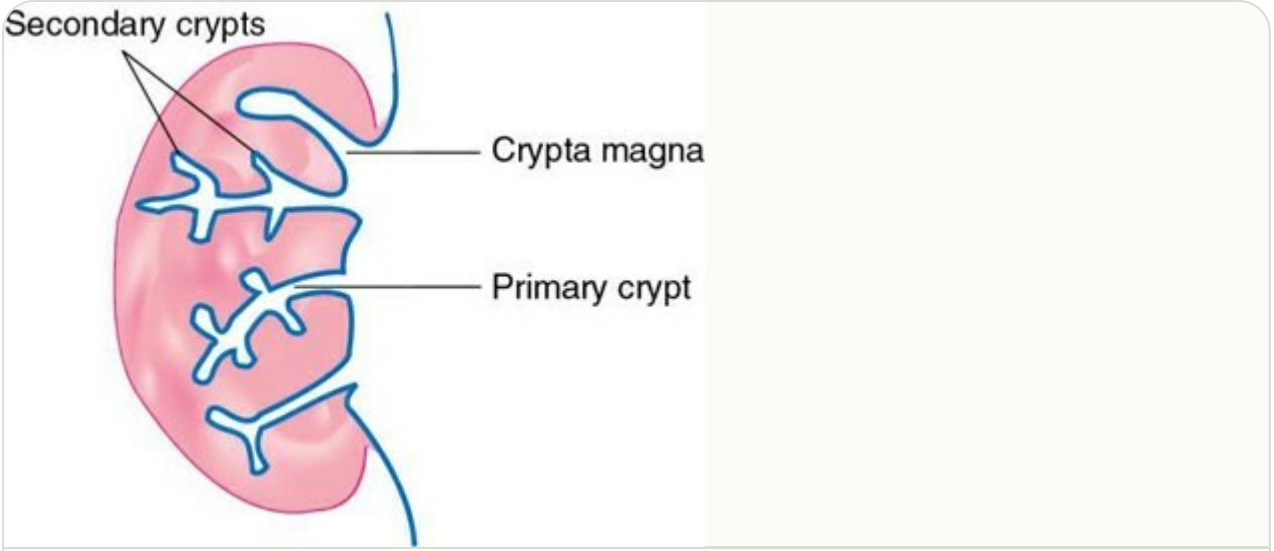


FIGURE 1 Primary and secondary crypts of the tonsil, with the crypta magna.
Dhingra, fig. 51.1

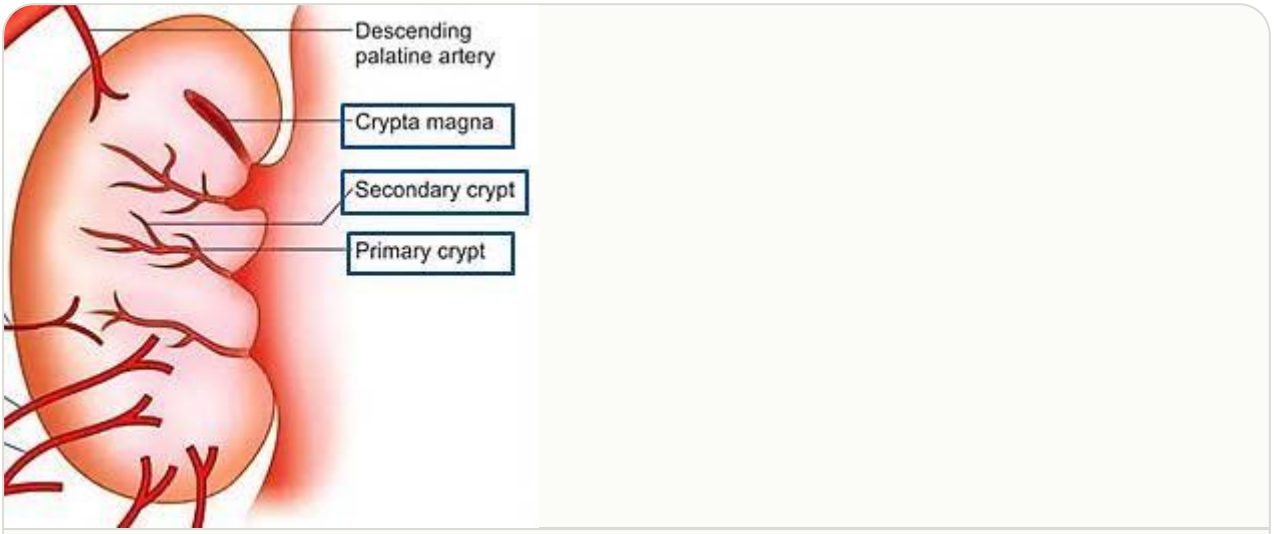


FIGURE 2 Tonsillar crypts and blood vessels.
Anatomy atlas · ENT Steps p. 214

1.2 Bed and relations

The bed is formed by the superior constrictor and styloglossus. The glossopharyngeal nerve and an elongated styloid process may lie in the lower part

of the fossa and can be reached through it after tonsillectomy. Beyond the constrictor lie the facial artery, submandibular gland, posterior belly of digastric, medial pterygoid and the angle of the mandible; the internal carotid artery lies about 2.5 cm behind and lateral to the tonsil.

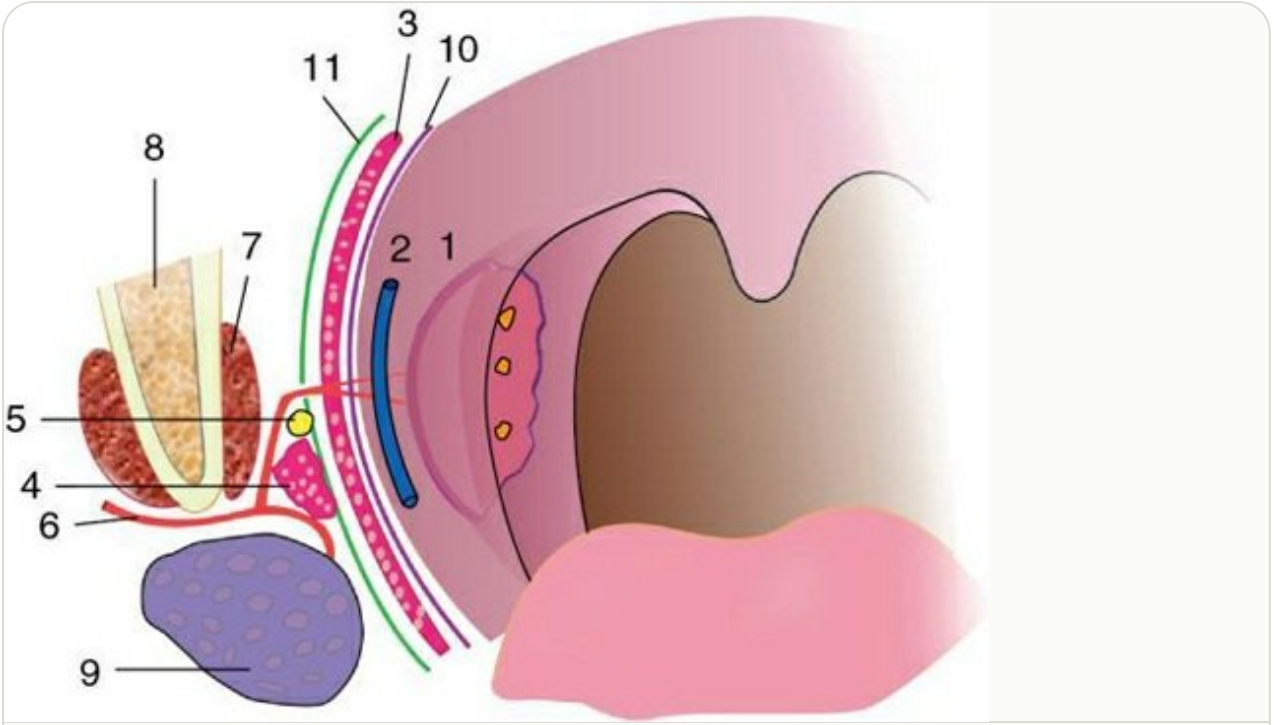


FIGURE 3 Relations of the tonsil: capsule (1), loose areolar tissue with the paratonsillar vein (2), superior constrictor (3), styloglossus (4), glossopharyngeal nerve (5), facial artery (6), medial pterygoid (7), angle of mandible (8), submandibular gland (9).

Dhingra, fig. 51.2

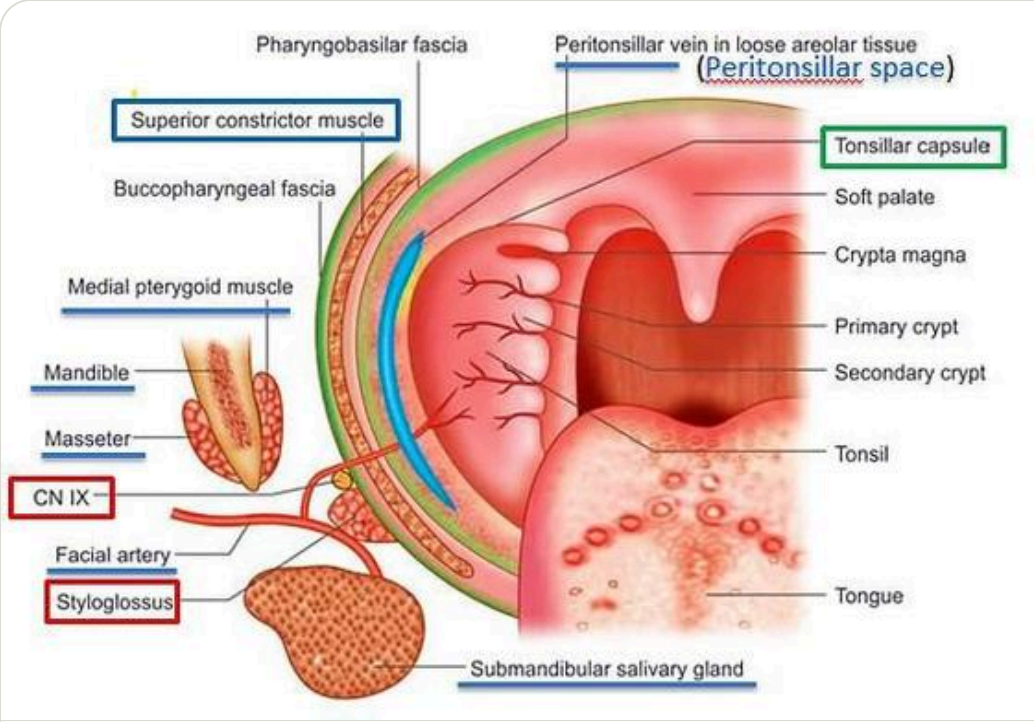


FIGURE 4 The bed of the tonsil in horizontal section.
Anatomy atlas · ENT Steps p. 215

1.3 Blood supply

Five arteries: the *tonsillar branch of the facial artery* (the main supply, entering at the lower pole), the ascending palatine (facial), the ascending pharyngeal, the dorsal lingual branches (lingual) and the descending palatine (maxillary). Veins drain to the *paratonsillar (external palatine) vein*, which descends from the soft palate across the lateral surface of the capsule to the common facial vein and pharyngeal plexus; it is the usual source of bleeding at tonsillectomy.

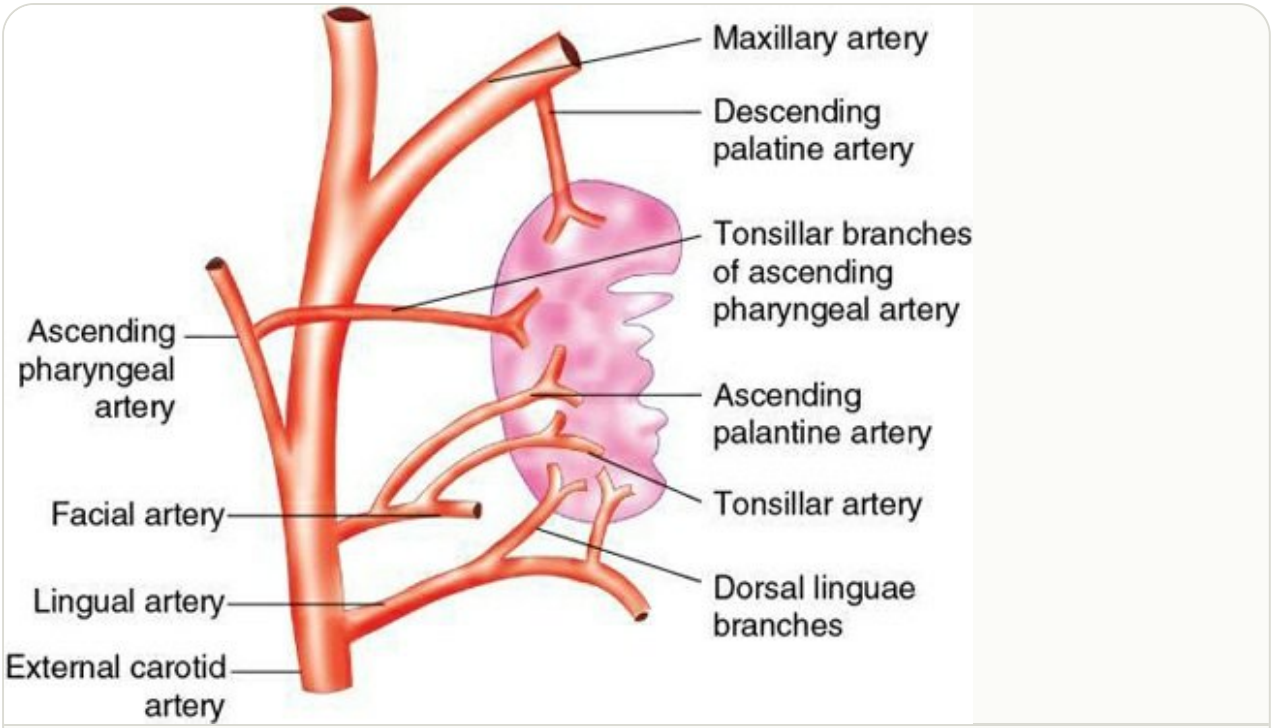


FIGURE 5 Arterial supply of the tonsil.
Dhingra, fig. 51.3

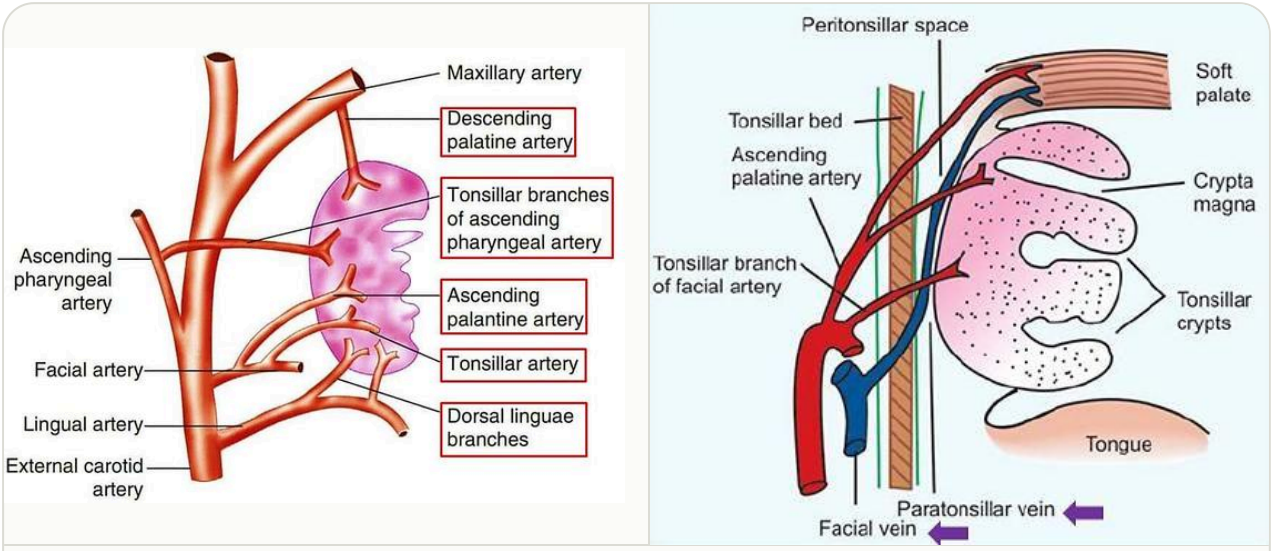


FIGURE 6 Arterial supply of the palatine tonsil, and the paratonsillar vein with the tonsillar arteries.
Anatomy atlas · ENT Steps p. 217, ENT Steps p. 218

1.4 Lymph and nerves

Lymphatics pierce the superior constrictor to the *jugulodigastric (tonsillar) node* below the angle of the mandible. Sensation comes from the glossopharyngeal nerve and

lesser palatine nerves; hence the earache of tonsillitis and after tonsillectomy.

1.5 Function

The crypts and their specialised epithelium (M cells, antigen-presenting cells) bring inhaled and swallowed antigens to B-cell follicles that make antibody, and alert the wider immune system. The tonsils are most active between four and ten years and involute after puberty. Removal has not been shown to impair immunity, but should still be done only for a clear indication.

2 CHAPTER TWO Acute tonsillitis

The common infection.

2.1 Types

- **Catarrhal (superficial):** part of a generalised, usually viral, pharyngitis.
- **Follicular:** pus in the crypts appears as yellow spots at their openings.
- **Parenchymatous:** the whole tonsil is uniformly swollen and red.
- **Membranous:** exudate from the crypts coalesces into a membrane over the surface.

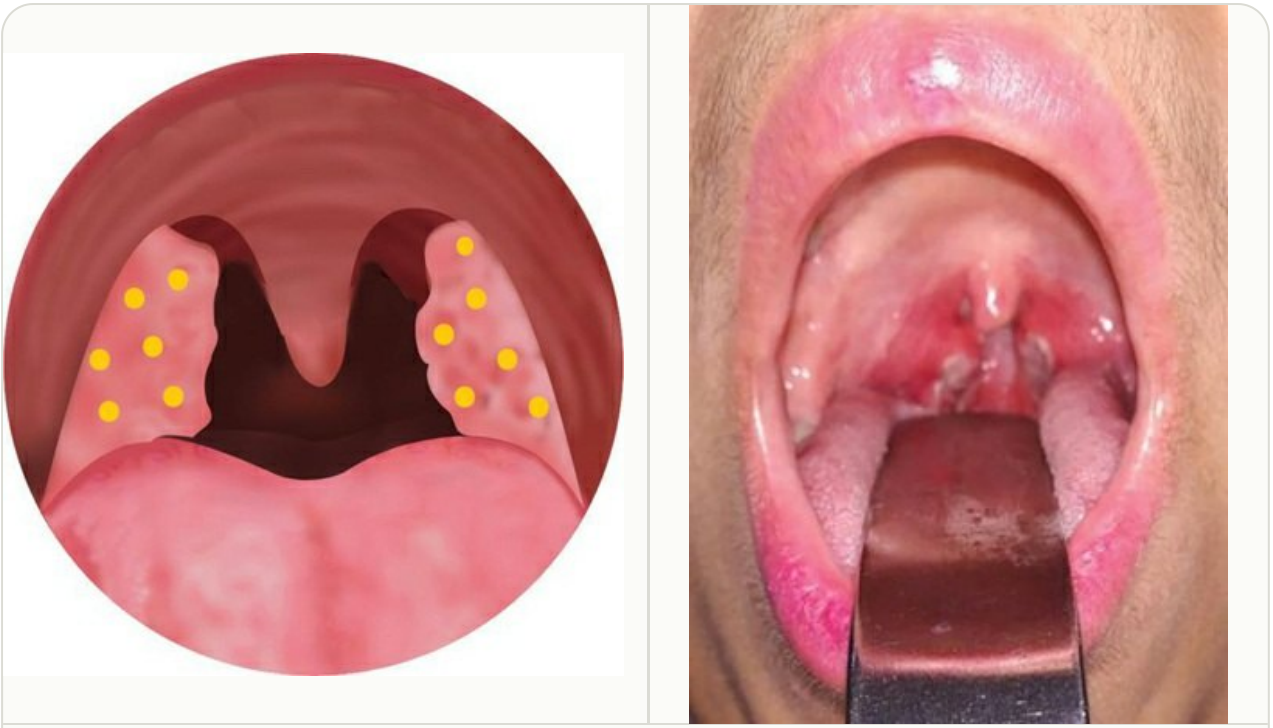


FIGURE 7 First, acute follicular tonsillitis. Second, acute membranous tonsillitis.
Dhingra, figs 51.4 and 51.5

2.2 Cause

School children mostly, and young adults; uncommon in infants and over fifty. Group A beta-haemolytic streptococcus is the commonest bacterial cause; staphylococci, pneumococci and *H. influenzae* also, often after a virus. Most episodes overall are viral.

2.3 Clinical features

Sore throat, painful swallowing (a child may refuse food), fever of 38 to 40°C with rigors, referred earache or true otitis media, malaise, and in children abdominal pain from mesenteric adenitis that can mimic appendicitis. The breath is foul, the tongue coated, the pillars and palate red; the tonsils are swollen with spots of pus or a membrane that wipes off easily; the jugulodigastric nodes are enlarged and tender.



FIGURE 8 Acute follicular tonsillitis: pus beads on the left tonsil; on the right they have coalesced into a membrane.

Dhingra, fig. 51.6

2.4 Treatment

Rest, fluids and analgesia (paracetamol or ibuprofen; aspirin is avoided in children). For bacterial tonsillitis, penicillin for ten days (erythromycin in allergy); amoxicillin is best avoided when glandular fever is possible.

2.5 Complications

- Chronic tonsillitis with recurrent attacks.
- Peritonsillar abscess, parapharyngeal abscess, cervical abscess from suppurating nodes.
- Acute otitis media.
- Rheumatic fever, acute glomerulonephritis, and endocarditis in valvular heart disease.

3 CHAPTER THREE
The membrane on the tonsil

A differential that matters.

MEMBRANE OVER THE TONSIL

CONDITION	CLUES
Membranous tonsillitis	Abrupt onset, high fever; membrane on the tonsil only, wipes off easily
Diphtheria	Slower onset, low fever, toxic child; dirty grey membrane spreading beyond the tonsil to palate and pharynx, adherent, bleeds when removed; bull neck
Vincent angina	Insidious, little fever; membrane usually on one tonsil, lifting off to leave an irregular ulcer; fusiform bacilli and spirochaetes on smear
Infectious mononucleosis	Young adult; both tonsils large and coated; posterior cervical nodes, spleen; atypical lymphocytes, positive Paul–Bunnell (Monospot) test
Agranulocytosis	Very ill; necrotic ulcers across the oropharynx; neutrophils profoundly reduced
Leukaemia	Anaemia, bleeding gums; blast cells in blood and marrow
Others	Aphthous ulcer, traumatic ulcer (membrane within 24 hours), candida, malignancy

Investigations: total and differential counts, blood film, throat swab and culture, Monospot test, marrow examination and biopsy as indicated.

3.1 Faucial diphtheria

Corynebacterium diphtheriae, spread by droplets, incubation two to six days; carriers exist. A grey-white tenacious membrane forms on the tonsils and spreads to the palate and posterior wall, bleeding when lifted; tender jugulodigastric nodes give a *bull neck*; the child is toxic though the fever is modest. The exotoxin damages the heart (myocarditis, arrhythmia, circulatory failure) and nerves (palatal, ocular and diaphragmatic paralysis, weeks later); laryngeal membrane obstructs the airway.

TREAT DIPHTHERIA ON SUSPICION

- Give **antitoxin** without waiting for culture: 20,000 to 40,000 units if under 48 hours and confined to the tonsils, 80,000 to 120,000 units if longer or more extensive, intravenously over an hour after a test dose, with adrenaline ready.
- **Penicillin or erythromycin** to kill the organism.
- **Isolate;** discharge after three negative swabs 24 hours apart; give a booster, since infection does not confer immunity.
- Immunise contacts and give them erythromycin for a week.

CHAPTER FOUR

4 Chronic tonsillitis

Infection that does not go away.

4.1 Causes

Incomplete resolution of acute attacks with walled-off microabscesses in the follicles, subclinical infection, and infected sinuses or teeth; mostly children and young adults.

4.2 Types

- **Chronic follicular:** crypts full of infected cheesy material, seen as yellow spots.
- **Chronic parenchymatous:** hyperplasia makes the tonsils very large; they may meet in the midline and obstruct speech, swallowing and breathing, with sleep apnoea and eventually cor pulmonale.
- **Chronic fibroid:** small, infected tonsils with recurrent sore throats.



FIGURE 9 Parenchymatous tonsillitis: the two tonsils almost touch.
Dhingra, fig. 51.7

4.3 Features

Recurrent sore throats, chronic irritation and cough, bad taste and halitosis, and with large tonsils thick speech, difficulty swallowing and choking at night. *Flushing of the anterior pillars* compared with the surrounding mucosa and persistent *enlargement of the jugulodigastric nodes* are reliable signs; pressure on the pillar may express pus.

TONSIL SIZE GRADING

Grade 0 tonsils are within the fossa; grade 1 occupy under 25% of the oropharyngeal airway; grade 2, 25 to 50%; grade 3, 50 to 75%; grade 4, more than 75% ("kissing" tonsils).

4.4 Local complications

- **Intratonsillar abscess:** a blocked crypt fills with pus; a red, swollen, very painful tonsil; antibiotics, drainage, and later tonsillectomy.
- **Tonsillolith:** a calcified crypt plug felt as a gritty mass, removed or treated by tonsillectomy.
- **Tonsillar cyst:** a yellow swelling from a blocked crypt, often symptomless, easily drained.



FIGURE 10 A tonsillar cyst on the left (arrow).
Dhingra, fig. 51.8

4.5 Lingual tonsil

Acute lingual tonsillitis gives a one-sided lump feeling and dysphagia; *compensatory hypertrophy* follows tonsillectomy and causes a lump sensation, thick voice and cough; a *lingual tonsil abscess* causes severe dysphagia, salivation, painful protrusion and risks laryngeal oedema.

4.6 Periodic fever (PFAPA)

In young children, recurrent episodes every three to eight weeks of fever, aphthous ulcers, pharyngitis and cervical adenitis lasting a few days, with complete health in between. Cyclic neutropenia must be excluded. A single dose of steroid aborts attacks; tonsillectomy is curative in most.

5 CHAPTER FIVE Peritonsillar abscess

Quinsy.

5.1 Definition and cause

Pus in the peritonsillar space between the tonsil capsule and the superior constrictor. Usually a crypt, often the crypta magna, becomes blocked and forms an intratonsillar abscess that bursts through the capsule; some arise from the mucous (Weber) glands of the supratonsillar fossa. Cultures are usually mixed aerobic (*Streptococcus pyogenes*, *Staphylococcus aureus*) and anaerobic.

5.2 Clinical features

Mostly young adults, usually one-sided.

- **General:** fever to 40°C, rigors, malaise, headache.
- **Local:** severe one-sided throat pain; odynophagia so marked that saliva dribbles from the mouth and the patient becomes dehydrated; a muffled *hot-potato voice*; foul breath; ipsilateral earache; and *trismus* from spasm of the medial pterygoid.

EXAMINATION FINDINGS IN QUINSY

SIGN	EXPLANATION
Bulging of the soft palate and anterior pillar above the tonsil	Pus in the supratonsillar part of the space
Uvula swollen and pushed to the opposite side	Mass effect
Tonsil buried in oedematous pillars, pushed down and medially	Displacement
Mucopus over the tonsil, tender jugulodigastric nodes	Infection
Neck tilted to the affected side (torticollis)	Muscle spasm



FIGURE 11 A left peritonsillar abscess.
Dhingra, fig. 52.4

TRISMUS IS THE KEY

Lee calls trismus the best single indicator of an abscess rather than peritonsillar cellulitis. If you are unsure, aspirate with a needle: pus settles the question and gives a culture.

5.3 Investigation

Diagnosis is clinical. Needle aspiration confirms pus. Contrast-enhanced CT when the diagnosis is uncertain, trismus prevents examination, or spread to the parapharyngeal space is suspected.

5.4 Treatment

1. Admit; intravenous fluids.
2. High-dose intravenous antibiotics covering aerobes and anaerobes (penicillin with metronidazole, or co-amoxiclav); a single dose of steroid reduces pain and trismus.
3. Analgesia; avoid aspirin. Saline or peroxide mouthwashes.
4. **Drain the pus:** needle aspiration, or incision at the point of maximum bulge, which lies just lateral to where a horizontal line through the base of the uvula meets a vertical line along the anterior pillar. A guarded blade makes a small stab and sinus forceps open the cavity; the forceps may be reinserted the next day.

5. **Tonsillectomy:** an *interval* tonsillectomy four to six weeks later is advised after a second quinsy in adults (Lee quotes recurrence of about 17%), and more readily in children or those with recurrent tonsillitis. *Quinsy (hot) tonsillectomy* during the acute illness is preferred by some, at the cost of more bleeding and the risk of rupture at induction.

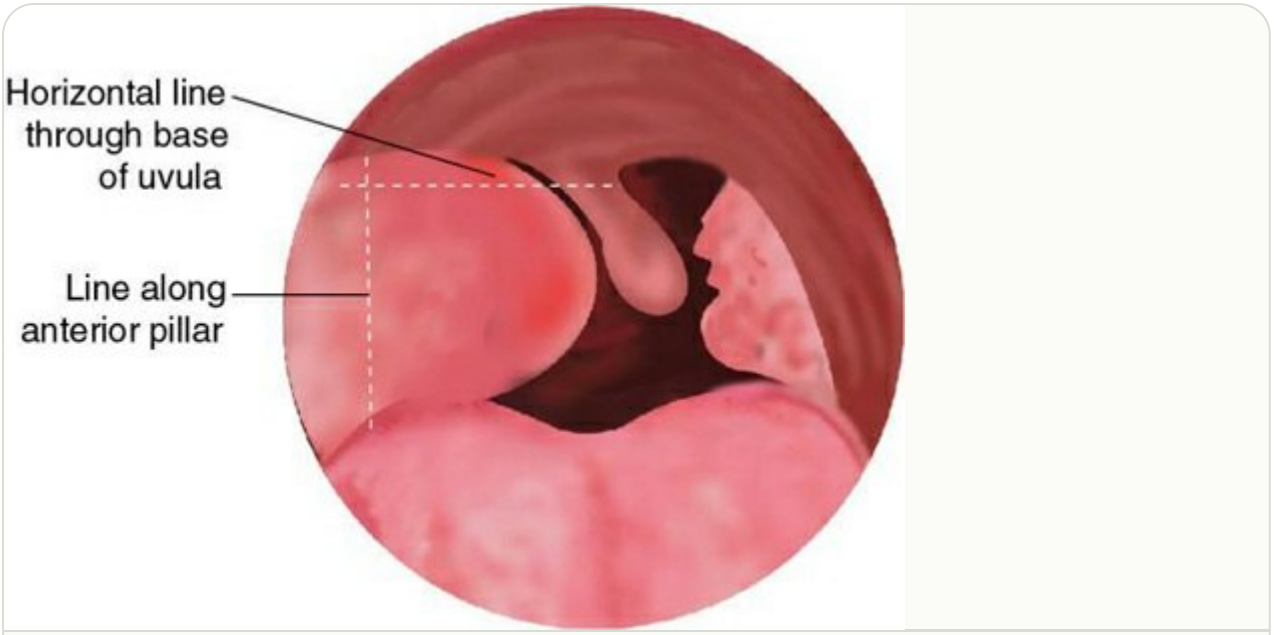
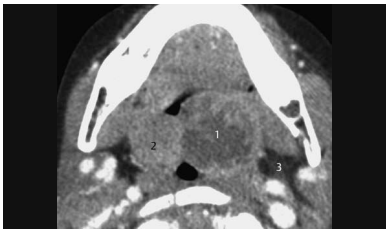


FIGURE 12 The site of drainage: just lateral to the junction of a vertical line along the anterior pillar and a horizontal line through the base of the uvula.
Dhingra, fig. 52.5

IMAGING IN THE APP 1 case

A tonsillar abscess on imaging.



Tonsillar Abscess
Infection & inflammation · 7 images

5.5 Complications

Parapharyngeal abscess (a quinsy is a potential parapharyngeal abscess), laryngeal oedema needing tracheostomy, aspiration pneumonia or lung abscess after spontaneous rupture, septicaemia with endocarditis, nephritis or brain abscess, internal jugular vein thrombosis (Lemierre syndrome), and haemorrhage from erosion of the carotid or jugular vessels.

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6 CHAPTER SIX Tonsillectomy

Indications, technique and complications.

6.1 Indications

6.1.1 Absolute (Dhingra)

- **Recurrent throat infection:** seven or more episodes in one year, five a year for two years, or three a year for three years (the Paradise criteria), or two weeks or more of school or work lost in a year.
- **Peritonsillar abscess:** in children four to six weeks after treatment; in adults after a second attack.
- **Tonsillitis causing febrile seizures.**
- **Hypertrophy** causing obstructive sleep apnoea, difficulty swallowing or interference with speech.
- **Suspicion of malignancy:** a one-sided enlarged tonsil may be lymphoma in a child or squamous carcinoma in an adult; excisional biopsy. Also in the search for an unknown primary.

6.1.2 Relative

Diphtheria or streptococcal carriers unresponsive to antibiotics, chronic tonsillitis with halitosis unresponsive to treatment, and recurrent streptococcal tonsillitis in valvular heart disease.

6.1.3 As part of another operation

Uvulopalatopharyngoplasty, glossopharyngeal neurectomy, and removal of an elongated styloid process.

CURRENT PRACTICE

Guidelines now recommend watchful waiting for recurrent sore throat that falls short of the Paradise criteria, and give obstructive sleep-disordered breathing as the commonest indication in children. A single intraoperative dose of dexamethasone reduces vomiting; routine postoperative antibiotics are not recommended; codeine is avoided in children.

6.2 Contraindications

Haemoglobin below 10 g/dL; acute upper respiratory infection, including acute tonsillitis; children under three (relative); overt or submucous cleft palate; bleeding disorders (von Willebrand disease, haemophilia, leukaemia, purpura, aplastic anaemia); uncontrolled diabetes, hypertension, heart disease or asthma; and, traditionally, during an epidemic of polio and during menstruation.

6.3 Technique (dissection and snare)

1. General anaesthesia with a tube; *Rose's position* (head extended on a shoulder roll, head ring), avoiding hyperextension.
2. The Boyle–Davis gag is opened and held by Draffin's bipods.
3. The tonsil is grasped and drawn medially; the mucosa is incised where it reflects from the tonsil onto the anterior pillar, and the incision carried round the upper pole.
4. The upper pole is freed with blunt scissors or a dissector in the plane outside the capsule, and dissection continues downwards to the lower pole.
5. A snare is passed over the pedicle at the lower pole, tightened and the tonsil removed.
6. A swab is pressed into the fossa; bleeding points are tied or coagulated. The other side is done in the same way.

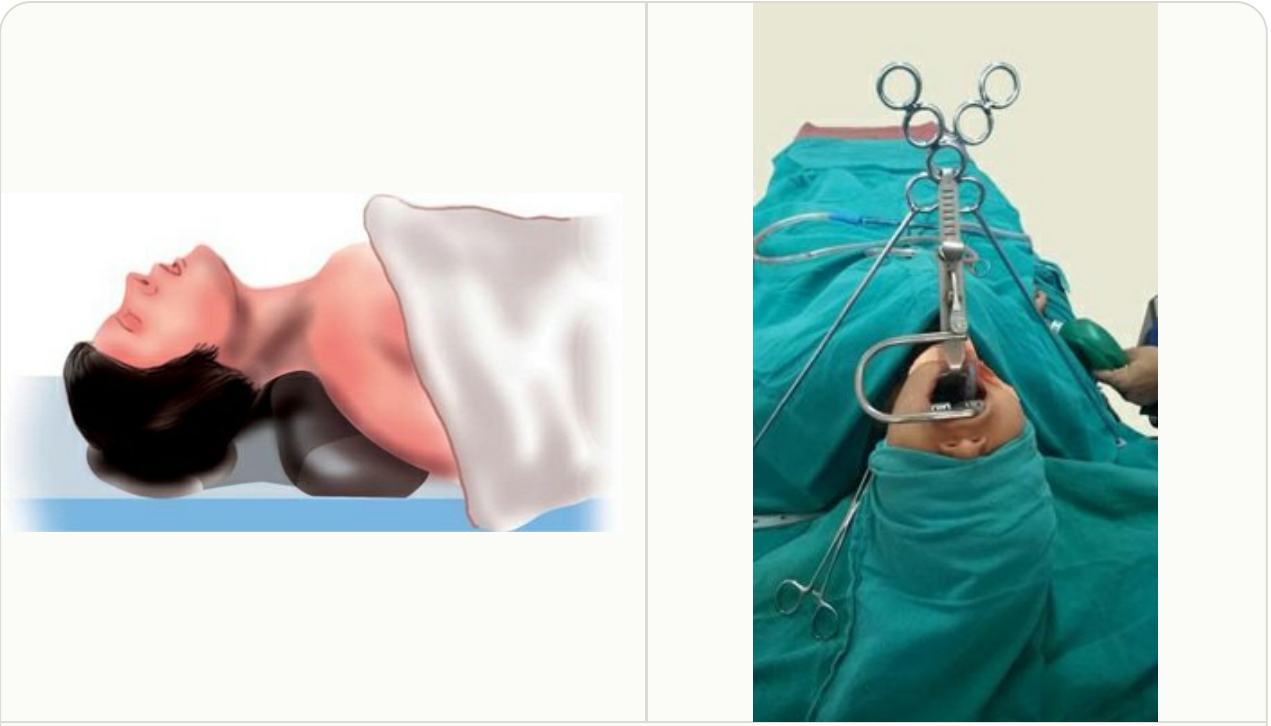


FIGURE 13 First, Rose's position: the neck extended over a sandbag and the head on a ring. Second, the Boyle–Davis gag held by Draffin's bipods.

Dhingra, figs 94.1 and 94.3

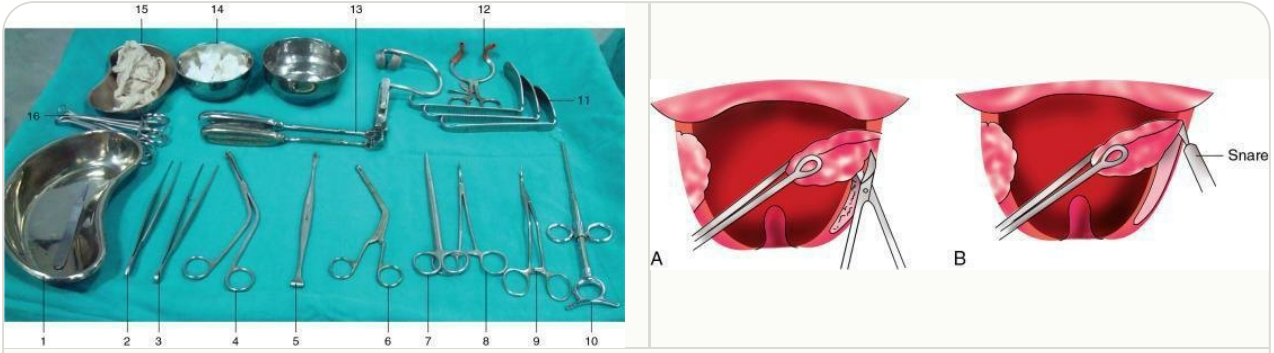


FIGURE 14 First, the instruments for tonsillectomy. Second, (A) the tonsil dissected from its bed and (B) the pedicle at the lower pole cut with a snare.

Dhingra, figs 94.2 and 94.4

6.4 Other techniques

- **Cold:** dissection and snare (the commonest), guillotine (largely abandoned), intracapsular tonsillotomy with a microdebrider (capsule preserved, less pain), harmonic scalpel, coblation (plasma-mediated, low temperature), cryosurgery.
- **Hot:** monopolar or bipolar electrocautery (less bleeding, more pain), KTP or CO2 laser tonsillectomy or staged laser tonsillotomy, radiofrequency.

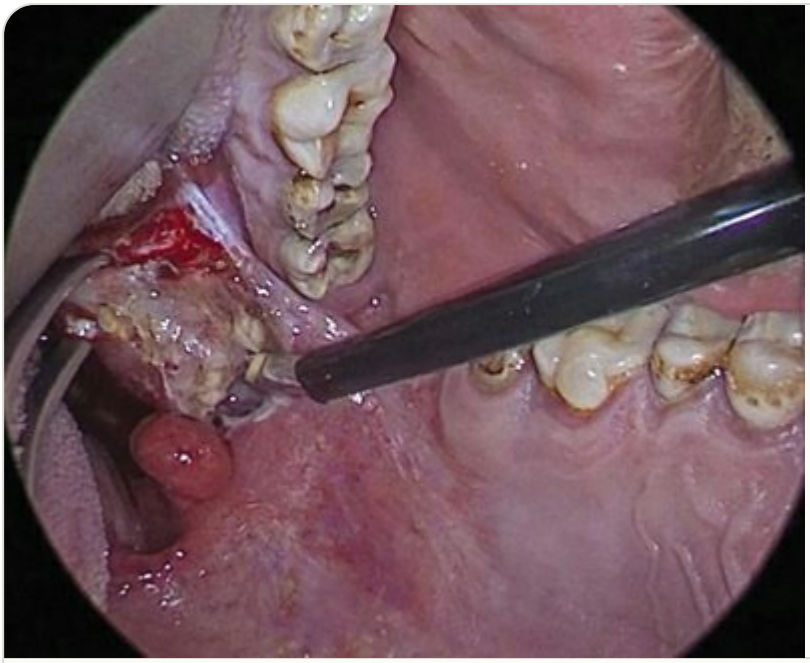


FIGURE 15 Tonsillectomy with coblation.
Dhingra, fig. 94.5

OPERATIVE GUIDES IN THEATRE 2 operations

Tonsillectomy and drainage of neck abscesses.

Tonsillectomy
Tonsillectomy for recurrent criteria or quinsy

Deep Neck Abscesses
Neck abscess

6.5 Aftercare

Recovery position until awake; watch pulse, breathing and blood pressure, and look for bleeding or repeated swallowing. Cold fluids first, then a soft diet, then normal food early, which keeps the fossae clean; regular analgesia before meals (paracetamol, ibuprofen; not aspirin); mouthwashes. Most go home the same or next day and return to work or school in about two weeks. Earache is referred pain, not otitis.

6.6 Complications

6.6.1 Immediate

- **Primary haemorrhage** at operation: pressure, ties, diathermy.

- **Reactionary haemorrhage** within 24 hours: remove the clot (which stops the constrictor from clamping the vessels), apply pressure or vasoconstrictor; if it persists, return to theatre.
- Injury to pillars, uvula, palate, tongue, constrictor or teeth; aspiration of blood; facial oedema; surgical emphysema.

6.6.2 Delayed

- **Secondary haemorrhage** from the fifth to tenth day, from infection and premature separation of the slough, often heralded by bloodstained saliva. Admit every patient; a clot alone is observed, with antibiotics and peroxide gargles; active bleeding returns to theatre under general anaesthesia; check haemoglobin and clotting, transfuse if needed; external carotid ligation is very rarely required.
- Infection of the fossa, parapharyngeal abscess, otitis media.
- Lung complications from aspiration: atelectasis, lung abscess.
- Scarring of the palate and pillars; velopharyngeal insufficiency.
- Tonsillar remnants that become infected; compensatory lingual tonsil hypertrophy.

POST-TONSILLECTOMY BLEEDING

A child who keeps swallowing, vomits dark blood or has a rising pulse is bleeding until proven otherwise. Children compensate until they suddenly do not: secure intravenous access, cross-match, and involve the anaesthetist early; induction in a bleeding child is a high-risk airway.

TECHNIQUE, PAIN AND SAFETY (OXFORD TEXTBOOK)

Techniques are either **extracapsular** (total tonsillectomy, by cold steel, bipolar or monopolar diathermy, or coblation) or **intracapsular** (partial tonsillectomy or tonsillotomy, by coblation or microdebrider, leaving the capsule as a biological dressing over the muscle). Hot techniques reduce operative bleeding but increase secondary haemorrhage and pain, and haemorrhage rises with the diathermy power used, so the lowest effective setting is advised. Intracapsular surgery, especially in young children having surgery for sleep-disordered breathing, gives less pain and fewer bleeds, at the cost of occasional regrowth. After a cluster of child deaths from haemorrhage in Austria, that country moved largely to intracapsular surgery for young children. There is still no consensus on the best method.

Haemorrhage is *primary* within 24 hours (usually from inadequate haemostasis) or *secondary* after 24 hours, most often on days 5 to 10. Every bleed should be assessed in hospital, and any needing intervention counts as significant; reported rates vary widely because definitions differ. Mortality in more than 300,000 English paediatric

tonsillectomies was about 1 in 27,000. Risk is higher in children under three with proven sleep apnoea and in those with craniofacial anomalies, Down syndrome, cerebral palsy, heart disease or bleeding disorders, who need surgery in a centre with paediatric high-dependency care.

Pain is near-universal and causes dehydration and readmission; regular paracetamol with ibuprofen is standard. **Codeine is contraindicated** in under-18s having tonsillectomy for sleep apnoea, because ultrarapid CYP2D6 metabolisers convert it to toxic levels of morphine, causing deaths from respiratory depression. **Adenoidectomy**, done with tonsillectomy in more than half of cases, is now often performed by suction diathermy or coblation under direct or mirror vision rather than with a blind curette.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Anatomy:** crypta magna, capsule, bed, blood supply and the paratonsillar vein.
- **Types of acute tonsillitis** and complications.
- **Differential diagnosis of a membrane** on the tonsil; **diphtheria** management.
- **Quinsy:** features, site of incision, and when to remove the tonsils.
- **Indications, contraindications** and **complications** of tonsillectomy; reactionary versus secondary haemorrhage.

7.2 Common pitfalls

COMMON PITFALLS

- Draining "quinsy" without trismus or a bulge: aspirate first.
- Giving amoxicillin to a young adult with glandular fever.
- Ignoring a one-sided enlarged tonsil in an adult.

- Sending home a child with a small "warning" bleed after tonsillectomy.

Written for this app from Oxford Textbook of Otolaryngology ch. 49, 87, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 51, 52, 94 and K. J. Lee ch. 36, 37, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Deep neck space infection

The fascia of the neck divides it into compartments that are tidy in the anatomy book and treacherous in practice: infection from a tooth, a tonsil or a node finds the planes between them and runs, sometimes down into the chest, sometimes around the airway. Deaths from deep neck infection come from three things: a lost airway, a spreading infection that reaches the mediastinum, and a great vessel eroded or thrombosed. Secure the airway, image the neck, give the right antibiotics and drain the pus.

AT A GLANCE

- Airway plan first.
- Contrast CT, antibiotics and drainage.
- Dental disease is the usual cause.
- Watch for mediastinitis.

1 CHAPTER ONE Fascia and spaces

The map.

1.1 Deep cervical fascia

- **Superficial (investing) layer:** surrounds the neck, splitting to enclose the sternomastoid, trapezius, submandibular and parotid glands.
- **Middle (visceral) layer:** surrounds the larynx, trachea, pharynx, oesophagus and thyroid; its buccopharyngeal part covers the constrictors.
- **Deep layer:** the *prevertebral fascia* in front of the vertebrae and muscles, with the thin *alar fascia* in front of it.

All three contribute to the *carotid sheath*.

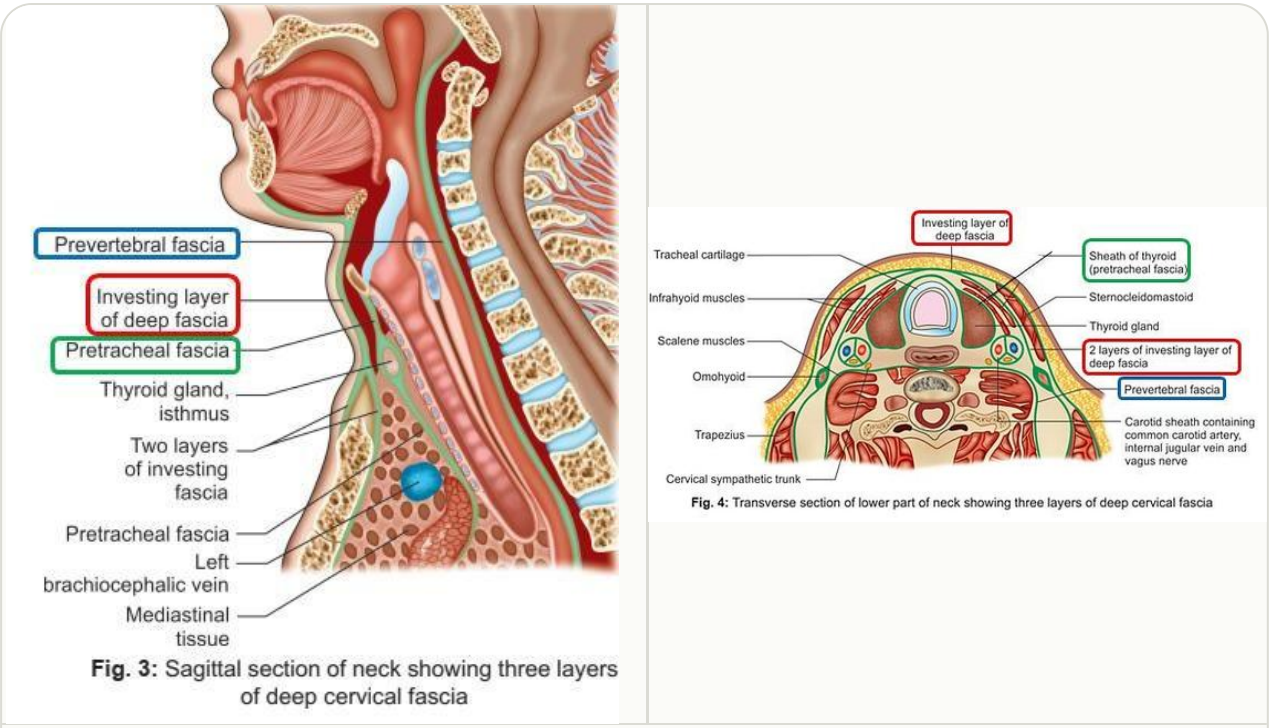


FIGURE 1 The layers of the deep cervical fascia in sagittal and transverse section.

Anatomy atlas · ENT Steps p. 274

1.2 Spaces

SPACES OF THE HEAD AND NECK

SPACE	EXTENT AND CONTENTS	USUAL SOURCE OF INFECTION
Parotid	Within the split superficial layer; gland, nodes, facial nerve, external carotid, retromandibular vein	Ascending infection via Stensen's duct in dehydrated patients
Submandibular (sublingual, submandibular, submental)	Above and below the mylohyoid, between mandible and hyoid	Teeth, sialadenitis, floor of mouth injury
Peritonsillar	Between the tonsil capsule and superior constrictor	Tonsillar crypts
Parapharyngeal	Inverted pyramid from skull base to hyoid, beside the pharynx	Tonsils, quinsy, teeth, parotid, Bezold abscess, other spaces

SPACE	EXTENT AND CONTENTS	USUAL SOURCE OF INFECTION
Retropharyngeal	Behind the buccopharyngeal fascia, in front of the alar fascia; skull base to about T2	Suppurating retropharyngeal nodes in children; trauma in adults
Danger	Between alar and prevertebral fascia; skull base to diaphragm	Spread from retropharyngeal or parapharyngeal spaces
Prevertebral	Behind the prevertebral fascia; skull base to coccyx	Tuberculosis and infection of the spine
Visceral	Around larynx, trachea, pharynx, oesophagus and thyroid, into the mediastinum	Perforation of the oesophagus, thyroiditis, surgery
Carotid sheath	Skull base to the thoracic inlet; carotid, jugular vein, vagus	Spread from adjacent spaces, drug injection
Masticator	Around the masseter, pterygoids and temporalis and the ramus	Second and third molars

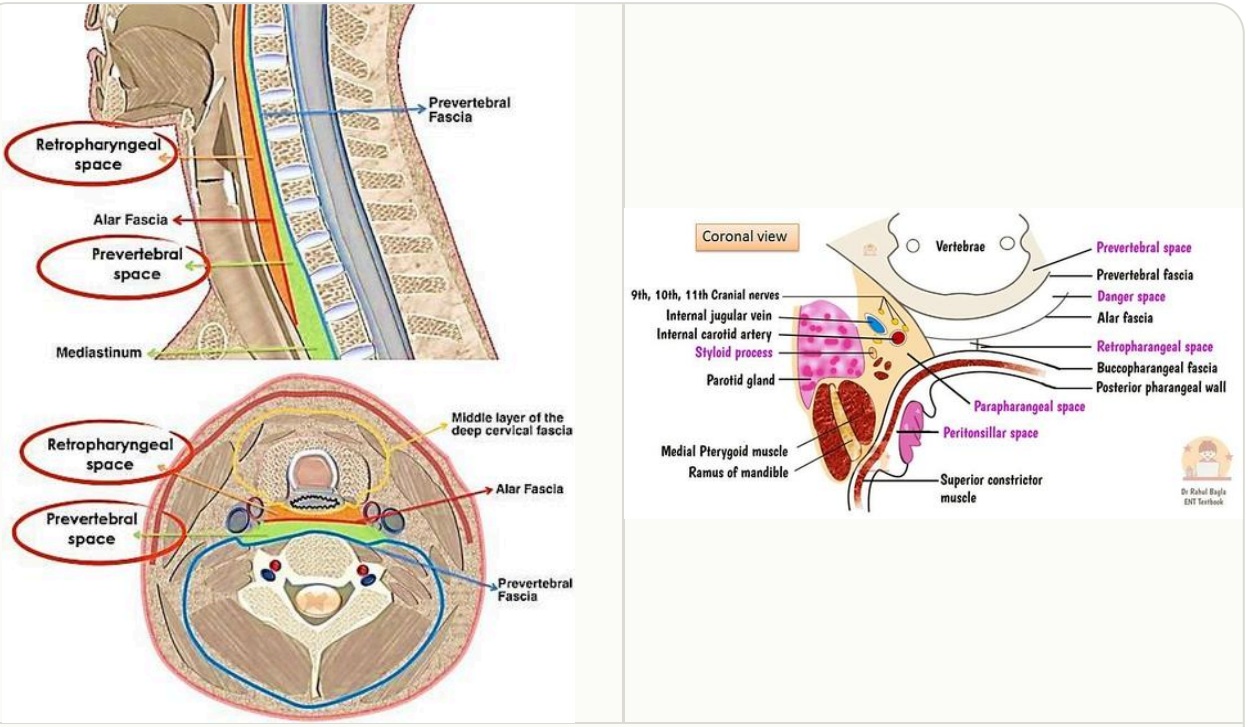


FIGURE 2 The retropharyngeal, danger and prevertebral spaces, and the parapharyngeal and retropharyngeal spaces.

Anatomy atlas · ENT Steps p. 277, ENT Steps p. 198

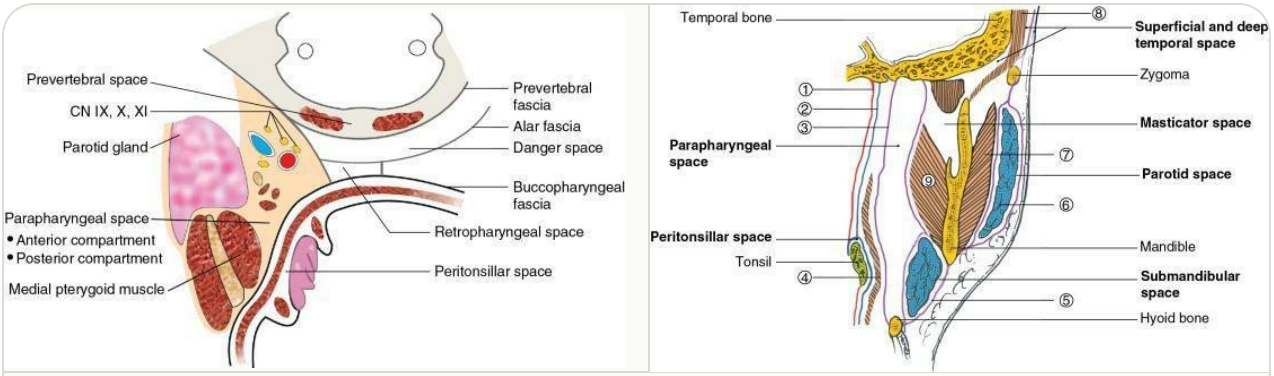


FIGURE 3 First, the spaces in relation to the pharynx where abscesses form. Second, the spaces of the head and neck in coronal section.

Dhingra, figs 52.6 and 52.9

ROUTES TO THE CHEST

The retropharyngeal space reaches the superior mediastinum; the danger space runs behind it to the diaphragm, and the carotid sheath, the "Lincoln highway of the neck", carries infection down to the root of the neck. The parapharyngeal space connects with the retropharyngeal, submandibular, parotid, carotid and visceral spaces, so it is the crossroads of spread.

2 CHAPTER TWO General principles

Before the details.

2.1 Sources

Dental infection is the commonest source in adults, and pharyngeal and tonsillar infection in children. Others: suppurating lymph nodes, sinusitis, mastoiditis (Bezold abscess), sialadenitis, trauma, intubation and endoscopy, foreign bodies, intravenous drug injection, and infected congenital cysts. In an adult, a "neck abscess" may be a necrotic metastasis from an oropharyngeal cancer.

2.2 Organisms

Usually mixed aerobic and anaerobic mouth flora: viridans and group A streptococci, peptostreptococci, staphylococci, *Bacteroides* and *Fusobacterium*. Methicillin-

resistant *Staphylococcus aureus* is common in infants; diabetics and the immunosuppressed acquire Gram-negative organisms; tuberculosis and actinomycosis cause chronic abscesses.

2.3 Evaluation

- **History:** pain, fever, swelling, odynophagia, drooling, voice change, trismus, breathing difficulty; recent dental work, sore throat, trauma or intubation; diabetes and immune status.
- **Examination:** the teeth, floor of mouth, tonsils and posterior wall; trismus; crepitus (gas-forming organisms); cranial nerves; and *flexible endoscopy of the airway* whenever there is voice change, dyspnoea, stridor or unexplained dysphagia. Oxygen saturation can be normal with a critical airway.
- **Blood:** count, glucose, renal function, cultures.
- **Imaging:** contrast-enhanced CT is the key study, showing which spaces are involved, abscess against phlegmon, and spread to the mediastinum; clinical examination alone misjudges the spaces in many cases. Lateral neck radiographs show prevertebral swelling (over about 7 mm at C2 in adults, or more than half a vertebral body in children); chest radiographs for mediastinitis and aspiration; ultrasound in children and for guided aspiration; MRI and MR angiography for intracranial or vascular complications.

2.4 Treatment

1. **Airway first.** Sit the patient up, give humidified oxygen; steroids and nebulised adrenaline buy time. If the airway is narrowed by more than half, secure it by awake fibreoptic intubation or tracheostomy under local anaesthesia; never paralyse a patient whose airway you cannot see.
2. **Fluids:** patients are dehydrated from not swallowing.
3. **Intravenous antibiotics** at once, broad-spectrum with anaerobic cover: co-amoxiclav or ampicillin–sulbactam, or a third-generation cephalosporin with metronidazole; clindamycin in penicillin allergy and in small children; vancomycin for MRSA; antipseudomonal agents in the immunosuppressed. Adjust to cultures.
4. **Drainage** when there is an abscess in a fascial space, gas, a threatened airway, or no improvement after 48 to 72 hours of antibiotics. Small contained abscesses and phlegmon, especially in children, often settle with antibiotics alone.
5. **Remove the source:** extract the tooth, remove the foreign body.

OPERATIVE GUIDES IN THEATRE 2 operations

Drainage of deep neck abscesses and tracheostomy.

Deep Neck Abscesses
Deep neck abscess
Open Tracheostomy
Tracheostomy

3 CHAPTER THREE Ludwig's angina

The floor of the mouth on fire.

3.1 Applied anatomy

Three primary spaces surround the mandible: *sublingual* (above the mylohyoid, under the oral mucosa), *submandibular* (below the mylohyoid, under the investing fascia, containing the gland and nodes) and *submental* (between the anterior bellies of digastric). The roots of the premolars and first molar lie above the mylohyoid line and drain into the sublingual space; those of the second and third molars lie below it and drain into the submandibular space. The spaces communicate around the back of the mylohyoid and across the midline.

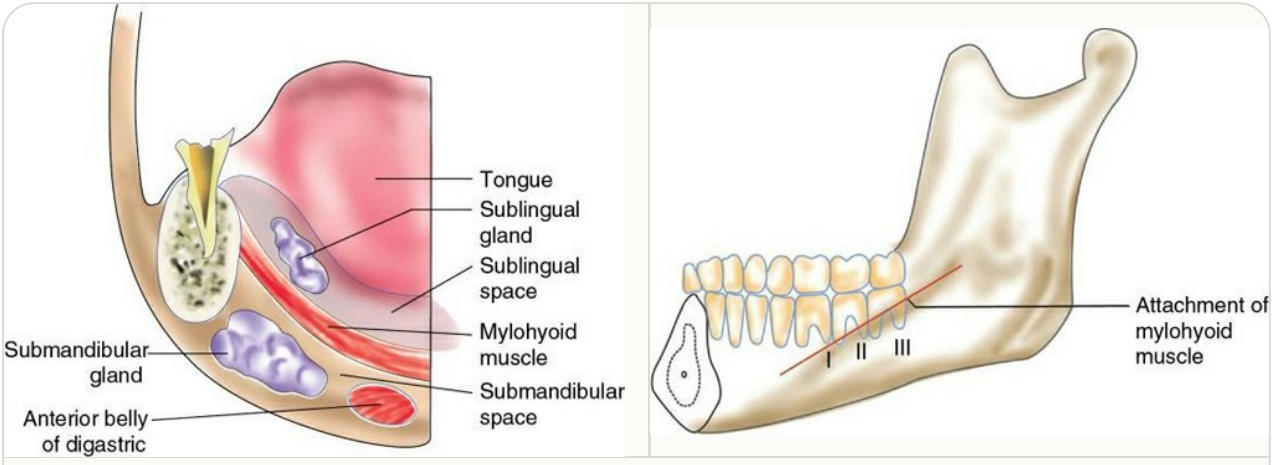


FIGURE 4 First, the submandibular space below the mylohyoid and the sublingual space above it. Second, the roots of the molars project below and those of the premolars above the attachment of the mylohyoid.
Dhingra, figs 52.1 and 52.2

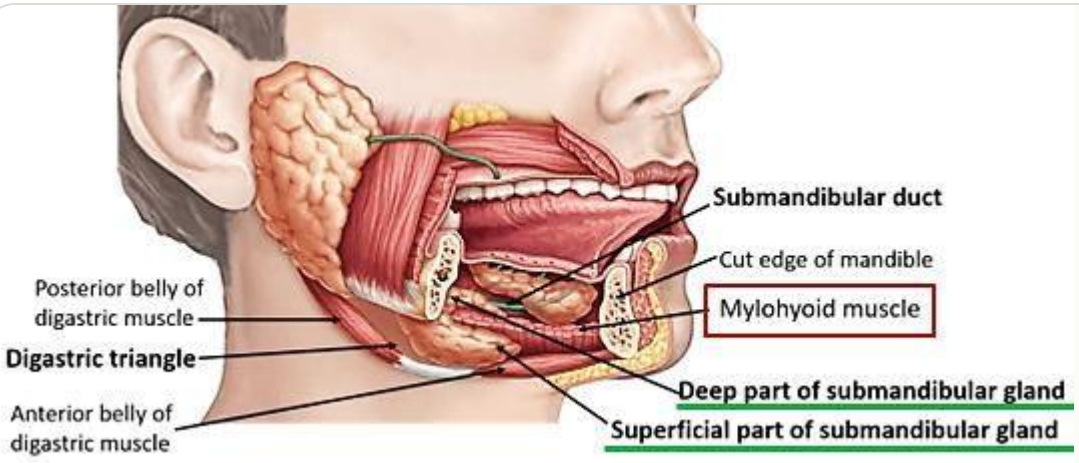


FIGURE 5 The submandibular gland around the mylohyoid.
Anatomy atlas · ENT Steps p. 189

3.2 Definition and cause

A rapidly spreading cellulitis of the submandibular, sublingual and submental spaces on both sides, usually without a discrete abscess. Dental infection accounts for about 80%; others are submandibular sialadenitis, floor-of-mouth injuries and mandibular fractures. Mixed aerobes and anaerobes.

3.3 Clinical features

Painful swallowing, drooling and trismus; a swollen floor of mouth that pushes the tongue up and back; a tense, *woody-hard*, tender swelling below the jaw on both sides; a muffled voice; the patient sits up, leaning forward. As the tongue is forced back and the larynx swells, the airway closes.



FIGURE 6 Ludwig's angina in a seven-year-old child.
Dhingra, fig. 52.3

3.4 Treatment

- Secure the airway: awake fibreoptic intubation or tracheostomy under local anaesthesia, before it is lost.
- High-dose intravenous antibiotics with anaerobic cover; steroids reduce swelling.
- Drainage: intraorally if confined to the sublingual space; externally for submandibular and submental involvement, through a transverse incision below the mandible with blunt opening of the mylohyoid. Often only serous fluid is found, yet decompression relieves the airway.
- Extract the offending tooth.

NEVER LIE DOWN A PATIENT WITH LUDWIG'S ANGINA

A sitting, drooling patient with a raised floor of mouth can obstruct completely if laid flat or sedated. Move to theatre with the airway team present.

3.5 Complications

Spread to the parapharyngeal and retropharyngeal spaces and mediastinum, airway obstruction, septicaemia and aspiration pneumonia.

4 CHAPTER FOUR Retropharyngeal abscess

Behind the pharynx.

4.1 Applied anatomy

The retropharyngeal space lies between the buccopharyngeal fascia in front and the alar fascia behind, from the skull base to the mediastinum. A midline raphe divides it into two lateral compartments (spaces of Gillette), each with retropharyngeal nodes that receive lymph from the nose, sinuses, adenoids, nasopharynx and middle ear and that regress after about four or five years of age.

4.2 Acute retropharyngeal abscess

4.2.1 Cause

Mostly children under three to five, from suppuration of the retropharyngeal nodes after infection of the adenoids, nasopharynx, nose or sinuses. In adults, penetrating injury of the posterior pharyngeal wall or oesophagus (fish bones, instruments); rarely mastoid pus tracking under the petrous bone.

4.2.2 Clinical features

- Fever, irritability, refusal to feed, dysphagia and drooling.
- Difficulty in breathing, stridor and a croupy cough.
- *Torticollis* and neck stiffness, with the head held extended; Lee names torticollis the best indicator in children.
- A bulge on one side of the midline of the posterior pharyngeal wall.

4.2.3 Investigation

A lateral soft tissue radiograph, taken with the neck extended, shows widening of the prevertebral shadow, sometimes with gas or a fluid level. Contrast CT shows the abscess, distinguishes it from phlegmon and shows extension below the hyoid or into the mediastinum.

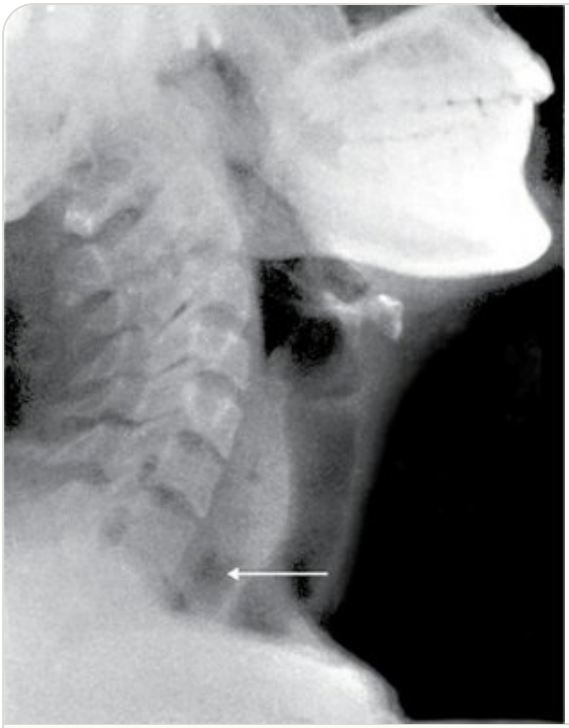


FIGURE 7 Retropharyngeal abscess: widening of the prevertebral space with gas (arrow) on a lateral radiograph.

Dhingra, fig. 52.7

4.2.4 Treatment

Intravenous antibiotics; many small abscesses and phlegmons in children resolve. A definite abscess is drained transorally with the child supine and head low, mouth held open by a gag and suction ready: aspirate first to relieve pressure, then make a vertical incision over the most fluctuant point, avoiding the carotid laterally. Under general anaesthesia, intubate gently to avoid rupture, and pack the pharynx. Tracheostomy if the airway is compromised. Low or extensive abscesses are drained through the neck.

4.3 Chronic retropharyngeal (prevertebral) abscess

Tuberculous, from caries of the cervical spine (in the midline, behind the prevertebral fascia) or from tuberculous retropharyngeal nodes (to one side, behind the buccopharyngeal fascia). Throat discomfort and mild dysphagia; a fluctuant swelling of the posterior wall; tuberculous neck nodes; radiographs show the spinal disease. Drain through the neck (along the anterior border of sternomastoid for a low abscess, the posterior border for a high one), never through the mouth when the spine is involved; antituberculous treatment, with the spinal surgeon.

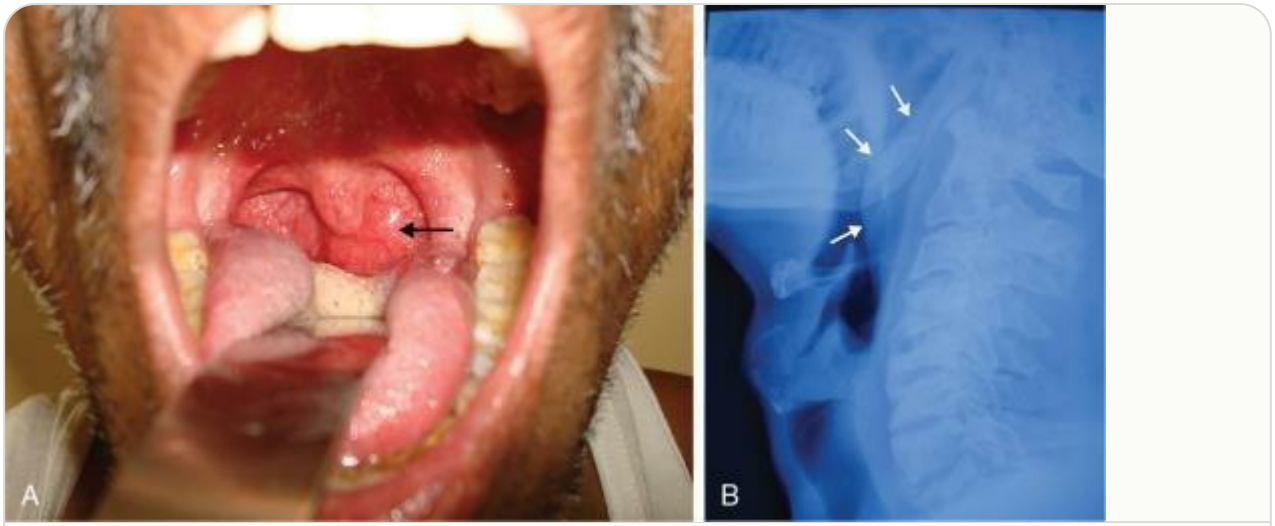


FIGURE 8 (A) A tuberculous prevertebral abscess seen in the oropharynx. (B) Radiograph of the same.

Dhingra, fig. 52.8

5 CHAPTER FIVE Parapharyngeal abscess

The lateral pharyngeal space.

5.1 Applied anatomy

An inverted pyramid with its base at the skull base and apex at the hyoid. Medially the buccopharyngeal fascia over the constrictors; laterally the medial pterygoid, ramus and deep lobe of the parotid; behind, the prevertebral fascia. The styloid process and its muscles divide it into:

- **Anterior (prestyloid) compartment:** related to the tonsillar fossa and medial pterygoid.
- **Posterior (poststyloid) compartment:** containing the internal carotid artery, internal jugular vein, cranial nerves IX to XII, the sympathetic trunk and upper deep cervical nodes.

5.2 Causes

The tonsils and adenoids, a ruptured quinsy, the last lower molar, Bezold abscess and petrositis, the parotid, retropharyngeal and submandibular spaces, and

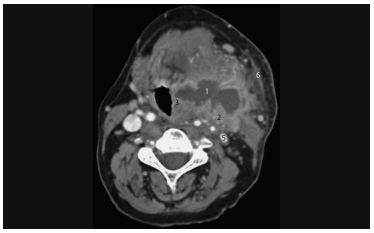
penetrating injury, including local anaesthetic injections.

5.3 Clinical features

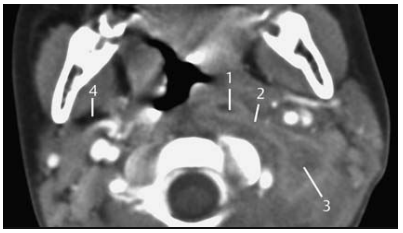
- **Anterior compartment:** the triad of medial prolapse of the tonsil and tonsillar fossa, marked *trismus* from medial pterygoid spasm, and a swelling behind the angle of the jaw; severe odynophagia.
- **Posterior compartment:** a bulge behind the posterior pillar, palsies of IX, X, XI and XII and Horner syndrome, and parotid swelling, with little trismus.
- **Both:** fever, toxaemia, sore throat and torticollis.

IMAGING IN THE APP 3 cases

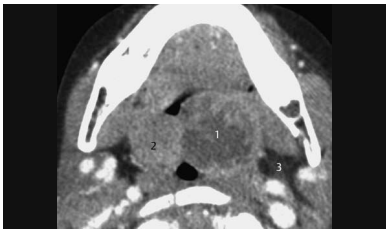
Deep neck abscesses on imaging.



Abscess of the Neck
Infection & inflammation · 1 image



Parapharyngeal Abscess
Infection & inflammation · 5 images



Tonsillar Abscess
Infection & inflammation · 7 images

5.4 Complications

Laryngeal oedema and airway obstruction; jugular thrombophlebitis and septicaemia; spread to the retropharyngeal space and down the carotid sheath to the mediastinum; mycotic aneurysm of the carotid and carotid blow-out.

5.5 Treatment

Intravenous antibiotics covering aerobes and anaerobes. Drainage under general anaesthesia (tracheostomy first if trismus is severe) through a horizontal incision 2 to 3 cm below the angle of the mandible, dissecting bluntly along the inner surface

of the medial pterygoid towards the styloid process; a drain is left. *Never drain a parapharyngeal abscess through the mouth*, for the great vessels lie in its wall.

6 CHAPTER SIX Other spaces

Parotid and masticator.

6.1 Parotid abscess

In dehydrated, postoperative or debilitated patients with poor oral hygiene, infection ascends Stensen's duct, usually *Staphylococcus aureus*. Five to seven days after an operation the parotid becomes swollen, red, indurated and tender; fluctuation is hard to feel through the thick capsule; pus can be expressed from the duct. Ultrasound or CT shows the loculi. Rehydrate, stimulate saliva, give intravenous antibiotics, and drain through a parotidectomy incision, opening the loculi bluntly in the line of the facial nerve branches. The thin deep fascia lets pus break into the parapharyngeal space.

6.2 Masticator space

Between the layers of the investing fascia around the masseter, pterygoids and temporalis, with the ramus, maxillary artery and inferior alveolar nerve; its three parts are masseteric, pterygomandibular and temporal. Infection of the second and third molars is the usual source; marked trismus is the cardinal sign. Drained through an incision lateral to the retromolar trigone, or above the zygomatic arch for the temporal space.

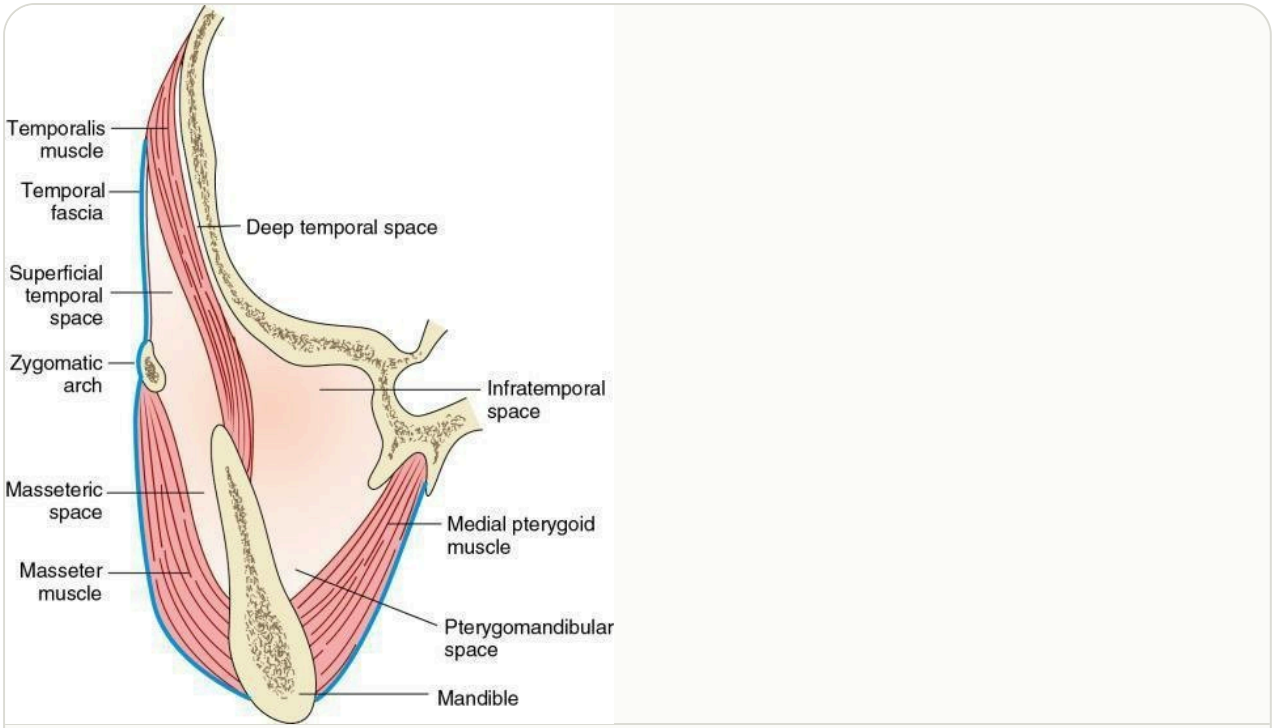


FIGURE 9 The masticator space and its three parts: masseteric, pterygomandibular and temporal.

Dhingra, fig. 52.10

7 CHAPTER SEVEN Dangerous complications

What kills.

7.1 Mediastinitis

Usually from the retropharyngeal or danger spaces: chest pain, dyspnoea, tachycardia and a widened mediastinum on the chest film. Mortality is high; combined cervical and thoracic drainage improves survival.

7.2 Lemierre syndrome

Septic thrombophlebitis of the internal jugular vein after pharyngitis, usually with *Fusobacterium necrophorum*: swinging fevers, tenderness along the sternomastoid, and septic emboli to the lungs and joints. Contrast CT shows a filling defect in the

vein. Two to three weeks of intravenous antibiotics; anticoagulation is debated; ligation if the patient deteriorates.

7.3 Carotid erosion

A mycotic pseudoaneurysm or rupture from retropharyngeal or parapharyngeal infection. Warnings: a *sentinel bleed* from the mouth, nose or ear, a pulsatile swelling, Horner syndrome or lower cranial nerve palsies. Angiography, then ligation or stenting.

7.4 Necrotising fasciitis

In the elderly, diabetic and immunosuppressed, usually dental in origin: spreading oedematous "peau d'orange" erythema with crepitus and gas on CT. Resuscitation, broad-spectrum antibiotics and repeated debridement to bleeding tissue.

7.5 Cavernous sinus thrombosis

Infection from the upper teeth or face tracking back through the valveless ophthalmic veins: fever, orbital pain, proptosis, ophthalmoplegia and a dilated pupil.

A NECK ABSCESS IN AN ADULT

After the infection settles, examine the upper aerodigestive tract. A cystic "abscess" in an adult may be a necrotic node from an HPV-related oropharyngeal cancer or an infected branchial cyst hiding one.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Layers of deep cervical fascia** and the spaces between them.
- **Ludwig's angina:** spaces, cause, airway management.
- **Retropharyngeal abscess:** acute versus chronic, radiograph, transoral versus external drainage.
- **Parapharyngeal abscess:** compartments, signs, and why it is never drained through the mouth.
- **Complications:** mediastinitis, Lemierre syndrome, carotid blow-out.

8.2 Common pitfalls

COMMON PITFALLS

- Sending a patient with a threatened airway to the CT scanner unaccompanied.
- Incising a parapharyngeal swelling through the mouth.
- Draining a tuberculous prevertebral abscess transorally.
- Calling a cystic neck mass in an adult an abscess and not looking for a primary.

Written for this app from Oxford Textbook of Otolaryngology ch. 62, Oxford Specialist Handbook ch. 12, Oxford Handbook of ENT ch. 25, Dhingra ch. 52 and K. J. Lee ch. 37. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Snoring and obstructive sleep apnoea

Every patient with obstructive sleep apnoea snores, but not every snorer has apnoea. The difference matters: primary snoring is a social nuisance, while apnoea, with its repeated nightly suffocation, raises blood pressure, strains the heart, causes strokes and puts sleepy drivers on the road. The bed partner gives the history, the sleep study gives the diagnosis, positive airway pressure is the first treatment, and surgery is chosen by finding where the airway collapses.

AT A GLANCE

- CPAP first in adults.
- Children usually need adenotonsillectomy.
- Find the site before surgery.
- Plan for post-op airway risk.
- CHAT trial: surgery improved behaviour and sleep but not attention; nearly half the waiting group normalised.

1 CHAPTER ONE Definitions

Words of the sleep study.

1.1 Terms

- **Apnoea:** cessation of airflow for 10 seconds or more during sleep.
- **Hypopnoea:** a reduction in airflow (commonly by 30 to 50%) with a fall in oxygen saturation of 3 to 4% or an arousal.
- **Apnoea–hypopnoea index (AHI):** apnoeas and hypopnoeas per hour of sleep; also called the respiratory disturbance index. Under 5 is normal.
- **Arousal:** a brief awakening on the EEG caused by respiratory effort; the arousal index is their number per hour.
- **Sleep efficiency:** minutes asleep divided by minutes in bed.
- **Multiple sleep latency test:** timed daytime naps to measure sleepiness and detect narcolepsy.

SEVERITY OF OBSTRUCTIVE SLEEP APNOEA (ADULTS)

GRADE	AHI (EVENTS PER HOUR)
Normal	Under 5
Mild	5 to 14
Moderate	15 to 29
Severe	30 or more

1.2 Types of apnoea

- **Obstructive:** the upper airway collapses while breathing effort continues.
- **Central:** the airway is open but the brain does not signal breathing; no effort is made.
- **Mixed:** a central start followed by obstruction.

1.3 Spectrum of sleep-disordered breathing

Primary snoring (no arousals, AHI under 5, no daytime sleepiness), **upper airway resistance** (effort-related arousals and sleepiness without frank apnoeas), and **obstructive sleep apnoea**.

2 CHAPTER TWO Mechanism

Why the throat closes in sleep.

2.1 Snoring

In sleep the pharyngeal muscles relax and the airway narrows; breathing through the narrowing sets the soft palate, uvula, pillars and tongue base vibrating, at up to 90 dB. Snoring affects about a quarter of men and fifteen per cent of women, and increases with age.

2.2 Obstruction

The pharynx has no bony support between the palate and the larynx; its patency depends on muscle tone against the suction of inspiration. Tone falls most in REM sleep and stage 2 sleep, when events are most frequent. Each apnoea ends with an arousal that restores tone, fragmenting sleep.

2.3 Sites

Obstruction may be at more than one level, and may differ from night to night:

- **Level I:** nose and nasopharynx: septal deviation, turbinate hypertrophy, valve collapse, polyps, adenoids, tumours.
- **Level II:** palate and tonsils: a long soft palate and uvula, large tonsils, bulky lateral walls.
- **Level III:** tongue base, hypopharynx and larynx: macroglossia, retrognathia, lingual tonsils, an omega-shaped or floppy epiglottis, tumours.

Obesity and a thick neck (collar over about 42 cm in men) narrow every level; alcohol, sedatives and hypnotics relax the muscles further.

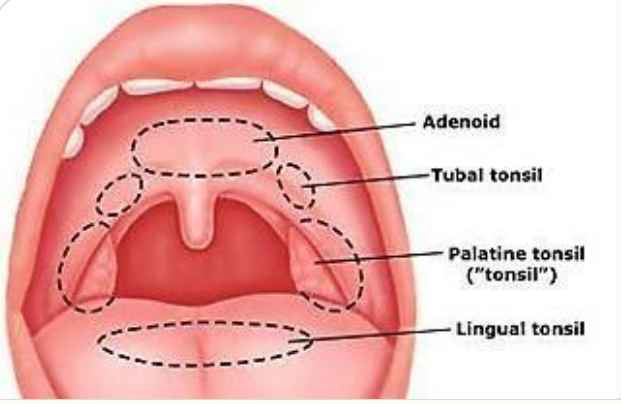


TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

FIGURE 1 Waldeyer's ring seen through the mouth, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 199, ENT Steps p. 208

2.4 Sleep architecture

An adult sleeps seven to eight hours in three to five cycles of 90 to 120 minutes. Non-REM sleep (75 to 80%) moves from light stages to deep slow-wave sleep; REM

sleep (20 to 25%) brings dreaming, irregular breathing and heart rate, and near-paralysis of skeletal muscle, which is why obstruction is worst then.

3 CHAPTER THREE Consequences

More than noise.

3.1 Personal

Loud snoring disturbs the partner (the snoring spouse syndrome); witnessed choking and gasping, restless unrefreshing sleep, excessive daytime sleepiness, morning headaches, fatigue, poor memory and concentration, irritability and depression, reduced libido, nocturia.

3.2 Medical

- **Hypertension:** apnoea is an independent risk factor; treatment lowers blood pressure, even resistant hypertension.
- **Heart:** arrhythmias, ischaemia, heart failure, pulmonary hypertension and cor pulmonale; untreated severe apnoea increases mortality.
- **Brain:** stroke.
- **Metabolic:** insulin resistance; polycythaemia.
- **Perioperative risk:** difficult airways and sensitivity to opioids and sedatives.

3.3 Social

Road traffic and workplace accidents. Ask every patient about sleepiness while driving; in some jurisdictions the doctor must advise the patient about notifying the licensing authority.

CHILDREN ARE DIFFERENT

In children the commonest cause is adenotonsillar hypertrophy. They show snoring, pauses and gasping, restless sleep with the neck hyperextended, sweating, bedwetting, morning headache, and by day hyperactivity, poor attention and school

performance, and sometimes failure to thrive, rather than sleepiness. An AHI above 1 is abnormal in a child, and adenotonsillectomy is first-line treatment.

4 CHAPTER FOUR Evaluation

Finding the cause and the level.

4.1 History

Take it from the bed partner as well as the patient: snoring, pauses, gasping, choking, restlessness and sweating at night; sleepiness, fatigue, headaches, memory and mood by day. Ask about sleep habits and shift work, alcohol, sedatives and caffeine, sleep position, nasal obstruction, menopause and weight change, and driving.

EPWORTH SLEEPINESS SCALE

Each situation is scored 0 (would never doze), 1 (slight chance), 2 (moderate chance) or 3 (high chance of dozing), for a total out of 24:

- Sitting and reading.
- Watching television.
- Sitting inactive in a public place, such as a theatre or meeting.
- As a passenger in a car for an hour without a break.
- Lying down to rest in the afternoon.
- Sitting and talking to someone.
- Sitting quietly after lunch without alcohol.
- In a car, stopped for a few minutes in traffic.

A total over 10 indicates excessive daytime sleepiness. The STOP-BANG and Berlin questionnaires screen for apnoea.

4.2 Examination

- **General:** body mass index (obese at 30 or more), neck circumference (over 42 cm in

men or 37.5 cm in women), blood pressure, signs of heart failure and hypothyroidism.

- **Nose:** septum, turbinates, valve, polyps; endoscopy for adenoids.
- **Mouth and oropharynx:** tonsil size, palate length and position, uvula, tongue size and scalloping, lateral walls, dental occlusion and retrognathia.
- **Hypopharynx and larynx:** flexible endoscopy of the tongue base, lingual tonsil and epiglottis.

FRIEDMAN GRADING

TONSILS	PALATE POSITION (TONGUE RELAXED IN THE MOUTH)
0: removed	I: tonsils, uvula and pillars all visible
1: hidden within the pillars	II: uvula visible, not the tonsils
2: reach the pillars	III: soft palate visible, not the uvula
3: beyond the pillars	IV: only the hard palate visible
4: meeting in the midline	

FRIEDMAN, NOT MALLAMPATI

For sleep surgery the palate is graded with the tongue relaxed inside the mouth (Friedman), not protruded as in the anaesthetist's Mallampati score. Large tonsils with a favourable palate predict success from palatal surgery.

4.3 Endoscopic tests

- **Müller manoeuvre:** with the nose pinched and mouth closed, the patient inhales forcefully while the endoscopist watches which level collapses; simple but poorly reproducible.
- **Drug-induced sleep endoscopy (DISE):** endoscopy under propofol or midazolam sedation shows collapse during sleep-like conditions; the *VOTE* system records the velum, oropharyngeal lateral walls, tongue base and epiglottis, with the degree and pattern (anteroposterior, lateral or concentric). Used before surgery, after failed surgery and when findings do not match the sleep study.

4.4 Imaging

Lateral cephalometry for craniofacial skeletal anomalies and the posterior airway space; a lateral neck radiograph for adenoids in children.

4.5 Sleep studies

Polysomnography is the gold standard. A full attended study (level 1) records EEG, electro-oculogram and chin and leg EMG (sleep staging), ECG, nasal and oral airflow, thoracic and abdominal effort, oxygen saturation, body position and snoring; it separates obstructive from central apnoea and primary snoring from apnoea. Home studies (level 3: airflow, effort, oximetry, position) are now widely used for uncomplicated adults; oximetry alone is insufficient. A split-night study diagnoses in the first half and titrates CPAP in the second, though events may be under-counted.

5 CHAPTER FIVE Non-surgical treatment

First-line.

5.1 Lifestyle

- Weight loss; bariatric surgery for morbid obesity.
- Avoid alcohol in the evening, sedatives and hypnotics; stop smoking.
- **Positional therapy:** sleeping on the side, with a device (or a ball sewn into the back of a shirt) when apnoea is supine-dependent.
- Treat nasal obstruction and allergy.

5.2 Positive airway pressure

Continuous positive airway pressure (CPAP) is a pneumatic splint that holds the airway open, usually at 5 to 20 cm of water, through a nasal, nasal-pillow or full-face mask. Bilevel (BiPAP) delivers a higher inspiratory and lower expiratory pressure for those who need high pressures; auto-titrating devices adjust continuously. It is the most

effective non-surgical treatment and the standard for moderate to severe apnoea, but only about half of patients use it adequately (at least four hours a night, five nights a week) in the long term, because of discomfort, claustrophobia, dryness, leaks and inconvenience. Coaching and follow-up improve adherence.

5.3 Oral appliances

A *mandibular advancement device* holds the jaw forward and draws the tongue and soft tissues away from the posterior wall; a *tongue-retaining device* holds the tongue forward. Effective for snoring and mild to moderate apnoea, and useful in retrognathia; side effects are jaw pain, dental movement and dryness.

6 CHAPTER SIX Surgery

Matched to the level.

6.1 Indications

Failure of or intolerance to CPAP, favourable anatomy, patient preference, and serious cardiovascular disease. Surgery is tailored to the level(s) of obstruction; multilevel procedures may be combined or staged.

6.2 Nasal surgery

Septoplasty, turbinate reduction, polypectomy, valve surgery and adenoidectomy. Nasal surgery seldom cures apnoea, because the nose is not a collapsing segment, but it improves sleep quality and CPAP tolerance and can lower the pressure needed.

6.3 Palate and oropharynx

- **Tonsillectomy (with adenoidectomy):** the main treatment in children; in adults with grade 3 or 4 tonsils, isolated tonsillectomy can be very effective.

- **Uvulopalatopharyngoplasty (UPPP):** introduced by Fujita in 1981; tonsillectomy with trimming of the uvula and free edge of the palate and suturing of the pillars to widen the oropharynx. Overall success is about 50%, rising above 80% in well-selected patients (Friedman tonsils 3 to 4, palate I to II). Complications: pain, bleeding, velopharyngeal insufficiency, nasopharyngeal stenosis and voice change.
- **Uvulopalatal flap:** the uvula is folded back onto the denuded palate, sparing the muscle.
- **Expansion sphincter pharyngoplasty:** the palatopharyngeus is rerouted superolaterally to widen the lateral dimension.
- **Uvulopalatoplasty (UPP, laser or radiofrequency assisted) and palatal stiffening (cautery-assisted or implants):** useful for snoring, much less for apnoea.

Dhingra notes that palatal surgery relieves snoring in about 80% but apnoea in only about half, with relapse when another level takes over.

6.4 Tongue base and hypopharynx

Radiofrequency reduction of the tongue base in repeated sessions, lingual tonsillectomy, midline glossectomy (coblation, laser, robotic), tongue suspension, hyoid suspension (to the mandible or thyroid cartilage), genioglossus advancement, and supraglottoplasty for epiglottic collapse.

6.5 Skeletal surgery

Maxillomandibular advancement (Le Fort I osteotomy of the maxilla and sagittal split of the mandible, fixed forwards with plates) enlarges the whole skeletal framework of the airway and is the most effective operation apart from tracheostomy, especially in retrognathic patients and after failed soft tissue surgery; it changes facial appearance and needs orthodontic care.

6.6 Hypoglossal nerve stimulation

An implanted stimulator fires the hypoglossal nerve with inspiration, protruding the tongue; for moderate to severe apnoea in patients who cannot use CPAP, with a suitable BMI and without concentric palatal collapse on DISE.

6.7 Tracheostomy

The only operation that reliably cures obstructive apnoea, by bypassing every level; reserved for life-threatening disease, severe cardiopulmonary complications and failure of everything else, because of its personal and social cost.

OPERATIVE GUIDES IN THEATRE 5 operations

Operations for sleep-disordered breathing.

Tonsillectomy
Tonsillectomy

Uvulopalatopharyngoplasty and Related Modifications
UPPP

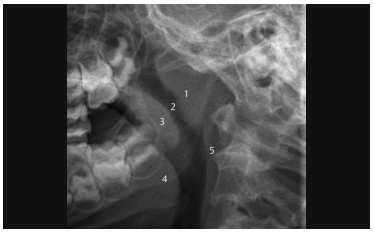
Drug-Induced Sedated Endoscopy
DISE

Hypoglossal Nerve Stimulation Therapy
Hypoglossal nerve stimulation

Maxillomandibular Advancement for Obstructive Sleep Apnea
Maxillomandibular advancement

IMAGING IN THE APP 1 case

Adenoid hypertrophy on imaging.



Adenoid Hypertrophy
Infection & inflammation · 1 image

6.8 Perioperative care

Patients with apnoea are sensitive to opioids and sedatives and may have difficult airways. Monitor oxygen saturation overnight after surgery in severe apnoea, with cardiovascular disease, after tongue base surgery, and in young children or those with comorbidities; restart CPAP after surgery when appropriate.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra

Calls tracheostomy the gold standard of surgical treatment, lists uvulopalatoplasty as the commonest procedure and quotes 80% success for snoring but 50% for apnoea.

Lee

Places CPAP as the gold standard of therapy, emphasises patient selection by Friedman stage and DISE, reporting UPPP success above 85% in favourable anatomy, and names maxillomandibular advancement the most effective operation short of tracheostomy.

They agree that surgery must be matched to the level of collapse.

SLEEP-DISORDERED BREATHING IN CHILDREN AND SELECTION FOR SURGERY (OXFORD TEXTBOOK)

About 10 to 15% of children snore, and 1 to 4% have obstructive sleep apnoea, more in obese and Black children. It usually begins between two and six years, when the tonsils and adenoids are largest relative to the airway; tonsillar hypertrophy is grade 3 or more on the Brodsky scale. Prematurity is a risk factor, and sleep apnoea can cause failure to thrive, poor behaviour and attention, and enuresis. Most children can be treated on a careful history and examination, ideally with a parental video; overnight **oximetry** helps triage (a normal trace does not exclude apnoea), and **polysomnography** is kept for uncertain cases and high-risk children: those under two, with obesity, Down syndrome, craniofacial or neuromuscular disease, or cardiac complications.

The **Childhood Adenotonsillectomy Trial (CHAT, 2013)** randomised children aged five to nine with non-severe apnoea to early adenotonsillectomy or watchful waiting for seven months. Surgery did not improve attention and executive function, but it improved symptoms, behaviour, quality of life and sleep study results; even so, about 45% of the waiting group normalised spontaneously, less often in obese and Black children and those with more severe disease. Watchful waiting is therefore reasonable for snoring and mild disease, with intranasal steroids and **montelukast** helping in the short term, especially with rhinitis or residual symptoms after surgery. Weight loss is recommended for obese children. Children with persistent apnoea after adenotonsillectomy, or without large tonsils, need drug-induced sleep endoscopy or MRI to find other levels of obstruction.

In adults, **transoral robotic** tongue base surgery requires a BMI below about 35 (ideally near 30) and a mouth opening of at least 2.5 cm; up to 30% of adults with apnoea are not overweight, and nasal surgery helps by easing CPAP use and reducing negative airway pressure.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definitions:** apnoea, hypopnoea, AHI and severity grades.
- **Obstructive versus central apnoea.**
- **Consequences** of untreated apnoea.
- **Evaluation:** Epworth score, Friedman staging, Müller manoeuvre, DISE, polysomnography.
- **CPAP** and its adherence problem.
- **Surgery by level,** UPPP selection, and maxillomandibular advancement.
- **Children:** adenotonsillectomy first.

7.2 Common pitfalls

COMMON PITFALLS

- Operating on the palate when the tongue base is the main site of collapse.
- Using oximetry alone to diagnose or exclude apnoea.
- Giving opioids freely after surgery to a patient with severe apnoea.
- Promising that nasal surgery will cure apnoea.

Written for this app from Oxford Textbook of Otolaryngology ch. 51, 75, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 16, Dhingra ch. 55 and K. J. Lee ch. 2. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Nasopharyngeal carcinoma

Nasopharyngeal carcinoma hides in the one part of the throat no one sees without an endoscope. It shows itself instead through its neighbours: a lump high in the neck, a blocked ear with fluid in an adult, a bloodstained nose, a squint from a sixth nerve palsy. It is linked to Epstein–Barr virus, common in southern China and South-East Asia, and treated not with the knife but with radiation and chemotherapy, to which it responds well.

AT A GLANCE

- Adult with unilateral effusion or a neck node.
- Biopsy the fossa of Rosenmüller.
- Radiotherapy is the backbone.
- Surgery is for salvage.
- Induction gemcitabine–cisplatin for high risk; plasma EBV DNA for monitoring; cumulative cisplatin about 200 mg/m².

1 CHAPTER ONE Overview

A cancer of geography and virus.

1.1 Tumours of the nasopharynx

TUMOURS OF THE NASOPHARYNX

BENIGN	MALIGNANT
Juvenile angiofibroma (the commonest benign tumour)	Nasopharyngeal carcinoma (the commonest malignancy)
Antrochoanal polyp, squamous papilloma	Lymphoma (mostly B-cell non-Hodgkin)
Thornwaldt cyst	Rhabdomyosarcoma (children)
Pleomorphic adenoma, craniopharyngioma	Chordoma, plasmacytoma
Paraganglioma, hamartoma	Adenoid cystic and other salivary carcinomas

BENIGN	MALIGNANT
Hairy polyp, teratoma (congenital)	Melanoma, haemangiopericytoma

Angiofibroma has its own topic; this one is about carcinoma.

1.2 Epidemiology

The highest incidence in the world is in southern China, Hong Kong and Taiwan, with high rates in South-East Asia, North Africa, the Mediterranean and among the Inuit and Aleut peoples. It is uncommon in India except in the North-East. Chinese born in North America have a lower risk than those born in China but still higher than other Americans, showing the interplay of genes and environment. Males outnumber females about three to one; the peak is in the fifth and sixth decades, but about one in five patients is under thirty, and these younger patients usually have undifferentiated tumours.

1.3 Aetiology

- **Genetic:** HLA haplotypes confer susceptibility; the risk persists after migration.
- **Epstein–Barr virus:** EBV DNA and EBV-encoded RNA are found in virtually every tumour cell of the non-keratinising types; clonal EBV in premalignant lesions suggests infection is an early, possibly initiating, event.
- **Environmental:** nitrosamines in salted, preserved fish eaten from childhood, with a diet low in vitamin C; smoke from burning incense and wood; tobacco, opium and occupational fumes.
- **Tobacco and alcohol:** linked to the keratinising (sporadic) type.

2 CHAPTER TWO Pathology

Three types.

2.1 WHO classification

WHO TYPES OF NASOPHARYNGEAL CARCINOMA

TYPE	DESCRIPTION	NORTH AMERICAN FREQUENCY	EBV AND RADIOSENSITIVITY
I	Keratinising squamous cell carcinoma	About 25%	Weaker association; least radiosensitive
II	Non-keratinising, differentiated	About 12%	Strong association
III	Non-keratinising, undifferentiated (lymphoepithelioma; Regaud and Schmincke types)	About 63%	Strongest association; most radiosensitive

The current WHO scheme groups types II and III as non-keratinising carcinoma (differentiated and undifferentiated). Under the electron microscope all are squamous; immunohistochemistry for epithelial (cytokeratin) and lymphoid markers separates undifferentiated carcinoma from lymphoma. In endemic areas the non-keratinising types predominate even more.

2.2 Gross forms

Proliferative (a polypoid mass causing nasal obstruction), ulcerative (bleeding), and infiltrative (submucosal, easily missed).

3 CHAPTER THREE Spread

Through every exit of the skull base.

3.1 Local

The commonest site of origin is the *fossa of Rosenmüller* in the lateral wall.

- **Forwards:** into the choanae and nasal cavity.
- **Laterally:** through the sinus of Morgagni into the parapharyngeal space and infratemporal fossa; into the pterygoid muscles (trismus).
- **Upwards:** through the foramen lacerum and foramen ovale into the middle cranial fossa and cavernous sinus (cranial nerves III, IV, V and VI); the *sixth nerve* is usually the first affected; into the sphenoid sinus and orbit (proptosis, optic nerve).
- **Backwards:** to the jugular foramen (IX, X, XI) and hypoglossal canal (XII), and the cervical sympathetic chain (Horner syndrome), directly or through enlarged retropharyngeal nodes.
- **Downwards:** to the oropharynx.

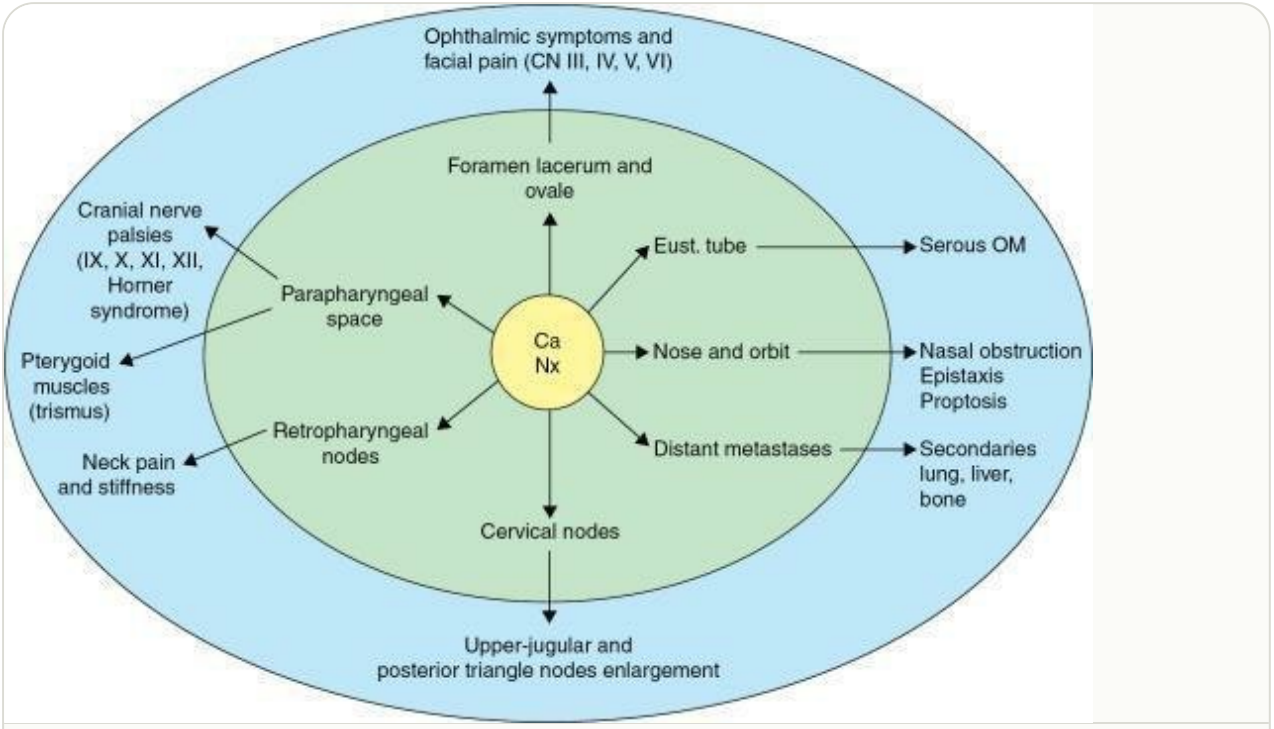


FIGURE 1 Routes of spread of nasopharyngeal carcinoma and the clinical features they produce. Dhingra, fig. 49.9

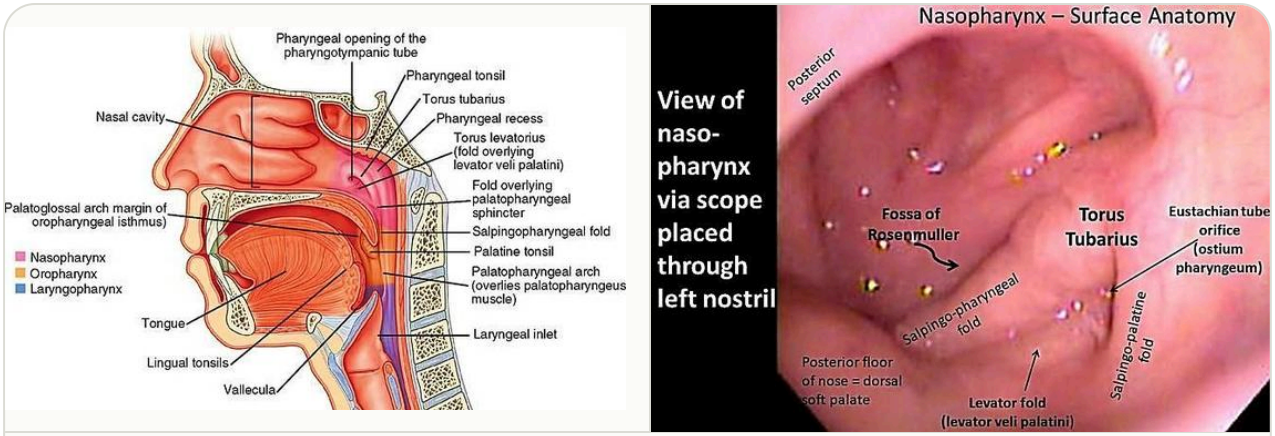


FIGURE 2 The lateral wall of the nasopharynx with the torus tubarius and fossa of Rosenmüller, and an endoscopic view.

Anatomy atlas · ENT Steps p. 203

3.2 Lymphatic

The nasopharynx is rich in lymphatics and nodal spread is early and often bilateral: to the lateral retropharyngeal node (of Rouvière), the upper jugular (level II) nodes high under the mastoid, and the spinal accessory chain in the posterior triangle (level V). Retropharyngeal nodes cause neck stiffness and press on the lower cranial nerves.

3.3 Distant

Bone, lung and liver. Nasopharyngeal carcinoma has the highest rate of distant metastasis of any head and neck cancer, present or developing in about a quarter to a third of patients.

4 CHAPTER FOUR Clinical features

The four groups of symptoms.

4.1 Cervical nodes

The commonest presentation (in 60 to 90%): a painless lump between the angle of the jaw and the mastoid, or in the posterior triangle. Nodes are present in about

three-quarters at diagnosis, half of them bilateral; often the node is the only complaint.

4.2 Ear

Eustachian tube obstruction causes conductive hearing loss, aural fullness and *otitis media with effusion*; tinnitus; rarely tumour grows up the tube.

UNILATERAL EFFUSION IN AN ADULT

An adult with fluid behind one drum and no obvious cause has a nasopharyngeal examination, and a biopsy if anything is seen. Inserting a grommet without looking at the nasopharynx misses cancers.

4.3 Nose

Nasal obstruction, bloodstained discharge or epistaxis, and hyponasal speech.

4.4 Eye and nerves

Diplopia and squint from a sixth nerve palsy; ophthalmoplegia (III, IV, VI); facial pain and numbness with loss of the corneal reflex (V); proptosis and visual loss; the jugular foramen syndrome (IX, X, XI); deviation of the tongue (XII); Horner syndrome; headache from skull base erosion.

TROTTER'S TRIAD

Conductive deafness from tubal blockage, ipsilateral temporoparietal neuralgia from the fifth nerve, and immobility of the soft palate on the same side from the vagus: the classic signs of a tumour invading the lateral nasopharynx and parapharyngeal space.

4.5 Others

Trismus, neck pain from retropharyngeal nodes, weight loss, and the effects of distant metastases.

PRESENTING FEATURES IN ORDER OF FREQUENCY

FEATURE	NOTES
Neck mass	Most common; often the only symptom
Hearing loss and aural fullness	Effusion from tubal obstruction
Nasal obstruction and epistaxis	Bloodstained post-nasal discharge
Cranial nerve palsy	Sixth nerve most often
Headache, earache, neck pain	Skull base and retropharyngeal involvement

5 CHAPTER FIVE Diagnosis and staging

See it, sample it, map it.

5.1 Endoscopy and biopsy

Rigid or flexible nasendoscopy under local anaesthesia shows a proliferative, ulcerative or submucosal lesion, and allows biopsy. If nodes suggest the diagnosis but nothing is seen, take biopsies from the fossa of Rosenmüller and posterior wall under general anaesthesia, sometimes through a transpalatal exposure. Fine-needle aspiration of a neck node, with EBV testing of the cells, points to a nasopharyngeal primary.

5.2 Imaging

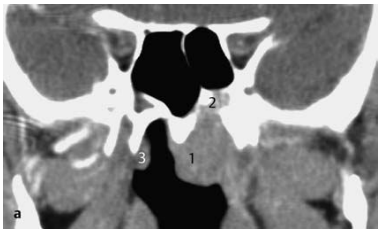
- **MRI:** the best study for soft tissue extent, the parapharyngeal and retropharyngeal spaces, perineural spread, skull base marrow, intracranial extension and nodes.
- **CT:** bone erosion of the skull base, clivus and pterygoid plates.
- **PET-CT:** distant metastases and detection of residual or recurrent disease after treatment; chest and abdominal imaging if PET is unavailable.

5.3 Other tests

A baseline *audiogram* (radiation and cisplatin both damage hearing), dental assessment before radiotherapy, and *EBV serology*: IgA antibodies to the early antigen are highly specific and those to the viral capsid antigen more sensitive, so both are used; plasma EBV DNA load is now used for screening in endemic areas, for prognosis and to monitor for recurrence.

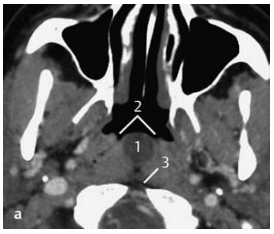
IMAGING IN THE APP 2 cases

Nasopharyngeal carcinoma and a Thornwaldt cyst on imaging.



Nasopharyngeal Carcinoma

Tumours · 11 images



Thornwaldt Cyst

Congenital · 2 images

5.4 Staging

TNM STAGING OF NASOPHARYNGEAL CARCINOMA (AJCC)

CATEGORY	DEFINITION
T1	Confined to the nasopharynx, or extending to the oropharynx or nasal cavity without parapharyngeal involvement
T2	Parapharyngeal extension, or involvement of the medial or lateral pterygoid or prevertebral muscles
T3	Invasion of bony structures of the skull base, cervical vertebrae or pterygoid structures, or the paranasal sinuses
T4	Intracranial extension, cranial nerves, hypopharynx, orbit, parotid, or soft tissue beyond the lateral pterygoid
N1	Unilateral cervical, or unilateral or bilateral retropharyngeal, nodes 6 cm or less, above the caudal border of the cricoid

CATEGORY	DEFINITION
N2	Bilateral cervical nodes 6 cm or less, above the caudal border of the cricoid
N3	Node over 6 cm, or extension below the caudal border of the cricoid

Dhingra quotes the older AJCC definitions, in which T1 was confined to the nasopharynx and N3b meant any node in the supraclavicular fossa (Ho's triangle, bounded by the medial and lateral ends of the clavicle and the point where neck meets shoulder). In every edition the nodal staging differs from other head and neck sites: low neck nodes carry the worst prognosis whatever their size.

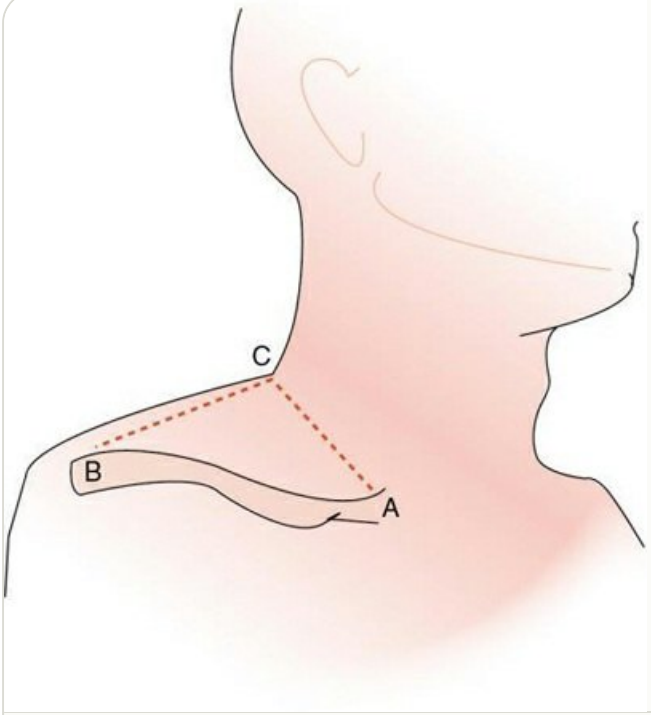


FIGURE 3 Ho's triangle (the supraclavicular fossa): the medial (A) and lateral (B) ends of the clavicle and the point where the neck meets the shoulder (C).
Dhingra, fig. 49.10

6 CHAPTER SIX Treatment

Radiation first.

6.1 Primary treatment

The nasopharynx is hard to reach surgically, and these tumours are radiosensitive, so *radiotherapy is the treatment of choice*.

- **Stage I:** radiotherapy alone to the nasopharynx and both sides of the neck, about 70 Gy.
- **Stage II to IVA:** concurrent chemoradiation with cisplatin, which improves survival over radiotherapy alone, with or without induction or adjuvant chemotherapy (cisplatin, fluorouracil, gemcitabine).
- **Intensity-modulated radiotherapy (IMRT)** delivers a high dose to the tumour while sparing the brainstem, spinal cord, optic apparatus, temporal lobes, cochleae and parotids.
- **Metastatic disease:** platinum-based chemotherapy, now with immunotherapy, and local treatment in selected cases.

6.2 Residual and recurrent disease

- **Neck:** persistent or recurrent nodes after radiation are treated by neck dissection; Dhingra advises a radical dissection because disease is often more extensive than it seems.
- **Nasopharynx:** after restaging with MRI and PET-CT, small recurrences are treated by *nasopharyngectomy*: endoscopic for limited central tumours, or open through a maxillary swing, transpalatal, Le Fort I, midfacial degloving or infratemporal fossa approach. Extension around the internal carotid, intracranially or widely into the parapharyngeal space contraindicates surgery.
- **Re-irradiation:** IMRT, proton therapy or brachytherapy (interstitial implants, gold grains), at greater risk of necrosis of the skull base and temporal lobe, cranial nerve palsies and carotid rupture.

OPERATIVE GUIDES IN THEATRE 3 operations

Salvage nasopharyngectomy approaches.

Endoscopic Nasopharyngectomy
Salvage nasopharyngectomy
Robotic-Assisted Nasopharyngectomy
Robotic salvage
Maxillary Swing/Transpalatal/Midface Degloving With Le Fort I Osteotomy Approach for Nasopharyngectomy
Maxillary swing

6.3 Effects of treatment

Xerostomia and dental caries, otitis media with effusion and sensorineural hearing loss, trismus, hypothyroidism and pituitary failure, osteoradionecrosis of the skull base, temporal lobe necrosis, cranial neuropathy, carotid stenosis, and second cancers. Grommets in irradiated ears often discharge chronically.

6.4 Outcome

With concurrent chemoradiation, local and regional control is about 70 to 80% and five-year survival about 70 to 80%; distant metastasis is now the commonest mode of failure.

7 CHAPTER SEVEN Other malignancies

Not all are carcinoma.

- **Lymphoma:** a bulky submucosal mass, usually B-cell non-Hodgkin, often with disease elsewhere in Waldeyer's ring; sinonasal NK/T-cell lymphoma in Asian patients.
- **Rhabdomyosarcoma:** the embryonal type in children, as a polypoid mass with rapid growth and cranial nerve involvement; chemotherapy and radiotherapy.
- **Chordoma:** from notochordal remnants in the clivus, destroying bone and presenting in the nasopharynx; surgery and proton therapy.
- **Plasmacytoma, adenoid cystic carcinoma and melanoma** are rare.

CHEMOTHERAPY AND FOLLOW-UP (OXFORD TEXTBOOK)

Nonkeratinising, EBV-associated carcinoma is the commonest type everywhere; EBV is shown in tissue by EBER in-situ hybridisation. Distant metastases are present in about 4% at diagnosis and develop in about a quarter after IMRT, which is now the main cause of

treatment failure. For locoregionally advanced disease, concurrent cisplatin-based chemoradiotherapy improves survival over radiotherapy alone; the cumulative cisplatin dose matters more than the schedule, with about 200 mg/m² as the effective threshold (160 mg/m² after induction). **Induction gemcitabine and cisplatin** before chemoradiotherapy improves survival in high-risk patients, and **adjuvant or metronomic capecitabine** improves failure-free survival in high-risk disease; the benefit of conventional adjuvant chemotherapy remains debated. **Plasma EBV DNA** is a promising marker for screening high-risk populations, risk stratification and surveillance, but assays still need standardisation. Proton and carbon-ion therapy have dosimetric advantages without proven better outcomes yet. Follow-up is every three to four months for two years, six-monthly to five years, then yearly, with yearly CT or MRI for five years. Isolated nodal recurrence is treated by salvage neck dissection, and small local recurrences by endoscopic nasopharyngectomy or re-irradiation.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Aetiology:** EBV, genetics and salted fish.
- **WHO types** and their link to EBV and radiosensitivity.
- **Routes of spread** and the cranial nerves involved; **Trotter's triad.**
- **Unilateral effusion in an adult.**
- **Investigations:** endoscopy and biopsy, MRI, EBV serology and DNA.
- **Treatment:** radiotherapy, concurrent chemoradiation, salvage surgery.

8.2 Common pitfalls

COMMON PITFALLS

- Inserting a grommet in an adult with a one-sided effusion without examining the nasopharynx.
- Excising a high level II neck node before endoscopy and needle aspiration.

- Missing a sixth nerve palsy as the sign of skull base invasion.

Written for this app from Oxford Textbook of Otolaryngology ch. 61, Oxford Specialist Handbook ch. 20, Dhingra ch. 49 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Oral and oropharyngeal cancer

Two cancers share this chapter and differ in almost everything except their histology. Oral cancer is visible, tied to tobacco, areca nut and alcohol, common in South Asia, and treated first by surgery. Oropharyngeal cancer hides behind the tongue and in the tonsil, presents with a lump in the neck, is increasingly caused by human papillomavirus in younger non-smokers, and is often treated by radiation. Both demand an examination that includes palpation, and both punish delay.

AT A GLANCE

- Non-healing ulcer over 2 weeks needs a biopsy.
- HPV has changed oropharyngeal disease.
- Oral cavity: surgery first.
- Treat the neck by depth.
- AJCC 8 stages p16-positive disease separately; cetuximab de-escalation worsened survival.

1 CHAPTER ONE Overview

Two regions, one disease.

1.1 Subsites

- **Oral cavity:** mucosal lip, buccal mucosa, oral tongue (anterior two-thirds), floor of mouth, upper and lower alveolus (gingiva), retromolar trigone and hard palate.
- **Oropharynx:** base of tongue, tonsil and tonsillar fossa with the pillars, soft palate and uvula (the faucial arch), and the posterior and lateral pharyngeal walls.

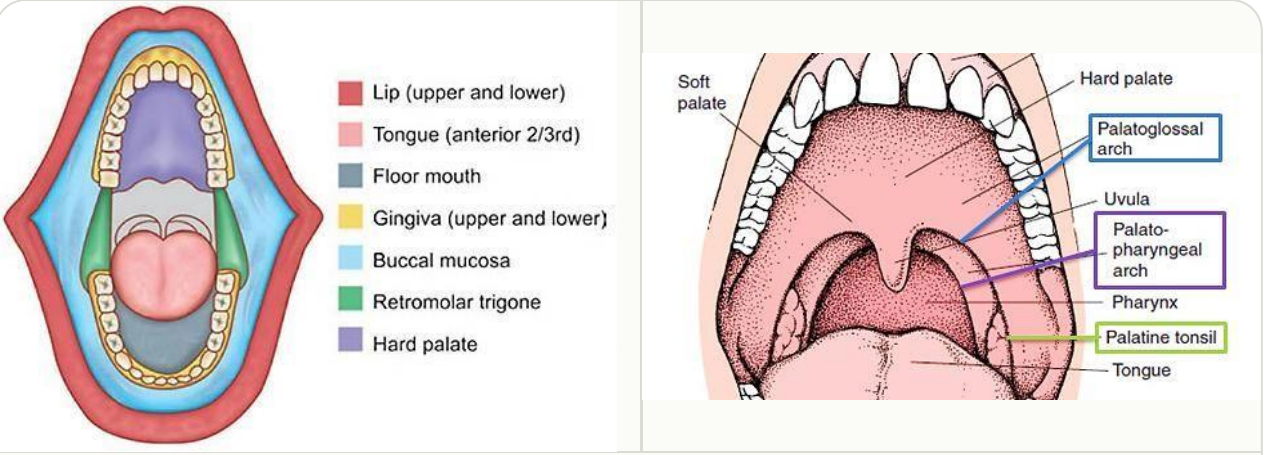


FIGURE 1 The subsites of the oral cavity, and the oropharynx through the open mouth.
Anatomy atlas · ENT Steps p. 165, ENT Steps p. 210

1.2 Epidemiology

India carries one of the highest burdens of oral cancer in the world: lip, tongue and mouth together make up about a fifth of all male cancers there, with buccal mucosa and tongue the leading sites, driven by tobacco chewing with lime, pan and areca nut, bidi smoking and alcohol. In the West the lip is the commonest oral subsite and tongue cancer is rising in young adults. Oropharyngeal cancer is rising rapidly in Europe and North America because of HPV.

1.3 Causes

- **Tobacco:** smoking increases oral cancer about sixfold; chewing tobacco places carcinogen at the site where the quid is held; *reverse smoking* (the burning end inside the mouth) causes palatal cancer.
- **Alcohol:** multiplies, rather than adds to, the effect of tobacco.
- **Areca nut and pan:** directly and through submucous fibrosis.
- **Human papillomavirus (HPV 16):** most cancers of the tonsil and base of tongue in North America; associated with oral sexual behaviour rather than tobacco.
- **Others:** chronic trauma from jagged teeth and dentures, poor oral hygiene, Plummer–Vinson syndrome, immunosuppression, sunlight (lower lip), and rare syndromes (Fanconi anaemia).

1.4 Field cancerisation

Tobacco and alcohol alter the whole mucosa of the upper aerodigestive tract, so dysplasia and cancer arise in more than one place. About 15% of patients have, or will develop, a second primary in the head and neck, oesophagus or lung.

1.5 Pathology

Squamous cell carcinoma in about 90%; minor salivary gland carcinomas (adenoid cystic, mucoepidermoid, adenocarcinoma), lymphoma (especially tonsil and base of tongue), melanoma (palate and gum) and Kaposi sarcoma make up the rest. Verrucous carcinoma is a slow, warty, well-differentiated variant of the buccal mucosa that rarely metastasises.

2 CHAPTER TWO Clinical picture

Look, feel, and feel the neck.

2.1 Gross forms

Exophytic, ulcerative (a non-healing ulcer with raised, rolled, everted edges, a shaggy base and induration), infiltrative, and superficially spreading (often arising in leukoplakia or erythroplakia).

2.2 Symptoms

- An ulcer or lump that does not heal; bleeding.
- Pain, often *referred to the ear* through the lingual and auriculotemporal nerves (V3) or the glossopharyngeal nerve.
- Difficulty in chewing, swallowing and speaking; a *hot-potato voice* with tongue base tumours; loosening of teeth or ill-fitting dentures; trismus when the pterygoids or masseter are invaded; numbness of the lip from inferior alveolar nerve invasion.
- A lump in the neck, sometimes the only complaint.
- Foul breath, weight loss and, late, a fungating mass or orocutaneous fistula.

REFERRAL NOW

A mouth ulcer or white or red patch that persists beyond three weeks, an unexplained lump in the mouth or neck, unexplained tooth mobility, persistent one-sided sore throat or earache with a normal ear: refer urgently for examination and biopsy.

2.3 Examination

Inspect every subsite with good light, palpate the lesion for induration and depth, palpate bimanually the floor of mouth and tongue, examine the tongue base and tonsils by palpation and endoscopy, and examine both sides of the neck level by level. Cranial nerves V, VII and XII; mouth opening; teeth.

3 CHAPTER THREE Oral cavity cancer

Site by site.

3.1 Lip

The commonest oral subsite in the West; more than 95% squamous, 95% on the *lower lip*, between the midline and the commissure, in older men with outdoor occupations; basal cell carcinoma is relatively commoner on the upper lip. Early blistering, crusting or ulceration on a background of actinic cheilitis. Nodes (submental and submandibular, bilateral near the midline) are late and occur in only about 10%.



FIGURE 2 Carcinoma of the upper lip and oral commissure, with leukoplakia of the buccal mucosa.

Dhingra, fig. 44.10

3.1.1 Treatment

Excision with a safe margin. Defects up to a third of the lip close primarily after a V or W excision; a third to a half with lip-switch flaps (Abbe, or Estlander at the commissure); a half to two-thirds with a Karapandzic flap; larger defects with Gillies fan, Bernard–Burow or free flaps, always realigning the vermilion border. Radiotherapy is an alternative for small lesions; neck dissection for nodes. Cure exceeds 90% in early disease; upper lip and commissure tumours do worse.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgery for cancer of the lip.

Cancer of the Lip

3.2 Buccal mucosa

Very common in India, equal in both sexes, at the angle of the mouth, along the occlusal line or in the gingivobuccal sulcus where the quid is kept; often multicentric because the whole cheek is "condemned". Exophytic or ulceroinfiltrative; it spreads deeply through buccinator and fat to the skin, back to the retromolar trigone and pterygoids (trismus), and up and down into the maxilla

and mandible. About half have nodes (submandibular, then upper jugular). Verrucous carcinoma favours this site.



FIGURE 3 Carcinoma of the buccal mucosa.
Dhingra, fig. 44.11

3.2.1 Treatment

T1 lesions by excision. Larger lesions, and any with muscle or bone involvement, by wide excision with marginal or segmental mandibulectomy or partial maxillectomy, reconstruction with local, pectoralis major or free flaps, neck dissection, and postoperative radiotherapy for stage III and IV.

3.3 Oral tongue

Usually men of fifty to seventy, though increasingly young adults; may arise in leukoplakia, at a chronic dental ulcer, or in smokers and chewers. The commonest site is the *middle of the lateral border* and the ventral surface; the tip and dorsum less often. It appears as an exophytic papillary growth, an indurated ulcer with rolled edges, or a submucosal indurated nodule. It infiltrates the tongue musculature (fixation, slurred speech), the floor of mouth and mandible. Nodes: the lateral tongue drains to the submandibular and upper jugular nodes (and can skip to level IV), the tip to the submental and jugulo-omohyoid nodes; midline and ventral lesions spread to both sides.

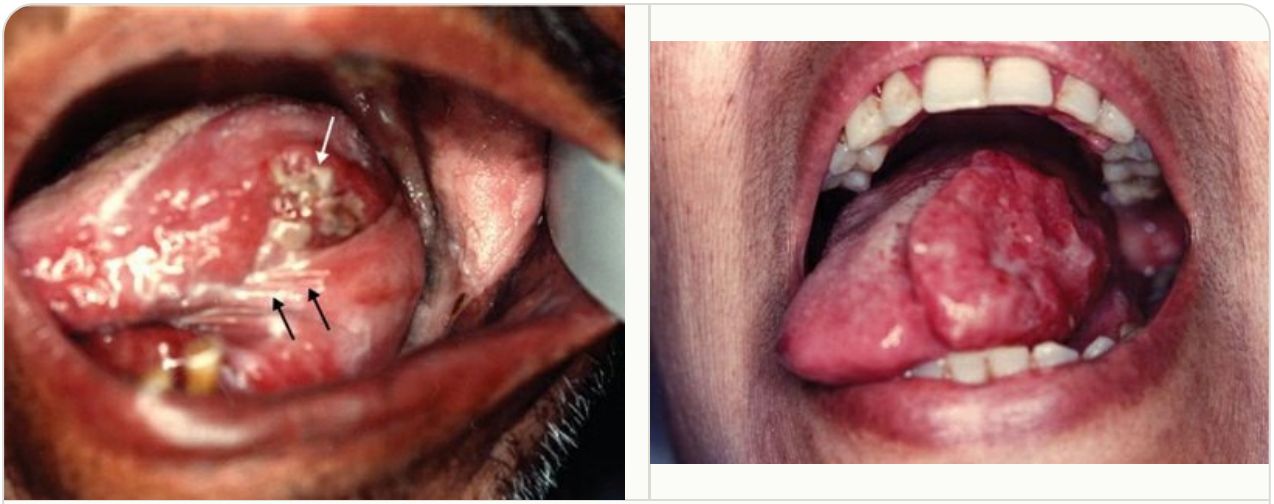


FIGURE 4 First, carcinoma of the lateral border of the tongue (arrow) with leukoplakia of the floor of mouth (double arrows). Second, an ulcerative squamous carcinoma of the tongue in a 40-year-old woman.

Dhingra, figs 44.12 and 44.13

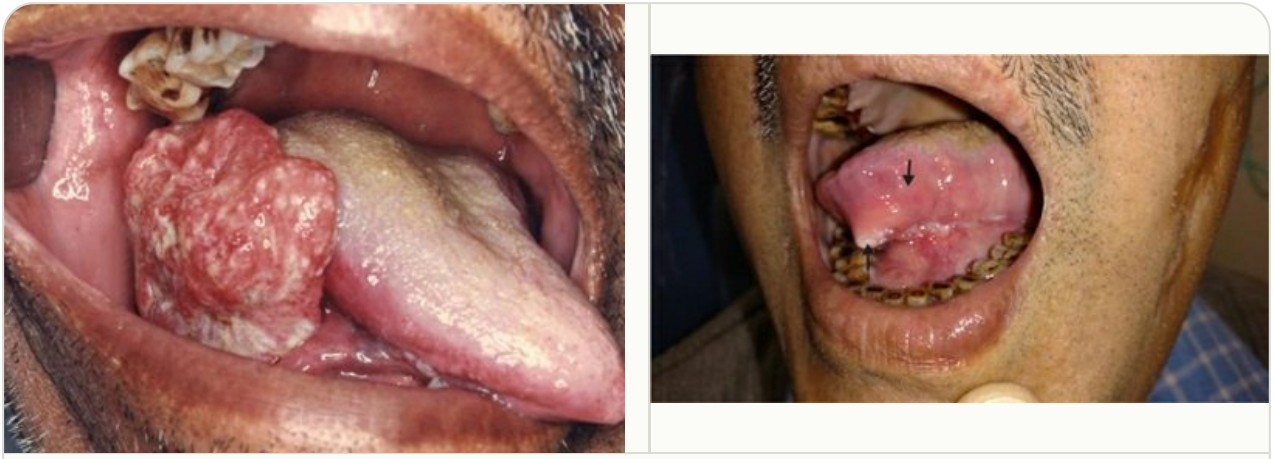


FIGURE 5 First, an exophytic growth on the right lateral border of the tongue. Second, cancer of the undersurface of the tongue.

Dhingra, figs 44.14 and 44.15

DEPTH OF INVASION

The thickness of a tongue cancer predicts nodal spread better than its width: occult metastases rise sharply beyond about 4 mm. Current staging (AJCC 8th edition) therefore includes depth of invasion in the T category, and elective neck dissection is advised for tumours deeper than about 3 to 4 mm.

3.3.1 Treatment

T1 and T2: partial glossectomy (transoral or with a pull-through), healing by secondary intention, primary closure, graft or free flap; radiotherapy (external beam with brachytherapy) is an alternative. Elective selective neck dissection of

> levels I to III (supraomohyoid) for an N0 neck when depth exceeds the threshold, bilateral for midline lesions; sentinel node biopsy is an alternative in early cancers. Advanced lesions: hemiglossectomy or total glossectomy with the floor of mouth, mandibulectomy and neck dissection (the *commando operation*), free flap reconstruction and postoperative radiotherapy or chemoradiation; total glossectomy may require laryngectomy to prevent aspiration.

3.4 Floor of mouth

Squamous carcinoma, men four times more than women, usually starting anteriorly near the submandibular duct openings, which may obstruct and enlarge the gland. Ulcerative or infiltrative, spreading to the ventral tongue, lingual gingiva, mandible and submental space; a submandibular swelling may be gland or node. Small lesions by excision or radiotherapy; larger ones by wide excision with marginal or segmental mandibulectomy; elective neck treatment even for N0 stage II disease, since occult nodes occur in about 40%; nodes often bilateral.



FIGURE 6 Ulcerative squamous carcinoma of the floor of mouth (arrows) in a 55-year-old man. Dhingra, fig. 44.16

3.5 Alveolus (gingiva)

Mostly men, on the lower jaw behind the first molar; early invasion of the bone and along the inferior alveolar nerve; nodes to the submandibular and upper jugular

groups. Surgery is preferred because radiotherapy risks osteoradionecrosis: marginal mandibulectomy for superficial lesions, segmental or hemimandibulectomy when the medulla is invaded; partial maxillectomy for upper alveolar lesions.

3.6 Retromolar trigone

Primary, or by extension from the gingiva, floor of mouth, buccal mucosa or faucial arch; it reaches the pterygomandibular raphe, pterygoids (trismus) and mandible early. Wide excision with mandibular resection and neck dissection.



FIGURE 7 Cancer of the right retromolar trigone with trismus in a 36-year-old man. Dhingra, fig. 44.17

3.7 Hard palate

Squamous or, more often, minor salivary gland carcinoma (adenoid cystic, mucoepidermoid, adenocarcinoma); squamous cancer is linked to reverse smoking. A painless irregularity felt with the tongue, a superficial ulcer, or a submucosal mass; it may invade the nasal floor and antrum and must be told apart from an antral carcinoma breaking through. Resection with the underlying bone or a partial maxillectomy, closed with an obturator prosthesis or flap.



FIGURE 8 A mixed salivary tumour of the palate.
Dhingra, fig. 44.18

OPERATIVE GUIDES IN THEATRE 6 operations

Glossectomy, floor of mouth excision, transoral inferior maxillectomy and mandibulectomy.

Partial Glossectomy
Partial glossectomy
Total Glossectomy
Total glossectomy
Excision of Cancer of the Floor of the Mouth
Floor of mouth
Transoral Inferior Maxillectomy
Marginal Mandibulectomy
Segmental Mandibulectomy

3.8 The mandible

Tumour reaches the mandible through the alveolar crest in the dentate jaw and through the occlusal surface in the edentulous jaw. Periosteal involvement needs at least a *marginal (rim) mandibulectomy*, preserving the lower border; invasion of the medulla, or an edentulous atrophic mandible, needs a *segmental* resection, reconstructed with a fibula free flap. Radiotherapy to a mandible already invaded leads to osteoradionecrosis.

4 CHAPTER FOUR Oropharyngeal cancer

Behind the tongue.

4.1 HPV and the changing disease

HPV-POSITIVE AND HPV-NEGATIVE OROPHARYNGEAL CANCER

FEATURE	HPV-POSITIVE	HPV-NEGATIVE
Typical patient	Younger, healthier, male (about 9:1), non-smoker	Older, smoker and drinker
Cause	HPV 16; oral sexual exposure	Tobacco and alcohol
Marker	p16 overexpression on immunohistochemistry	p16 negative
Primary	Often small, in the tonsil or tongue base crypts	Often larger, infiltrative
Nodes	Early, often large and *cystic*	Solid
Prognosis	Much better; staged separately (AJCC 8th)	Worse

The viral E6 and E7 oncoproteins inactivate p53 and Rb, and p16 is overexpressed as a result. Vaccination against HPV is expected to prevent most of these cancers.

THE CYSTIC NECK LUMP IN AN ADULT

A cystic lump in the upper neck of an adult is a metastasis from an HPV-related tonsil or tongue base cancer until proven otherwise, not a branchial cyst. Examine and image the oropharynx, aspirate for p16 or HPV, and if no primary is seen, perform tonsillectomy and tongue base biopsy or mucosectomy.

4.2 Base of tongue

Silent for a long time; early symptoms (soreness, a lump feeling, mild dysphagia) are attributed to the lingual tonsil. Many present with nodes, often bilateral: about 70% have neck metastases at presentation, so the tongue base is a classic site of the occult primary. Late: referred otalgia, dysphagia, bleeding, a hot-potato voice and tongue fixation. Deeply infiltrative into the tongue, vallecula, pre-epiglottic space,

epiglottis and tonsil. It is larger on palpation than it looks in the mirror: always palpate, preferably under anaesthesia.

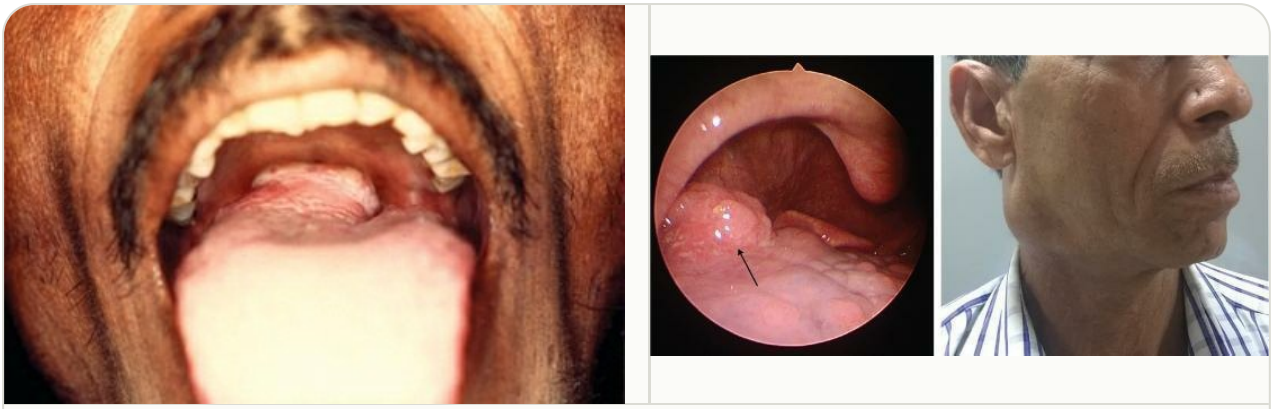


FIGURE 9 First, an exophytic growth at the base of the tongue. Second, carcinoma of the right base of tongue: (A) endoscopic view, (B) secondaries in the jugulodigastric nodes.

Dhingra, figs 53.1 and 53.2

4.3 Tonsil and tonsillar fossa

Squamous carcinoma presents as an ulcer with a necrotic base, or as a smooth, enlarged tonsil when it lies within the crypts; lymphoma as a smooth, one-sided enlargement, sometimes ulcerated. Persistent sore throat, dysphagia, otalgia and a neck lump, later bleeding, fetor and trismus. It spreads to the palate, pillars, tongue base, pharyngeal wall, pterygoids and mandible, and the parapharyngeal space; about half have nodes at presentation, the jugulodigastric node first.

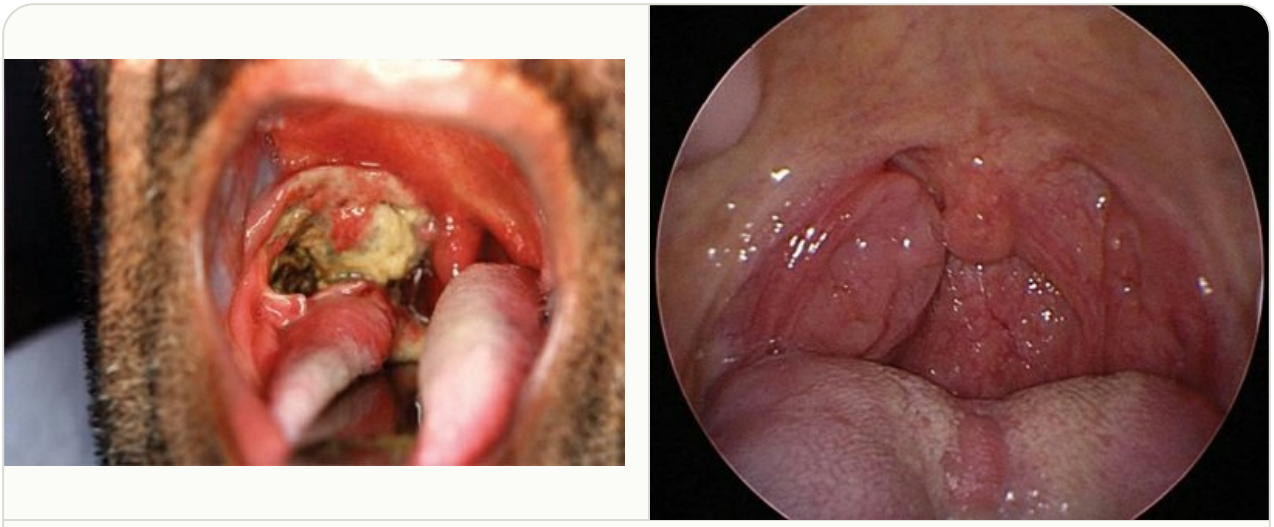


FIGURE 10 First, squamous cell carcinoma involving the tonsil, pillars and soft palate. Second, lymphoma of the right tonsil presenting as one-sided tonsillar enlargement.

Dhingra, figs 53.3 and 53.4

THE ONE-SIDED ENLARGED TONSIL

CAUSE	CLUE
Normal asymmetry or a medially placed tonsil	Long-standing, smooth, no symptoms
Unilateral tonsillitis, peritonsillitis, quinsy	Pain, fever, trismus
Tonsillolith, cyst	Gritty or yellow
Parapharyngeal tumour	Tonsil pushed medially, normal mucosa
Carcinoma or lymphoma	Progressive, ulcerated, firm, nodes, otalgia; biopsy or tonsillectomy

4.4 Faucial arch (soft palate, uvula, anterior pillar)

Usually well-differentiated, superficially spreading squamous carcinoma on the oral surface, noticed early and slow to metastasise, behaving more like oral cancer; often accompanied by other primaries. Radiotherapy or transoral excision.



FIGURE 11 An ulcerative lesion of the palatine arch: a well-differentiated squamous cell carcinoma.

Dhingra, fig. 53.5

4.5 Posterior and lateral pharyngeal walls

Silent, spreading submucosally up to the nasopharynx and down to the hypopharynx, and back to the prevertebral fascia; nodes in about 60%, often bilateral and retropharyngeal. Radiotherapy or excision through a lateral pharyngotomy, often with mandibulotomy, and neck dissection.

4.6 Benign tumours of the oropharynx

Papilloma (pedunculated, on the tonsil, palate or pillars; excised), haemangioma (treated only if growing or bleeding), pleomorphic adenoma of the palate (excised completely), vallecular mucous cysts (excised or marsupialised), and rarely lipoma, fibroma and neuroma. Parapharyngeal tumours (deep-lobe parotid, schwannoma, paraganglioma, lipoma, carotid aneurysm) push the tonsil and palate medially and mimic oropharyngeal tumours: never biopsy through the mouth without imaging.

5 CHAPTER FIVE Evaluation and staging

Before treatment.

5.1 Investigations

- **Biopsy** of the primary; fine-needle aspiration of nodes; p16 immunohistochemistry (and HPV testing) for every oropharyngeal cancer.
- **CT and MRI** of the primary and neck: depth, bone invasion (CT and orthopantomogram for the mandible), extension into the pterygoids, parapharyngeal space, pre-epiglottic space and skull base; nodes, including retropharyngeal.
- **Chest CT or PET-CT** for distant metastases and second primaries; PET-CT also for the unknown primary.
- **Panendoscopy** under anaesthesia (laryngoscopy, pharyngoscopy, oesophagoscopy) to map the tumour and look for synchronous cancers.
- Dental assessment before radiotherapy; nutrition and swallowing assessment.

5.2 TNM essentials

T STAGING (AJCC 8TH EDITION)

CATEGORY	ORAL CAVITY	OROPHARYNX
T1	2 cm or less, depth of invasion 5 mm or less	2 cm or less
T2	2 cm or less with depth over 5 mm, or 2 to 4 cm with depth 10 mm or less	Over 2 cm, up to 4 cm
T3	Over 4 cm, or depth over 10 mm	Over 4 cm, or extension to the lingual surface of the epiglottis
T4a	Cortical bone, maxillary sinus, skin of face (and, for lip, inferior alveolar nerve or floor of mouth)	Larynx, extrinsic tongue muscle, medial pterygoid, hard palate or mandible
T4b	Masticator space, pterygoid plates, skull base, or encasing the internal carotid	Lateral pterygoid, pterygoid plates, lateral nasopharynx, skull base, or encasing the carotid

For oral cavity and HPV-negative oropharyngeal cancer, N staging follows node size, number and side, and *extranodal extension* (which upstages); for HPV-positive oropharyngeal cancer the N categories are simpler and the stages lower, reflecting the better prognosis. Dhingra prints the older (6th edition) tables, based on size alone.

6 CHAPTER SIX Treatment

By site, stage and the person.

6.1 Principles

Treatment is planned by a multidisciplinary team (surgeons, radiation and medical oncologists, radiologists, pathologists, reconstructive surgeons, dentists, speech and swallowing therapists, dietitians), balancing cure against speech, swallowing and appearance.

- **Oral cavity:** surgery first. Early (stage I and II): excision with a margin, with elective neck treatment according to depth. Advanced (III and IV): resection, reconstruction and neck dissection, followed by radiotherapy, or chemoradiation when margins are positive or there is extranodal extension.
- **Oropharynx:** early: radiotherapy (IMRT) or transoral surgery (robotic or laser) with neck dissection, both treating the neck. Advanced: concurrent chemoradiation with cisplatin (organ preservation), or surgery with adjuvant (chemo)radiation; surgery through a mandibulotomy is mostly reserved for salvage. Radiosensitive lymphoepithelioma and lymphoma are treated by radiotherapy or chemotherapy.
- **Systemic agents:** cisplatin is the standard radiosensitiser; cetuximab (an EGFR antibody) is an alternative for those unfit for cisplatin, though inferior in HPV-positive disease; immunotherapy for recurrent and metastatic disease.

6.2 The neck

- **N0:** elective selective neck dissection (levels I to III for oral cavity, II to IV for oropharynx) or elective radiotherapy when the risk of occult metastasis exceeds about 20%; sentinel node biopsy in early oral cancer.
- **N+:** comprehensive (modified radical) neck dissection, followed by radiotherapy for multiple nodes or extranodal extension.

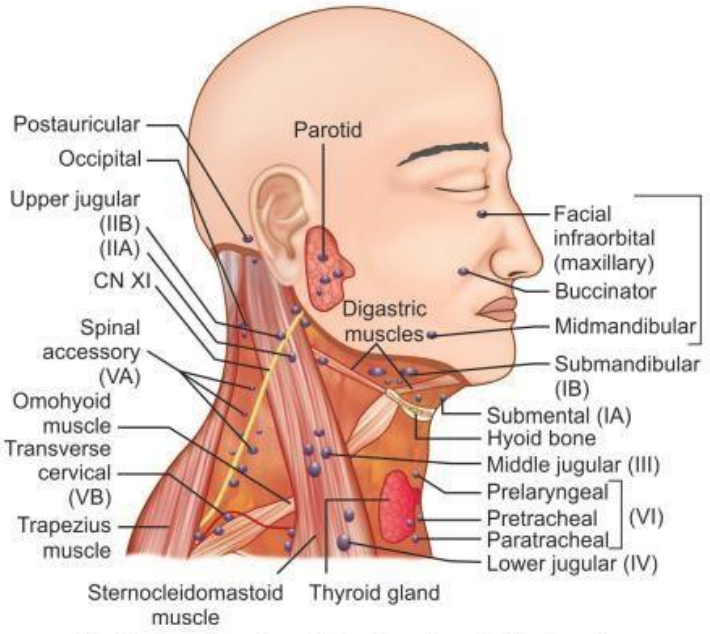


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

FIGURE 12 The lymph node levels of the neck.

Anatomy atlas · ENT Steps p. 285

6.3 Reconstruction

Primary closure, skin grafts, local flaps (nasolabial, tongue, facial artery musculomucosal), regional flaps (pectoralis major, deltopectoral) and free flaps (radial forearm for the tongue and floor, anterolateral thigh for bulk, fibula for the mandible). Tracheostomy protects the airway after major resections.

OPERATIVE GUIDES IN THEATRE 7 operations

Glossectomy, floor of mouth and tonsil surgery, base of tongue resection, neck dissection and free flaps.

Partial Glossectomy
Partial glossectomy

Total Glossectomy
Total glossectomy

Excision of Cancer of the Floor of the Mouth
Floor of mouth

Transoral and Robotic Tonsil Surgery for Cancers of the Base of the Tongue and Pharynx
Tonsil surgery

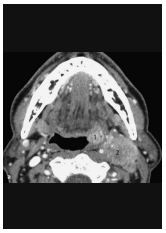
Transoral Robotic Surgery for Neoplasms of the Base of Tongue
Base of tongue

Selective Neck Dissection
Neck dissection

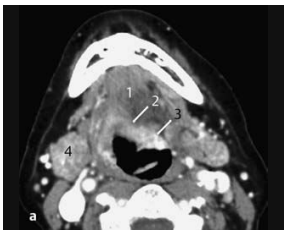
Microvascular Reconstruction of the Head and Neck
Free flaps

IMAGING IN THE APP 4 cases

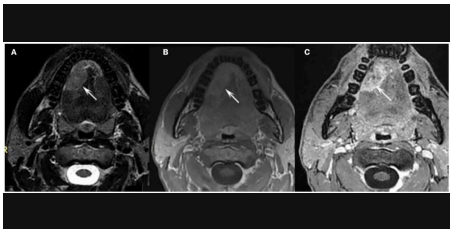
Tonsillar carcinoma, base of tongue tumour, oral and oropharyngeal squamous carcinoma, and synchronous malignancies.



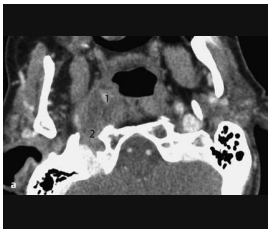
Tonsillar Carcinoma
Tumours · 8 images



Oropharyngeal Tumor, Base of Tongue
Tumours · 11 images



Squamous cell carcinoma: oral cavity, oropharynx and larynx
After surgery · 14 images



Multiple Synchronous Malignancies
Tumours · 3 images

6.4 Palliation

Radiotherapy or chemotherapy for pain and bleeding, tracheostomy and gastrostomy for airway and nutrition, and good analgesia and palliative care.

6.5 Prognosis

Early oral cancer is cured in most patients; advanced disease in roughly a third to a half. Extranodal extension, nodes low in the neck, positive margins and perineural invasion worsen survival. HPV-positive oropharyngeal cancer has markedly better survival than HPV-negative. Lifelong follow-up looks for recurrence (most in the first two years) and second primaries, and stopping smoking improves outcome.

6.6 Chemoprevention

Retinoids, beta-carotene and vitamin E can induce regression of leukoplakia and reduce second primaries while taken, but no agent has replaced stopping tobacco, areca nut and alcohol.

7 CHAPTER SEVEN
Non-squamous cancers

The others.

- **Minor salivary gland carcinoma:** 80 to 90% of minor salivary tumours are malignant in some series; the palate is the commonest site; adenoid cystic (about 40%, with perineural spread), adenocarcinoma and mucoepidermoid. Wide excision with neck dissection for nodes, and radiotherapy.
- **Mucosal melanoma:** palate and gum, sixth decade, men twice as often; pigmented, later ulcerated, sometimes amelanotic; wide excision including bone; five-year survival around 15%.
- **Lymphoma:** mostly non-Hodgkin, in the tonsil and tongue base, as a smooth bulky submucosal mass, with neck nodes in many; chemotherapy and radiotherapy.
- **Kaposi sarcoma:** reddish-purple palatal plaques, mainly in AIDS; responds to antiretroviral treatment, radiotherapy or chemotherapy.

HPV, STAGING AND DE-ESCALATION (OXFORD TEXTBOOK)

HPV-positive oropharyngeal cancer, mostly HPV-16, is now a distinct disease with a better prognosis. Smoking worsens prognosis whatever the HPV status. Local and regional failure is less common in HPV-positive disease, but distant metastases, mostly to the lung, occur at similar rates, so chest surveillance still matters. The **AJCC eighth edition** stages p16-positive and p16-negative tumours separately, with a separate pathological staging after surgery, so that stage groups reflect the better outlook; but guidelines do not yet change treatment on HPV status alone. Neck aspirates and biopsies are tested for p16, and dental extractions needed before radiotherapy are best done at the staging anaesthetic to reduce the risk of osteoradionecrosis.

Treatment. Transoral surgery (laser or robotic) with neck dissection as the only treatment is appropriate for T1–T2 tumours with no nodes or a single node, when margins are clear and there are no adverse pathological features; patients must accept that adjuvant treatment may be needed. Transoral resection with adjuvant therapy gives similar control and survival to chemoradiotherapy, so expected morbidity and function guide the choice. Chemotherapy is added to postoperative radiotherapy for extranodal extension or positive margins, and patients with gross extranodal extension on imaging are better treated with primary chemoradiotherapy to avoid triple therapy. **De-escalation** (lower radiation dose, surgery alone, or replacing cisplatin) aims to reduce long-term toxicity in HPV-positive patients; the De-ESCALaTE and RTOG 1016 trials showed that replacing cisplatin with cetuximab *worsened* survival, so de-escalation belongs only within trials.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Risk factors,** field cancerisation and second primaries.
- **Commonest sites** in India and in the West.
- **Tongue cancer:** site, spread, depth of invasion, referred otalgia and treatment, including the commando operation.
- **Mandible:** marginal versus segmental resection.
- **HPV-related oropharyngeal cancer** and the cystic neck node.
- **Management of the N0 neck.**
- **Differential diagnosis of a one-sided enlarged tonsil.**

8.2 Common pitfalls

COMMON PITFALLS

- Treating a persistent ulcer with mouthwashes rather than a biopsy.
- Judging a tongue base cancer by looking without feeling.
- Excising a cystic neck lump as a branchial cyst in an adult without looking for a primary.
- Irradiating a mandible already invaded by tumour.

Written for this app from Oxford Textbook of Otolaryngology ch. 56, Oxford Specialist Handbook ch. 6, Dhingra ch. 44, 53 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Dysphagia and pharyngeal pouch

Swallowing is a reflex so well rehearsed that we notice it only when it fails. Dysphagia is a symptom, never a diagnosis, and in an adult it is cancer until proven otherwise. The elderly patient who gurgles when he drinks, brings back yesterday's food and coughs at night has a pharyngeal pouch, the classic ENT cause, through the weak spot between the two parts of the inferior constrictor.

AT A GLANCE

- Regurgitation and aspiration in the elderly suggest a pouch.
- Barium before scopes.
- Button batteries are an emergency.
- Fever or crepitus means perforation.
- Cricopharyngeal bars are seen in a third of normal people; stapling is less effective for small pouches.

1 CHAPTER ONE Dysphagia

Where is the hold-up?

1.1 Definitions

Dysphagia is difficulty in swallowing; *odynophagia* is pain on swallowing; *globus* is a lump feeling in the throat between meals without difficulty swallowing food; *aspiration* is passage of material below the vocal folds.

1.2 Types by level

- **Oral:** poor lip seal, chewing or bolus control; drooling and food left in the mouth (stroke, facial palsy, tongue surgery, neuromuscular disease).
- **Oropharyngeal:** difficulty starting a swallow, coughing or choking as soon as one swallows, nasal regurgitation, food sticking in the throat (stroke, motor neurone disease, myasthenia, Parkinson disease, head and neck cancer and its treatment, cricopharyngeal dysfunction, pouch).

- **Oesophageal:** food sticking behind the sternum seconds after swallowing (strictures, rings and webs, achalasia and spasm, tumours, extrinsic compression).

CAUSES OF DYSPHAGIA

GROUP	EXAMPLES
Lumen	Foreign body, food bolus
Wall: inflammation	Pharyngitis, tonsillitis, quinsy, oesophagitis (reflux, candida, herpes, eosinophilic, pill, caustic)
Wall: tumour	Carcinoma of the oropharynx, hypopharynx or oesophagus
Wall: structural	Pouch, webs (Plummer–Vinson), Schatzki ring, peptic and caustic strictures
Wall: motility	Achalasia, diffuse spasm, presbyoesophagus, scleroderma, polymyositis
Outside the wall	Goitre, cervical osteophytes, mediastinal nodes or tumour, aortic aneurysm, aberrant right subclavian artery (dysphagia lusoria)
Neurological	Stroke, motor neurone disease, multiple sclerosis, Parkinson disease, myasthenia, bulbar and pseudobulbar palsy, vagal palsy

RED FLAGS

Progressive dysphagia for solids then liquids, weight loss, odynophagia, haematemesis or haemoptysis, voice change, referred otalgia, a neck mass, or age over fifty with new dysphagia: urgent endoscopy for malignancy.

1.3 Evaluation

- **History:** level, solids or liquids, progression, regurgitation, cough with eating, weight, voice, heartburn, neurological symptoms.
- **Examination:** mouth, oropharynx, flexible laryngoscopy (pooling, vocal fold movement, sensation), neck, cranial nerves.
- **Videofluoroscopy (modified barium swallow):** the oral and pharyngeal phases, penetration and aspiration, and the effect of postures and textures.
- **Fibreoptic endoscopic evaluation of swallowing (FEES):** swallowing coloured food under endoscopic view, with or without sensory testing.
- **Barium swallow:** pouches, webs, rings, strictures, motility.

- **Endoscopy:** rigid or flexible oesophagoscopy, or unsedated transnasal oesophagoscopy, with biopsy.
- **Manometry:** motility disorders (achalasia, spasm).
- **CT** for tumours and extrinsic compression.

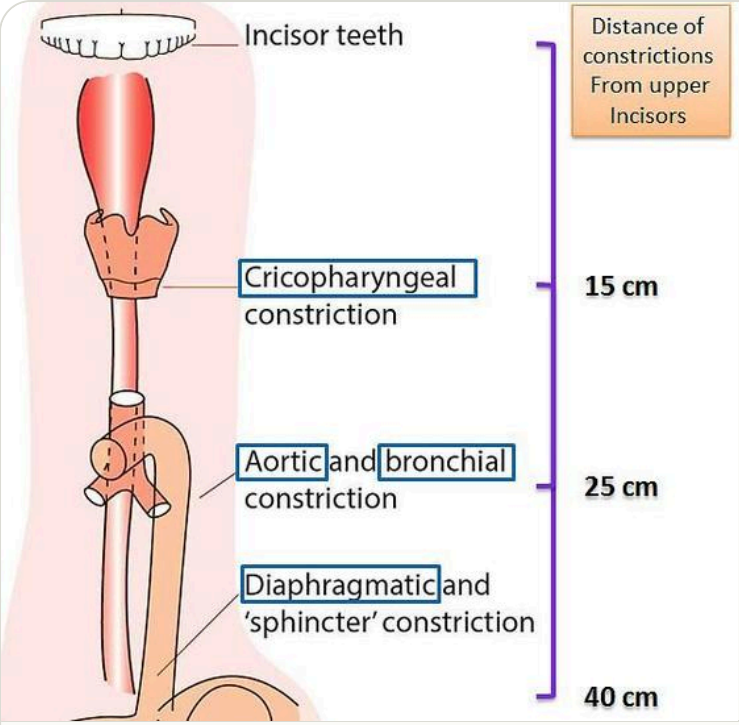


FIGURE 1 The constrictions of the oesophagus and their distance from the incisors.
Anatomy atlas · ENT Steps p. 226

1.4 Oropharyngeal dysfunction and its treatment

- **Tongue base weakness:** residue in the valleculae; chin tuck, tongue base exercises.
- **Pharyngeal sensory loss:** silent aspiration, worse with thin liquids; thickened fluids, head turned to the weak side.
- **Vocal fold palsy:** aspiration of thin liquids and a breathy voice; medialisation.
- **Poor laryngeal elevation:** residue and failure of the sphincter to open; the Mendelsohn manoeuvre, exercises and stimulation.
- **Failure of upper oesophageal sphincter opening:** a cricopharyngeal bar (present in many asymptomatic elderly people) or a pouch; dilatation, botulinum toxin or cricopharyngeal myotomy.

Speech and language therapists guide postures, textures and exercises; a gastrostomy feeds those who cannot swallow safely.

2 CHAPTER TWO Pharyngeal pouch

Zenker diverticulum.

2.1 Definition

A *pulsion diverticulum* in which pharyngeal mucosa herniates through *Killian's dehiscence*, the weak triangle in the posterior pharyngeal wall between the oblique *thyropharyngeus* above and the transverse *cricopharyngeus* below, the two parts of the inferior constrictor. Also called a hypopharyngeal diverticulum.

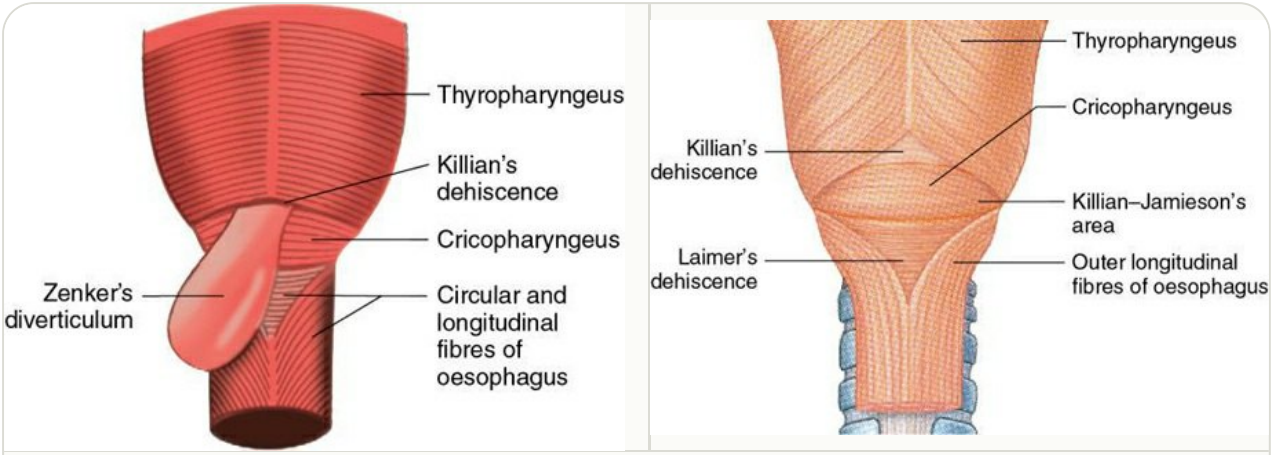


FIGURE 2 First, a hypopharyngeal (Zenker) diverticulum herniating through Killian's dehiscence. Second, the potential sites of diverticula: Killian's dehiscence, the Killian–Jamieson area and Laimer's dehiscence.

Dhingra, figs 54.1 and 54.2

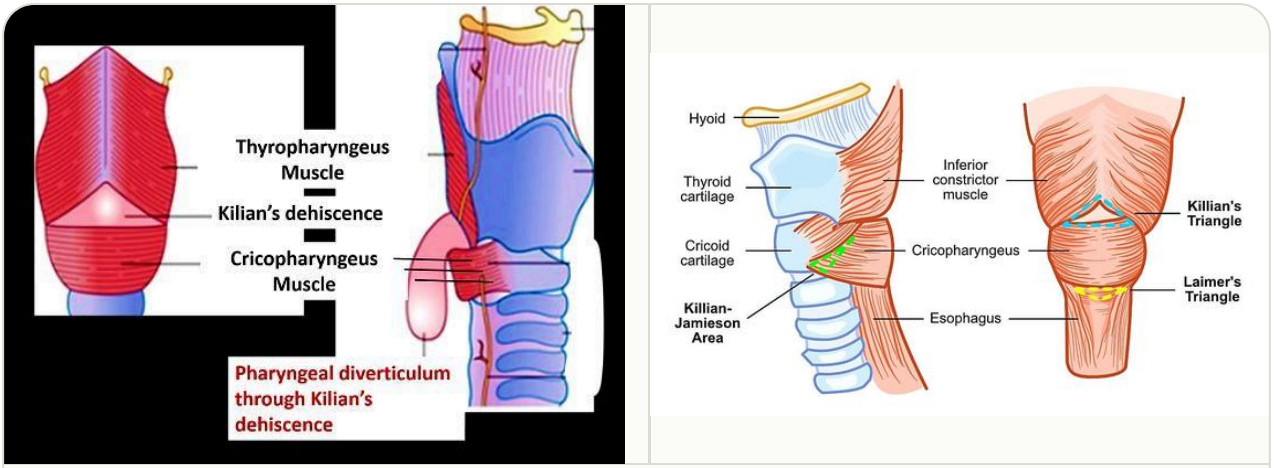


FIGURE 3 Killian's dehiscence and the pharyngeal pouch, and Killian's and Laimer's triangles.
Anatomy atlas · ENT Steps p. 198, ENT Steps p. 225

2.2 Cause

Not fully known. Incoordination between pharyngeal contraction and cricopharyngeal relaxation, failure of the sphincter to open (fibrosis of cricopharyngeus, poor laryngeal elevation) and raised pressure above it force mucosa out through the weak area. Reflux may contribute. It is a disease of the elderly, usually over sixty, men more than women.

2.3 Pathology

The pouch begins in the midline, lies first behind the oesophagus, then enlarges downwards and comes to lie to its left. As it grows, its mouth comes to lie in line with the pharynx while the oesophageal opening is pushed forward, so food enters the sac preferentially.

2.4 Clinical features

- **Dysphagia:** often appearing after a few swallows, once the pouch fills and presses on the oesophagus; progressive.
- **Gurgling** in the neck on swallowing (*borborygmi*); a swelling in the left side of the neck that empties with pressure.
- **Regurgitation** of undigested food, sometimes hours later and at night when lying down, with *nocturnal cough* and *aspiration pneumonia*.

- **Halitosis**, weight loss and malnutrition.
- Associated hiatus hernia; rarely carcinoma within a long-standing pouch.

2.5 Diagnosis

A *barium swallow* shows the sac and its size, best in the lateral view. Endoscopy carries a risk of entering and perforating the sac, but is needed if carcinoma is suspected; nasogastric tubes are passed with care.



FIGURE 4 Barium swallow showing an early Zenker's diverticulum (arrow) at the level of the sixth cervical vertebra in a 31-year-old woman.

Dhingra, fig. 54.3

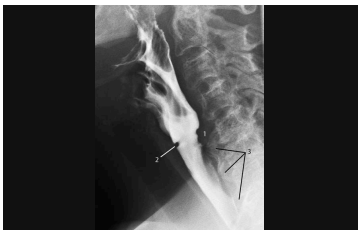
IMAGING IN THE APP 3 cases

Zenker diverticulum, oesophageal passage disorders and aspiration.



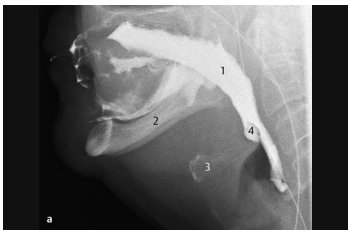
Zenker Diverticulum (Pharyngeal Pouch)

Infection & inflammation · 9 images



Esophageal Passage Disorders

Infection & inflammation · 2 images



Aspiration

Infection & inflammation · 3 images

2.6 Treatment

Small asymptomatic pouches can be watched. For symptoms:

2.6.1 Endoscopic

- **Endoscopic stapling diverticulotomy:** the party wall (the *diverticular bar*, containing cricopharyngeus) between the pouch and the oesophagus is divided and sealed with a linear stapler through a Weerda diverticuloscope; the pouch becomes part of the oesophagus and the muscle is divided at the same time. Now the commonest operation.
- **Dohlman's procedure / CO2 laser diverticulotomy:** the bar is divided by diathermy or laser through the diverticuloscope; suited to poor-risk and elderly patients.

Endoscopic methods are quick, with rapid return to eating, but are less effective for very small pouches (too little bar to staple) and may be impossible when the neck cannot be extended or the mouth opened.

2.6.2 External (cervical)

Through an incision along the anterior border of sternomastoid, usually on the left, the pouch is exposed, and either excised, inverted or suspended (diverticulopexy), *always with cricopharyngeal myotomy* to prevent recurrence.

MYOTOMY IS THE KEY

Whatever is done to the sac, dividing cricopharyngeus treats the cause. Excision without myotomy invites recurrence and fistula.

2.6.3 Complications

> Mediastinitis from perforation, pharyngocutaneous fistula, recurrent laryngeal nerve palsy, recurrence, and, with endoscopic methods, failure to complete the division.

OPERATIVE GUIDES IN THEATRE 4 operations

Upper oesophageal sphincter surgery and oesophagoscopy.

Dysfunction of the Upper Esophageal Sphincter
Upper oesophageal sphincter
Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy
Partial Pharyngectomy
Partial pharyngectomy
Total Laryngopharyngectomy
Total laryngopharyngectomy

3 CHAPTER THREE Other causes

Webs, rings and motility.

3.1 Plummer–Vinson syndrome

Iron deficiency anaemia, dysphagia from *postcricoid webs*, glossitis, angular cheilitis and koilonychia, mostly in women; a predisposition to postcricoid carcinoma. Barium swallow shows the web; iron replacement and dilatation of the web; surveillance.

3.2 Rings and strictures

A *Schatzki ring* at the gastro-oesophageal junction causes intermittent dysphagia for solids and food impaction; dilatation. Strictures from reflux, pills, caustics, radiotherapy or surgery are dilated and the cause treated.

3.3 Motility disorders

- **Achalasia:** loss of the myenteric plexus; progressive dysphagia for solids and liquids, regurgitation and chest pain; a dilated oesophagus tapering to a *bird's beak*; aperistalsis with failure of the lower sphincter to relax on manometry. Pneumatic

dilatation, surgical or peroral endoscopic myotomy, or botulinum toxin; exclude a tumour at the cardia (pseudoachalasia).

- **Diffuse oesophageal spasm:** chest pain and dysphagia, a "corkscrew" oesophagus; nitrates or calcium channel blockers.
- **Scleroderma, polymyositis, diabetes and hypothyroidism** also impair peristalsis.

3.4 Eosinophilic oesophagitis

Young atopic adults with dysphagia and food bolus obstruction; rings on endoscopy and eosinophils on biopsy; elimination diets, swallowed steroids and dilatation.

3.5 Extrinsic compression

Goitre, cervical osteophytes, mediastinal masses, and *dysphagia lusoria* from an aberrant right subclavian artery passing behind the oesophagus (often with a non-recurrent right laryngeal nerve).

CRICOPHARYNGEAL DYSFUNCTION AND NEWER OPTIONS (OXFORD TEXTBOOK)

The upper oesophageal sphincter is formed by the lower inferior constrictor, the cricopharyngeus and the upper oesophagus. In **cricopharyngeal dysfunction** the sphincter opens too little or relaxes poorly; videofluoroscopy may show a *cricopharyngeal bar* at C5–C6, but a bar is seen in about a third of people without symptoms, so it must be matched to the complaint. **High-resolution pharyngeal manometry** with videofluoroscopy shows whether the pharynx contracts normally against a high residual sphincter pressure, which predicts benefit from dilatation, botulinum toxin or myotomy. **Transnasal oesophagoscopy** under local anaesthetic examines the oesophagus in clinic and allows balloon dilatation of strictures and placement of feeding tubes. For pouches, endoscopic stapling remains first line for medium and large pouches in fit patients; it may not divide enough of the septum in small pouches (under about 2 to 3 cm), where laser or open myotomy with diverticulectomy or inversion is preferred. Flexible endoscopic septotomy, including the submucosal tunnelling technique (Z-POEM), is increasingly offered by gastroenterologists, especially to frail patients who cannot extend the neck for rigid access.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of dysphagia** by lumen, wall and outside, and the red flags.
- **Investigations:** videofluoroscopy, FEES, barium swallow, endoscopy, manometry.
- **Killian's dehiscence** and the anatomy of a pouch.
- **Clinical features** of a pharyngeal pouch.
- **Treatment:** endoscopic stapling, Dohlman, excision with cricopharyngeal myotomy.
- **Plummer–Vinson syndrome.**

4.2 Common pitfalls

COMMON PITFALLS

- Passing a rigid oesophagoscope blindly in a patient with a pouch.
- Excising a pouch without cricopharyngeal myotomy.
- Attributing progressive dysphagia in an older smoker to "age" without endoscopy.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, Oxford Specialist Handbook ch. 10, Oxford Handbook of ENT ch. 13, Dhingra ch. 54 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Applied anatomy of the oral cavity

The mouth is a small room with many tenants: lip, cheek, gum, teeth, tongue, palate and floor, each with its own nerve, vessel and lymphatic route. Know where each subsite begins and ends, and where its lymph goes, and the examination of the mouth, the staging of its cancers and the planning of its neck dissections follow naturally.

AT A GLANCE

- The oral cavity ends at the circumvallate papillae, palate junction and anterior pillars.
- Tongue: lingual nerve and chorda tympani in front, glossopharyngeal behind.
- Lateral tongue may skip to level IV; floor of mouth crosses the midline.
- Referred otalgia comes through V3 and IX.

1 CHAPTER ONE Boundaries

Where the mouth ends and the pharynx begins.

1.1 Extent

The oral cavity runs from the vermilion border of the lips in front to the *oropharyngeal isthmus* behind: the junction of the hard and soft palate above, the line of the circumvallate papillae below, and the anterior tonsillar pillars and retromolar trigones at the sides. Everything behind this ring (soft palate, tonsils, base of tongue, valleculae) is oropharynx, a distinction that matters because the two regions are staged and treated differently.

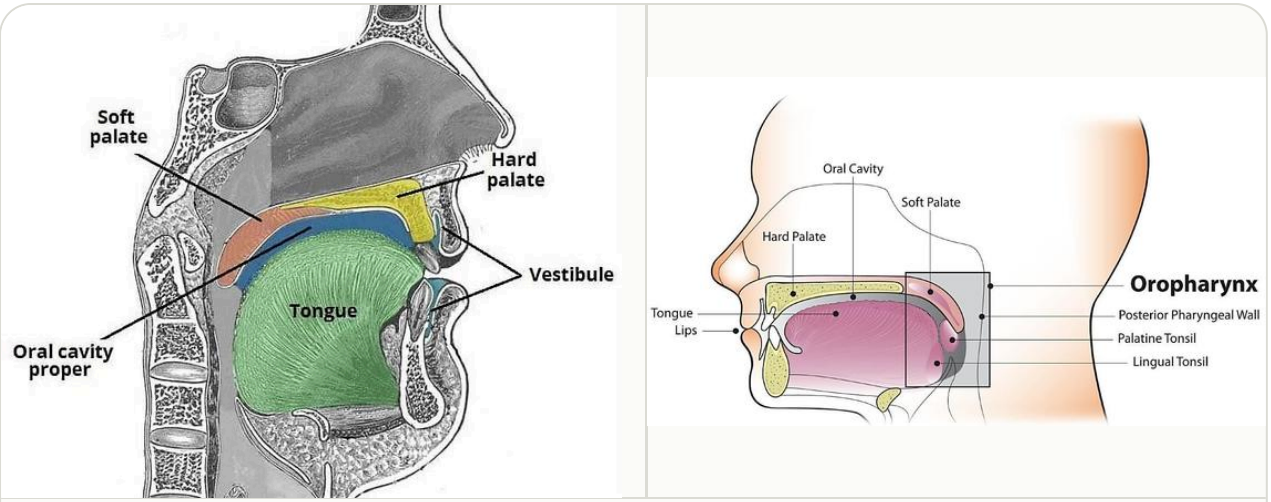


FIGURE 1 The vestibule and the oral cavity proper, and the boundary between the oral cavity and the oropharynx.

Anatomy atlas · ENT Steps p. 164, ENT Steps p. 165

1.2 Subsites

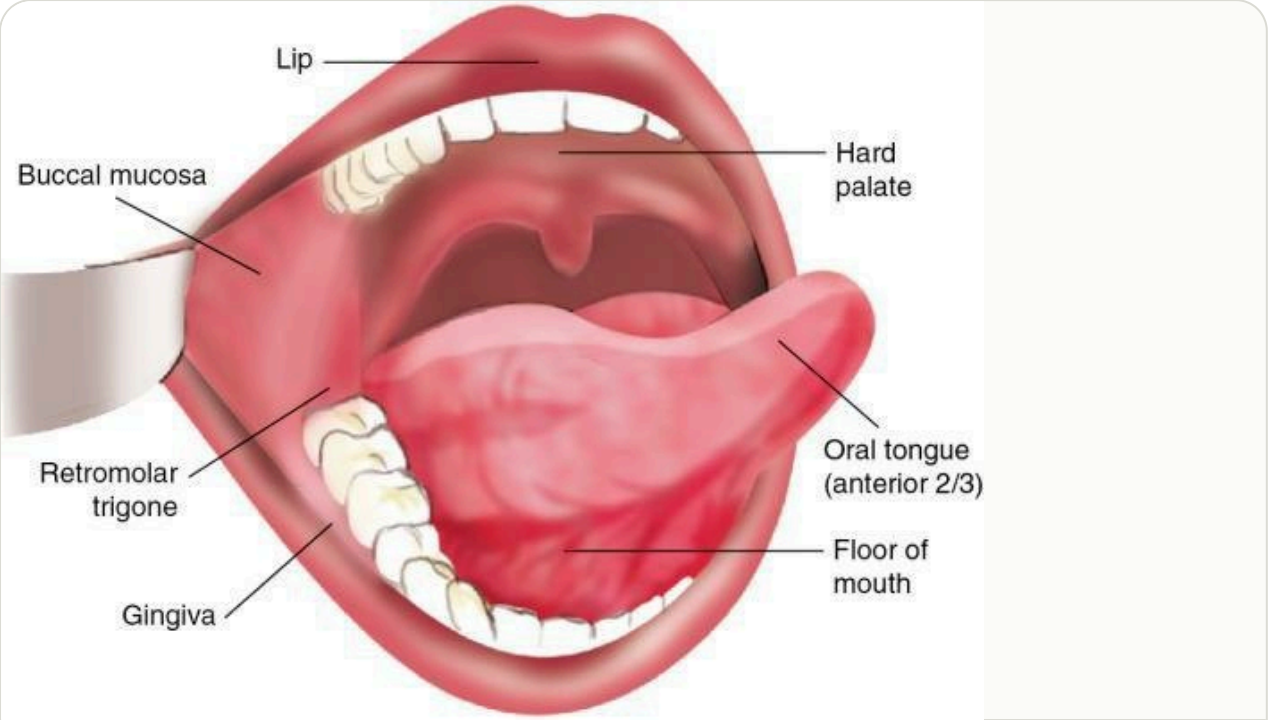


FIGURE 2 The subsites of the oral cavity: lip, buccal mucosa, gingiva, retromolar trigone, hard palate, oral tongue and floor of mouth.

Dhingra, fig. 42.1

- **Lips:** the mucosal lip, from the vermilion border to the line where the lips meet.
- **Buccal mucosa:** the lining of the cheeks and inner lips, back to the pterygomandibular raphe.

- **Upper and lower alveolar ridges (gingivae):** the mucosa over the tooth-bearing bone.
- **Retromolar trigone:** the triangle of mucosa over the front of the ascending ramus, its base behind the last molar and its apex at the maxillary tuberosity.
- **Hard palate:** the roof of the mouth.
- **Oral tongue:** the anterior two-thirds, in front of the circumvallate papillae, with its tip, lateral borders, dorsum and undersurface.
- **Floor of mouth:** the crescent between the lower gingiva and the undersurface of the tongue, with the frenulum and the sublingual papillae where the submandibular ducts open.

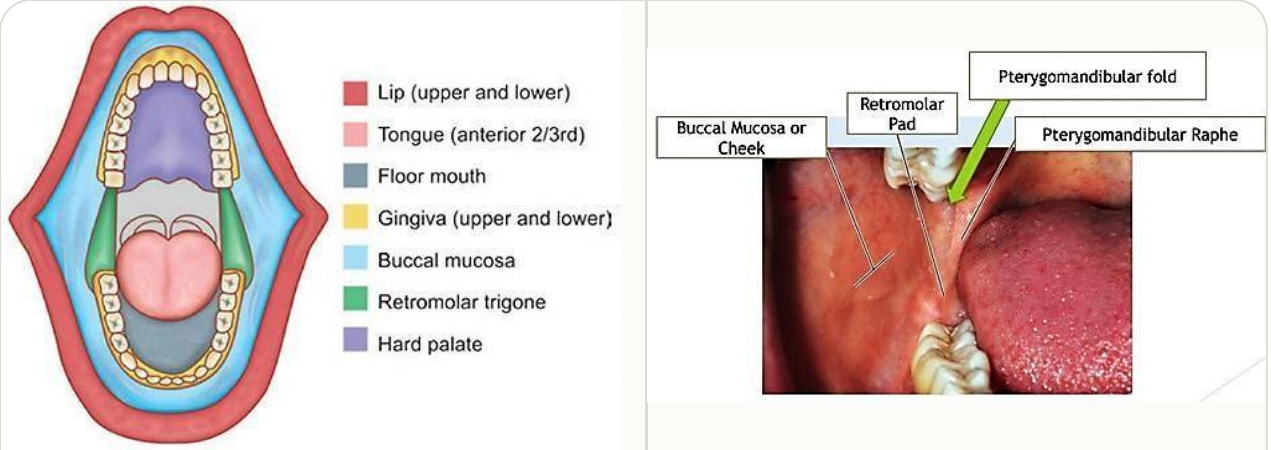


FIGURE 3 The subsites of the oral cavity, and the retromolar trigone with the pterygomandibular raphe.

Anatomy atlas · ENT Steps p. 165, ENT Steps p. 171

EXAMINING THE FLOOR OF MOUTH

The anterior floor is seen best when the patient lifts the tongue tip to the palate; the lateral floor when a tongue depressor pushes the tongue to the other side. Always palpate the floor bimanually: small cancers here are felt before they are seen.

1.3 Functions

Intake and chewing of food, the oral phase of swallowing, the shaping of speech, and protection against noxious substances through saliva, taste and the gag reflex.

2 CHAPTER TWO The lips

The gateway.

2.1 Surface anatomy

The *vermilion* is non-keratinised squamous epithelium without hair or sebaceous glands; its colour comes from the capillaries beneath. The vermilion border marks the change from skin to mucosa; the philtral ridges and Cupid's bow shape the upper lip; the two lips meet at the oral commissures.

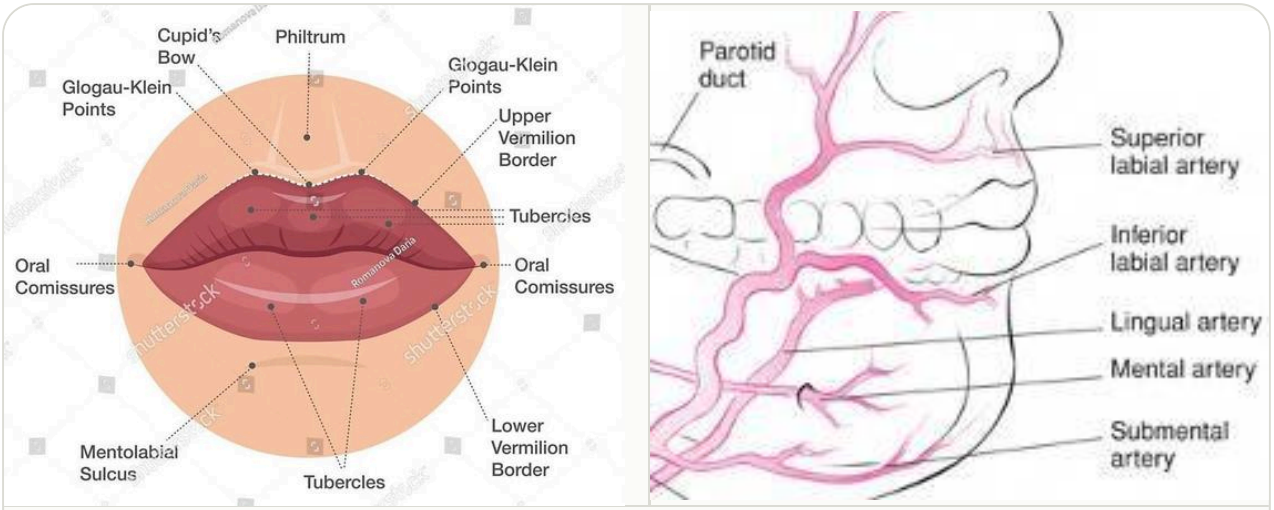


FIGURE 4 Lip landmarks, and the arterial supply of the lips from the superior and inferior labial arteries.

Anatomy atlas · ENT Steps p. 166, ENT Steps p. 168

2.2 Muscles

All are supplied by the facial nerve. The *orbicularis oris* forms the sphincter. At each commissure several muscles meet in a tendinous knot, the *modiolus*: orbicularis oris, buccinator, levator and depressor anguli oris, zygomaticus major and risorius. The lip elevators (levator labii superioris, levator labii superioris alaeque nasi, zygomaticus minor) and depressors (depressor labii inferioris, mentalis, platysma) complete the set.

2.3 Nerves, vessels and lymph

- **Sensation:** upper lip from the infraorbital nerve (V2), lower lip from the mental nerve (V3).
- **Arteries:** the superior and inferior labial branches of the facial artery.
- **Lymph:** the lower lip drains to the submandibular nodes, its midline also to the submental nodes, and it may cross the midline; the upper lip drains to the submandibular, preauricular and infraparotid nodes.

CHAPTER THREE

3 Teeth, gums and jaws

What the dentist sees.

3.1 The tooth

- **Enamel:** the hardest outer layer, laid down by ameloblasts.
- **Dentine:** beneath it, formed throughout life by odontoblasts, and sensitive.
- **Pulp:** the inner core of vessels and nerves.
- **Cementum:** covers the root and anchors it through the periodontal ligament to the alveolar bone.

3.2 Numbers

Twenty deciduous teeth (no premolars) and thirty-two permanent teeth: in each quadrant two incisors, one canine, two premolars and three molars, the third molar being the wisdom tooth.

3.3 Gingiva

The attached gingiva is bound firmly to the alveolar bone; the looser alveolar mucosa lies beyond it.

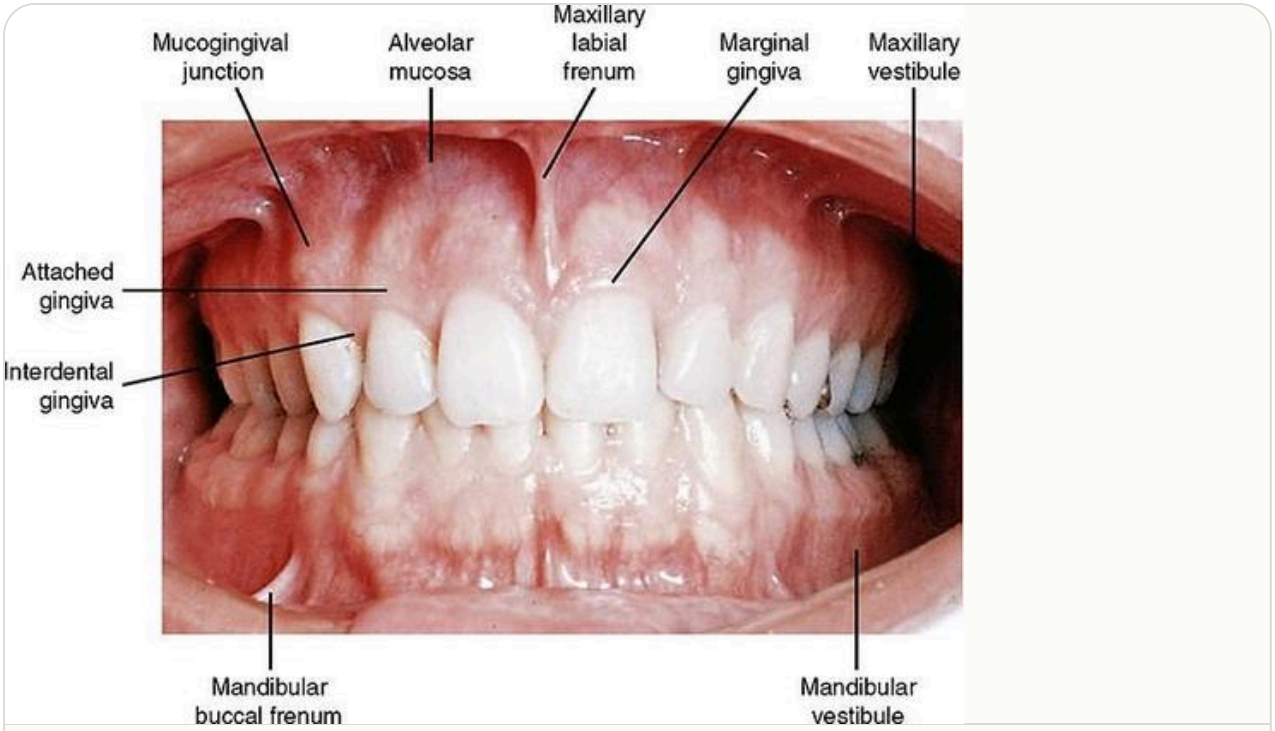


FIGURE 5 Gingiva and alveolar mucosa.
Anatomy atlas · ENT Steps p. 168

3.4 Muscles of mastication

All supplied by the mandibular nerve (V3):

- **Masseter:** from the zygomatic arch to the angle and lateral ramus; elevates the mandible.
- **Temporalis:** from the temporal fossa to the coronoid process; elevates and retracts.
- **Lateral pterygoid:** from the greater wing of the sphenoid and lateral pterygoid plate to the condyle; opens the mouth, protrudes and moves the jaw from side to side.
- **Medial pterygoid:** from the medial side of the lateral pterygoid plate and the maxillary tuberosity to the medial angle; elevates and moves the jaw from side to side.

Infiltration of these muscles by infection or cancer produces *trismus*.

4 CHAPTER FOUR The tongue

The busiest muscle in the head.

4.1 Surface

The *sulcus terminalis* and the V of circumvallate papillae divide the oral tongue in front from the base of tongue behind. At the apex of the V lies the *foramen caecum*, from which the thyroid descends in the embryo; a lingual thyroid or thyroglossal cyst begins here. Behind lie the lingual tonsil and the valleculae, both oropharynx.

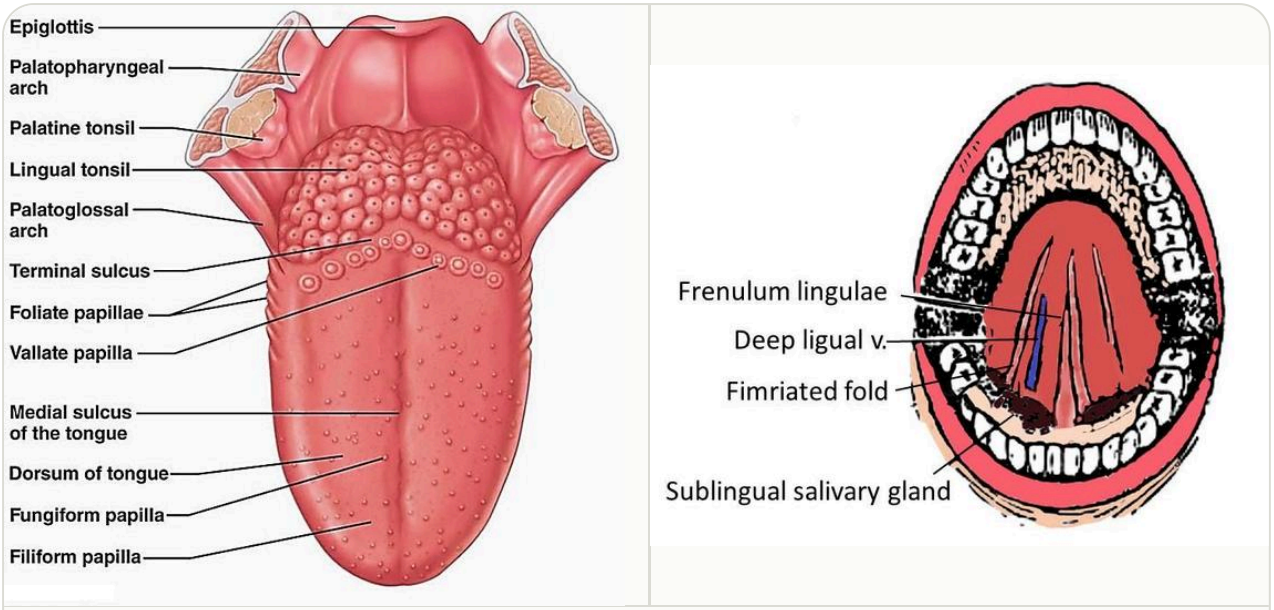


FIGURE 6 The dorsum of the tongue with its papillae, sulcus terminalis and lingual tonsil, and its undersurface with the floor of mouth.

Anatomy atlas · ENT Steps p. 173, Doss p. 177

4.2 Papillae

- **Filiform:** the most numerous; texture, temperature and pain; no taste.
- **Fungiform:** mostly at the tip; taste buds on their upper surface.
- **Foliate:** on the lateral border; taste buds on their sides.
- **Circumvallate:** eight to twelve in a V in front of the sulcus terminalis; taste buds in their walls.

4.3 Muscles

The intrinsic muscles (longitudinal, transverse and vertical) change its shape; the extrinsic muscles move it.

EXTRINSIC MUSCLES OF THE TONGUE

MUSCLE	FROM	ACTION	NERVE
Genioglossus	Mental spine of the mandible	Protrudes and depresses; the bulk of the tongue	Hypoglossal
Hyoglossus	Hyoid bone	Depresses and retracts	Hypoglossal
Styloglossus	Styloid process	Retracts and elevates	Hypoglossal
Palatoglossus	Palatine aponeurosis	Raises the back of the tongue, closes the isthmus	Vagus (pharyngeal plexus)

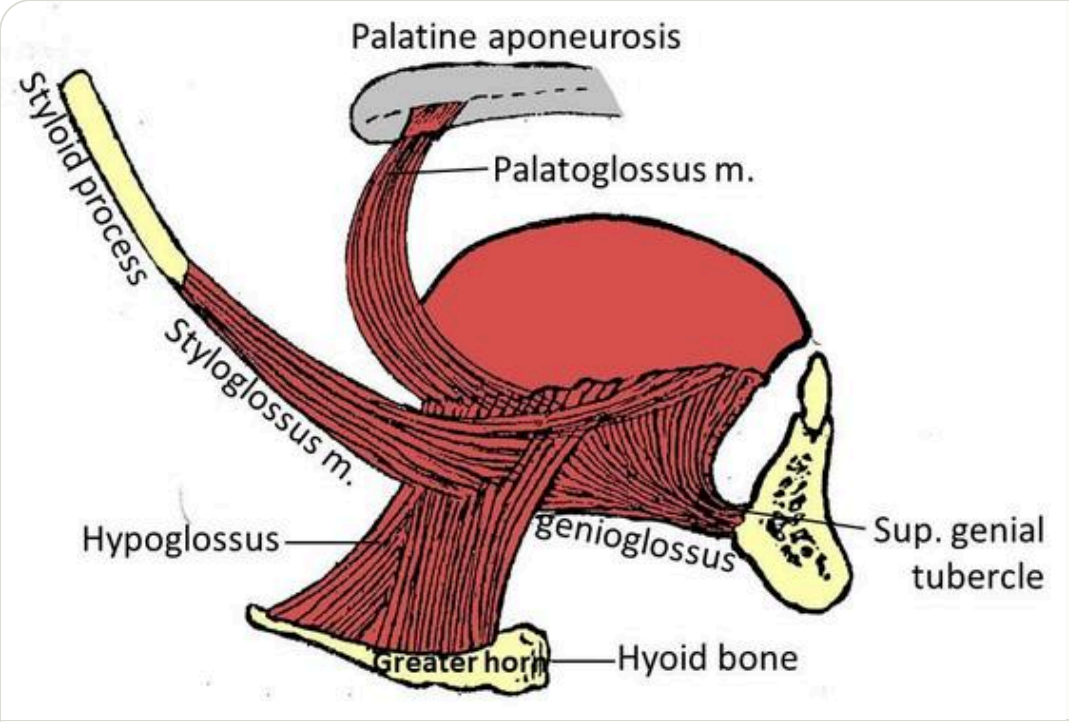


FIGURE 7 The extrinsic muscles of the tongue.
Anatomy atlas · Doss p. 179

TESTING THE HYPOGLOSSAL NERVE

With a lower motor neurone lesion the protruded tongue deviates towards the paralysed side, pushed across by the healthy genioglossus, and the paralysed half wastes and fasciculates.

4.4 Nerves and taste

- **Oral tongue:** touch, pain and temperature by the lingual nerve (V3); taste by the chorda tympani (VII) to the geniculate ganglion and nucleus solitarius.

- **Base of tongue:** both general sensation and taste by the glossopharyngeal nerve (IX).
- **Epiglottis and valleculae:** a little taste by the internal laryngeal branch of the vagus (X).

The five tastes are sweet, sour, salty, bitter and umami; flavour is taste combined with smell.

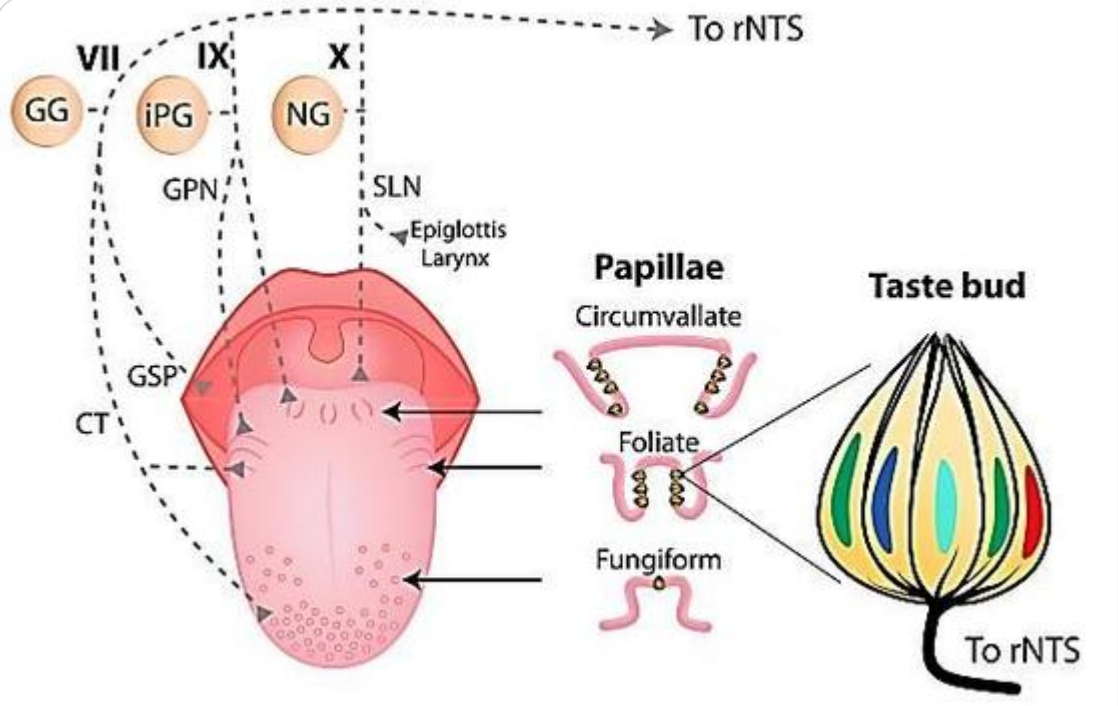


FIGURE 8 The nerve supply and taste pathways of the tongue.
Anatomy atlas · ENT Steps p. 174

REFERRED OTALGIA

The lingual nerve and the auriculotemporal nerve are both branches of V3, and the glossopharyngeal nerve also supplies the middle ear through Jacobson's nerve. A cancer of the tongue or tonsil therefore often presents as earache with a normal ear.

4.5 Vessels and lymph

The lingual artery, second branch of the external carotid, supplies the tongue; the lingual vein and the vein accompanying the hypoglossal nerve drain it. Lymph from the tip goes to the submental and jugulo-omohyoid nodes, from the lateral border to the ipsilateral submandibular and deep cervical nodes (sometimes skipping

straight to level IV), and from the central tongue and base to the deep cervical nodes of both sides.

5 CHAPTER FIVE Palate and floor

Roof and floor.

5.1 Hard palate

The palatine processes of the maxillae in front and the horizontal plates of the palatine bones behind, covered by mucoperiosteum firmly bound to bone.

- **Incisive foramen:** in the midline behind the incisors; nasopalatine nerve and vessels.
- **Greater palatine foramen:** medial to the second or third molar; greater palatine nerve and vessels, the pedicle of palatal flaps.
- **Lesser palatine foramina:** behind it, to the soft palate.

Sensation is from the nasopalatine and greater palatine nerves (V2); blood from the greater palatine artery; lymph to the upper deep cervical and lateral retropharyngeal nodes, the front of the palate also to the submandibular nodes.

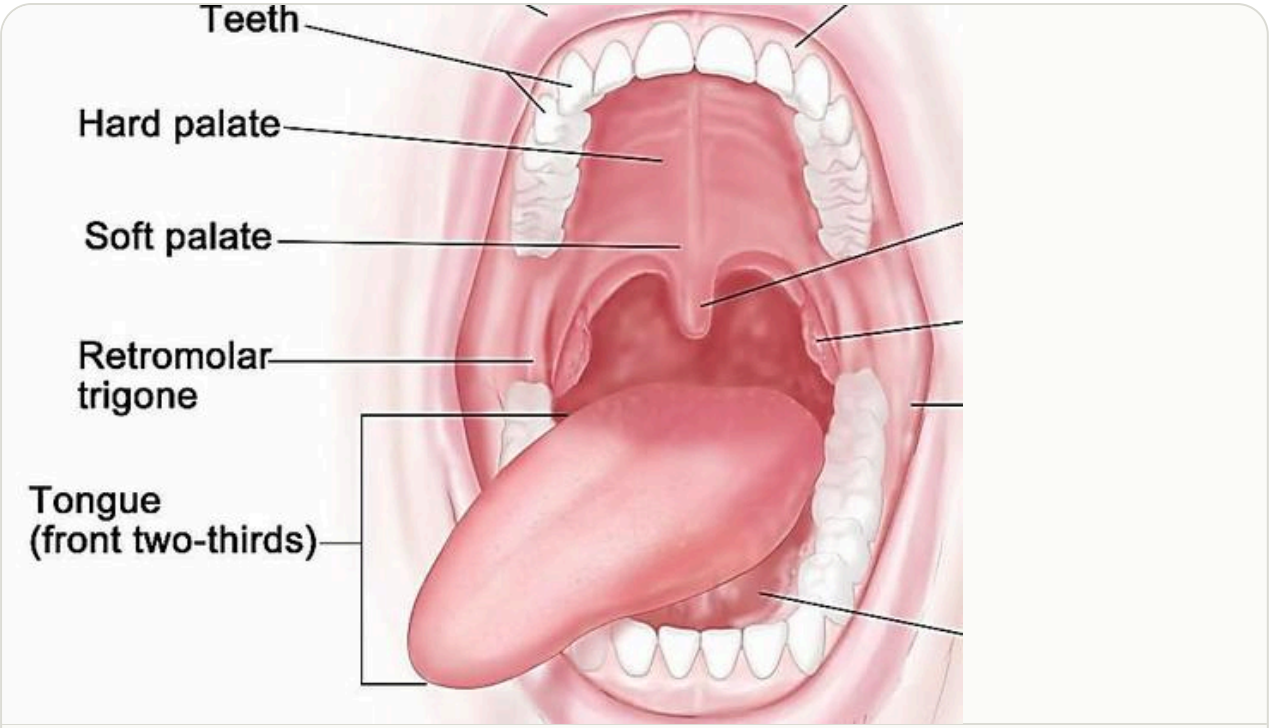


FIGURE 9 The hard palate, soft palate, retromolar trigone and oral tongue.
Anatomy atlas · ENT Steps p. 172

5.2 Floor of mouth

The mylohyoid muscle forms the diaphragm of the floor. Above it lie the sublingual glands, the submandibular duct, the lingual nerve looping beneath the duct, and the hypoglossal nerve; below it, the submandibular space. The anterior floor drains to the submandibular nodes and crosses the midline freely; the posterior floor drains to the upper deep cervical nodes.

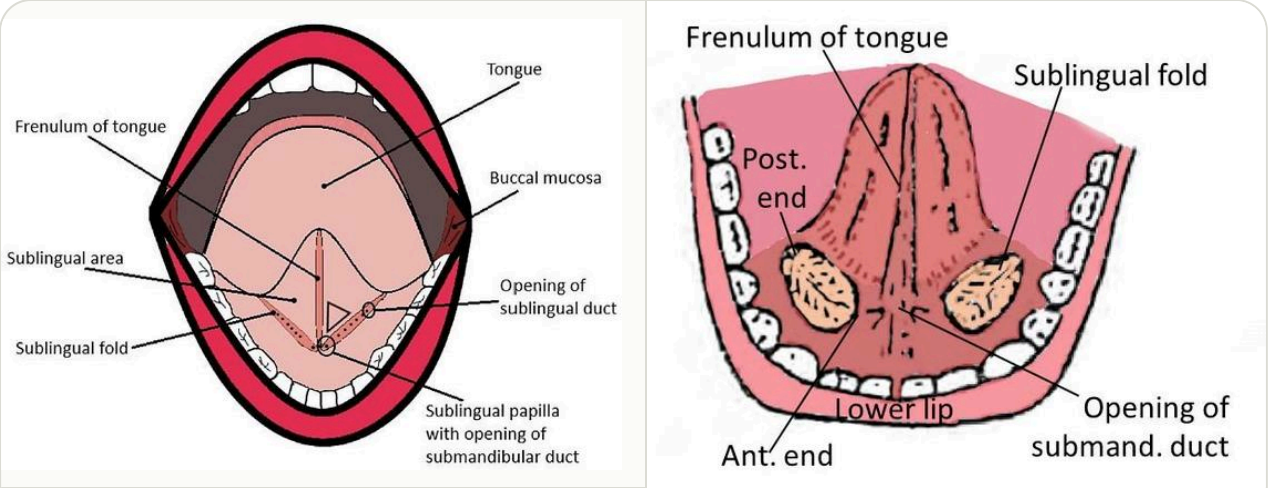


FIGURE 10 The floor of the mouth with the sublingual papillae and folds, and the openings of the ducts.

Anatomy atlas · ENT Steps p. 169, Doss p. 121

5.3 Saliva

One to one and a half litres a day. At rest about two-thirds comes from the submandibular glands; when stimulated, more than half comes from the parotids. The parotid (Stensen) duct opens opposite the upper second molar; the submandibular (Wharton) duct beside the frenulum; the sublingual glands by many small ducts into the floor. Minor salivary glands line the lips, cheeks, palate and tongue, and are the source of the palate's salivary tumours.

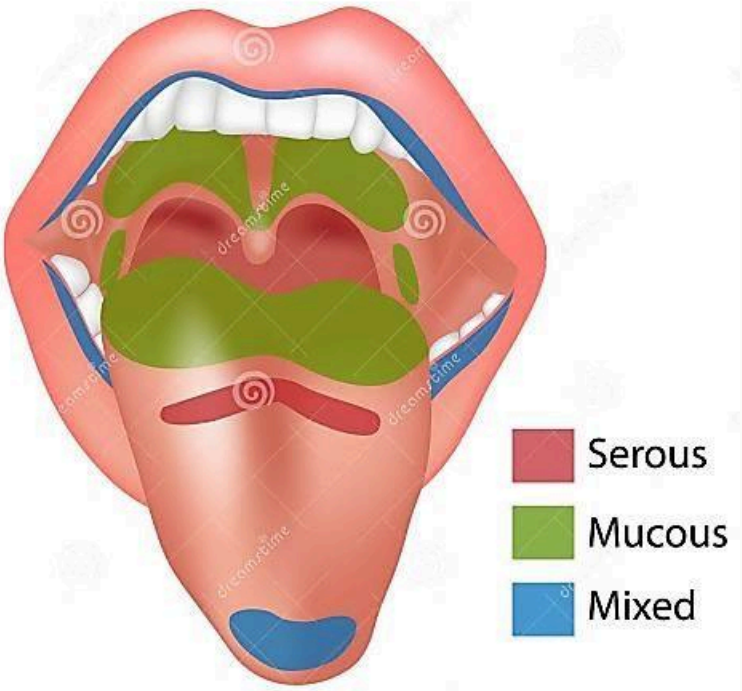


FIGURE 11 Distribution of serous and mucous minor salivary glands.
Anatomy atlas · ENT Steps p. 193

6 CHAPTER SIX Lymphatic drainage

The map for the neck.

LYMPHATIC DRAINAGE OF THE ORAL CAVITY

SUBSITE	FIRST NODES
Lower lip	Submental (midline) and submandibular (lateral); may cross the midline
Upper lip	Submandibular, preauricular and infraparotid
Buccal mucosa	Submental and submandibular
Alveolar ridges	Submental and submandibular
Hard palate	Upper deep cervical and lateral retropharyngeal; front of palate to submandibular
Floor of mouth	Anterior to submandibular, crossing the midline; posterior to upper deep cervical

SUBSITE	FIRST NODES
Tongue tip	Submental and jugulo-omohyoid
Lateral tongue	Ipsilateral submandibular and deep cervical, sometimes skipping to level IV
Central tongue and base	Deep cervical nodes of both sides

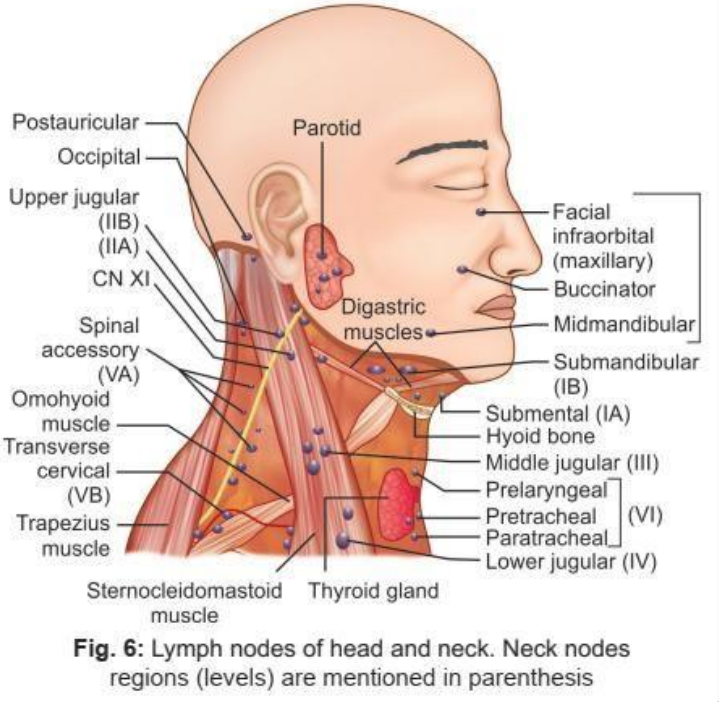


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

FIGURE 12 The lymph node levels of the neck.

Anatomy atlas · ENT Steps p. 285

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Boundaries** of the oral cavity and its subsites.
- **Nerve supply of the tongue:** general sensation and taste, front and back.
- **Extrinsic muscles** and the test of the hypoglossal nerve.

- **Lymphatic drainage** of the lip, tongue and floor of mouth.
- **Referred otalgia** from the mouth and pharynx.

7.2 Common pitfalls

COMMON PITFALLS

- Calling the base of tongue or the tonsil "oral cavity".
- Forgetting that the lateral tongue can skip to level IV nodes.
- Inspecting the floor of mouth without palpating it.

Written for this app from Oxford Specialist Handbook ch. 6, Dhingra ch. 42 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Oral ulcers, white patches and submucous fibrosis

Most mouths that hurt have a harmless reason: an aphthous ulcer, a sharp tooth, a crop of herpes or a patch of thrush. The skill is in sorting the many benign ulcers and white patches from the few that matter, and in remembering the one rule that keeps patients alive: an ulcer that has not healed in three weeks, once the cause has been removed, is biopsied.

AT A GLANCE

- Aphthous ulcers recur on movable mucosa and heal in two weeks.
- A white patch that will not wipe off is not thrush: biopsy.
- Submucous fibrosis: areca nut, blanching, bands and trismus; premalignant.
- Three-week rule for any non-healing ulcer.

1 CHAPTER ONE Overview

Ulcers, patches and the tongue.

1.1 Approach

Take a history of duration, recurrence, pain, habits (tobacco, areca nut, alcohol), dentures, drugs, skin, eye and genital lesions, and general illness. Examine every subsite, palpate any lesion for induration, and feel the neck.

CAUSES OF ORAL ULCERS

GROUP	EXAMPLES
Viral	Herpangina, primary and recurrent herpes simplex, hand, foot and mouth disease
Bacterial	Vincent infection (acute necrotising ulcerative gingivitis), tuberculosis, syphilis
Fungal	Candidiasis
Immune	Aphthous ulcers, Behçet syndrome

GROUP	EXAMPLES
Trauma	Cheek bite, jagged tooth, ill-fitting denture; chemical (aspirin, phenol, silver nitrate) and thermal burns
Neoplasm	Squamous cell carcinoma, minor salivary gland carcinoma, lymphoma
Skin disease	Erythema multiforme, lichen planus, pemphigus, mucous membrane pemphigoid, lupus
Blood disease	Leukaemia, agranulocytosis, cyclic neutropenia, pancytopenia
Other	Drug reactions, vitamin B12 and folate deficiency, radiation and chemotherapy mucositis, diabetes, uraemia

THE NON-HEALING ULCER

An ulcer persisting beyond three weeks after any cause (a sharp tooth, a denture) has been removed, an indurated or rolled edge, a fixed lesion, numbness, or a hard neck node: biopsy for carcinoma.

2 CHAPTER TWO Infections

Viruses, bacteria and fungi.

2.1 Viral

2.1.1 Herpangina

Coxsackie A infection of children: small vesicles on the pillars, tonsils, soft palate and uvula rupture into 2 to 4 mm ulcers with a yellow base and red areola, and heal within a week.

2.1.2 Herpetic gingivostomatitis

Primary herpes simplex in children: fever, malaise and clusters of vesicles anywhere in the mouth, with swollen, bleeding gums, healing in one to two weeks. *Recurrent* herpes in adults reactivates from the trigeminal ganglion after stress, fever, sunlight or immunosuppression, usually as *herpes labialis* at the vermilion

border, sometimes on the hard palate and attached gingiva. Treatment is symptomatic; aciclovir shortens recurrent attacks if started early.

2.1.3 Hand, foot and mouth disease

A childhood viral infection with ulcers on the palate, tongue and cheeks and vesicles on the hands and feet.

2.2 Bacterial

2.2.1 Vincent infection

Acute necrotising ulcerative gingivitis, caused by a fusiform bacillus and the spirochaete *Borrelia vincentii*: it begins at the interdental papillae, spreads along the gum margins under a grey necrotic slough, and smells foul; on the tonsil it is *Vincent angina*. A smear confirms it. Penicillin or erythromycin with metronidazole, mouthwashes and dental hygiene. In malnourished children it can spread into the face as *noma*.

2.2.2 Specific infections

Tuberculosis (a painful, undermined ulcer, often on the tongue), syphilis and actinomycosis (sulphur granules) cause chronic ulcers; biopsy distinguishes them from cancer.

2.3 Fungal: candidiasis

Candida albicans takes hold in infants, diabetics, the immunosuppressed, denture wearers, after antibiotics, inhaled steroids or radiotherapy, and in a dry mouth.

- **Thrush (pseudomembranous):** white plaques that wipe off to leave a raw red base.
- **Erythematous (atrophic):** red patches, often under dentures.
- **Chronic hyperplastic (candidal leukoplakia):** a white patch that will not wipe off, usually at the commissures; it can harbour dysplasia, so biopsy it.
- **Angular cheilitis** at the corners of the mouth, and **median rhomboid glossitis** on the midline dorsum.

Topical nystatin or clotrimazole; fluconazole for severe disease; remove the cause.

ORAL HAIRY LEUKOPLAKIA

Corrugated white streaks on the lateral borders of the tongue that do not rub off, caused by Epstein–Barr virus in HIV or immunosuppression. It is harmless in itself but a marker of immune failure; it regresses with antiretroviral treatment.

3 CHAPTER THREE Immune and skin disorders

Ulcers that come back.

3.1 Aphthous ulcers

The commonest non-traumatic ulcers: recurrent, shallow, painful, on the *movable* mucosa (lips, cheeks, tongue, floor, soft palate) and sparing the hard palate and attached gingiva.



FIGURE 1 Multiple aphthous ulcers on the uvula and faucial pillars (arrowheads). Dhingra, fig. 43.1

- **Minor:** the common form, 2 to 10 mm, multiple, with a necrotic centre and red halo; heal in about two weeks without a scar.
- **Major:** over 1 cm, deep, often at the back of the mouth, lasting six weeks or more and healing with a scar.
- **Herpetiform:** crops of tiny ulcers.

The cause is unknown; associations include deficiency of vitamin B12, folate or iron, stress, hormones, food sensitivity, coeliac and Crohn's disease, and drugs such as

NSAIDs and nicorandil. Unlike viral ulcers, they recur, spare fixed mucosa and come without fever or nodes. Treat with topical steroids, chlorhexidine or tetracycline mouthwashes and topical anaesthetic gels, and correct deficiencies.

3.2 Behçet syndrome

Recurrent aphthous-like oral ulcers with genital ulcers and uveitis, sometimes with skin, joint and nervous system disease; the ulcers are punched out.

3.3 Erythema multiforme

A hypersensitivity reaction, often to herpes simplex or drugs (sulphonamides, anticonvulsants): bullae rupture into ulcers under a pseudomembrane, the lips form *haemorrhagic crusts*, and target lesions appear on the skin; the severe form merges into Stevens–Johnson syndrome. Supportive care; steroids for severe disease.

3.4 Lichen planus

A T-cell mediated disease. The *reticular* form shows lace-like white Wickham striae on both buccal mucosae and needs no treatment; the *erosive* form produces painful ulcers with a keratotic edge and responds to topical steroids. Erosive and atrophic forms carry a small risk of carcinoma, so they are followed up.

3.5 Pemphigus and pemphigoid

- **Pemphigus vulgaris:** autoantibodies to desmoglein 3 split the epithelium above the basal layer; fragile blisters break into painful erosions, often before skin lesions appear, in patients of fifty to seventy. Systemic steroids and immunosuppression.
- **Mucous membrane pemphigoid:** the split is beneath the epithelium; blisters on the gums, cheek and palate, and scarring of the conjunctiva. Steroids and dapsone.

3.6 Blood disorders and drugs

Leukaemia swells and ulcerates the gums; agranulocytosis and cyclic neutropenia cause ulceration as defences fail. A blood film, counts and marrow examination are needed.



FIGURE 2 Palatal ulceration in a patient whose white cell count fell to 2,500 per cubic millimetre on cisplatin and fluorouracil chemotherapy.

Dhingra, fig. 43.4

3.7 Mucositis

Radiotherapy and chemotherapy (methotrexate, fluorouracil, bleomycin) redden, then ulcerate the mucosa from about the second week; treatment is supportive.

4 CHAPTER FOUR Trauma

A sharp tooth can mimic cancer.

A jagged tooth or ill-fitting denture ulcerates the lateral tongue or cheek; a pencil or toothbrush injures the palate; acids, alkalis, hot fluids and an aspirin held against a painful tooth burn the mucosa.



FIGURE 3 First, an exophytic ulcerated mass on the left side of the tongue caused by sharp teeth (white arrows) in a man who chewed tobacco and pan; it healed after the teeth were extracted. Second, an ulcer on the lateral border of the tongue that simulated carcinoma (A) and healed completely after extraction of the jagged tooth (B).

Dhingra, figs 43.2 and 43.3

REMOVE THE CAUSE, THEN REVIEW

Smooth the tooth or adjust the denture and look again in two to three weeks. A traumatic ulcer heals; one that does not is a biopsy.

5 CHAPTER FIVE The tongue

Variants and deficiencies.

5.1 Variants

- **Geographic tongue (migratory glossitis):** red, smooth areas without papillae, outlined by a white keratotic margin, that change shape from day to day; harmless.
- **Fissured tongue:** congenital, or with deficiency states; with facial palsy and lip swelling it forms *Melkersson–Rosenthal syndrome*.
- **Hairy tongue:** elongated filiform papillae stained brown or black by bacteria, with smoking and antibiotics; scrape the tongue and treat the cause.
- **Median rhomboid glossitis:** a red, smooth, rhomboid area in front of the foramen caecum, now attributed to chronic candida.
- **Fordyce spots:** yellow ectopic sebaceous glands in the buccal and labial mucosa; normal.

- **Leukoedema:** a milky film on the buccal mucosa that disappears when the cheek is stretched, unlike leukoplakia.

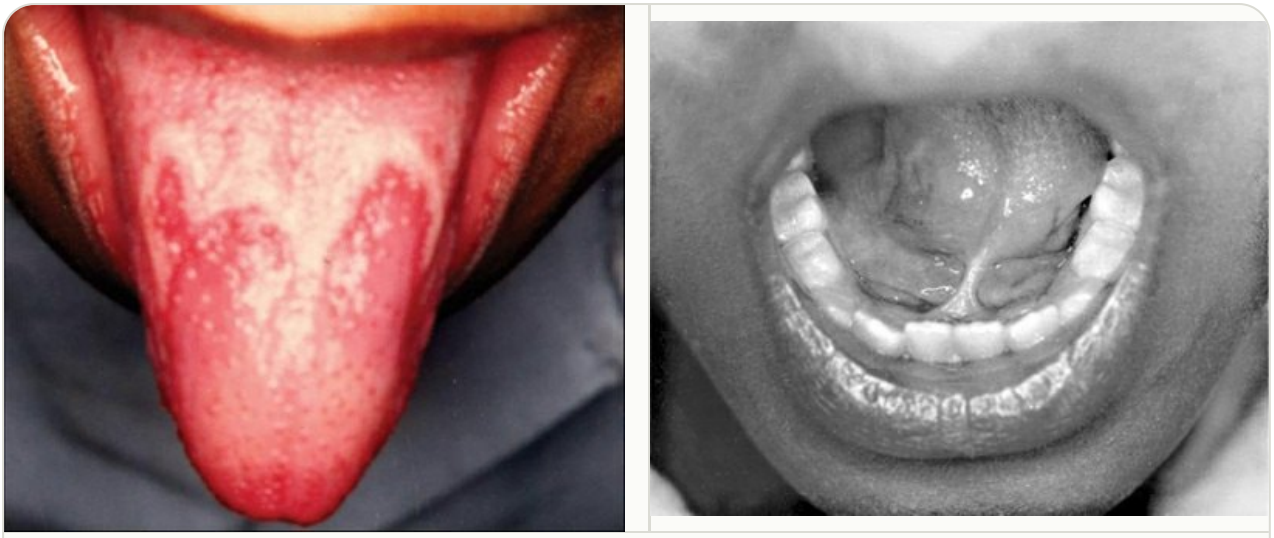


FIGURE 4 First, geographic tongue. Second, tongue-tie.
Dhingra, figs 43.5 and 43.6

5.2 Tongue-tie (ankyloglossia)

A short frenulum. If the tongue can be protruded beyond the lower incisors it rarely affects speech; symptomatic ties are released (frenotomy, or transverse release with vertical closure).

5.3 Systemic clues

- **Smooth, red tongue:** iron, B12 or folate deficiency; ash-grey mucosa and atrophy in iron deficiency (Plummer–Vinson syndrome).
- **Angular cheilitis:** riboflavin or iron deficiency, candida.
- **Gingival hyperplasia and bleeding:** scurvy, leukaemia, phenytoin.
- **Strawberry tongue:** scarlet fever and Kawasaki disease; **Koplik spots:** measles.

5.4 Pigmentation

Racial melanosis, amalgam tattoo, Peutz–Jeghers syndrome, Addison disease, Kaposi sarcoma and heavy metals (the blue line of lead) all pigment the mouth; a

new or changing pigmented lesion, especially on the palate, is biopsied to exclude melanoma.

5.5 Macroglossia

Lymphangioma, haemangioma, hypothyroidism, acromegaly, amyloidosis, Down syndrome and storage diseases.

5.6 Nicotine stomatitis

Red pin-points in the centre of white papules on the palate of pipe and reverse smokers, from inflamed minor salivary gland ducts; it regresses when smoking stops. The heat, not the nicotine, is to blame.

6 CHAPTER SIX Submucous fibrosis

Areca nut and the mouth that will not open.

6.1 Definition and cause

Oral submucous fibrosis is a chronic, insidious fibrosis beneath the epithelium of the mouth and pharynx, first described in India by Joshi in 1953 and common across South and South-East Asia. *Areca (betel) nut* is the essential cause, chewed alone, with tobacco, or as pan; tobacco, alcohol, poor nutrition and an immune reaction to arecoline add to it.

6.2 Pathogenesis

Activated T-lymphocytes and macrophages release fibrogenic cytokines that drive fibroblasts to make more collagen while collagenase falls, so collagen is laid down faster than it is removed. The lamina propria becomes fibroelastic and avascular, the epithelium atrophies, and the fibrosis can reach the muscle.

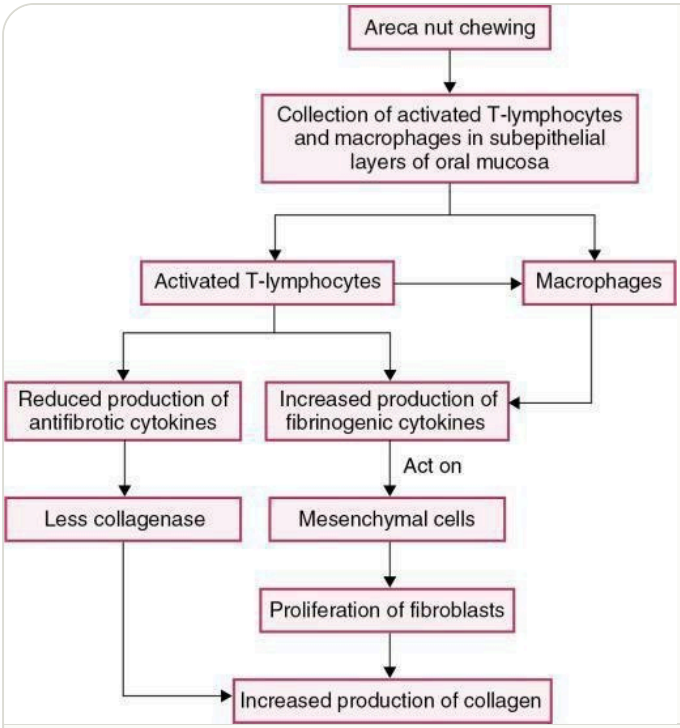


FIGURE 5 The cellular immune response to areca nut that leads to increased collagen production in submucous fibrosis.

Dhingra, fig. 43.7

6.3 Clinical features

Usually between twenty and forty. Intolerance of chillies and spicy food, a burning mouth, recurrent vesicles on the palate and pillars, then difficulty in opening the mouth and protruding the tongue. The soft palate, pillars and buccal mucosa blanch and lose their suppleness, with palpable fibrous bands; trismus becomes progressive until the patient cannot brush the teeth.



FIGURE 6 Submucous fibrosis: (A) blanching of the soft palate and faucial pillars; (B) marked trismus.

Dhingra, fig. 43.8

A PREMALIGNANT CONDITION

Carcinoma develops in about 3 to 7.6% (some series report more). Trismus makes examination difficult, so look carefully for leukoplakia and early cancer, and biopsy anything suspicious.

6.4 Treatment

- Stop areca nut, pan, tobacco and spicy food; correct anaemia and vitamin deficiency.
- Intralesional dexamethasone with hyaluronidase, given twice a week for eight to ten weeks, softens the bands and relieves trismus.
- Jaw-opening exercises.
- **Surgery** for advanced trismus: release of the bands with cover by skin graft, buccal fat pad, nasolabial or tongue flaps, a palatal island flap or radial forearm free flaps, with coronoidectomy and temporalis myotomy where needed; recurrence is common.

7 CHAPTER SEVEN Odontogenic cysts

Cysts in the jaws.

7.1 Dentigerous cyst

A developmental cyst around the crown of an unerupted or impacted tooth, usually the third molar or the upper canine, presenting in the second or third decade as a painless swelling of the jaw. Radiographs show a unilocular cyst containing the crown; in the maxilla the tooth may be pushed up to the orbital floor. Small cysts are removed with the tooth; large ones are first marsupialised to avoid a fracture. Ameloblastoma or carcinoma can arise in the wall, so the lining is always examined.

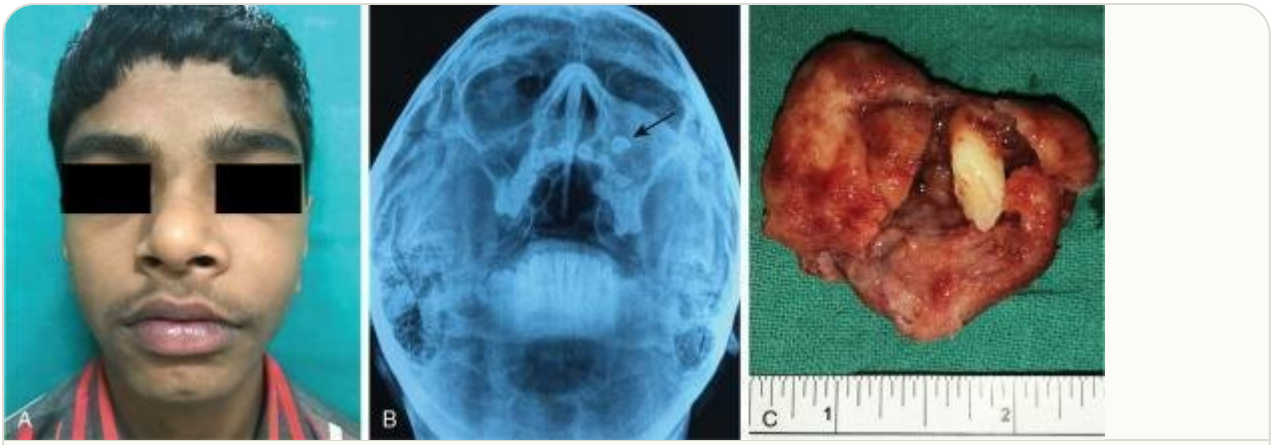


FIGURE 7 Dentigerous cyst in a 14-year-old boy: (A) swelling of the left cheek; (B) radiograph; (C) the specimen with its tooth.

Dhingra, fig. 43.9

7.2 Dental (radicular) cyst

An inflammatory cyst at the apex (periapical) or side (lateral radicular) of a dead tooth. Enucleated when the tooth is extracted; lining left behind forms a residual cyst.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra

Groups the commoner ulcers, tongue lesions and submucous fibrosis in one chapter, with a strong Indian emphasis on areca nut and reverse smoking.

Lee

Adds the immunofluorescence patterns that separate pemphigus (intercellular "chicken-wire") from pemphigoid (linear basement membrane), and quotes a higher malignant risk for submucous fibrosis (7 to 30%).

The figures differ because the series differ; both treat submucous fibrosis as premalignant.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of oral ulceration** and the three-week rule.
- **Aphthous versus herpetic ulcers.**
- **Forms of candidiasis,** and why the hyperplastic form is biopsied.
- **Lichen planus,** pemphigus and pemphigoid.
- **Submucous fibrosis:** cause, features, premalignancy and treatment.
- **Dentigerous cyst.**

8.2 Common pitfalls

COMMON PITFALLS

- Prescribing repeated mouthwashes for a persistent single ulcer instead of a biopsy.
- Treating a white patch that will not rub off as thrush.
- Missing an early cancer in a mouth with submucous fibrosis and trismus.

Written for this app from Oxford Textbook of Otolaryngology ch. 54, 57, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 43 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Benign oral lesions and premalignant patches

Lumps in the mouth are common and mostly benign: a fibrous polyp where the cheek is bitten, a pyogenic granuloma on the gum, a mucocele on the lower lip, a ranula under the tongue. Between these and cancer sits a group of patches, white and red, that are not yet malignant but may become so. Recognise each, excise what needs excising, and keep the premalignant patient under watch.

AT A GLANCE

- Leukoplakia is a clinical term: a white patch not otherwise explained.
- Erythroplakia carries far the higher malignant risk.
- Remove the ranula with its sublingual gland.
- Every palatal lump is biopsied: many minor salivary tumours are malignant.

1 CHAPTER ONE Overview

Three groups.

1.1 Classification

- **Benign tumours:** solid (papilloma, fibroma, haemangioma, lymphangioma, torus, pyogenic granuloma, granular cell tumour, minor salivary gland adenoma) and cystic (mucocele, ranula, dermoid).
- **Premalignant lesions and conditions:** leukoplakia, erythroplakia, submucous fibrosis and others.
- **Malignant lesions:** squamous cell carcinoma and the non-squamous cancers (see the oral and oropharyngeal cancer topic).

1.2 The rule

Any lump that is growing, ulcerated, indurated or fixed, or any patch that is red, speckled or nodular, is biopsied.

CHAPTER TWO

2 Solid benign tumours

The common lumps.

2.1 Papilloma

Common, usually in the third to fifth decades, on the palate, uvula, tongue and lips: a small (under 1 cm), white, pedunculated growth with finger-like projections, linked to HPV 6 and 11. Excisional biopsy; recurrence is rare.

2.2 Fibroma (fibroepithelial polyp)

A smooth, mucosa-covered, pedunculated or sessile lump about 1 cm across, firm or soft, along the bite line of the cheek or on the tongue, caused by chronic irritation. Conservative excision.



FIGURE 1 A fibroepithelial polyp in the left cheek (arrow).
Dhingra, fig. 44.1

2.3 Haemangioma

Mostly in children, on the lip, tongue and cheek: a soft, red or blue, compressible lump that may bleed when bitten. Capillary, cavernous and mixed types are described. Lesions present at birth are observed, since many regress; propranolol and steroids help proliferating haemangiomas. Large, persistent or growing lesions are treated by embolisation (alone or before surgery), laser or excision;

sclerotherapy works poorly in diffuse lesions. In older adults, dilated veins (*phlebectasia*) can look similar.

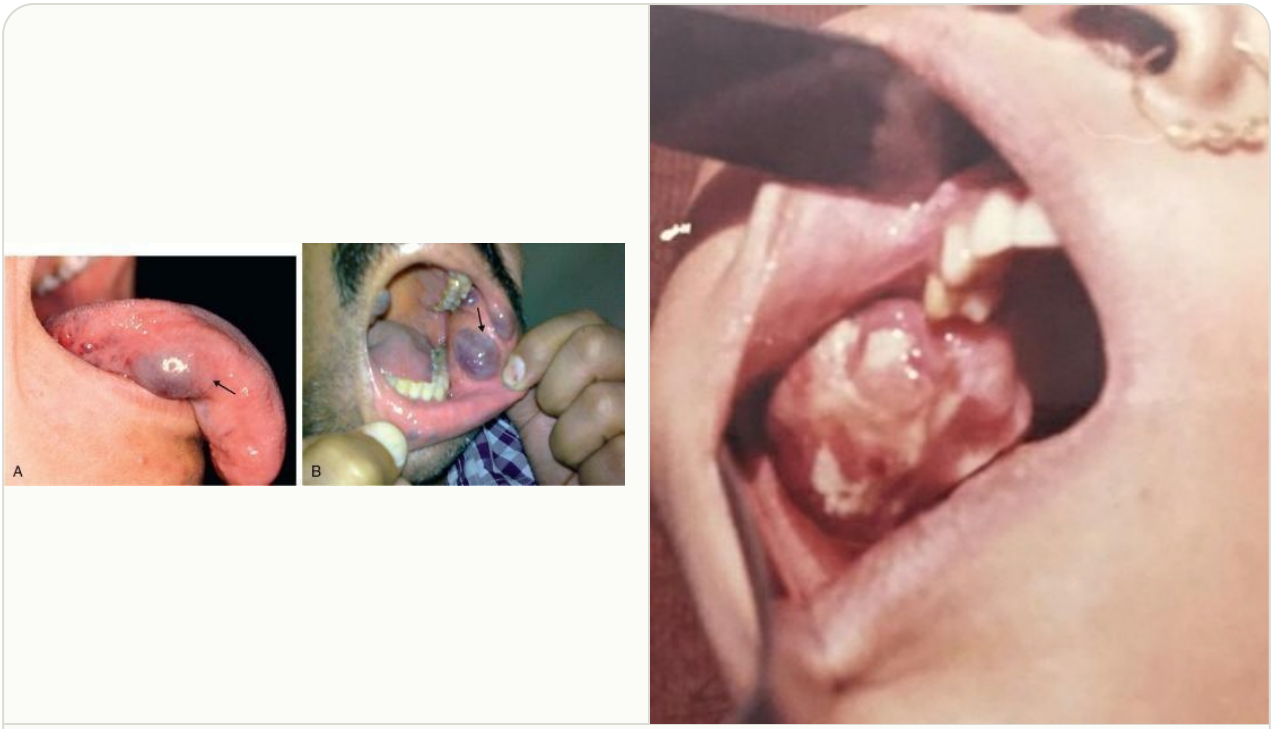


FIGURE 2 First, (A) a haemangioma on the lateral border of the tongue and (B) multiple haemangiomas of the lips, cheek and tongue in a man who bled when he bit them. Second, a haemangioma of the right cheek in a young girl.
Dhingra, figs 44.2 and 44.3

2.4 Lymphangioma

Usually in the anterior two-thirds of the tongue, as a localised compressible swelling or a diffuse enlargement causing *macroglossia*. It does not regress. Small lesions are excised; large symptomatic ones are debulked, since complete removal is rarely possible.

2.5 Torus

A bony outgrowth covered by mucosa: *torus palatinus* in the midline of the hard palate (the commoner), *torus mandibularis* on the lingual surface of the mandible near the premolars, usually on both sides. Harmless; removed only if it interferes with speech, chewing or dentures.

2.6 Pyogenic granuloma

A reactive overgrowth of granulation tissue after trauma or irritation, usually on the anterior gums, sometimes the tongue, lips or cheek: a soft, red to purple lump that bleeds on touch. Excision; recurrence is unlikely after complete removal.



FIGURE 3 A pyogenic granuloma.
Dhingra, fig. 44.4

2.6.1 Pregnancy granuloma

Identical clinically and histologically, arising in the first trimester (*epulis gravidarum*) and regressing after delivery; excised only if it persists, since it recurs if removed during pregnancy.

2.7 Granular cell tumour

A firm submucosal nodule, usually on the dorsum of the tongue, once called a myoblastoma but derived from Schwann cells; its histology shows pseudoepitheliomatous hyperplasia that can be mistaken for carcinoma. Excision. *Congenital epulis* is a related tumour on the gums of newborn girls.

2.8 Minor salivary gland tumours

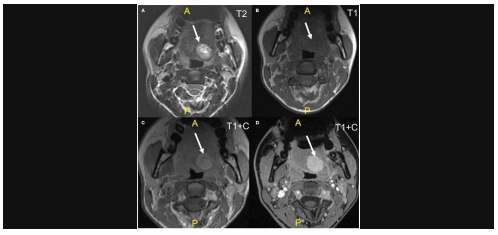
Pleomorphic adenoma is the commonest benign tumour, usually a painless submucosal lump on the hard or soft palate. Because a large proportion of minor salivary gland tumours are malignant, every palatal lump is biopsied, and adenomas are widely excised to prevent recurrence.

2.9 Other benign tumours

Neurofibroma and schwannoma (the tongue is a favoured site), lipoma (soft, yellow, in the cheek or floor), solitary fibrous tumour and verruciform xanthoma, all treated by excision.

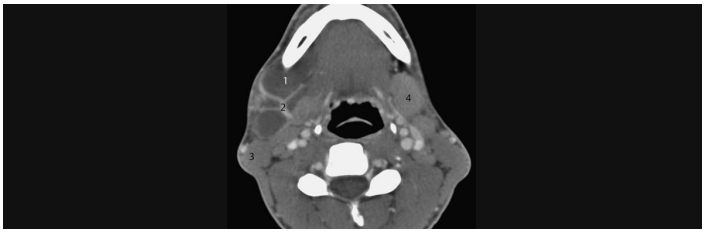
IMAGING IN THE APP 2 cases

Schwannoma of the oral cavity, and lymphangioma.



Schwannomas of the oral cavity and neck

After surgery · 3 images



Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

Congenital · 14 images

3 CHAPTER THREE Cysts

Mucocele, ranula, dermoid.

3.1 Mucocele

A retention or extravasation cyst of a minor salivary gland, almost always on the *lower lip*, after a bite: a soft, bluish, translucent swelling that may burst and refill. Excision with the underlying gland.



FIGURE 4 (A) A mucocele of the lower lip. (B) A mucocele on the ventral surface of the tongue tip.
Dhingra, fig. 44.5

3.2 Ranula

A cystic, translucent, bluish swelling in the floor of mouth to one side of the frenulum, pushing the tongue up, from obstruction of the ducts of the *sublingual gland*. A *plunging ranula* passes through or behind the mylohyoid into the neck and presents as a soft submandibular or submental swelling.



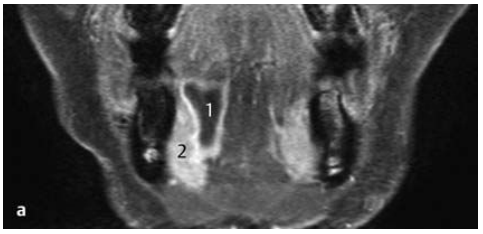
FIGURE 5 A ranula: a translucent swelling under the tongue.
Dhingra, fig. 44.6

3.2.1 Treatment

Dhingra describes excision for small ranulas and marsupialisation for large ones, noting that the thin wall makes complete removal difficult. Recurrence after marsupialisation alone is common; excision of the ranula *with the sublingual gland* through the mouth, protecting the submandibular duct and lingual nerve, is now preferred and also cures most plunging ranulas.

IMAGING IN THE APP 1 case

Ranula of the sublingual gland.



Ranula of the Sublingual Gland

Infection & inflammation · 10 images

OPERATIVE GUIDES IN THEATRE 1 operation

Ranula surgery.

Surgical Management of Ranula

Ranula and sublingual gland excision

3.3 Dermoid

A sublingual dermoid lies above the mylohyoid, in or beside the midline, and shines white through the mucosa, unlike the translucent ranula; a submental dermoid lies below the mylohyoid and bulges under the chin. Excised through the mouth or the neck according to its relation to the muscle.

SWELLINGS IN THE FLOOR OF MOUTH

FEATURE	RANULA	SUBLINGUAL DERMOID	SUBMANDIBULAR CALCULUS
Colour	Bluish, translucent	White, opaque	Normal mucosa over the duct
Consistency	Soft, fluctuant	Doughy	Hard, palpable along the duct

FEATURE	RANULA	SUBLINGUAL DERMOID	SUBMANDIBULAR CALCULUS
Position	One side of the frenulum	Midline or near it	Along Wharton's duct
Other	May plunge into the neck	Present from youth	Pain and swelling with meals

4 CHAPTER FOUR Premalignant lesions

White and red patches.

4.1 Definitions

A *premalignant lesion* is altered tissue in which cancer is more likely to develop (leukoplakia, erythroplakia); a *premalignant condition* is a general state with increased risk (submucous fibrosis, lichen planus, iron deficiency).

PRECANCEROUS LESIONS AND CONDITIONS OF THE ORAL CAVITY

- Leukoplakia and erythroplakia.
- Oral submucous fibrosis.
- Chronic hyperplastic candidiasis.
- Lichen planus (erosive and atrophic).
- Reverse-smoking palate and actinic cheilitis.
- Sideropenic anaemia (Plummer–Vinson syndrome).
- Pigmented lesions and melanosis; discoid lupus.

4.2 Leukoplakia

4.2.1 Definition

The WHO defines leukoplakia as a white patch that cannot be characterised clinically or pathologically as any other disease. It is a clinical term that says nothing about histology; lichen planus, candidiasis, leukoedema and white sponge naevus must be excluded.

4.2.2 Causes and sites

Smoking, tobacco and areca chewing, alcohol (especially with smoking), chronic trauma from teeth or dentures, candida, submucous fibrosis and Plummer–Vinson syndrome. In India the buccal mucosa and commissures are the commonest sites, at the place where the quid is held; the floor of mouth, tongue, gingivobuccal sulcus and lips are also affected. Mostly men in their forties.

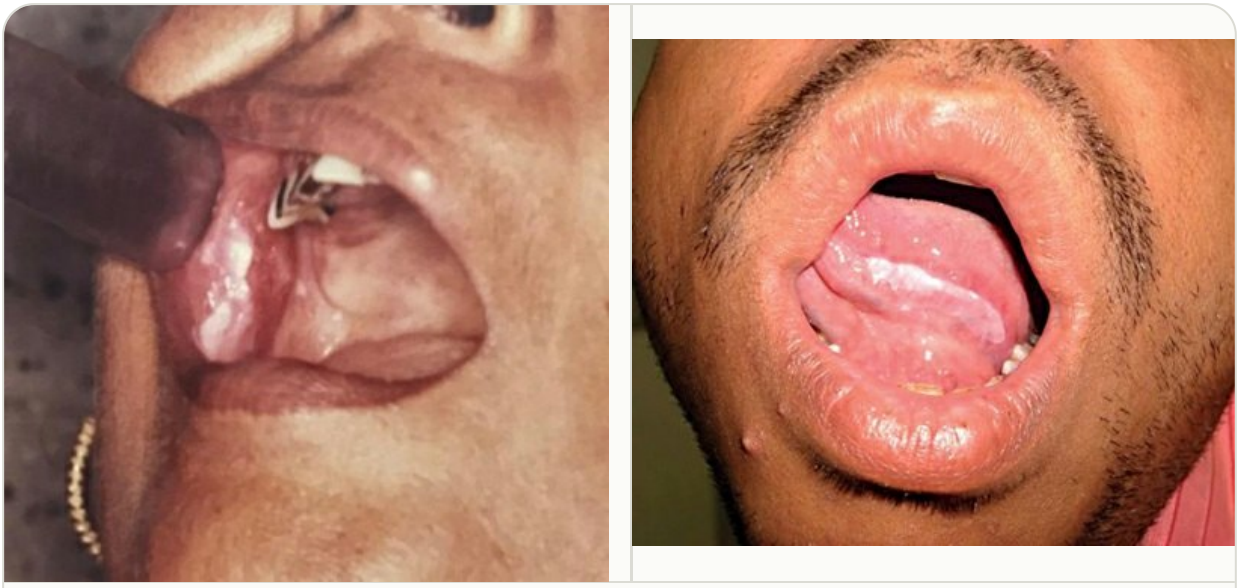


FIGURE 6 First, leukoplakia of the cheek after pan and tobacco use. Second, leukoplakia on the lateral border of the tongue.
Dhingra, figs 44.7 and 44.8

4.2.3 Types

- **Homogeneous:** a uniform, smooth or wrinkled white plaque; the lowest risk.
- **Speckled (nodular):** white nodules on a red base.
- **Erosive (erythroleukoplakia):** white mixed with red, with erosions and fissures.
- **Proliferative verrucous leukoplakia:** multifocal, warty and relentlessly progressive, often in older women without the usual risk factors; a very high risk of carcinoma.



FIGURE 7 Leukoplakia at the site where a quid of tobacco and lime was held.
Dhingra, fig. 44.9

4.2.4 Malignant potential

About a quarter show some dysplasia, and the higher the grade the greater the risk. Quoted rates of malignant change range from under 1% to 17.5%, averaging about 5%; speckled, erosive and verrucous forms, the floor of mouth and ventral tongue, and long follow-up carry the highest risk.

4.2.5 Management

- Stop tobacco, areca and alcohol, and remove sharp teeth: many lesions regress.
- Biopsy, from the reddest or most nodular area, to grade any dysplasia.
- Mild or no dysplasia: review regularly, and excise if it persists or changes.
- Moderate dysplasia or worse: excise, or ablate with laser or cryotherapy.
- Retinoids and beta-carotene can induce remission, but lesions tend to return when they are stopped.

4.3 Erythroplakia

A velvety red patch or plaque that cannot be characterised as any other disease, often slightly depressed with a sharp edge, on the floor of mouth, ventral tongue, soft palate and gingivobuccal sulcus. The redness is the vascular submucosa shining through thin, poorly keratinised epithelium. Most already show severe dysplasia, carcinoma in situ or invasive cancer when first seen, and the risk is many times that of leukoplakia. Excision biopsy and close follow-up.

RED IS WORSE THAN WHITE

Erythroplakia and the red parts of a speckled leukoplakia are where carcinoma is found. Biopsy the red.

4.4 Pigmented lesions

Benign melanotic macules and naevi are common; about a quarter of mucosal melanomas look benign at first, so a new, enlarging or irregular pigmented lesion on the palate or gum is biopsied.

4.5 Chemoprevention

Vitamin A, beta-carotene, vitamin E and retinoids such as 13-cis-retinoic acid can make leukoplakia regress in a quarter to a half of patients and may reduce second primary cancers, but the benefit lasts only as long as the drug is taken; stopping tobacco remains the most effective prevention.

5 CHAPTER FIVE Clinical comparison

Telling them apart.

ORAL WHITE PATCHES

LESION	WIPES OFF?	CLUE
Thrush	Yes, leaving a red base	Infant, antibiotics, steroids, diabetes
Leukoedema	Disappears on stretching	Bilateral buccal mucosa, a normal variant
Lichen planus	No	Lace-like Wickham striae, both cheeks
Hairy leukoplakia	No	Lateral tongue borders, HIV
Hyperplastic candidiasis	No	Commissures; responds partly to antifungals
Leukoplakia	No	Tobacco site; a diagnosis of exclusion

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition of leukoplakia,** its types and malignant potential.
- **Erythroplakia** and why it is worse.
- **List of premalignant lesions and conditions.**
- **Ranula:** origin, the plunging type, and treatment by excision of the sublingual gland.
- **Mucocele, pyogenic granuloma** and **torus.**

6.2 Common pitfalls

COMMON PITFALLS

- Using "leukoplakia" as a histological diagnosis.
- Marsupialising a ranula and leaving the sublingual gland.
- Biopsying the white part of a speckled leukoplakia instead of the red.
- Excising a pregnancy granuloma during pregnancy.

Written for this app from Oxford Textbook of Otolaryngology ch. 57, Oxford Specialist Handbook ch. 6, Dhingra ch. 44 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Applied anatomy and physiology of the pharynx

The pharynx is a shared corridor: air passes through it to the larynx and food through it to the oesophagus, and many times a minute it switches between the two without error. Its walls hold the lymphoid ring that guards both portals, its spaces carry infection down into the chest, and its three parts each give rise to their own cancers. The anatomy explains the symptoms: the earache of a tonsil cancer, the nasal voice of a palatal palsy, the pouch at Killian's dehiscence.

AT A GLANCE

- Fossa of Rosenmuller: commonest site of nasopharyngeal carcinoma.
- Killian's dehiscence lies between thyropharyngeus and cricopharyngeus.
- All pharyngeal muscles are vagal except stylopharyngeus (IX) and tensor palati (V3).
- The piriform sinus drains richly to the neck.

1 CHAPTER ONE The pharynx in general

A fibromuscular funnel.

1.1 Extent

A conical fibromuscular tube 12 to 14 cm long, from the base of the skull (basisphenoid and basiocciput) to the lower border of the cricoid cartilage at the level of the sixth cervical vertebra, where it becomes the oesophagus. It is about 3.5 cm wide at the top and narrows to 1.5 cm at the pharyngo-oesophageal junction, the narrowest part of the gut after the appendix, which is why swallowed foreign bodies lodge there.

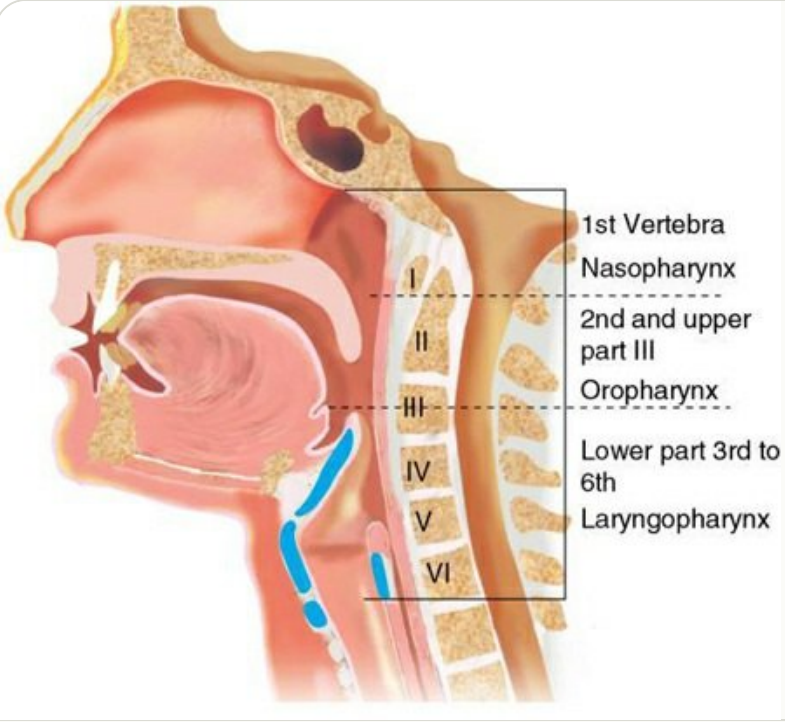
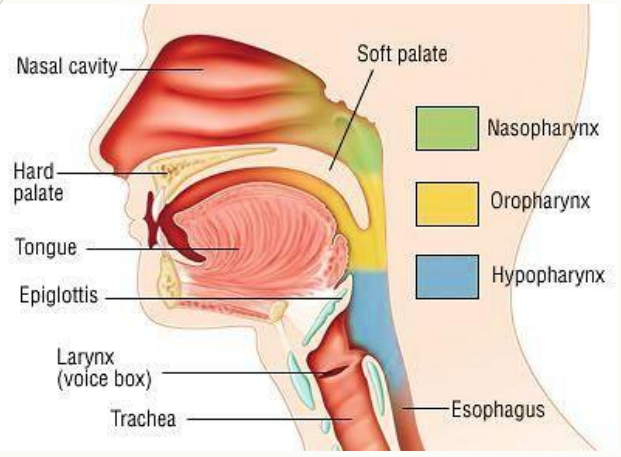


FIGURE 1 Divisions of the pharynx and the vertebrae related to the posterior wall: nasopharynx (C1), oropharynx (C2 and upper C3), and laryngopharynx (C3 to C6).

Dhingra, fig. 47.3



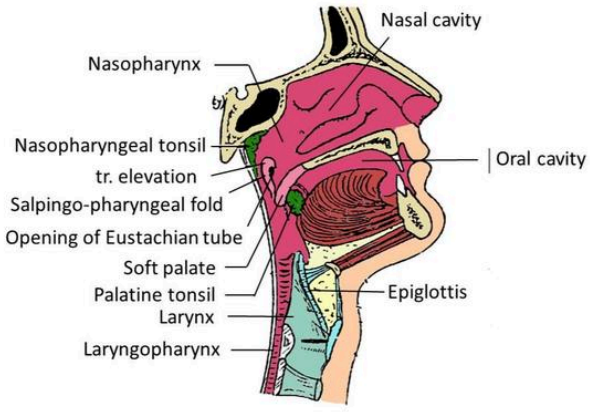


FIGURE 2 The divisions of the pharynx, and its cavity in sagittal section.

Anatomy atlas · ENT Steps p. 194, Doss p. 168

1.2 Layers of the wall

From within outwards:

- **Mucous membrane:** ciliated columnar in the nasopharynx, stratified squamous below; continuous with the lining of the Eustachian tubes, nose, mouth, larynx and oesophagus.

- **Pharyngobasilar fascia (pharyngeal aponeurosis):** thick at the skull base, where it fills the gap above the superior constrictor, thin below.
- **Muscular coat:** an outer circular layer (the three constrictors) and an inner longitudinal layer (stylopharyngeus, salpingopharyngeus, palatopharyngeus).
- **Buccopharyngeal fascia:** covers the constrictors and continues forwards over the buccinator.

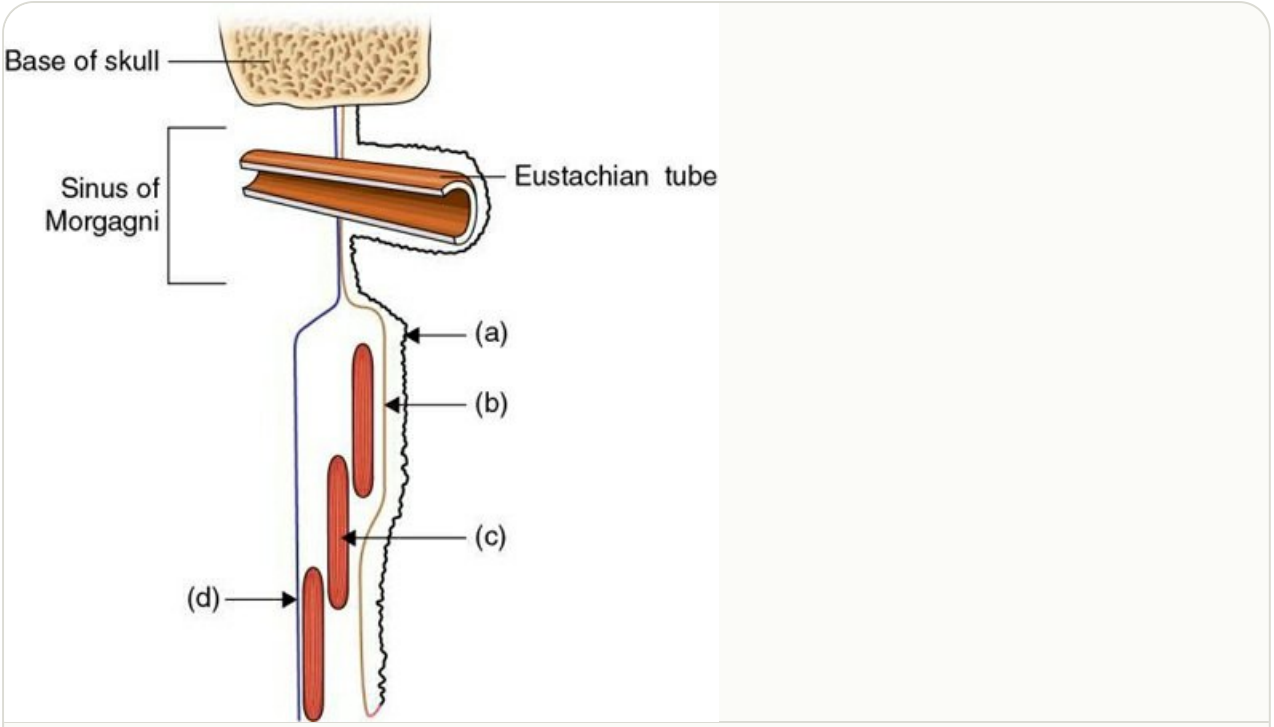


FIGURE 3 Structure of the pharyngeal wall: (a) mucous membrane, (b) pharyngobasilar fascia, © muscular coat, (d) buccopharyngeal fascia; the sinus of Morgagni lies above the superior constrictor.

Dhingra, fig. 47.1

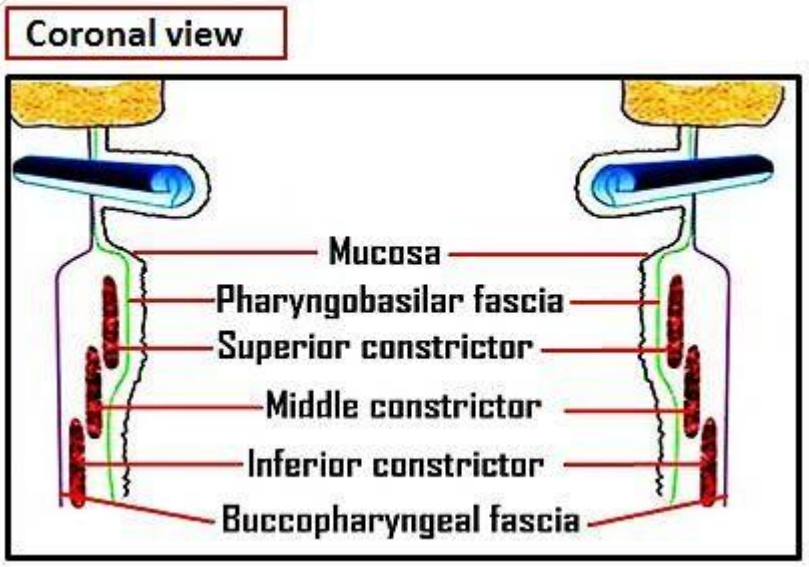


FIGURE 4 The layers of the pharyngeal wall.
Anatomy atlas · ENT Steps p. 197

2 CHAPTER TWO Muscles and gaps

Three constrictors, three elevators.

2.1 Constrictors

Each passes backwards to insert into the midline raphe, the lower overlapping the upper like stacked cups. All are supplied by the pharyngeal plexus (vagus).

- **Superior:** from the pterygoid hamulus, pterygomandibular raphe and mandible.
- **Middle:** from the hyoid bone and stylohyoid ligament.
- **Inferior:** from the thyroid and cricoid cartilages, in two parts: the oblique *thyropharyngeus* and the transverse *cricopharyngeus*, which is the upper oesophageal sphincter, tonically closed at rest and relaxing on swallowing.

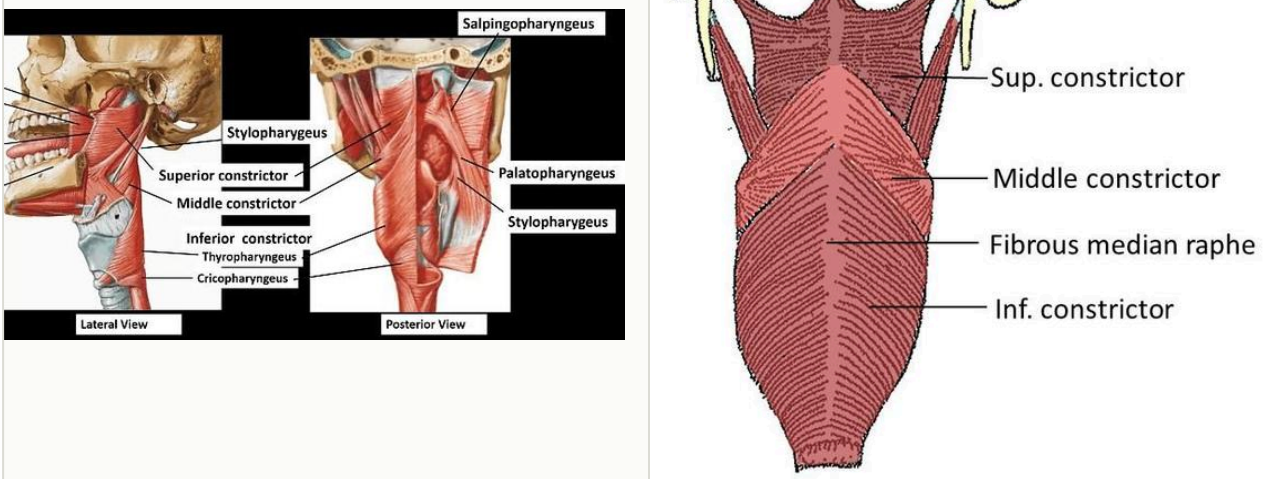


FIGURE 5 The pharyngeal constrictors in lateral and posterior views.
Anatomy atlas · ENT Steps p. 197, Doss p. 166

2.2 Longitudinal muscles

- **Stylopharyngeus:** from the styloid process; the only pharyngeal muscle supplied by the glossopharyngeal nerve.
- **Salpingopharyngeus:** from the Eustachian tube cartilage; raises the salpingopharyngeal fold.
- **Palatopharyngeus:** from the palate; forms the posterior pillar and, with fibres of the superior constrictor, *Passavant's ridge*.

All elevate the pharynx and larynx during swallowing.

2.3 Gaps and what passes through them

- **Above the superior constrictor (sinus of Morgagni):** the Eustachian tube, levator and tensor veli palatini, and the ascending palatine artery.
- **Between superior and middle constrictors:** stylopharyngeus and the glossopharyngeal nerve.
- **Between middle and inferior constrictors:** the internal laryngeal nerve and superior laryngeal vessels.
- **Below the inferior constrictor:** the recurrent laryngeal nerve and inferior laryngeal vessels.

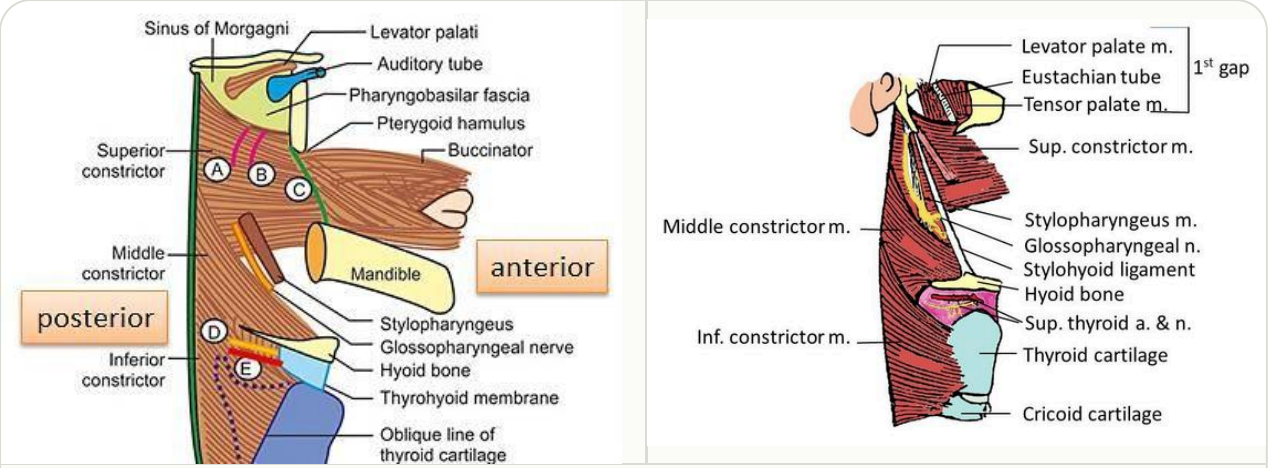


FIGURE 6 The gaps between the constrictors and the structures passing through them.
Anatomy atlas · ENT Steps p. 204, Doss p. 167

2.4 Killian's dehiscence

A weak triangle in the posterior wall between thyropharyngeus above and cricopharyngeus below. Mucosa herniates through it to form a *pharyngeal pouch* (Zenker diverticulum), and an oesophagoscope can perforate it, hence its old name, the *gateway of tears*. Below cricopharyngeus lie Laimer's triangle and the lateral Killian–Jamieson area.

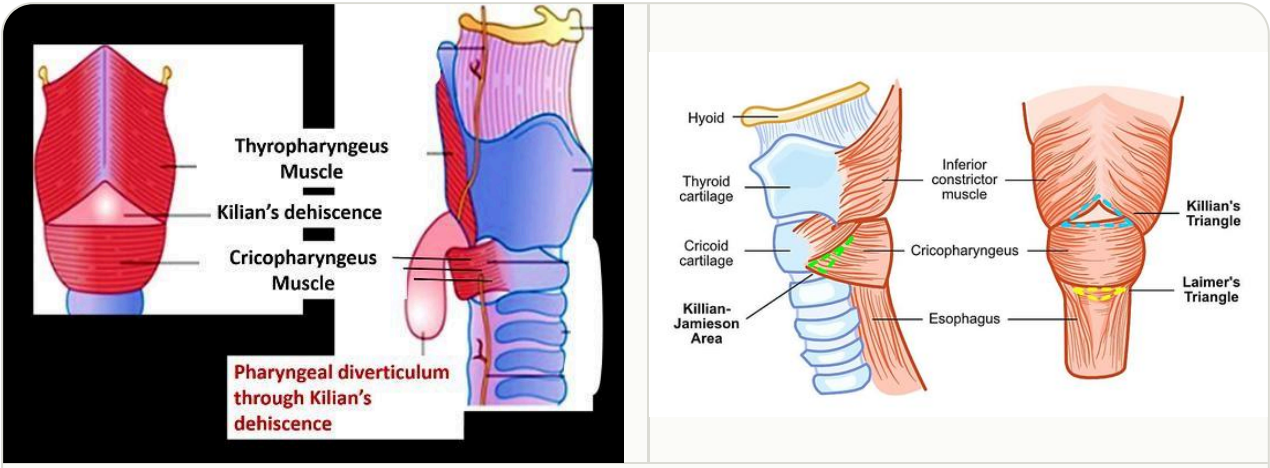


FIGURE 7 Killian's dehiscence and the pharyngeal pouch, and Killian's and Laimer's triangles.
Anatomy atlas · ENT Steps p. 198, ENT Steps p. 225

3 CHAPTER THREE Waldeyer's ring

Sentinels at the portals.

3.1 Components

Lymphoid tissue lies throughout the pharyngeal submucosa and gathers into masses that ring the entrances of the air and food passages:

- **Nasopharyngeal tonsil (adenoids)** at the junction of roof and posterior wall.
- **Palatine tonsils** between the pillars.
- **Lingual tonsil** on the base of tongue.
- **Tubal tonsils** around the Eustachian tube openings.
- **Lateral pharyngeal bands** behind the posterior pillars.
- **Nodules** scattered on the posterior wall.

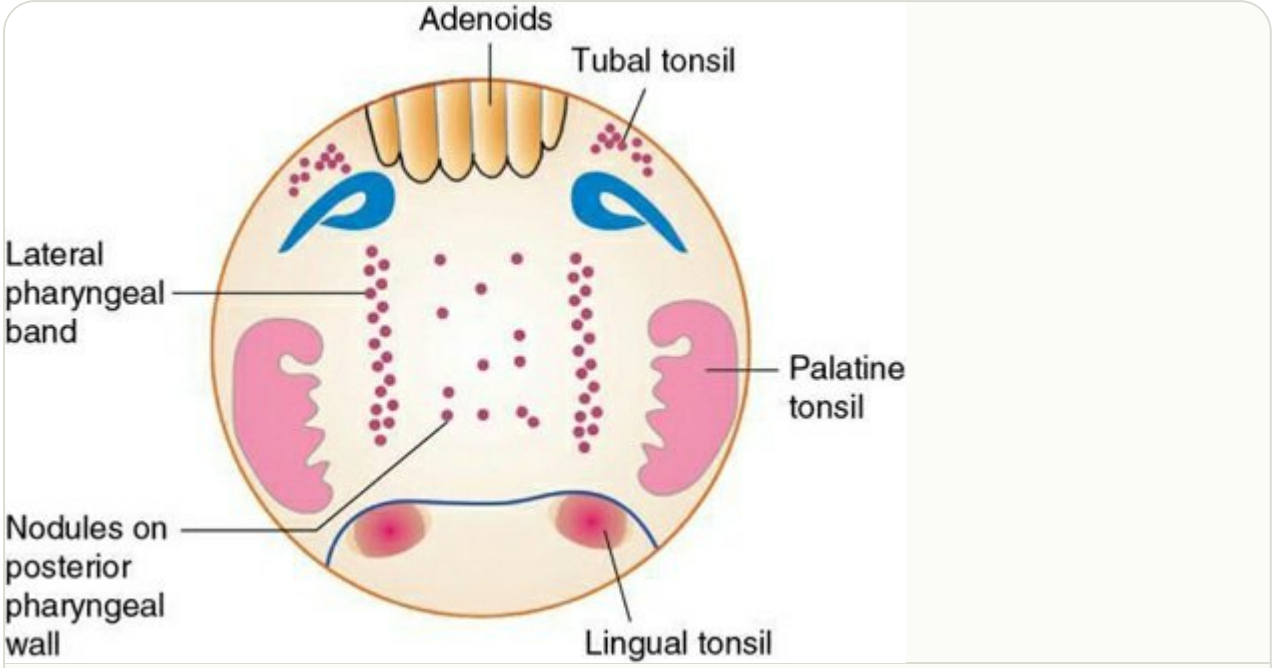


FIGURE 8 Waldeyer's ring.
Dhingra, fig. 47.2

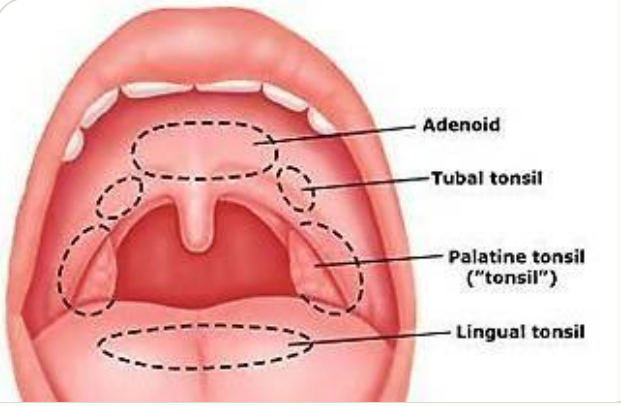


TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

FIGURE 9 Waldeyer's ring seen through the mouth, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 199, ENT Steps p. 208

3.2 Immunology

B-lymphocytes in the germinal centres produce secretory IgA, T-lymphocytes in the parafollicular areas give cell-mediated immunity, and plasma cells secrete IgM and IgG against organisms that enter the crypts. The tissue is most active in childhood. No lasting immune deficit has been shown after removal of the tonsils and adenoids.

4 CHAPTER FOUR Nasopharynx

Behind the nose.

4.1 Boundaries

Behind the nasal cavities, from the skull base to the level of the hard palate.

- **Roof and posterior wall:** basisphenoid and basiocciput, curving down onto the arch of the atlas.
- **Floor:** the upper surface of the soft palate in front; behind it the *nasopharyngeal isthmus*, closed by the palate in swallowing and speech.
- **Anterior wall:** the choanae and the posterior ends of the turbinates.
- **Lateral wall:** the Eustachian tube opening, about 1.25 cm behind the inferior turbinate, capped above and behind by the *torus tubarius*; behind the torus, the *fossa of*

Rosenmüller, the commonest site of nasopharyngeal carcinoma; and the salpingopharyngeal fold descending from it.

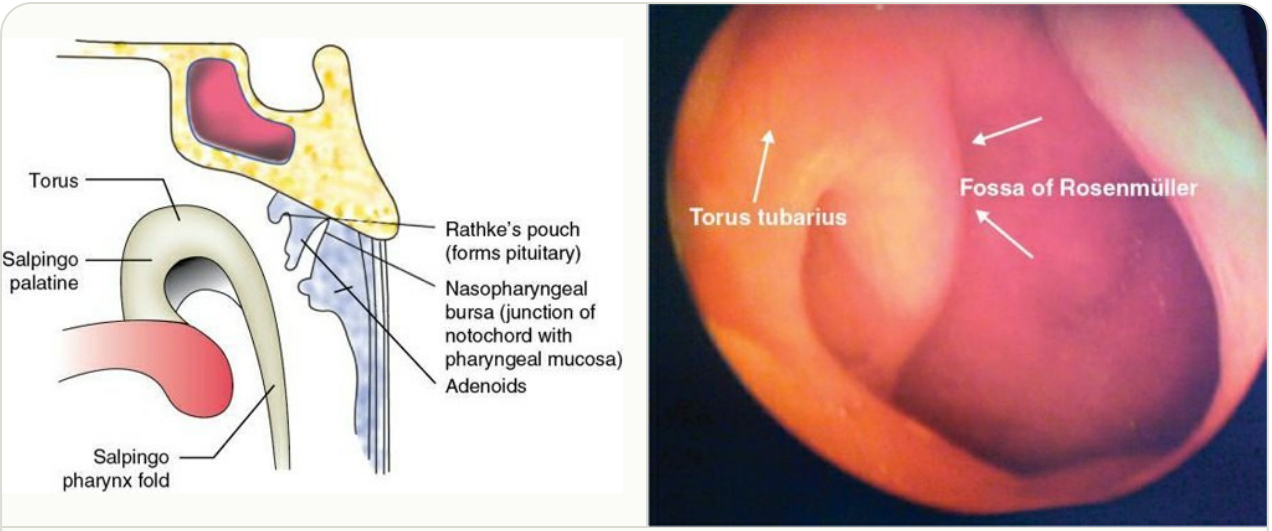


FIGURE 10 First, the nasopharynx in sagittal section with the adenoids, nasopharyngeal bursa and Rathke's pouch. Second, an endoscopic view of the torus tubarius and the fossa of Rosenmüller behind it.

Dhingra, figs 47.4 and 47.5

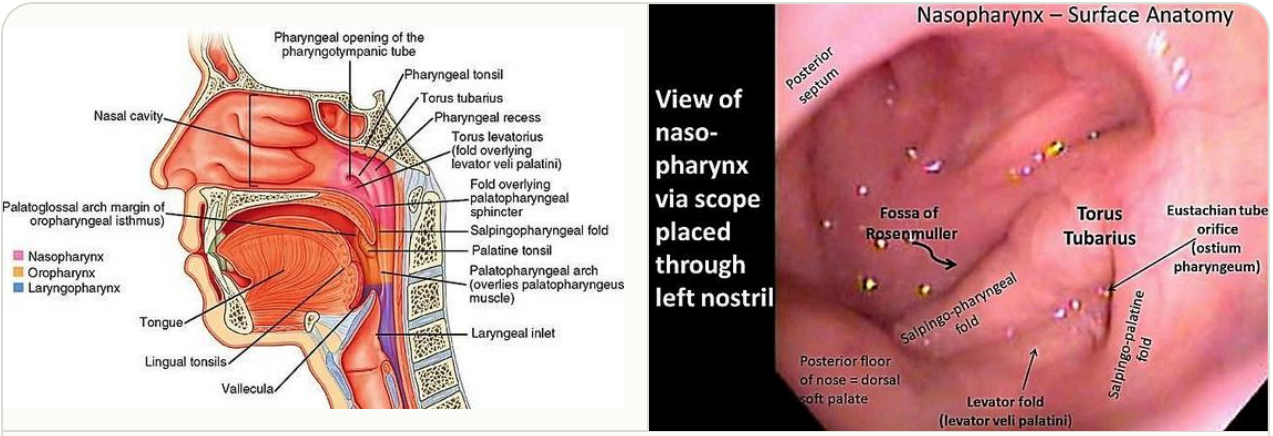


FIGURE 11 The lateral wall of the nasopharynx with the torus tubarius and fossa of Rosenmüller, and an endoscopic view.

Anatomy atlas · ENT Steps p. 203

4.2 Special features

- **Adenoids:** grow until about six years and then atrophy; the overlying mucosa is thrown into radiating folds.
- **Nasopharyngeal bursa:** a midline recess within the adenoid, marking where the notochord touched the pharyngeal endoderm; infection causes postnasal discharge, and an abscess or cyst is *Thornwaldt disease*.

- **Rathke's pouch:** a dimple above the adenoids, remnant of the invagination that forms the anterior pituitary; a craniopharyngioma can arise along its course.
- **Passavant's ridge:** a ridge around the isthmus against which the soft palate closes.

4.3 Lymphatic drainage

To the retropharyngeal (node of Rouvière) and parapharyngeal nodes, then the upper deep cervical and spinal accessory (posterior triangle) chains, often to both sides.

4.4 Functions

Air conduit; ventilation and pressure equalisation of the middle ear; velopharyngeal closure in swallowing, vomiting and speech; a resonator for the voice (hyponasal with obstruction, hypernasal with velopharyngeal insufficiency); and drainage of nasal mucus.

5 CHAPTER FIVE Oropharynx

Behind the mouth.

5.1 Boundaries

From the plane of the hard palate to the plane of the hyoid bone, opening forwards into the mouth through the oropharyngeal isthmus (soft palate above, tongue below, anterior pillars at the sides).

- **Anterior wall:** the base of tongue with the lingual tonsil, and the *valleculae*, each between the median and lateral glossoepiglottic folds, a site for retention cysts.
- **Lateral wall:** the palatine tonsil in its fossa between the anterior (palatoglossal) and posterior (palatopharyngeal) pillars.
- **Posterior wall:** in front of C2 and C3, with the retropharyngeal space behind.
- **Superior wall:** the undersurface of the soft palate.

The upper border of the epiglottis and the pharyngoepiglottic folds separate it from the hypopharynx.

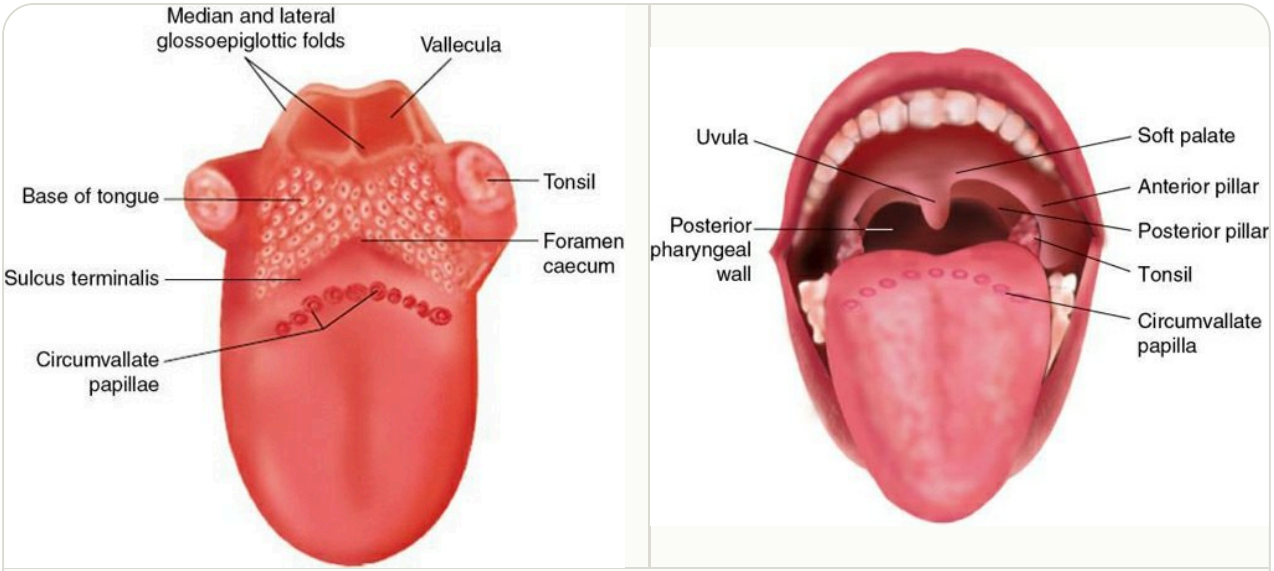


FIGURE 12 First, the base of tongue and valleculae. Second, the structures seen in the oropharynx: uvula, soft palate, anterior and posterior pillars, tonsil and posterior pharyngeal wall.
Dhingra, figs 47.6 and 47.7

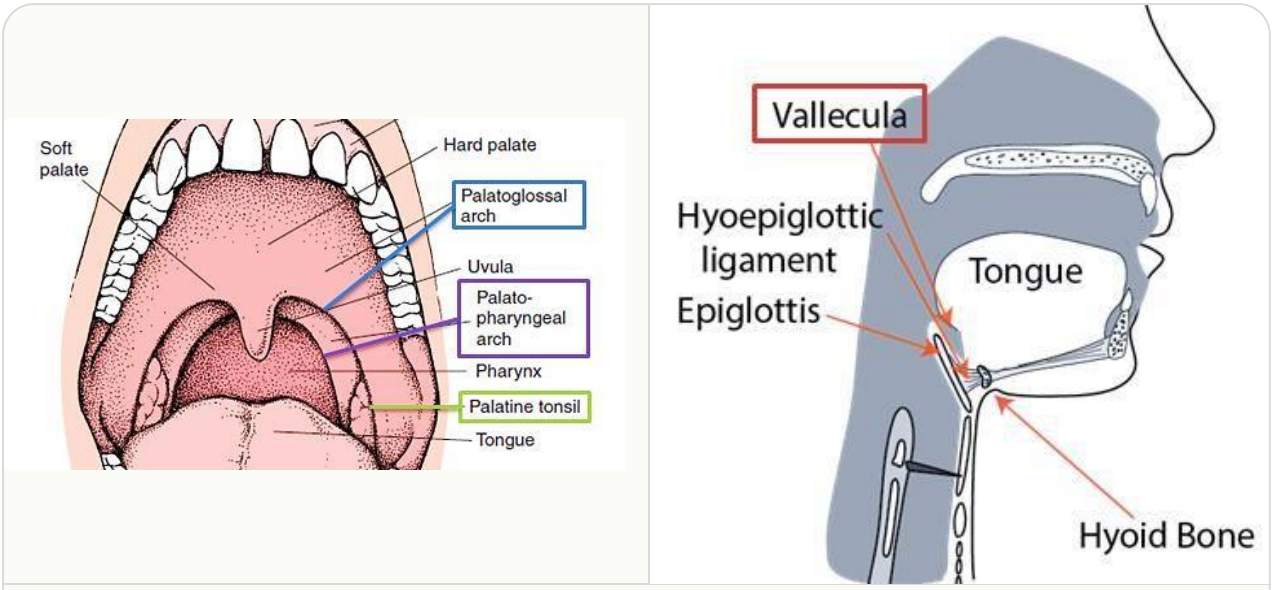


FIGURE 13 The oropharynx through the open mouth, and the vallecula with the hyoepiglottic ligament.
Anatomy atlas · ENT Steps p. 210

5.2 Soft palate

A mobile flap hung from the back of the hard palate. When it rises it seals the nasopharynx from the oropharynx; when it falls against the tongue it seals the

mouth, so we can chew and breathe together.

MUSCLES OF THE SOFT PALATE

MUSCLE	ACTION	NERVE
Levator veli palatini	Raises the palate	Vagus (pharyngeal plexus)
Tensor veli palatini	Tightens the palate; opens the Eustachian tube	Mandibular nerve (V3)
Palatoglossus	Narrows the isthmus; forms the anterior pillar	Vagus
Palatopharyngeus	Raises the pharynx; forms the posterior pillar	Vagus
Musculus uvulae	Shortens the uvula	Vagus

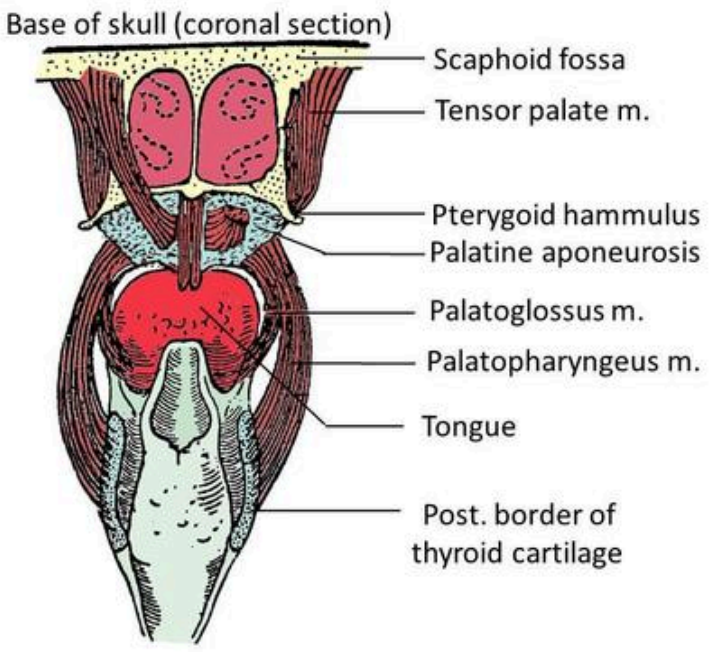


FIGURE 14 The muscles of the soft palate.
Anatomy atlas · Doss p. 181

THE UVULA IN A VAGAL PALSY

With a unilateral vagal (pharyngeal plexus) palsy the uvula and palate are pulled towards the healthy side on saying "aah". The tensor alone is supplied by the trigeminal nerve.

5.3 The palatine tonsil

Its anatomy, blood supply and bed are described with tonsillitis; it drains to the *jugulodigastric (tonsillar) node*.

5.4 Lymphatic drainage

To the upper jugular chain, especially the jugulodigastric node; the soft palate, pharyngeal walls and base of tongue also drain through the retropharyngeal and parapharyngeal nodes, and the base of tongue drains to both sides.

5.5 Functions

A passage for air and food, the pharyngeal phase of swallowing, shaping of speech sounds, taste (taste buds on the base of tongue, soft palate, pillars and posterior wall), and immune defence through Waldeyer's ring.

6 CHAPTER SIX Hypopharynx

Behind and beside the larynx.

6.1 Boundaries and subsites

From the plane of the hyoid bone to the lower border of the cricoid, in front of C3 to C6.

- **Piriform sinus (fossa):** on each side of the larynx, from the pharyngoepiglottic fold to the oesophagus; bounded laterally by the thyrohyoid membrane and thyroid cartilage, medially by the aryepiglottic fold, arytenoid and cricoid. It is the lateral food channel where fish bones lodge. The *internal laryngeal nerve* runs under its mucosa, so it can be anaesthetised there, and through it a piriform cancer refers pain to the ear.
- **Postcricoid region:** the anterior wall behind the cricoid lamina; the site of carcinoma in women with Plummer–Vinson syndrome.
- **Posterior pharyngeal wall:** from the hyoid to the level of the cricoarytenoid joints.

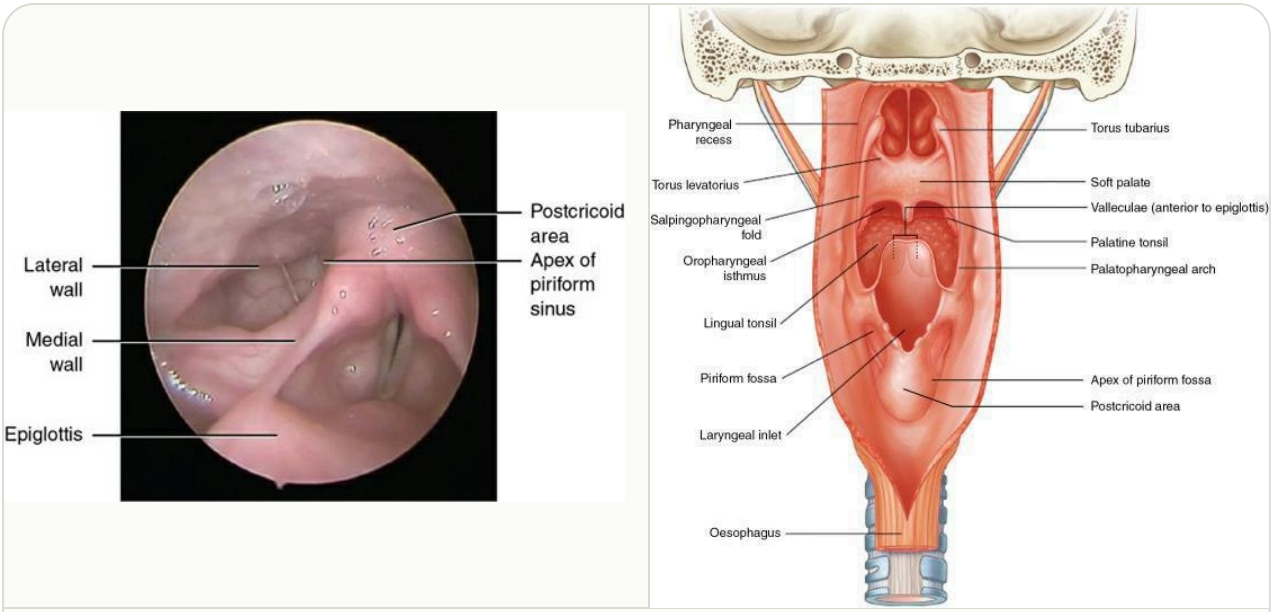


FIGURE 15 First, the right piriform fossa with its lateral and medial walls, apex and the postcricoid area. Second, the pharynx opened from behind, showing the nasopharynx, oropharynx and laryngopharynx.

Dhingra, figs 47.8 and 47.9

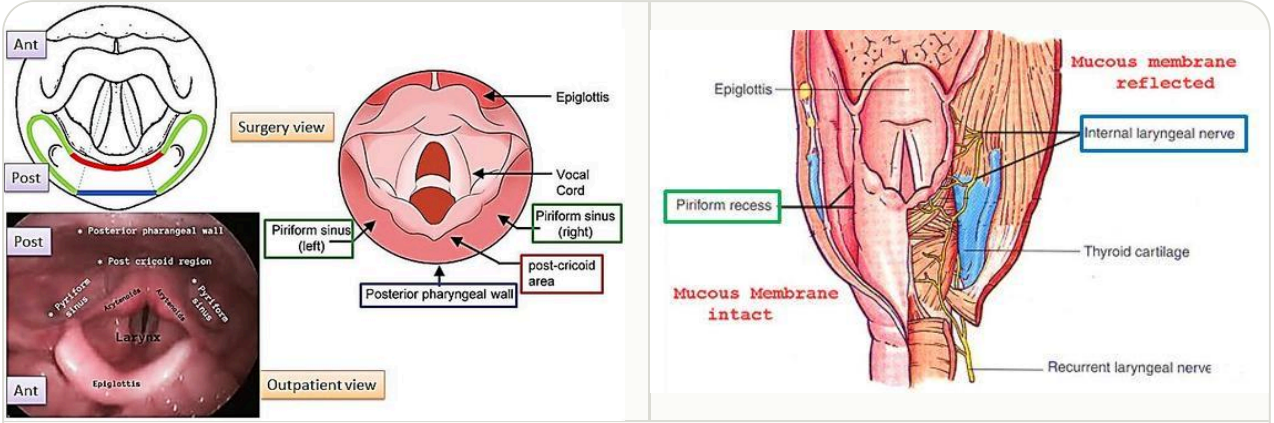


FIGURE 16 The three subsites of the hypopharynx, and the internal laryngeal nerve under the mucosa of the piriform fossa.

Anatomy atlas · ENT Steps p. 220, ENT Steps p. 221

6.2 Lymphatic drainage

The piriform sinus is richly supplied; its lymphatics pierce the thyrohyoid membrane to the upper jugular nodes, which is why its cancers so often present with neck nodes. The posterior wall drains to the retropharyngeal and parapharyngeal nodes; the postcricoid region to the parapharyngeal, paratracheal and supraclavicular nodes.

7 CHAPTER SEVEN
Nerves, vessels and spaces

The supply lines.

7.1 Nerve supply

- **Motor:** the pharyngeal plexus (vagus) supplies all pharyngeal and palatal muscles except stylopharyngeus (glossopharyngeal) and tensor veli palatini (mandibular nerve).
- **Sensory:** nasopharynx by the maxillary nerve (V2); oropharynx by the glossopharyngeal nerve, the afferent limb of the gag reflex; hypopharynx by the internal laryngeal branch of the vagus.

REFERRED OTALGIA FROM THE THROAT

The glossopharyngeal nerve supplies the middle ear through Jacobson's nerve, and the vagus supplies the ear canal through Arnold's nerve. Tonsillitis, a quinsy, a post-tonsillectomy throat and cancers of the tonsil, base of tongue or piriform sinus all present with earache.

7.2 Blood supply

The ascending pharyngeal artery, the tonsillar and ascending palatine branches of the facial artery, branches of the maxillary and lingual arteries, and below, the inferior thyroid artery. Veins form a pharyngeal plexus draining to the pterygoid plexus and internal jugular vein.

7.3 Pharyngeal spaces

- **Retropharyngeal space:** behind the pharynx, in front of the prevertebral fascia, from the skull base to the bifurcation of the trachea; contains the retropharyngeal nodes, which regress after childhood.
- **Parapharyngeal space:** beside the pharynx, an inverted pyramid from the skull base to the hyoid; its post-styloid part contains the carotid vessels, internal jugular vein, the last four cranial nerves and the sympathetic chain.

Their infections are described with deep neck space infection.

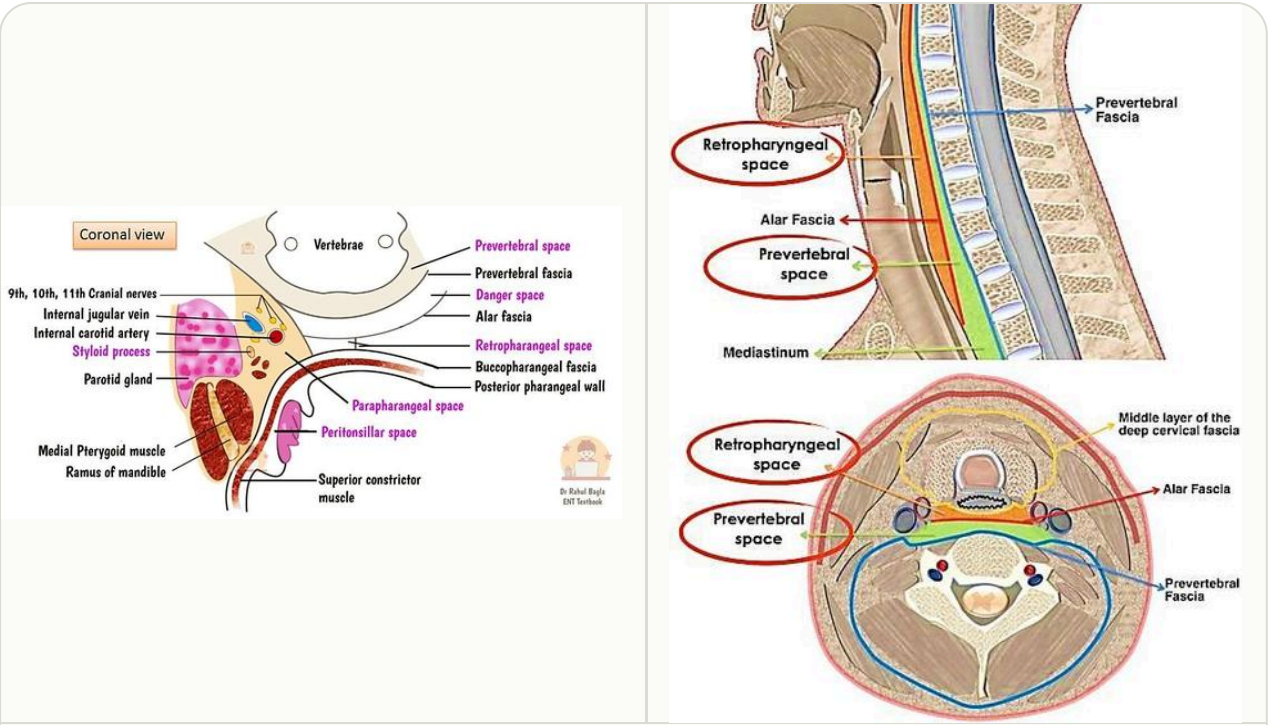


FIGURE 17 The parapharyngeal and retropharyngeal spaces, and the retropharyngeal, danger and prevertebral spaces.

Anatomy atlas · ENT Steps p. 198, ENT Steps p. 277

8 CHAPTER EIGHT Swallowing

Three phases.

8.1 Oral phase

Voluntary. Food is chewed, mixed with saliva and formed into a bolus that the tongue pushes back. When the bolus touches the anterior pillars, receptors supplied by the glossopharyngeal and vagus nerves trigger the pharyngeal phase.

8.2 Pharyngeal phase

Involuntary and lasting under a second, while breathing stops:

1. The soft palate rises and meets Passavant's ridge, closing off the nasopharynx.

2. The vocal folds, false folds and arytenoids close the glottis, and the epiglottis tilts down.
3. The larynx and hyoid rise and move forwards under the suprahyoid and thyrohyoid muscles.
4. The tongue base drives back and the constrictors contract from above down.
5. Cricopharyngeus relaxes and is pulled open by the forward movement of the larynx, the most important factor in opening the sphincter.

8.3 Oesophageal phase

A primary peristaltic wave carries the bolus to the stomach in three to six seconds, and the lower oesophageal sphincter relaxes.

SWALLOWING NERVES

Sensation by V, IX and X; muscles by V (jaw, mylohyoid, tensor), VII (lips, face), IX (stylopharyngeus), X (palate, pharynx, larynx, cricopharyngeus) and XII (tongue). Incoordination between pharyngeal contraction and cricopharyngeal relaxation is thought to underlie the pharyngeal pouch.

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Layers of the pharyngeal wall** and the sinus of Morgagni.
- **Killian's dehiscence.**
- **Waldeyer's ring.**
- **Boundaries of the nasopharynx** and the fossa of Rosenmüller.
- **Subsites of the hypopharynx** and the internal laryngeal nerve.
- **Nerve supply of the pharynx** and referred otalgia.

- **Phases of swallowing.**

9.2 Common pitfalls

COMMON PITFALLS

- Forgetting that stylopharyngeus (IX) and tensor veli palatini (V3) are the exceptions to vagal supply.
- Placing the valleculae in the hypopharynx; they belong to the oropharynx.
- Looking only at the ear in a patient with otalgia and a normal drum.

Written for this app from Oxford Specialist Handbook ch. 6, Dhingra ch. 47 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Adenoids and nasopharyngitis

A child who breathes through the mouth, snores, has a constant wet nose and a dull, hearing-impaired look is the classic child with adenoids. Size alone does not matter; what matters is the adenoid relative to the nasopharynx it fills, and what it does to the nose, the ears and sleep. Most children outgrow the problem; those who do not, and whose ears or breathing suffer, are helped by a short operation.

AT A GLANCE

- Adenoids have no crypts and no capsule.
- Nasal, aural and general symptoms; adenoid facies.
- Submucous cleft is a contraindication to adenoidectomy.
- Grisel syndrome and atlantoaxial instability in Down syndrome.

1 CHAPTER ONE The adenoid

The nasopharyngeal tonsil.

1.1 Anatomy

The adenoid lies at the junction of the roof and posterior wall of the nasopharynx, as vertical ridges of lymphoid tissue separated by deep clefts. It is covered by ciliated columnar, stratified squamous and transitional epithelium and, unlike the palatine tonsil, has *no crypts and no capsule*.



FIGURE 1 An adenoid mass after removal with a curette: ridges of lymphoid tissue separated by deep clefts.

Dhingra, fig. 48.1

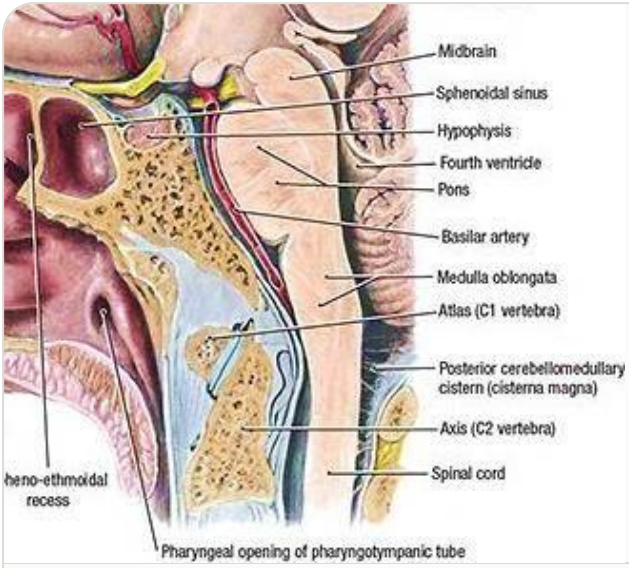


TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

FIGURE 2 The nasopharynx in sagittal section, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 200, ENT Steps p. 208

1.2 Growth and regression

Present at birth, it enlarges physiologically to a peak at about three to seven years, regresses from puberty and has largely gone by twenty; allergy or infection can

keep it large beyond fifteen.

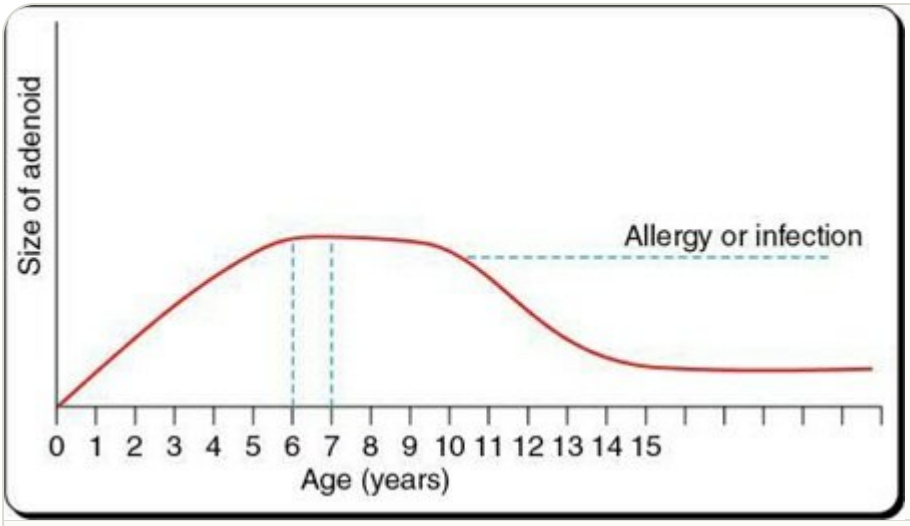


FIGURE 3 Size of the adenoid with age: maximum at six to seven years, regressing at puberty; persistence beyond fifteen with allergy or infection.

Dhingra, fig. 48.2

1.3 Supply

- **Arteries:** ascending palatine (facial), ascending pharyngeal, the pharyngeal branch of the maxillary artery, and the ascending cervical branch of the inferior thyroid.
- **Lymph:** to the upper jugular nodes, directly or through the retropharyngeal and parapharyngeal nodes.
- **Nerves:** glossopharyngeal and vagus, which explains the earache of adenoiditis.

2 CHAPTER TWO Adenoid hypertrophy

Too big for the space.

2.1 Causes

Physiological enlargement in childhood, a general tendency to lymphoid hyperplasia, recurrent rhinitis, sinusitis and tonsillitis, and allergy.

2.2 Clinical features

2.2.1 Nasal

- **Obstruction:** the commonest symptom, with mouth breathing; infants struggle to feed because they cannot breathe and suck at once, and fail to thrive.
- **Discharge:** secretions cannot drain back through the blocked choanae, and chronic rhinitis adds to them: the "wet, bubbly nose".
- **Sinusitis:** chronic maxillary sinusitis and adenoid infection feed each other.
- **Epistaxis** when the adenoid is inflamed.
- **Voice:** hyponasal (*rhinolalia clausa*), losing the nasal consonants.

2.2.2 Aural

- **Eustachian tube obstruction:** a retracted drum and conductive deafness.
- **Recurrent acute otitis media** from spread of infection up the tube.
- **Otitis media with effusion:** the adenoid is an important cause; the fluctuating size gives a fluctuating hearing loss.
- **Chronic suppurative otitis media** that will not settle while the adenoid is infected.

2.2.3 General

- **Adenoid facies:** an elongated face with a dull expression, open mouth, prominent crowded upper teeth, short upper lip, pinched nose from disuse of the alae, and a high-arched palate because the tongue no longer moulds it.
- **Sleep-disordered breathing:** snoring and obstructive sleep apnoea, with daytime tiredness and poor concentration (*aprosexia*).
- **Pulmonary hypertension and cor pulmonale** with long-standing severe obstruction.



FIGURE 4 Adenoid facies: the open mouth of the mouth breather.
Dhingra, fig. 48.3

SIZE IS RELATIVE

A large adenoid in a large nasopharynx may cause no symptoms; a modest one in a small nasopharynx can block it completely. Judge the airway, not the adenoid.

2.3 Diagnosis

- **Flexible nasendoscopy** in a cooperative child: the best view of the adenoid, the choanae and the tube openings, without radiation.
- **Lateral soft tissue radiograph** of the nasopharynx: shows the adenoid and the remaining airway; useful in children who will not tolerate a scope, though crying lifts the palate and can exaggerate the size.
- Mirror examination is possible in some older children.
- Examine the whole nose for other causes of obstruction, and the ears with tympanometry and audiometry.

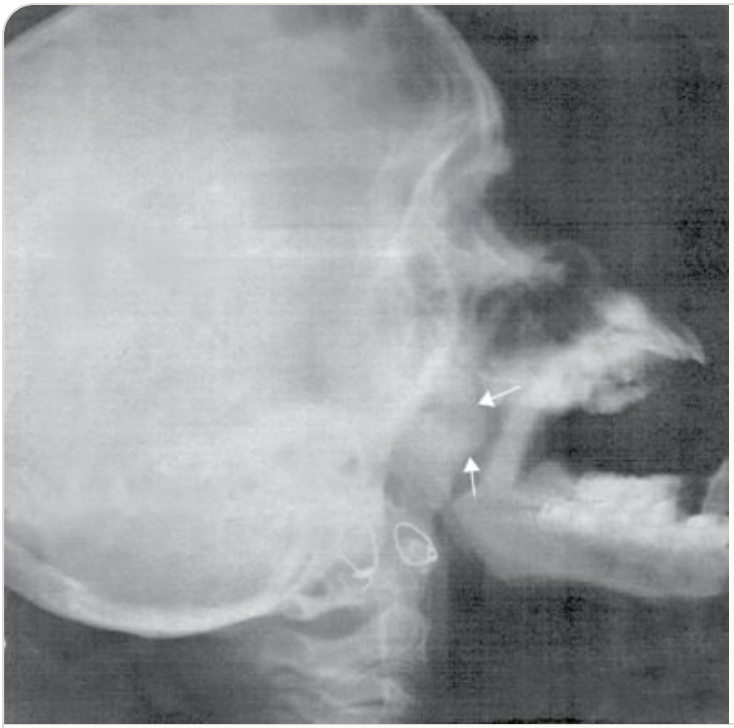


FIGURE 5 Enlarged adenoids (arrows) on a lateral radiograph of a seven-year-old girl, leaving very little airway.

Dhingra, fig. 48.4

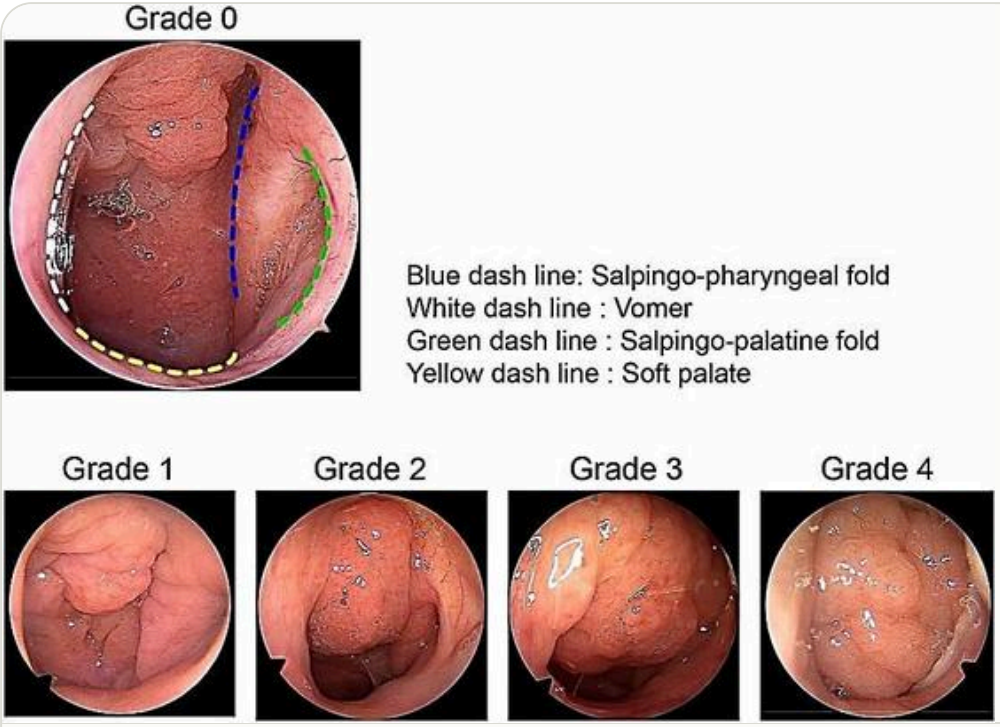
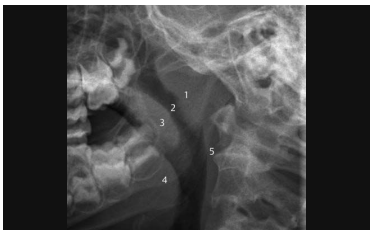


FIGURE 6 Endoscopic grading of adenoids.

Anatomy atlas · ENT Steps p. 207

IMAGING IN THE APP 1 case

Adenoid hypertrophy on imaging.



Adenoid Hypertrophy

Infection & inflammation · 1 image

2.4 Treatment

2.4.1 Medical

For milder symptoms: intranasal steroid sprays, with montelukast in some children, treatment of allergy, saline and decongestant drops for short periods, and breathing exercises. Many children improve as the adenoid regresses.

2.4.2 Adenoidectomy

For marked symptoms. Dhingra's indications:

- Adenoid hypertrophy causing snoring, mouth breathing, sleep apnoea or hyponasal speech.
- Recurrent rhinosinusitis.
- Otitis media with effusion associated with adenoid hyperplasia (usually with grommets).
- Recurrent ear discharge in chronic otitis media associated with adenoiditis.
- Dental malocclusion: adenoidectomy does not correct it but helps prevent relapse after orthodontic treatment.

CONTRAINDICATIONS

- **Cleft palate or submucous cleft:** removing the adenoid that helps close the velopharynx causes velopharyngeal insufficiency. Look for a bifid uvula, a notch in the hard palate and a translucent midline zone (*zona pellucida*).
- **Bleeding disorders.**
- **Acute upper respiratory infection.**

3 CHAPTER THREE The operation

Adenoidectomy.

3.1 Technique

General anaesthesia with an oral tube; the same position as tonsillectomy, but never hyperextend the neck. With the Boyle–Davis gag in place, the nasopharynx is examined by retracting the palate and by palpation (feeling also for a submucous cleft). An adenoid curette with a guard is passed until it touches the back of the septum, pressed back and swept down, with the head slightly flexed to protect the atlas; lateral tags are removed with smaller curettes or punch forceps, taking care of the Eustachian tube openings. Bleeding is stopped by packing, with diathermy for persistent points; rarely a postnasal pack is left for 24 hours. When the tonsils are also to come out, the adenoid is removed first and the nasopharynx packed.

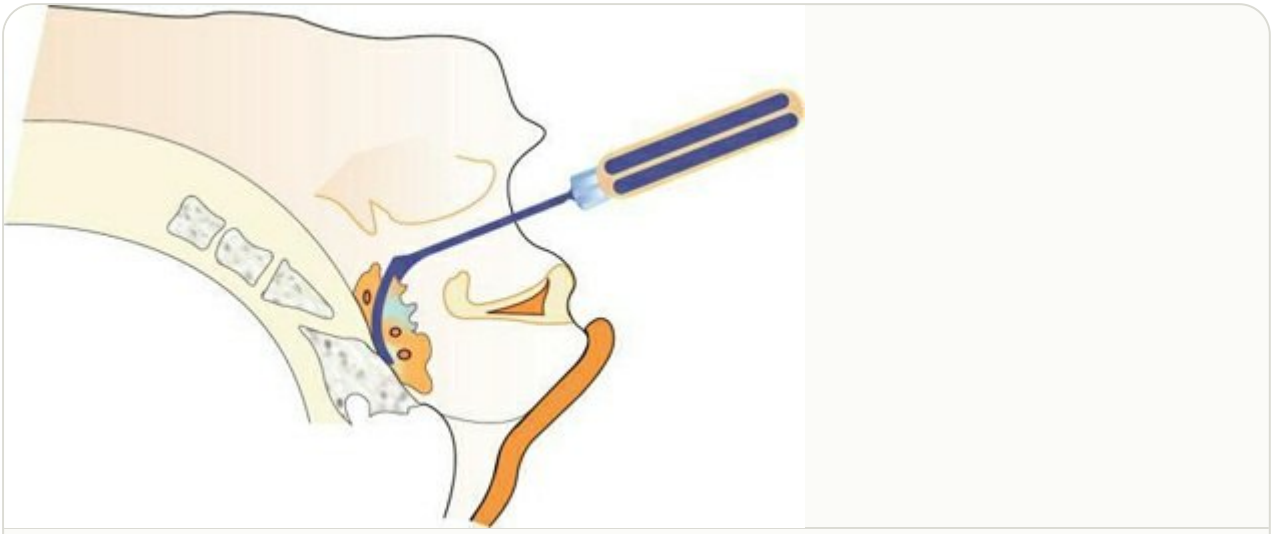


FIGURE 7 Adenoidectomy with the curette.
Dhingra, fig. 95.1

3.2 Endoscopic techniques

Suction diathermy, a microdebrider or coblation under endoscopic or mirror vision remove the adenoid more completely, including tissue around the tube openings and choanae.

OPERATIVE GUIDES IN THEATRE 1 operation

Adenoidectomy.

Adenoidectomy
Adenoidectomy

3.3 Complications

- **Haemorrhage:** usually primary; watch for a rising pulse and the vomiting of dark swallowed blood. A postnasal pack under general anaesthesia may be needed.
- **Injury to the Eustachian tube openings.**
- **Injury to the pharyngeal muscles and cervical spine** from hyperextension or pressure; 10 to 20% of children with Down syndrome have atlantoaxial instability.
- **Grisel syndrome:** neck pain and torticollis from inflammatory subluxation of the atlantoaxial joint, needing a collar or traction.
- **Velopharyngeal insufficiency:** hypernasal speech and nasal regurgitation, usually transient, but persistent with an unrecognised submucous cleft.
- **Nasopharyngeal stenosis** from scarring.
- **Regrowth** of tissue left behind; Lee advises biopsy of recurrent regrowth to exclude lymphoma.

4 CHAPTER FOUR Other nasopharyngeal inflammations

Nasopharyngitis and Thornwaldt disease.

4.1 Acute nasopharyngitis

Part of a common cold or influenza, or a bacterial infection (streptococcus, pneumococcus, *Haemophilus*). Dryness and burning behind the nose, then pain and difficulty in swallowing; in severe cases fever and nodes. The mucosa is red, swollen and covered in exudate. Most settle with analgesia; antibiotics for severe bacterial infection; decongestant drops for the associated adenoiditis of children.

4.2 Chronic nasopharyngitis

Adults with chronic nasal, sinus and pharyngeal infection, heavy smokers and drinkers, and those exposed to dust and fumes. Postnasal discharge and crusting, constant hawking and snorting back of secretions; congested mucosa with mucopus or crusts. Treat the source, stop smoking and drinking, and use alkaline douches and steam inhalation.

4.3 Thornwaldt disease

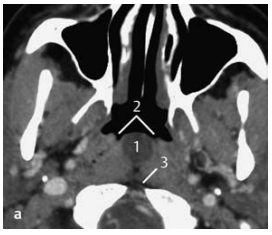
Infection of the *pharyngeal bursa*, the midline recess within the adenoid marking the embryonic attachment of the notochord.

- Persistent postnasal discharge with crusting.
- Nasal obstruction, Eustachian tube blockage and otitis media with effusion.
- A dull occipital headache, recurrent sore throat and low fever.

A cystic, fluctuant midline swelling sits on the posterior nasopharyngeal wall. Antibiotics, then marsupialisation and removal of the lining. Asymptomatic Thornwaldt cysts found on MRI need no treatment.

IMAGING IN THE APP 1 case

A Thornwaldt cyst.



Thornwaldt Cyst

Congenital · 2 images

THE ADULT WITH A ONE-SIDED NASOPHARYNGEAL PROBLEM

An adult with unilateral otitis media with effusion, a nasopharyngeal mass or a neck node is examined and, if needed, biopsied for nasopharyngeal carcinoma. Adenoid tissue in an adult is unusual; consider HIV and lymphoma.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Adenoid versus palatine tonsil:** no crypts, no capsule.
- **Symptoms** under nasal, aural and general headings; **adenoid facies.**
- **Lateral radiograph** and its limits.
- **Indications and contraindications** for adenoidectomy, especially submucous cleft.
- **Complications,** including Grisel syndrome and atlantoaxial instability in Down syndrome.
- **Thornwaldt disease.**

5.2 Common pitfalls

COMMON PITFALLS

- Removing the adenoid without feeling the palate for a submucous cleft.
- Hyperextending the neck of a child with Down syndrome.
- Calling a nasopharyngeal mass in an adult "adenoids" without a biopsy.

Written for this app from Oxford Textbook of Otolaryngology ch. 87, 88, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 48, 95 and K. J. Lee ch. 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Acute and chronic pharyngitis

A sore throat is one of the commonest reasons anyone sees a doctor, and almost always it is a virus that will pass in a week. The task is to recognise the few that are not: the streptococcal throat that can damage heart and kidney, glandular fever that must not be given amoxicillin, diphtheria, the immunosuppressed throat with candida, and the "chronic pharyngitis" that is really a cancer, a reflux or a smoker's warning.

AT A GLANCE

- Viral causes predominate; group A streptococcus matters for its complications.
- No amoxicillin in glandular fever.
- Chronic pharyngitis: remove nasal, dental, smoking and reflux causes.
- Eagle syndrome: an elongated styloid process causing throat and ear pain.

1 CHAPTER ONE Overview

One symptom, many causes.

1.1 Definition

Pharyngitis is inflammation of the pharyngeal mucosa, acute or chronic; when the tonsils are chiefly involved it is called tonsillitis, and the two overlap.

CAUSES OF ACUTE PHARYNGITIS

GROUP	ORGANISMS AND NOTES
Viral (most)	Rhinovirus, coronavirus, influenza and parainfluenza, adenovirus (pharyngoconjunctival fever), coxsackie A (herpangina, hand, foot and mouth), Epstein–Barr virus (infectious mononucleosis), cytomegalovirus, herpes simplex, acute HIV infection, measles, chickenpox
Bacterial	Group A beta-haemolytic *Streptococcus pyogenes*; groups C and G streptococci; mixed anaerobes (Vincent angina); *Corynebacterium diphtheriae*; *Neisseria gonorrhoeae*; *Arcanobacterium*; *Chlamydia* and *Mycoplasma*; syphilis
Fungal	*Candida albicans*
Protozoal	Toxoplasmosis (rare)

WHY GROUP A STREPTOCOCCUS MATTERS

It is the cause of rheumatic fever and post-streptococcal glomerulonephritis, and of peritonsillar abscess and scarlet fever. Most of the reasoning about swabs and antibiotics is about finding it.

CHAPTER TWO

2 Acute pharyngitis

The common sore throat.

2.1 Clinical features

- **Mild:** throat discomfort, malaise, low fever; a congested pharynx without nodes.
- **Moderate to severe:** pain, painful swallowing, headache, high fever; red pharynx with exudate, enlarged tonsils and lymphoid follicles on the posterior wall.
- **Very severe:** oedema of the soft palate and uvula, and tender cervical nodes.

Viral and bacterial infections cannot be separated reliably by examination. Cough, rhinorrhoea, conjunctivitis and hoarseness point to a virus; fever, tonsillar exudate, tender anterior cervical nodes and the absence of cough point to streptococcus.

CENTOR CRITERIA

One point each for fever, tonsillar exudate, tender anterior cervical nodes and absence of cough (the McIsaac modification adds a point under 15 years and subtracts one over 45). With low scores streptococcus is unlikely and antibiotics are not needed; with high scores, test or treat. Criteria are a guide, not a diagnosis.

2.2 Diagnosis

A throat swab culture detects about 90% of group A streptococci; a rapid antigen test gives an answer in minutes. Diphtheria needs special media and gonococcal swabs must be plated at once. No growth suggests a viral cause. Monospot or EBV serology and a blood film when glandular fever is suspected.

2.3 Treatment

2.3.1 General

Rest, plenty of fluids, warm saline gargles and analgesics; viscous lidocaine before meals for severe pain.

2.3.2 Streptococcal pharyngitis

Penicillin V orally for ten days, or a single intramuscular injection of benzathine penicillin; erythromycin or another macrolide for penicillin allergy. The full course is what prevents rheumatic fever.

2.3.3 Others

Diphtheria with antitoxin and penicillin or erythromycin; gonococcal pharyngitis with appropriate antibiotics according to local guidance.

3 CHAPTER THREE Viral pharyngitis

Recognisable patterns.

3.1 Infectious mononucleosis

Epstein–Barr virus in older children and young adults: fever, severe sore throat with a grey-white membranous exudate on the tonsils, palatal petechiae, large posterior as well as anterior cervical nodes, splenomegaly and hepatitis. Atypical lymphocytes on the film and a positive Monospot (heterophile) test, or EBV serology, confirm it.

GLANDULAR FEVER: THREE CAUTIONS

- **Amoxicillin or ampicillin** produces a widespread rash in most patients; avoid them.
- **Airway:** massive tonsillar enlargement can obstruct breathing; steroids, and rarely a nasopharyngeal airway or tonsillectomy.
- **Spleen:** avoid contact sports for several weeks because of the risk of rupture.

3.2 Others

- **Cytomegalovirus:** a mononucleosis-like illness in transplant and immunosuppressed patients, with a negative heterophile test.
- **Herpangina:** coxsackie A; fever, sore throat and small vesicles with a red halo on the soft palate and pillars.
- **Pharyngoconjunctival fever:** adenovirus; sore throat, fever and conjunctivitis, sometimes abdominal pain.
- **Acute lymphonodular pharyngitis:** coxsackie; white-yellow nodules on the posterior wall.
- **Measles:** Koplik spots on the buccal mucosa opposite the molars, three or four days before the rash.
- **Acute HIV infection:** a mononucleosis-like illness with rash and mouth ulcers.

4 CHAPTER FOUR Other acute throats

Fungi, uvula and swelling.

4.1 Candida

Oropharyngeal thrush extending from the mouth in the immunosuppressed, debilitated, diabetic or those on steroids, inhaled steroids or antibiotics: pain and dysphagia with white plaques; hyphae on microscopy. Nystatin, clotrimazole, fluconazole or itraconazole.

4.2 Quincke disease

Acute swelling of the uvula, with bacterial tonsillitis or after trauma (thermal burns, heavy snoring).

4.3 Angioedema

Rapid, painless, watery swelling of the lips, tongue, floor of mouth, uvula and larynx. Allergic angioedema is histamine-mediated; *ACE inhibitor* and hereditary (C1

esterase inhibitor deficiency) angioedema are bradykinin-mediated and respond poorly to antihistamines. Adrenaline, steroids and antihistamines for allergy; stop the ACE inhibitor; C1 inhibitor concentrate or icatibant for hereditary attacks; secure the airway by awake fibreoptic intubation or tracheostomy when the swelling threatens it.

AIRWAY WARNING SIGNS

Drooling, inability to swallow saliva, a muffled voice, stridor or a patient who must sit up and lean forward: do not lie them down or examine with a tongue depressor; call for senior airway help.

5 CHAPTER FIVE Chronic pharyngitis

The irritated throat.

5.1 Types

- **Chronic catarrhal pharyngitis:** congestion of the posterior wall with engorged vessels and excess mucus.
- **Chronic hypertrophic (granular) pharyngitis:** a thick, oedematous wall studded with red nodules of hypertrophied lymphoid follicles, hypertrophied lateral pharyngeal bands, and sometimes an elongated, oedematous uvula.
- **Atrophic pharyngitis:** dry, glazed mucosa with crusts, usually with atrophic rhinitis.



FIGURE 1 Granular pharyngitis: reddish nodules on the posterior pharyngeal wall.
Dhingra, fig. 50.1

5.2 Causes

- **Infection nearby:** chronic rhinitis and sinusitis with postnasal drip; dental and tonsillar sepsis.
- **Mouth breathing:** from nasal obstruction (polyps, rhinitis, septal deviation, turbinate hypertrophy), adenoids, protruding teeth or habit, so the throat receives unfiltered, unwarmed, dry air.
- **Irritants:** smoking, tobacco and pan chewing, alcohol, highly spiced food, dust and industrial fumes.
- **Laryngopharyngeal reflux.**
- **Faulty voice use and habitual throat clearing**, which keep the hypertrophic form going.

5.3 Symptoms

Discomfort, worse in the morning; a foreign body or lump sensation with a constant urge to swallow or clear the throat; a voice that tires and cracks; an irritable cough; gagging on opening the mouth.

5.4 Treatment

1. Find and remove the cause: treat the nose, sinuses, teeth and reflux; stop smoking, chewing and drinking.
2. Voice rest and speech therapy for faulty voice use; stop habitual hawking and throat clearing.
3. Warm saline gargles, especially in the morning.
4. Mandl's paint to the mucosa.
5. Cautery of prominent granules with silver nitrate, or diathermy.

For atrophic pharyngitis: alkaline sprays and irrigations to remove crusts, Mandl's paint, and potassium iodide by mouth to promote secretion, with treatment of the atrophic rhinitis.

CHRONIC SORE THROAT IN AN ADULT SMOKER

A persistent one-sided sore throat, earache, dysphagia, weight loss, haemoptysis or a neck lump is not "chronic pharyngitis" until the tongue base, tonsils, hypopharynx and larynx have been examined endoscopically and the neck palpated.

5.5 Keratosis pharyngis

White or yellow horny spikes on the tonsils, posterior wall or lingual tonsils, firmly adherent, from hypertrophy and keratinisation of the epithelium. There is no inflammation and no fever, which separates it from follicular tonsillitis. It is harmless and often regresses; reassurance is all that is needed.



FIGURE 2 Keratosis pharyngis: keratotic excrescences that resemble follicular tonsillitis.
Dhingra, fig. 50.2

CHAPTER SIX

6 Throat pain without inflammation

Eagle syndrome and others.

6.1 Eagle syndrome (styalgia)

An *elongated styloid process* (normal about 25 mm; over 30 mm is elongated) or a calcified stylohyoid ligament causes pain in the tonsillar fossa and upper neck radiating to the ear, worse on swallowing and turning the head, often after tonsillectomy. The tip can be felt through the tonsillar fossa; an open-mouth or lateral radiograph or CT with 3D reconstruction shows it. Many people have long styloids without symptoms. Symptomatic processes are shortened through the tonsillar fossa or through the neck.

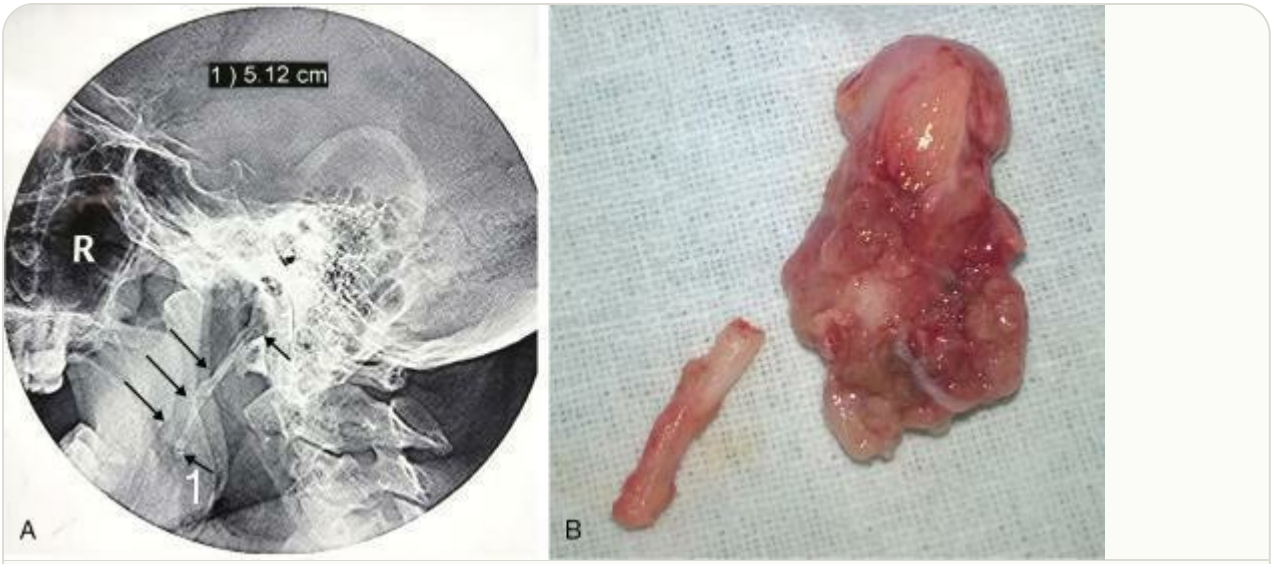


FIGURE 3 An elongated styloid process: (A) on radiograph, (B) excised after tonsillectomy.
Dhingra, fig. 53.6

OPERATIVE GUIDES IN THEATRE 1 operation

Transoral removal of an elongated styloid process.

Transoral Removal of Elongated Styloid Process
Removal of an elongated styloid process

6.2 Glossopharyngeal neuralgia

Brief, severe, electric pain in the tonsil, tongue base and ear, triggered by swallowing or yawning; carbamazepine, and microvascular decompression for resistant cases.

6.3 Globus

A persistent lump feeling in the throat between meals, with normal swallowing of food. Examine fully and endoscopically; reflux and anxiety are common associations, and red flags (pain, dysphagia, weight loss, voice change, neck mass) call for investigation.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of acute pharyngitis,** and why group A streptococcus matters.
- **Centor criteria,** swabs and rapid antigen tests.
- **Infectious mononucleosis:** features and the amoxicillin rash.
- **Chronic pharyngitis:** types, causes and treatment.
- **Keratosis pharyngis** versus follicular tonsillitis.
- **Eagle syndrome.**

7.2 Common pitfalls

COMMON PITFALLS

- Giving amoxicillin to a young adult with an exudative throat and big posterior nodes.
- Stopping penicillin after three days because the throat feels better.
- Labelling an adult smoker's persistent throat pain as chronic pharyngitis without endoscopy.
- Treating ACE inhibitor angioedema with antihistamines alone.

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CLINICAL · MOUTH & THROAT

Carcinoma of the hypopharynx

Cancer of the hypopharynx is the quiet killer of the throat. It grows in a wide, hidden gutter beside the larynx, gives little more than a pricking on swallowing until it is large, spreads early through a rich web of lymphatics and submucosal channels, and so presents late, often with a lump in the neck. It carries the worst prognosis of the head and neck squamous cancers, and its treatment usually involves the larynx.

AT A GLANCE

- Piriform sinus, postcricoid region, posterior wall.
- Plummer-Vinson syndrome predisposes to postcricoid cancer.
- Three-quarters have nodes at presentation.
- Look for a synchronous oesophageal cancer.

1 CHAPTER ONE Overview

The gutter beside the larynx.

1.1 Subsites

- **Piriform sinus:** on each side of the larynx, from the pharyngoepiglottic fold to the oesophagus.
- **Postcricoid region:** the anterior wall behind the cricoid lamina.
- **Posterior pharyngeal wall:** from the level of the hyoid to the cricoarytenoid joints.

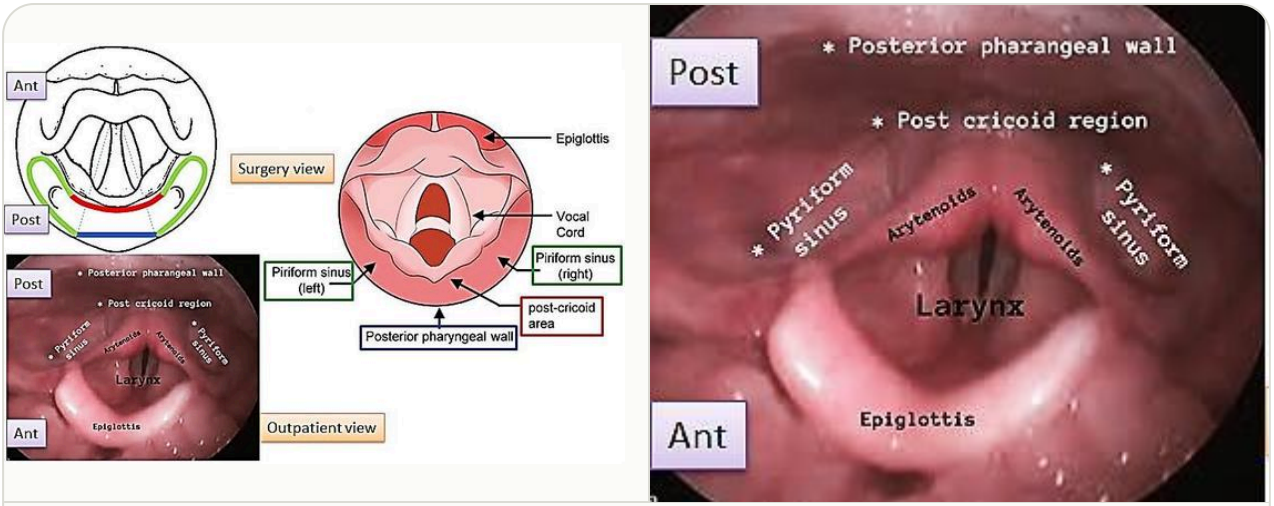


FIGURE 1 The three subsites of the hypopharynx, and an endoscopic view of the hypopharynx and larynx.

Anatomy atlas · ENT Steps p. 220, ENT Steps p. 221

1.2 Frequency and causes

Common in India. More than 90% are squamous cell carcinomas. In Dhingra's series the piriform sinus accounts for about 60%, the postcricoid region 30% and the posterior wall 10%; piriform sinus cancers predominate in North America, postcricoid cancers in parts of Europe. Tobacco and alcohol are the main causes; *Plummer–Vinson (Paterson–Brown–Kelly) syndrome* (iron deficiency anaemia, dysphagia, upper oesophageal webs, glossitis and koilonychia, in women of thirty to fifty) predisposes to postcricoid cancer.

1.3 Benign tumours

Exceptionally rare: papilloma, adenoma, lipoma, fibroma, leiomyoma, appearing as smooth, sometimes pedunculated masses.

1.4 Behaviour

- **Silent growth:** the piriform sinus is roomy, so tumours become large before they obstruct.
- **Submucosal spread:** tumours extend well beyond their visible edge, so resection margins must be generous.

- **Early lymphatic spread:** 60 to 75% have palpable nodes at presentation, often bilateral, including retropharyngeal and paratracheal nodes.
- **Second primaries:** a synchronous oesophageal cancer is found in a significant minority; always examine the oesophagus.
- **Distant metastasis** in about a fifth.

2 CHAPTER TWO Carcinoma of the piriform sinus

The commonest.

2.1 Clinical features

Mostly men over forty. Early symptoms are few: a feeling of *something sticking in the throat* or a pricking on swallowing, localised to one side. Then pain on swallowing, referred otalgia (through the internal laryngeal and vagus nerves), progressive dysphagia, and a lump high in the neck that may be the first sign. Hoarseness and stridor indicate spread to the larynx, fixation of the hemilarynx or laryngeal oedema.

2.2 Spread

- **Up** to the vallecula and base of tongue.
- **Down** to the postcricoid region and oesophagus.
- **Medially** to the aryepiglottic fold, ventricle and paraglottic space (fixing the vocal fold).
- **Laterally** through the thyroid cartilage and thyrohyoid membrane into the thyroid gland and neck.
- **Nodes:** about three-quarters at presentation, half bilateral, to the upper and middle jugular nodes; nodes may appear long after the primary has been controlled.

POOLING OF SALIVA

On flexible laryngoscopy, saliva pooling in one piriform sinus is a sign of obstruction there; ask the patient to puff the cheeks out to open the sinus and show the tumour.

2.3 Treatment

Early tumours without nodes: radiotherapy (or chemoradiation), preserving the larynx, or selected transoral laser or robotic resection, or Ogura's extended partial laryngopharyngectomy, with neck treatment. Tumours confined to the piriform sinus but advanced: total laryngectomy with partial pharyngectomy, the remaining pharynx closed primarily, with neck dissection. Extension to the postcricoid region: total laryngopharyngectomy with neck dissection and reconstruction. Postoperative radiotherapy routinely, or chemoradiation for positive margins or extranodal spread. Many now treat advanced disease with concurrent chemoradiation, or with induction chemotherapy followed by chemoradiation in responders, to try to save the larynx, keeping surgery for salvage.

3 CHAPTER THREE Carcinoma of the postcricoid region

The woman's cancer.

3.1 Clinical features

Often women, sometimes young (in their twenties and thirties) with Plummer–Vinson syndrome, in about a third of cases. Progressive dysphagia, first for solids, with weight loss and malnutrition; voice change or aphonia when the tumour reaches the recurrent laryngeal nerve, the posterior cricoarytenoid muscle or the cricoarytenoid joint.

3.2 Spread

Circumferentially (annular), causing tight stenosis; down into the cervical oesophagus; forwards into the arytenoids and cricoid; to the paratracheal nodes, which are not palpable, often on both sides.

3.3 Signs

The tumour may be hidden on mirror examination; oedema and redness of the arytenoid region, pooling of saliva, and *loss of laryngeal crepitus* (the grating felt when

the larynx is moved across the spine) suggest it. A lateral soft tissue radiograph shows a widened prevertebral shadow.

3.4 Treatment

Prognosis is poor. Small tumours may be treated by radiotherapy or chemoradiation to preserve the larynx; most present late and need *total laryngopharyngo-oesophagectomy* with reconstruction by gastric pull-up or colon transposition, or a free jejunal graft if the oesophagus can be preserved below.

4 CHAPTER FOUR Posterior pharyngeal wall

The least common.

Mostly men over fifty. Usually exophytic, sometimes ulcerative; it stays localised for a long time, then invades the prevertebral fascia, muscles and vertebrae. Nodal spread is often bilateral and retropharyngeal; about half have nodes at presentation. Dysphagia, spitting of blood, or a neck mass without other symptoms. Early lesions are treated by radiotherapy, or by excision through a lateral pharyngotomy (or transhyoid approach, or transorally) with repair, preserving the larynx; advanced lesions by laryngopharyngectomy and neck dissection. Fixation to the prevertebral fascia makes a tumour unresectable (T4b).

5 CHAPTER FIVE Evaluation

Mapping a hidden tumour.

5.1 Examination

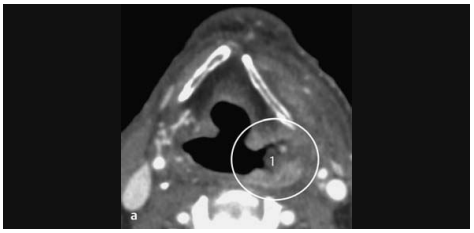
Flexible laryngoscopy of the piriform sinuses, postcricoid region and posterior wall (puffing the cheeks opens the sinuses); vocal fold mobility; palpation of the larynx for crepitus and the neck for nodes, including the thyroid and the central compartment.

5.2 Investigations

- **Barium swallow:** shows the lower limit of postcricoid and oesophageal disease, webs, and a second oesophageal tumour.
- **Contrast CT** of the neck and chest: the primary, cartilage and prevertebral invasion, and nodes; **MRI** for cartilage and soft tissue extent.
- **PET-CT:** distant metastases (lungs, liver, bone, mediastinal nodes) and second primaries.
- **Panendoscopy** under anaesthesia: biopsy, mapping of the extent (including submucosal spread) and a search for synchronous primaries, including oesophagoscopy.
- Thyroid function, haemoglobin and iron studies; nutrition and swallowing assessment.

IMAGING IN THE APP 1 case

Supraglottic and pharyngeal tumours on imaging.



Supraglottic and Pharyngeal Tumors

Tumours · 4 images

5.3 T staging

T STAGING OF HYPOPHARYNGEAL CANCER (AJCC 8TH EDITION)

CATEGORY	DEFINITION
T1	One subsite and 2 cm or less
T2	More than one subsite or an adjacent site, or over 2 cm up to 4 cm, without fixation of the hemilarynx
T3	Over 4 cm, or fixation of the hemilarynx, or extension to the oesophageal mucosa

CATEGORY	DEFINITION
T4a	Thyroid or cricoid cartilage, hyoid, thyroid gland, oesophageal muscle or central compartment soft tissue
T4b	Prevertebral fascia, encasement of the carotid artery, or mediastinal structures

6 CHAPTER SIX Treatment

Larynx and life.

6.1 Choosing

The decision balances cure against the larynx and swallowing: subsite, stage, nodal disease, the patient's fitness and wishes, and local expertise all count.

- **T1 to T2, N0:** radiotherapy (treating both sides of the neck and the retropharyngeal nodes), or larynx-preserving surgery (transoral laser or robotic resection, partial pharyngectomy, extended partial laryngopharyngectomy) with neck dissection.
- **Larger T2 and T3, or any node-positive disease:** concurrent chemoradiation; induction chemotherapy with response-guided treatment; or total laryngopharyngectomy with neck dissection and adjuvant therapy.
- **T4a:** usually total laryngopharyngectomy with neck dissection and postoperative (chemo)radiation; chemoradiation where the patient declines laryngectomy.
- **Postcricoid tumours** usually need laryngopharyngectomy, with oesophagectomy if the oesophagus is involved.

6.2 Reconstruction

- **Partial pharyngeal defects:** primary closure, or a pectoralis major, radial forearm or anterolateral thigh flap patch.
- **Circumferential defects:** a tubed fasciocutaneous free flap or a free jejunal graft.
- **With oesophagectomy:** gastric pull-up (or colon interposition).

Voice is restored by tracheoesophageal puncture and valve after laryngectomy.

OPERATIVE GUIDES IN THEATRE 5 operations

Total laryngopharyngectomy, pharyngectomy, partial pharyngectomy, reconstruction of hypopharyngeal defects and oesophagoscopy.

Total Laryngopharyngectomy
Total laryngopharyngectomy
Pharyngectomy
Partial Pharyngectomy
Partial pharyngectomy
Reconstruction of Hypopharyngeal Defects
Reconstruction of hypopharyngeal defects
Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy

6.3 Outcome

Five-year survival is poor: roughly 20 to 50% for piriform sinus cancer, about a third for postcricoid cancer, and around a fifth for the posterior wall. Dhingra quotes a gross five-year cure rate of only 19% for posterior wall tumours. Distant metastases and second primaries cause many late deaths.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra
Advises total laryngectomy with partial pharyngectomy for piriform tumours, laryngopharyngectomy when the postcricoid region is involved, with routine postoperative radiotherapy, and radiotherapy for early tumours.

Lee
Gives a wider menu following the NCCN guidelines (induction chemotherapy, chemoradiation, transoral surgery, Ogura's partial laryngopharyngectomy) and notes that many prefer non-surgical treatment for larynx preservation.

Both agree on the late presentation, early nodal spread and poor prognosis.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Subsites** of the hypopharynx and their relative frequency.
- **Plummer–Vinson syndrome** and postcricoid carcinoma.
- **Why it presents late,** submucosal spread and early nodes.
- **Signs:** pooling of saliva, loss of laryngeal crepitus, hemilarynx fixation.
- **Investigations,** including barium swallow and oesophagoscopy for a second primary.
- **Treatment options** and reconstruction (jejunum, gastric pull-up, flaps).

7.2 Common pitfalls

COMMON PITFALLS

- Dismissing a localised, one-sided pricking on swallowing as globus.
- Missing a synchronous oesophageal cancer.
- Taking narrow margins in a tumour that spreads submucosally.

Written for this app from Oxford Textbook of Otolaryngology ch. 60, Oxford Specialist Handbook ch. 10, Dhingra ch. 54 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Oesophagus and dysphagia

Dysphagia is a symptom with a long list of causes, from a stroke to a stricture to a cancer. Most are sorted by asking three questions. Is the problem getting food out of the mouth and pharynx (oropharyngeal), or getting it down the oesophagus (oesophageal)? Is it worse for solids or for liquids? Is it progressive or intermittent? Progressive dysphagia to solids, then liquids, with weight loss in an older smoker is cancer until endoscopy proves otherwise.

AT A GLANCE

- Constrictions at 15, 25 and 40 cm from the incisors.
- Achalasia: liquids and solids, bird beak, exclude a cardia cancer.
- Caustics: alkali penetrates deep; stricture from week 3.
- Pain after oesophagoscopy is perforation until proved otherwise.

1 CHAPTER ONE Applied anatomy

A 25-centimetre muscular tube.

1.1 Course and narrowings

The oesophagus is about 25 cm long. It runs from the cricopharyngeus at C6 to the cardia at T11, deviating slightly to the left in the neck and again in the lower chest. Along the way it has three normal constrictions, measured at endoscopy from the upper incisors:

1. The pharyngo-oesophageal junction at C6, about *15 cm*.
2. The crossing of the aortic arch and left main bronchus at T4, about *25 cm*.
3. The diaphragmatic hiatus at T10, about *40 cm*, with the cardia at about 40 to 42 cm.

These narrowings are where foreign bodies lodge, where caustic burns are worst and where instruments perforate, so their distances are worth knowing by heart.

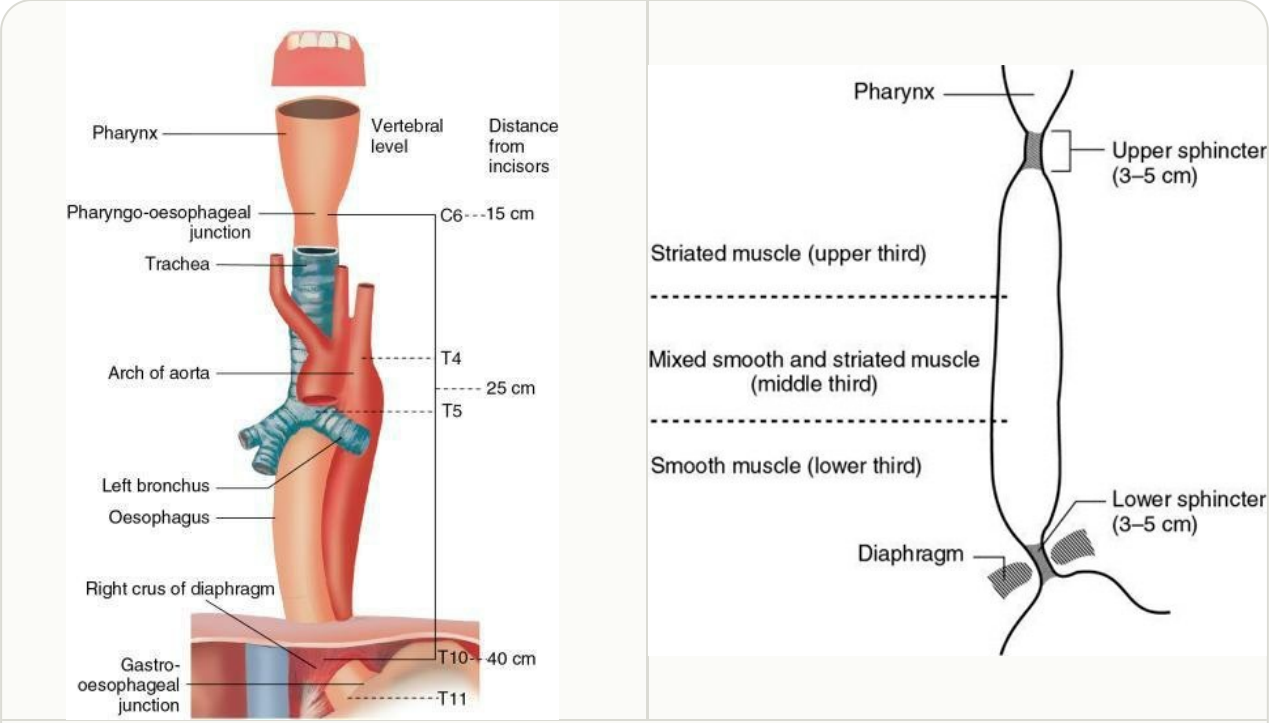


FIGURE 1 First, the oesophagus and its normal constrictions, with distances from the upper incisors. Second, the oesophagus with its sphincters and the change from striated to smooth muscle.

Dhingra, figs 67.1 and 67.2

1.2 Wall and supply

The wall has a mucosa of non-keratinised stratified squamous epithelium, a submucosa carrying *Meissner's* plexus, and inner circular and outer longitudinal muscle with *Auerbach's (myenteric)* plexus between them. Outside there is only adventitia. The absence of a *serosa* is the reason leaks spread freely through the mediastinum and anastomoses are fragile.

The muscle is striated in the upper third, mixed in the middle third and smooth in the lower third, which explains why diseases of skeletal muscle affect the top and scleroderma the bottom. The upper third is supplied by the inferior thyroid artery, the middle by branches of the aorta, and the lower by the left gastric and inferior phrenic arteries. The lower oesophagus is a portosystemic anastomosis and the site of *varices*. Lymph drains to deep cervical, mediastinal and coeliac nodes, and a long submucosal lymphatic network allows *skip lesions* in cancer. The nerve supply is from the vagus and the sympathetic chain.

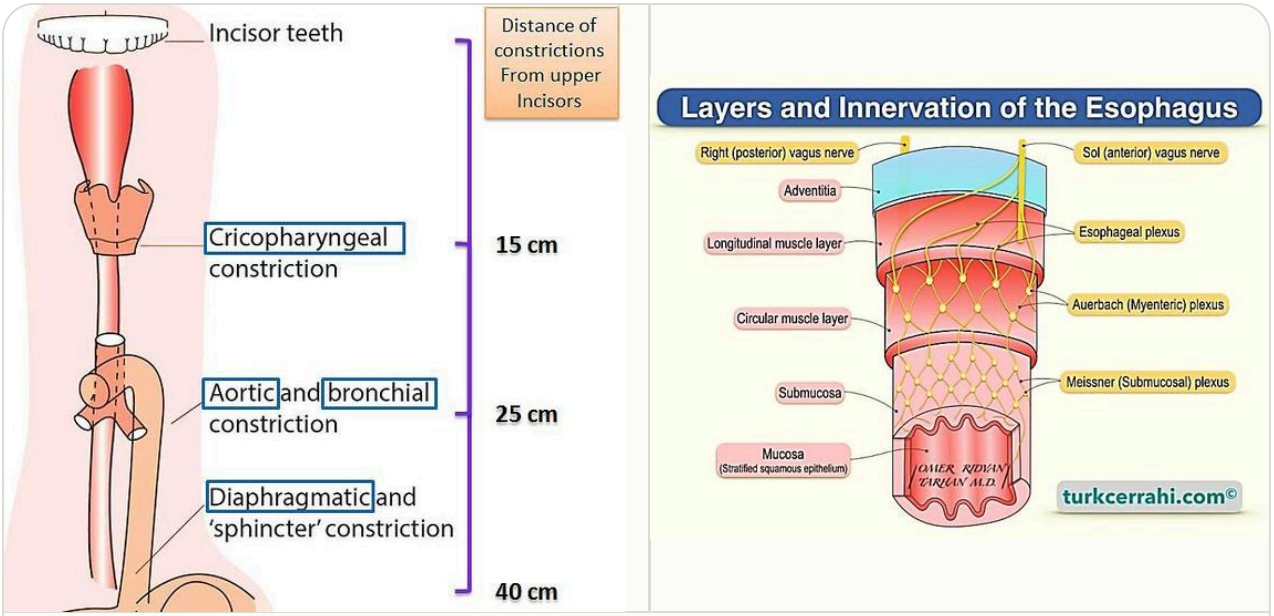


FIGURE 2 Constrictions of the oesophagus and their distances from the incisors, and the layers and innervation of the oesophagus.

Anatomy atlas · ENT Steps p. 226, ENT Steps p. 227

1.3 Sphincters

The **upper oesophageal sphincter** is mainly the *cricopharyngeus*, a zone 3 to 5 cm long that stays tonically closed at rest to stop air entering the oesophagus and refluxate entering the pharynx. On swallowing it relaxes and is pulled open by elevation of the larynx. Above it, between the thyropharyngeus and the cricopharyngeus, lies *Killian's dehiscence*, the site of the pharyngeal pouch; below it, between the cricopharyngeus and the oesophageal muscle, lies *Laimer's triangle*.

The **lower oesophageal sphincter** is a physiological high-pressure zone rather than an anatomical sphincter. It is helped by the "pinch-cock" of the diaphragmatic crura, the acute *angle of His*, the mucosal folds, and the positive intra-abdominal pressure acting on the abdominal segment.

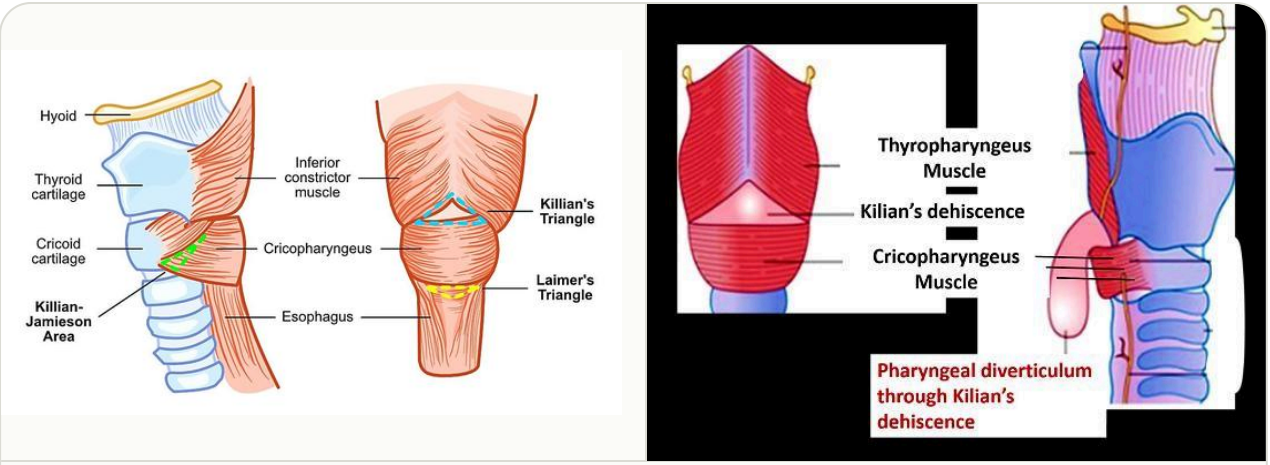


FIGURE 3 Killian and Laimer triangles, and Killian dehiscence with a pharyngeal pouch.
Anatomy atlas · ENT Steps p. 225, ENT Steps p. 198

2 CHAPTER TWO Swallowing

Three phases.

2.1 The phases

The **oral phase** is voluntary: food is chewed, mixed with saliva, formed into a bolus and pushed back by the tongue against the palate. It ends when the bolus touches the faucial pillars and triggers the reflex swallow.

The **pharyngeal phase** is a reflex lasting under a second, a sequence of protective closures and propulsion. The soft palate rises against *Passavant's ridge* to close the nasopharynx, and the tongue presses against the palate to close the mouth. Breathing stops and the larynx closes from below upwards, at the true folds, then the false folds, then the arytenoids against the epiglottis. The larynx rises and moves forward under the tongue base, which tips the epiglottis down and *pulls the upper sphincter open*; laryngeal elevation is the single most important factor in opening it. The constrictors then contract in sequence while the cricopharyngeus relaxes.

In the **oesophageal phase**, peristalsis carries the bolus to the stomach in about 3 to 8 seconds, with the lower sphincter relaxing ahead of it.

2.2 Peristalsis

Primary peristalsis is the wave started by a swallow. **Secondary peristalsis** is set off locally by distension and clears what the first wave left behind. **Tertiary contractions** are non-propulsive and uncoordinated; they appear in the elderly (*presbyoesophagus*) and in spasm.

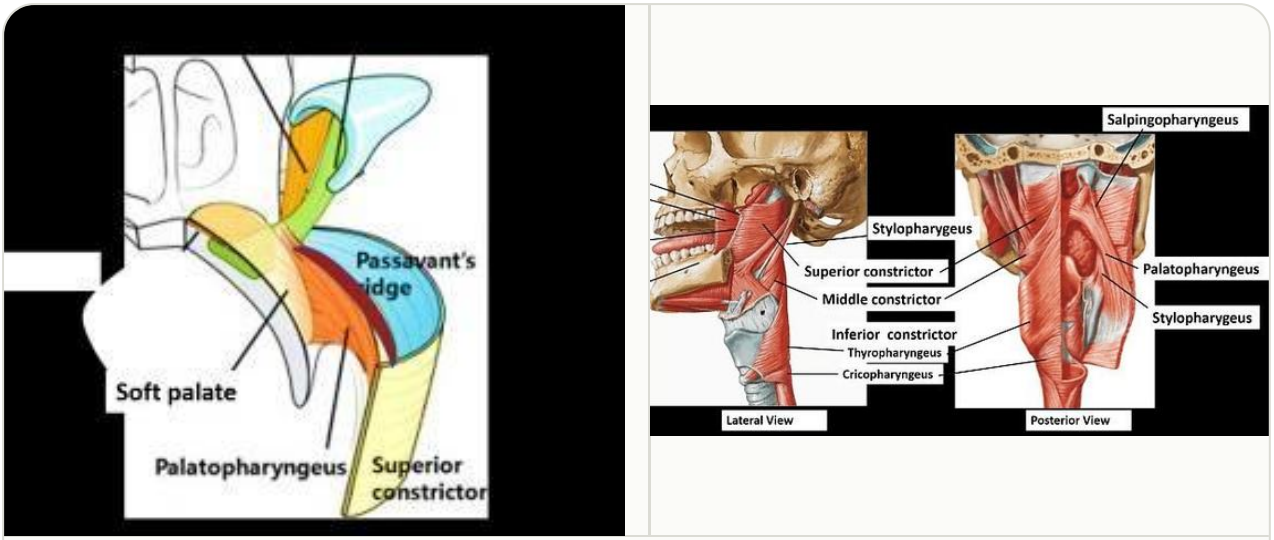


FIGURE 4 Passavant ridge, and the pharyngeal constrictors.
Anatomy atlas · ENT Steps p. 205, ENT Steps p. 197

3 CHAPTER THREE Approach to dysphagia

Where, what and how fast.

3.1 Terms

Dysphagia is difficulty swallowing. **Odynophagia** is painful swallowing and suggests ulceration, inflammation or infection. **Globus** is a persistent feeling of a lump in the throat, worst between meals, with no true difficulty in swallowing food.

3.2 Oropharyngeal versus oesophageal

TWO KINDS OF DYSPHAGIA

FEATURE	OROPHARYNGEAL (PRE-OESOPHAGEAL)	OESOPHAGEAL
Complaint	Difficulty starting a swallow; food sticks in the neck	Food sticks behind the sternum (though it may be felt in the neck)
Associated	Coughing or choking on swallowing, nasal regurgitation, drooling, dysarthria, a wet voice	Regurgitation, heartburn, chest pain, weight loss
Typical causes	Stroke, Parkinson disease, motor neurone disease, myasthenia, myositis, cranial nerve palsies, head and neck cancer and its treatment, pharyngeal pouch, cricopharyngeal dysfunction	Stricture, cancer, achalasia, spasm, eosinophilic oesophagitis, rings and webs, extrinsic compression
First test	Examination, flexible endoscopy, videofluoroscopy or FEES	Endoscopy (upper GI), barium swallow, manometry

3.3 Causes by site

It helps to think of the tube from the inside out. **In the lumen** lies a foreign body or food bolus. **In the wall** are inflammation (reflux, infection, caustic, eosinophilic, radiation or pill oesophagitis), strictures, webs and rings, tumours, and motility disorders. **Outside the wall** are the pharyngeal pouch, goitre and thyroid tumours, cervical *osteophytes* in diffuse idiopathic skeletal hyperostosis, mediastinal nodes and tumours, an enlarged left atrium, aortic aneurysm, and vascular rings causing *dysphagia lusoria*.

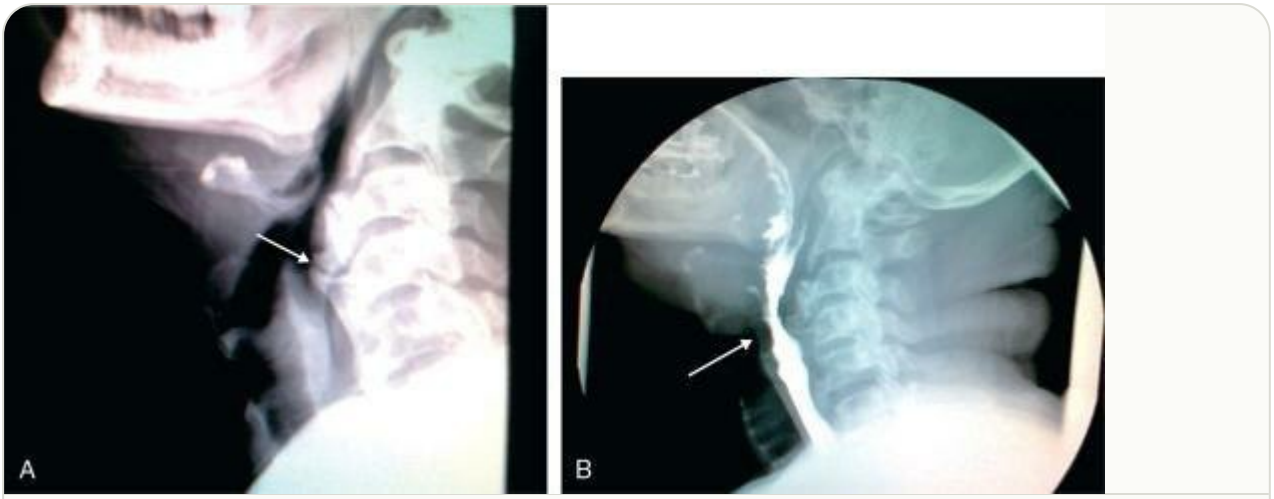


FIGURE 5 (A and B) Large cervical osteophytes pressing on the oesophagus on barium swallow. Dhingra, fig. 69.1

3.4 History

The **tempo** points to the cause. Sudden dysphagia means a foreign body or bolus, often stuck on an existing stricture or tumour; progression over weeks to months means cancer; intermittent symptoms suggest a ring or spasm; and long-standing dysphagia suggests achalasia or a web.

The **consistency** separates mechanical from motor disease. Solids first and liquids later indicate a mechanical obstruction such as a stricture or cancer, whereas liquids as bad as or worse than solids from the start indicate a motility disorder such as achalasia, or a neuromuscular cause.

Other clues complete the picture. Night-time regurgitation of undigested food with gurgling and halitosis suggests a pharyngeal pouch; heartburn suggests reflux and a peptic stricture; nasal regurgitation suggests palatal weakness; and coughing on swallowing means aspiration. Odynophagia suggests candida or herpes in the immunosuppressed, a pill or a caustic. Weight loss, hoarseness or a neck mass suggests cancer, and recurrent bolus impaction in a young atopic man suggests eosinophilic oesophagitis.

3.5 Examination and tests

Examine the mouth and oropharynx, and look at the larynx and hypopharynx with a flexible endoscope for pooling in the pyriform sinuses and for cord movement; then

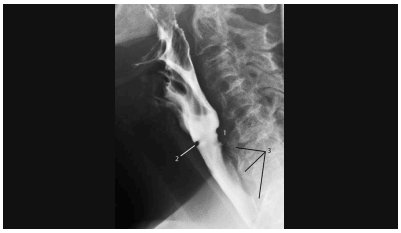
examine the neck, the cranial nerves and the chest. A blood count and iron studies look for iron deficiency and Plummer–Vinson syndrome.

The **barium swallow** shows pouches, strictures, webs, rings, achalasia, spasm and extrinsic compression. **Videofluoroscopy**, the modified barium swallow, studies the oral and pharyngeal phases with different consistencies and demonstrates aspiration, and **flexible endoscopic evaluation of swallowing (FEES)** shows residue, penetration and aspiration at the bedside. A chest film, a lateral soft-tissue neck film and CT are used for masses.

Endoscopy, either oesophagogastroduodenoscopy or transnasal oesophagoscopy, sees and biopsies the mucosa, and *every new oesophageal dysphagia in an adult needs it.* For function, **high-resolution manometry** is the test for motility disorders, classified by the *Chicago classification*, and **24-hour pH and impedance monitoring** quantifies reflux.

IMAGING IN THE APP 1 case

Disorders of the oesophageal passage on imaging.



Esophageal Passage Disorders

Infection & inflammation · 2 images

4 CHAPTER FOUR Inflammation and injury

Reflux, infection, caustics, perforation.

4.1 Gastro-oesophageal reflux disease

Reflux disease is a failure of the barrier, through weakness or transient relaxation of the lower sphincter, hiatus hernia, obesity, pregnancy, smoking, alcohol, fatty meals,

scleroderma, and drugs that relax smooth muscle such as calcium-channel blockers, nitrates and anticholinergics. Its typical symptoms are heartburn and regurgitation.

For the ENT surgeon the extra-oesophageal effects matter as much. *Laryngopharyngeal reflux* causes hoarseness, throat clearing, globus and chronic cough, with posterior laryngitis, contact ulcers and granulomas; reflux also contributes to subglottic stenosis, asthma, dental erosion and, in children, otitis media with effusion.

The complications are erosive oesophagitis, peptic stricture and *Barrett's oesophagus*, in which columnar *intestinal metaplasia* replaces the squamous lining of the distal oesophagus. It appears as salmon-pink mucosa, is premalignant for *adenocarcinoma*, and needs endoscopic surveillance.

Treatment starts with lifestyle change: weight loss, raising the head of the bed, no meals in the three hours before bed, stopping smoking, and less alcohol, caffeine, chocolate and fat. Proton pump inhibitors, H2 blockers and alginates follow, and antireflux surgery such as a *Nissen fundoplication* is reserved for refractory disease.

4.2 Infective oesophagitis

Infection occurs in the immunosuppressed, after chemotherapy or steroids, in HIV and in diabetes, and presents with odynophagia. **Candida** produces white plaques and is treated with fluconazole; **herpes simplex** produces small punched-out ulcers and is treated with aciclovir; and **cytomegalovirus** produces large shallow ulcers and is treated with ganciclovir.

4.3 Eosinophilic oesophagitis

Eosinophilic oesophagitis typically affects young atopic men with asthma, eczema or food allergy, and presents with dysphagia and *recurrent food bolus impaction*. Endoscopy shows rings ("trachealisation"), furrows and white exudates, and biopsy shows *15 or more eosinophils per high-power field*. It is treated with a PPI, swallowed topical steroids such as budesonide or fluticasone, elimination diets and careful dilatation.

4.4 Pill and radiation oesophagitis

Tetracyclines and doxycycline, potassium chloride, iron, NSAIDs and bisphosphonates can lodge at the aortic narrowing and ulcerate it, especially when taken at bedtime without water. Radiation causes acute ulceration and, later, stricture and dysmotility.

4.5 Caustic (corrosive) injury

Two patterns are seen: a toddler who swallows a household product by accident, usually in small amounts with less severe injury, and an adolescent or adult who swallows a large amount on purpose, with severe injury. **Alkalis** such as drain cleaner and lye cause *liquefactive necrosis* and penetrate deeply into the oesophagus. **Acids** cause *coagulative necrosis*, which limits their penetration, and damage the stomach more, leading to pyloric stenosis or gastric perforation.

The injury evolves in four stages:

1. Acute necrosis in the first days.
2. Sloughing and granulation over two to three weeks, when the wall is weakest and perforation most likely.
3. Fibrosis and *stricture* from about the third week, over months.
4. A lifelong increased risk of *squamous carcinoma*, decades later.

Assessment begins with the airway: stridor, hoarseness and drooling mean laryngeal injury, which may need intubation or tracheostomy. Burns of the lips and mouth are unreliable, because their absence does not exclude oesophageal injury. Look for perforation, mediastinitis or peritonitis, and take chest and abdominal films, with CT if perforation is suspected. Endoscopy is then done gently within 12 to 48 hours, going no further than a severe circumferential burn, and graded by Zargar's system: grade 0 normal, 1 oedema and erythema, 2a superficial ulcers and exudates, 2b deep or circumferential ulcers, 3a focal necrosis and 3b extensive necrosis. Grades 0 to 2a heal without stricture; grade 2b and above often stricture.

Management follows a sequence:

1. Resuscitate and protect the airway.
2. *No induced vomiting, no neutralising agents and no blind nasogastric tube* (most current guidance).

3. Analgesia and acid suppression.
4. Nil by mouth at first; a nasogastric tube placed under vision, or a gastrostomy, for feeding in deep burns.
5. Antibiotics for deep burns or perforation.
6. Steroids are controversial and not used routinely.
7. Emergency oesophagectomy or gastrectomy for full-thickness necrosis or perforation.
8. Follow-up contrast studies for stricture.

Strictures are treated by repeated dilatation, retrograde dilatation through a gastrostomy over a swallowed string when the stricture cannot be passed from above, and oesophageal replacement with colon or stomach when dilatation fails.

WHERE THE BOOKS DIFFER · CORROSIVE BURNS: NEUTRALISATION AND STEROIDS

Dhingra	Lee
Neutralisation with a weak acid or alkali within six hours, a nasogastric tube to keep the lumen open, and steroids from 48 to 96 hours for four to six weeks to prevent stricture.	Supportive care, early endoscopic grading, airway protection when the larynx is involved, and management of later strictures, with no routine neutralisation or steroids.

Current practice avoids neutralisation, because the heat of the reaction adds injury and vomiting re-exposes the oesophagus. Steroids are not routinely given, because trials have not shown that they prevent stricture.

4.6 Perforation

Instrumentation is the commonest cause of perforation: rigid oesophagoscopy, dilatation and the removal of foreign bodies, usually just above or at the cricopharyngeus at Killian's dehiscence, or in the distal oesophagus. Foreign bodies, trauma and malignancy are other causes, and *Boerhaave syndrome* is a spontaneous full-thickness rupture after vomiting, usually of the left lower oesophagus. It must be distinguished from the **Mallory–Weiss** tear, a mucosal tear at the gastro-oesophageal junction after retching that presents with haematemesis and is not a perforation.

A **cervical perforation** causes neck pain, dysphagia, fever, tenderness and *surgical emphysema* in the neck. A **thoracic perforation** causes severe chest or *interscapular*

pain, fever, tachycardia and shock, with surgical emphysema, *Hamman's sign* (a mediastinal crunch with each heartbeat), pneumothorax and pleural effusion. *Mackler's triad* of vomiting, chest pain and subcutaneous emphysema is the classic picture of Boerhaave syndrome.

Neck and chest films show air in the prevertebral space, mediastinum or pleura, and a *water-soluble contrast swallow* and CT confirm the leak. Every patient is kept nil by mouth on intravenous fluids, broad-spectrum antibiotics and antifungals, with nutritional support. Small contained cervical perforations are often managed conservatively, with cervical drainage if an abscess forms. Thoracic perforations need early surgical repair, ideally within about 24 hours, with drainage, or endoscopic stents and clips in selected cases; late perforations need drainage, diversion or resection.

PAIN AFTER AN ENDOSCOPY

Neck, chest or back pain, fever, tachycardia or crepitus after oesophagoscopy or dilatation means perforation until proved otherwise. Stop oral intake, image with water-soluble contrast and CT, and start antibiotics. Every hour of delay raises mortality.

5 CHAPTER FIVE Motility disorders

Manometry makes the diagnosis.

5.1 Achalasia

Achalasia results from loss of the inhibitory neurons of the *myenteric (Auerbach's) plexus*, producing *aperistalsis* and failure of the lower sphincter to relax, usually with a high resting pressure. A secondary form occurs in *Chagas disease* from *Trypanosoma cruzi*, and *pseudoachalasia* is produced by a cancer at the cardia, which must always be excluded by endoscopy, especially in an older patient with rapid weight loss.

The patient has dysphagia to *both solids and liquids* from the outset, regurgitates undigested food especially at night with cough and aspiration, has chest pain, and loses weight slowly. The barium swallow shows a dilated oesophagus with a fluid

level tapering smoothly to a "*bird's beak*" at the cardia, and a chest film may show a widened mediastinum and an absent gastric bubble. Manometry is the gold standard, and the Chicago classification divides achalasia into types I to III, which predicts the response to treatment.

The options are *pneumatic dilatation*, laparoscopic *Heller myotomy* with a partial fundoplication, and *peroral endoscopic myotomy (POEM)*. Botulinum toxin injected into the sphincter gives temporary relief in the frail, and calcium-channel blockers and nitrates have only a weak effect. The long-term risk of squamous carcinoma is increased.

5.2 Spastic disorders

Distal (diffuse) oesophageal spasm produces premature, non-propulsive contractions with normal relaxation of the lower sphincter. It causes intermittent dysphagia and chest pain that mimics angina, and the barium swallow shows a "*corkscrew*" or "*rosary bead*" oesophagus. It is treated with nitrates, calcium-channel blockers, botulinum toxin and sometimes myotomy. The **hypercontractile** ("nutcracker" or "jackhammer") oesophagus has very strong contractions that remain peristaltic, and causes chest pain.

5.3 Systemic causes

In **scleroderma** (systemic sclerosis), smooth-muscle atrophy and fibrosis give *aperistalsis of the lower two-thirds* and a *weak lower sphincter*, with severe reflux, strictures and Barrett's oesophagus, as part of the CREST syndrome. Polymyositis and dermatomyositis affect the striated upper oesophagus, and myasthenia gravis, motor neurone disease, diabetes, hypothyroidism and presbyoesophagus also disturb motility.

5.4 Cricopharyngeal dysfunction

Here the upper sphincter fails to relax or open, from fibrosis of the muscle, neurological disease such as stroke, Parkinson disease, motor neurone disease or muscular dystrophy, or reflux. Videofluoroscopy shows a posterior *cricopharyngeal bar*, though a bar is also seen in up to a third of asymptomatic elderly people, so it

must be interpreted with the symptoms. It is treated by dilatation, botulinum toxin injection or *cricopharyngeal myotomy*, endoscopic or external; myotomy is avoided in severe reflux.

OPERATIVE GUIDES IN THEATRE 2 operations

Dysfunction of the upper oesophageal sphincter, and oesophagoscopy and hypopharyngoscopy.

Dysfunction of the Upper Esophageal Sphincter
Upper sphincter dysfunction

Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy

CHAPTER SIX

6 Structural disorders

Webs, rings, pouches, hernias.

6.1 Plummer–Vinson (Paterson–Brown-Kelly) syndrome

This syndrome affects middle-aged women and combines iron-deficiency anaemia with dysphagia from a *postcricoid web*, glossitis, angular stomatitis, koilonychia, atrophic mucosa and sometimes achlorhydria. It carries an increased risk of *postcricoid* and oesophageal carcinoma, about 10% in the older literature. It is treated with iron, with B12 and B6 if deficient, dilatation of the web, and surveillance.

6.2 Schatzki ring

A Schatzki ring is a thin mucosal ring at the squamocolumnar junction, often with a small hiatus hernia, in people over 50. It causes intermittent dysphagia to solids and, once the lumen is under 13 mm, the "steakhouse syndrome" of bolus impaction. It is treated by dilatation and a PPI.

6.3 Diverticula

The **pharyngeal (Zenker's) pouch** is a pulsion diverticulum through Killian's dehiscence and is covered with the other pharyngeal conditions. A **mid-**

oesophageal traction diverticulum is pulled out by inflamed mediastinal nodes, classically tuberculous, and an **epiphrenic diverticulum** is a pulsion diverticulum just above the lower sphincter, usually secondary to a motility disorder.

IMAGING IN THE APP 1 case

A pharyngeal (Zenker) diverticulum on barium swallow.



Zenker Diverticulum (Pharyngeal Pouch)

Infection & inflammation · 9 images

6.4 Hiatus hernia

In the **sliding** hernia, about 90% of the total, the gastro-oesophageal junction slides up into the chest, and reflux is common. In the **para-oesophageal (rolling)** hernia the junction stays below the diaphragm while the fundus rolls up beside it; reflux is slight, but the hernia can bleed, obstruct or strangulate, and repair is recommended.

6.5 Strictures

Benign strictures may be peptic, caustic, radiation-induced, anastomotic after surgery, pill-induced, the result of a foreign body, eosinophilic or congenital. Every stricture is biopsied to exclude cancer. Treatment is dilatation with bougies, often over a guidewire, or balloons, together with treatment of the cause; stents or surgery are reserved for refractory cases.

6.6 Congenital anomalies

Oesophageal atresia with tracheo-oesophageal fistula is commonest (about 85%) as *proximal atresia with a distal fistula*. The baby drools, chokes and becomes

cyanosed at the first feed, the abdomen distends, and a nasogastric tube coils in the upper pouch at 9 to 13 cm. It is associated with the *VACTERL* anomalies. The **H-type fistula** (about 4%) presents later with coughing on feeds and recurrent pneumonia. **Vascular rings and an aberrant right subclavian artery** cause dysphagia lusoria, and the latter brings a non-recurrent right laryngeal nerve that is at risk in thyroid surgery.

7 CHAPTER SEVEN Tumours

Benign and malignant.

7.1 Benign tumours

The **leiomyoma** accounts for about two-thirds of benign tumours. It is a smooth, intramural submucosal mass that causes dysphagia only when large and is best seen with endoscopic ultrasound; mucosal biopsy is avoided, because it scars the plane needed for enucleation. **Fibrovascular polyps** of the upper oesophagus can be regurgitated into the mouth and obstruct the larynx, and papillomas, lipomas, haemangiomas and granular cell tumours also occur.

7.2 Oesophageal cancer

Squamous cell carcinoma is the commonest type worldwide and in India, and arises mostly in the *middle and upper* oesophagus. Its risk factors are tobacco, smoked and chewed, and alcohol; very hot drinks; dietary nitrosamines and poor diet; *achalasia*; *caustic strictures*; Plummer–Vinson syndrome; tylosis (palmoplantar keratoderma); and a previous head and neck cancer through field change. **Adenocarcinoma** is the commonest type in Western countries, arises in the *distal* oesophagus and cardia, and develops in *Barrett's oesophagus* with reflux and obesity.

The tumour spreads directly into the trachea, bronchi, aorta, pericardium and recurrent laryngeal nerves; through lymphatics to cervical, mediastinal and coeliac nodes, including skip lesions; and through the blood to liver, lung and bone.

The cardinal symptoms are *progressive dysphagia, first to solids and then to liquids*, and *weight loss*. Odynophagia and back pain suggest spread outside the wall, hoarseness means recurrent nerve involvement, and coughing on swallowing means a *tracheo-oesophageal fistula*. A sudden bolus obstruction may be the first sign.

Diagnosis is by endoscopy and biopsy. The barium swallow shows an irregular "*apple-core*" narrowing with shouldering, and staging uses CT of chest and abdomen, *PET-CT* and endoscopic ultrasound for depth and nodes, with bronchoscopy for tumours near the airway.

Early tumours are resected endoscopically. Resectable locally advanced disease is treated with neoadjuvant chemoradiotherapy (the CROSS regimen) or perioperative chemotherapy followed by oesophagectomy; definitive chemoradiotherapy is a standard option for squamous carcinoma, particularly in the upper and cervical oesophagus. The stomach is brought up as a gastric pull-up to reconstruct, while the cervical oesophagus after pharyngo-oesophagectomy is replaced with a free jejunal flap or a tubed radial forearm or anterolateral thigh flap. **Cervical oesophageal cancer** is often an extension of postcricoid or hypopharyngeal cancer and may need a *pharyngolaryngo-oesophagectomy* with gastric pull-up.

For palliation, self-expanding metal *stents*, brachytherapy, laser or argon plasma ablation and a feeding gastrostomy or jejunostomy relieve dysphagia, and a covered stent seals a tracheo-oesophageal fistula. The prognosis is poor: overall five-year survival is about 15 to 20%, because more than half of patients have unresectable or metastatic disease at diagnosis.

OPERATIVE GUIDES IN THEATRE 1 operation

Cancer of the cervical oesophagus.

Cancer of the Cervical Esophagus
Cervical oesophageal cancer

8 CHAPTER EIGHT Oesophagoscopy

Rigid, flexible and transnasal.

8.1 Rigid oesophagoscopy

Rigid oesophagoscopy is still the instrument for sharp or impacted foreign bodies, for biopsy of cricopharyngeal and upper oesophageal lesions, for dilatation, for endoscopic pouch surgery and for panendoscopy in head and neck cancer. It is contraindicated by cervical spine disease or injury, trismus, a receding jaw, aortic aneurysm and severe systemic illness.

1. General anaesthesia with the tube to the left of the mouth. Supine with the neck flexed and the head extended (the "sniffing" position), so the mouth, pharynx and oesophagus align. Protect the upper teeth.
2. Pass the scope down the right side of the tongue to the arytenoids, then behind the cricoid in the midline.
3. Open the cricopharyngeus by *gentle, sustained lifting*, never force. This is where perforation happens.
4. Keep the lumen in view at all times. Aortic pulsation marks the 25 cm level. The red, velvety gastric mucosa marks the cardia.
5. Inspect again on withdrawal.

The complications are injury to the teeth, lips and arytenoids, tracheal compression in children, and *perforation*, which is why pain, fever and emphysema after the procedure are taken seriously.

8.2 Flexible and transnasal oesophagoscopy

Flexible oesophagoscopy is done under topical anaesthesia or sedation in the left lateral position through a mouth guard. It sees the oesophagus, stomach and duodenum; allows biopsy, dilatation, banding, stents and removal of soft foreign bodies; is safe in neck and jaw disease; and has largely replaced the rigid scope for diagnosis. **Transnasal oesophagoscopy** is an unsedated office examination with a thin scope passed through the nose, used for dysphagia, reflux and Barrett's screening, the search for a second primary, and tracheo-oesophageal puncture after laryngectomy.



FIGURE 6 Flexible oesophagoscopy in the left lateral position, through a mouth guard, viewing on a monitor.

Dhingra, fig. 93.1

RIGID VERSUS FLEXIBLE OESOPHAGOSCOPY

FEATURE	RIGID	FLEXIBLE
Anaesthesia	General	Topical with or without sedation
Neck and jaw disease	Difficult or contraindicated	Possible
Reach	Oesophagus	Oesophagus, stomach, duodenum
Foreign bodies	Best for sharp and impacted objects; can sheath sharp points	Soft, blunt objects; needs an overtube for sharp ones
Biopsy of the upper sphincter region	Excellent	Can miss the postcricoid area
Setting	Inpatient theatre	Day case or office

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three constrictions** and their distances; Killian dehiscence.
- **The three phases of swallowing** and the role of laryngeal elevation.
- **Oropharyngeal versus oesophageal dysphagia;** solids versus liquids; the investigation sequence.
- **Achalasia:** bird's beak, manometry, pseudoachalasia; dilatation, Heller myotomy, POEM.
- **Plummer–Vinson syndrome** and postcricoid cancer.
- **Caustic injury:** alkali versus acid, the stages, grading, and the stricture and cancer risk.
- **Perforation:** causes and signs; Boerhaave syndrome.
- **Barrett oesophagus** and adenocarcinoma; risk factors for squamous carcinoma.
- **Rigid oesophagoscopy:** position, steps, complications.

9.2 Common pitfalls

COMMON PITFALLS

- Calling progressive dysphagia in an older patient "globus" without endoscopy.
- Forcing a rigid scope through the cricopharyngeus.
- Missing perforation in a patient with pain after endoscopy.
- Treating achalasia without excluding a cancer at the cardia.
- Inducing vomiting or neutralising after a caustic ingestion.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, Oxford Specialist Handbook ch. 10, Oxford Handbook of ENT ch. 13, Dhingra ch. 67, 68, 69, 93 and K. J. Lee ch. 36, 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Swallowed foreign bodies

A swallowed coin in a toddler, a fish bone in an adult's throat, a meat bolus in an elderly man, a denture after a seizure. Most swallowed objects pass harmlessly. The ENT surgeon's job is to find the ones that will not: those stuck in the pharynx or oesophagus, and those that are sharp, large, multiple magnets or button batteries. Remove them under direct vision before they ulcerate, perforate or block the airway. A button battery in the oesophagus is an emergency measured in hours.

AT A GLANCE

- Commonest site: just below the cricopharyngeus.
- Oesophageal coin is flat on the PA film; tracheal coin is edge-on.
- Button battery: halo sign, remove immediately, honey or sucralfate en route.
- Never use a Foley catheter or push it down blind.

1 CHAPTER ONE Where they lodge

Pharynx and oesophagus.

1.1 The pharynx

In the pharynx, fish bones and needles lodge in the **tonsil**, often in a crypt, where they can be seen and lifted out with forceps. Fish bones also catch in the **tongue base and vallecula**, where they are found with a mirror or endoscope and removed under topical anaesthesia with curved forceps; a needle buried in the tongue base may migrate and need imaging and an open pharyngotomy. Wires, pins and staples stick to the **posterior pharyngeal wall**. In the **pyriform fossa**, fish and chicken bones, needles and dentures are common; small objects can be removed in clinic, but large or impacted ones, and all objects in children, are removed under general anaesthesia.

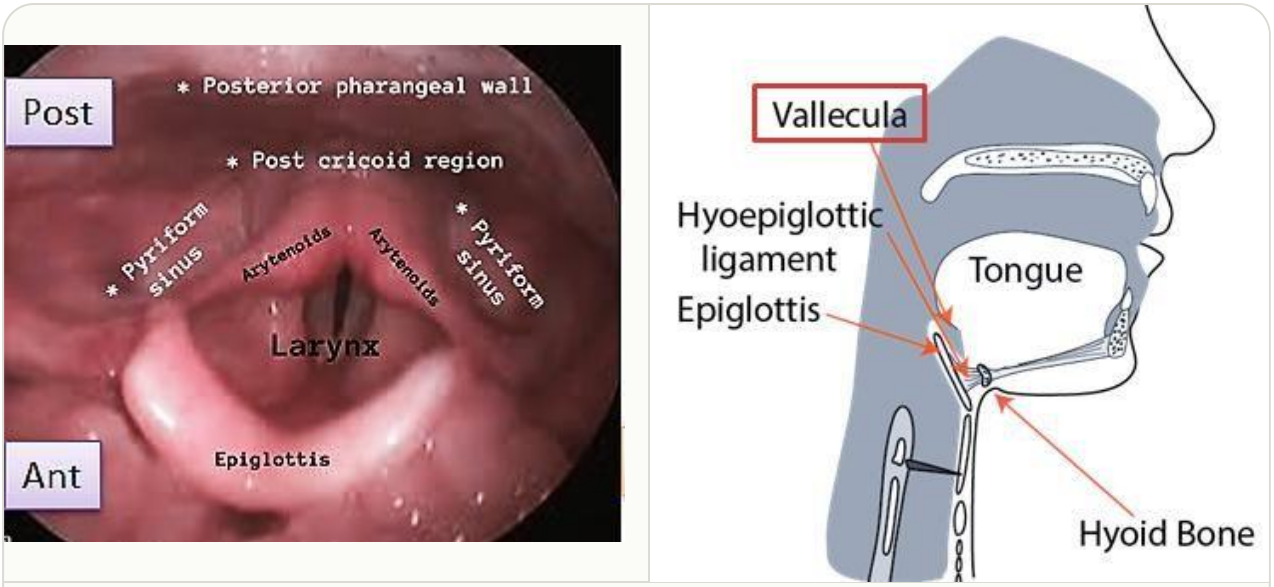


FIGURE 1 The hypopharynx and larynx as seen endoscopically, and the vallecula with the hyoepiglottic ligament.

Anatomy atlas · ENT Steps p. 221, ENT Steps p. 210

1.2 The oesophagus

By far the commonest site, about 70%, is *at or just below the cricopharyngeus*, and coins are typically held at the sphincter itself. Objects that pass it may stop at the crossing of the aorta and left bronchus (about 20%) or at the cardia (about 10%), while sharp objects catch anywhere. An object that lodges below the cricopharyngeus in an adult suggests an underlying stricture, ring, eosinophilic oesophagitis or cancer, and the oesophagus should be endoscoped or imaged once the oedema settles.

1.3 Who swallows things

About 80% of patients are children under five, who swallow coins, toys, beads, *button batteries* and magnets. Adults swallow food boluses, especially meat, and fish and chicken bones; they swallow dentures, partly because an upper denture dulls palatal sensation; and they are predisposed by hasty eating, poor chewing, alcohol, sleep, seizures and unconsciousness. Psychiatric patients and prisoners may swallow objects deliberately. In an older adult, a sudden bolus obstruction may be the first sign of a *narrowed oesophagus* from stricture or carcinoma.

2 CHAPTER TWO Assessment

History, signs, films.

2.1 Symptoms

The patient usually describes choking or gagging at the time, followed by pain or a pricking feeling localised in the neck, often beside the trachea, and worse on swallowing. Pain localises well in the pharynx and cervical oesophagus but poorly lower down. Dysphagia may begin as partial and become complete as oedema develops, and drooling with inability to swallow saliva means complete obstruction and a risk of aspiration.

In small children the respiratory tract may dominate: stridor or cough from compression of the soft posterior tracheal wall or from laryngeal oedema, or wheeze and recurrent chest infection from a long-standing oesophageal object. Chest, back or abdominal pain, fever and emphysema warn of perforation. Some patients, especially children, have few symptoms at all, so normal eating and a normal X-ray do not exclude a foreign body.

2.2 Signs

The signs are tenderness low in the neck beside the trachea (*Jackson's sign*), pain on rocking the larynx against the spine, *pooling of saliva in the pyriform fossae* that does not clear with swallowing, and sometimes the object itself in the postcricoid region. Surgical emphysema, fever and tachycardia mean perforation.

2.3 Imaging

Plain posteroanterior and lateral films are taken of the neck, chest and abdomen, and in children from mouth to anus so that every object is found. A coin in the *oesophagus* lies *flat (en face) on the PA view* and edge-on on the lateral; a coin in the *trachea* lies the other way, edge-on on the PA view, because the glottis is a sagittal slit. A button battery shows a *double ring ("halo") on the PA view* and a *step-off* on the lateral, so every apparent coin must be looked at carefully.

Many fish bones, wood, plastic, aluminium and glass are invisible on X-ray. On the lateral neck film their presence is suggested indirectly by air in the cervical oesophagus, *prevertebral soft-tissue thickening*, loss of the cervical lordosis, or air in the tissues from perforation. CT is the most sensitive test for bones, perforation and abscess, and should be used freely when symptoms persist despite a negative film. Barium is avoided because it coats the object and mucosa, can be aspirated and makes endoscopy harder; a water-soluble study is used only if perforation is suspected and CT is unavailable.

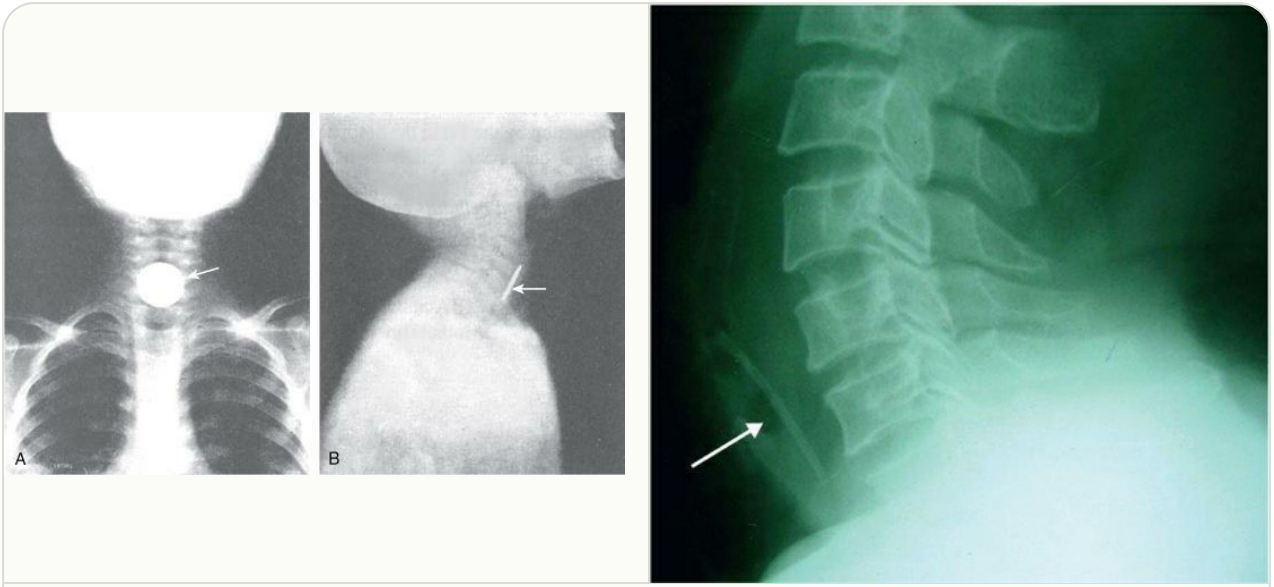


FIGURE 2 First, a coin in the upper oesophagus: (A) flat in the PA view, (B) edge-on in the lateral. Second, a fish bone in the cervical oesophagus with a bubble of air (arrow).

Dhingra, figs 70.1 and 70.2

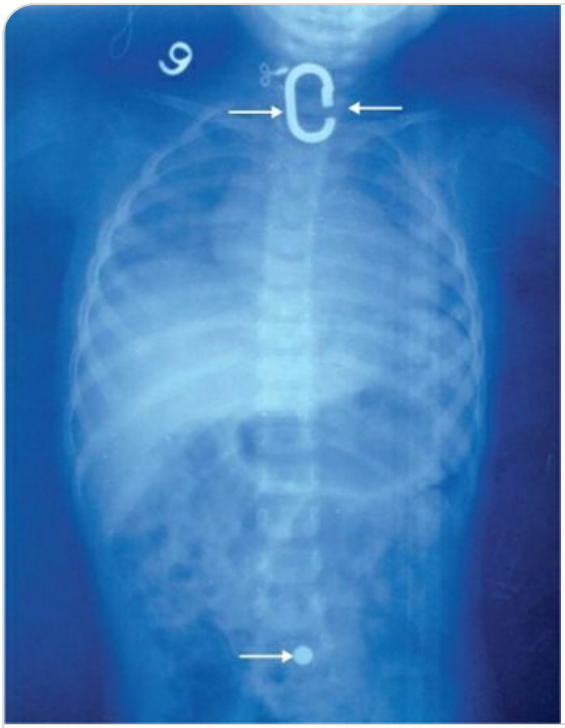


FIGURE 3 Two metallic foreign bodies: a large ring at the cricopharyngeus and a small piece in the abdomen (arrows).

Dhingra, fig. 70.3

IMAGING IN THE APP 1 case

Foreign bodies of the food and air passages on imaging.



Foreign Bodies

Trauma · 3 images

3 CHAPTER THREE Removal

Under direct vision.

3.1 Timing

Some objects must come out **within hours**: a *button battery in the oesophagus*, any object causing complete obstruction with drooling or airway compromise, sharp or pointed objects in the oesophagus, and any case with signs of perforation. Other oesophageal foreign bodies, including coins and food boluses that do not pass, should come out **within 24 hours**, because prolonged impaction increases ulceration, perforation and stricture. A smooth coin in the lower oesophagus of a well child can be **observed** for 12 to 24 hours with a repeat film, since it may pass into the stomach.

3.2 Methods

Rigid oesophagoscopy under general anaesthesia is the traditional ENT approach and the best for sharp or impacted objects and for objects at the cricopharyngeus. Sharp points are drawn into the end of the scope and withdrawn with it ("*ensheathing*"), open safety pins are closed or turned so that the point trails, and a hypopharyngoscope or long-bladed laryngoscope suits objects at the upper sphincter.

Flexible endoscopy under sedation, usually by gastroenterologists, suits food boluses, coins and blunt objects, with an overtube to protect the oesophagus from sharp ones; it also reveals any underlying eosinophilic oesophagitis, stricture or cancer, which should be biopsied. **Open surgery** is needed occasionally: a *cervical oesophagotomy* for impacted hooked objects such as partial dentures above the thoracic inlet, and *thoracotomy or thoracoscopy* for impacted lower objects and perforations.

OPERATIVE GUIDES IN THEATRE 1 operation

Oesophagoscopy and hypopharyngoscopy.

Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy

METHODS TO AVOID

- Blind removal with a Foley balloon catheter: the object can be aspirated as it is pulled into the pharynx.
- Pushing an object into the stomach blind.
- Meat tenderisers (papain): they can digest the oesophageal wall.
- Emetics.
- Glucagon, which relaxes the LES but does not help when there is a stricture or ring, and has little proven benefit.

3.3 After removal

After removal the mucosa is re-examined for ulceration, perforation and an underlying cause. Oral fluids start if the procedure was uneventful, the patient is watched for pain, fever and emphysema, and follow-up endoscopy or a contrast study is arranged if a stricture or tumour is suspected.

4 CHAPTER FOUR Special objects

Batteries, magnets, sharps.

4.1 Button (disc) batteries

A button battery lodged against moist mucosa passes a current that *hydrolyses tissue fluid and generates hydroxide at the negative pole*, producing an alkaline, liquefactive burn, to which leakage of the contents and pressure add. The injury is fast: the mucosa is damaged within an hour, the muscle within two to four hours, and the wall can perforate within 8 to 12 hours. Lithium cells of 20 mm or more are the most dangerous. Late complications include tracheo-oesophageal fistula, mediastinitis, stricture, vocal fold palsy and an *aorto-oesophageal fistula with fatal haemorrhage*, which can appear days or weeks later.

A battery in the oesophagus is removed *endoscopically at once*, at any hour. If ingestion was within 12 hours, honey (in children over one year) or sucralfate is given on the way to theatre to coat and protect the mucosa, provided it does not delay removal. Afterwards CT or MRI looks for deep injury, and significant injuries are observed in hospital.

Once past the oesophagus, most batteries in the stomach pass. They are removed if the child has symptoms, if a large battery has been swallowed by a young child, or if it remains in the stomach on a repeat film at about 48 hours; Dhingra's threshold is a child under six with a battery of 1.5 cm or more. Batteries in the nose and ear are emergencies too.

THE HALO SIGN

Any round radio-opaque object in the oesophagus of a child is a button battery until a careful look excludes it. Look for the double contour (halo) on the frontal film and the step at the edge on the lateral. Remove it now, not on tomorrow's list.

4.2 Magnets

A single magnet usually passes. Two or more magnets, or a magnet with a metal object, attract each other across loops of bowel and cause pressure necrosis, fistulae and perforation. Remove them while they can be reached endoscopically, and seek early surgical advice if they have passed beyond.

4.3 Sharp and long objects

Pins, needles, bones, open safety pins and denture fragments with hooks carry the highest risk of perforation and are removed from the oesophagus urgently. Beyond the stomach, sharp objects and long objects (over about 5 cm in a small child and 6 cm or more in adults) may not negotiate the duodenum; they are removed endoscopically if reachable, or followed with daily films, and surgery is needed for pain, tenderness, failure to progress or perforation.

4.4 Objects beyond the stomach

Most objects that reach the stomach pass within a week on a normal diet, without purgatives, while the stools are checked and films repeated. Operation is needed for abdominal pain or tenderness, an object that does not progress on serial films, a sharp object likely to perforate, a long object in a small child, and pyloric stenosis or other obstruction.

4.5 Food bolus impaction

If the patient can swallow saliva and is comfortable, a short wait may allow the bolus to pass. If saliva cannot be swallowed, or impaction lasts beyond 12 to 24 hours, the bolus is removed endoscopically, gently pushed on if there is no stricture or removed piecemeal. Almost every adult then needs investigation, usually *biopsy for eosinophilic oesophagitis* and a search for rings, strictures and cancer.

5 CHAPTER FIVE Complications

Why we do not wait.

5.1 What a neglected object does

A foreign body can obstruct the airway through laryngeal oedema or, in children, tracheal compression. Perforation leads to *retropharyngeal or parapharyngeal abscess* and *mediastinitis*, and a sharp object can erode into the *aorta* to form an aorto-oesophageal fistula, announced by a sentinel bleed before exsanguination; pericarditis and empyema can also follow. Periesophageal cellulitis and neck abscess, tracheo-oesophageal fistula, ulceration and stricture from a neglected object, and migration of needles and bones into the neck, sometimes needing open exploration, complete the list.

PHARYNGEAL AND OESOPHAGEAL FOREIGN BODIES AT A GLANCE

SITUATION	ACTION
Fish bone seen in tonsil or tongue base	Remove under topical anaesthesia with forceps

SITUATION	ACTION
Persistent pricking with a normal examination	Flexible endoscopy; CT if still suspected; oesophagoscopy if seen or symptoms persist
Coin in the upper oesophagus	Rigid or flexible removal within 24 hours
Coin in the lower oesophagus, well child	Observe up to 24 hours; repeat film; remove if still there
Button battery in the oesophagus	Immediate removal; honey or sucralfate en route if it does not delay
Food bolus, cannot swallow saliva	Urgent endoscopy; biopsy for eosinophilic oesophagitis
Sharp object in the oesophagus	Urgent rigid oesophagoscopy, sheathing the point
Two or more magnets	Remove while reachable; surgical review
Pain, fever, emphysema	Suspect perforation: nil by mouth, antibiotics, CT, surgical opinion

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Commonest site** (just below the cricopharyngeus) and the three narrowings.
- **Coin orientation:** oesophagus versus trachea.
- **Button battery:** halo sign, mechanism and timing of injury, immediate removal, honey or sucralfate.
- **Signs of an oesophageal foreign body** and of perforation.
- **Radiolucent objects;** why barium is avoided; the role of CT.
- **Rigid oesophagoscopy technique** for sharp objects.

- **What not to do:** Foley catheter, pushing down, papain.

6.2 Common pitfalls

COMMON PITFALLS

- Mistaking a button battery for a coin.
- Sending home a patient with a persistent foreign body sensation and a "normal X-ray".
- Using barium as the first study.
- Forgetting to look for an underlying stricture, eosinophilic oesophagitis or cancer in an adult with bolus impaction.

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CLINICAL · MOUTH & THROAT

Cleft lip and palate and velopharyngeal insufficiency

Cleft lip and palate is the commonest congenital malformation of the head and neck, and its care lasts from the antenatal scan to the end of growth. The ENT surgeon's part is large and long: feeding and airway in the newborn, glue ear in nearly every child with a cleft palate, speech and velopharyngeal insufficiency, and the submucous cleft that is missed until the adenoids come out. Care is given by a multidisciplinary team, to a timetable.

AT A GLANCE

- Tensor veli palatini is the only palatal muscle supplied by V3.
- Submucous cleft: bifid uvula, zona pellucida, palpable notch.
- Lip repair at about 3 months, palate at 6 to 12 months.
- Pharyngeal flap for central gaps; sphincter pharyngoplasty for lateral gaps.

1 CHAPTER ONE Basics

Anatomy, embryology, causes.

1.1 The lip

The lip is built around the *orbicularis oris*, the sphincter of the mouth supplied by the facial nerve, with the superior labial artery running deep to it. Its surface landmarks are what the surgeon must restore: the *white roll* at the vermilion border, where a step of only 1 mm is visible; *Cupid's bow*; and the two *philtral columns*, the lines of embryological fusion, with the philtral dimple between them.

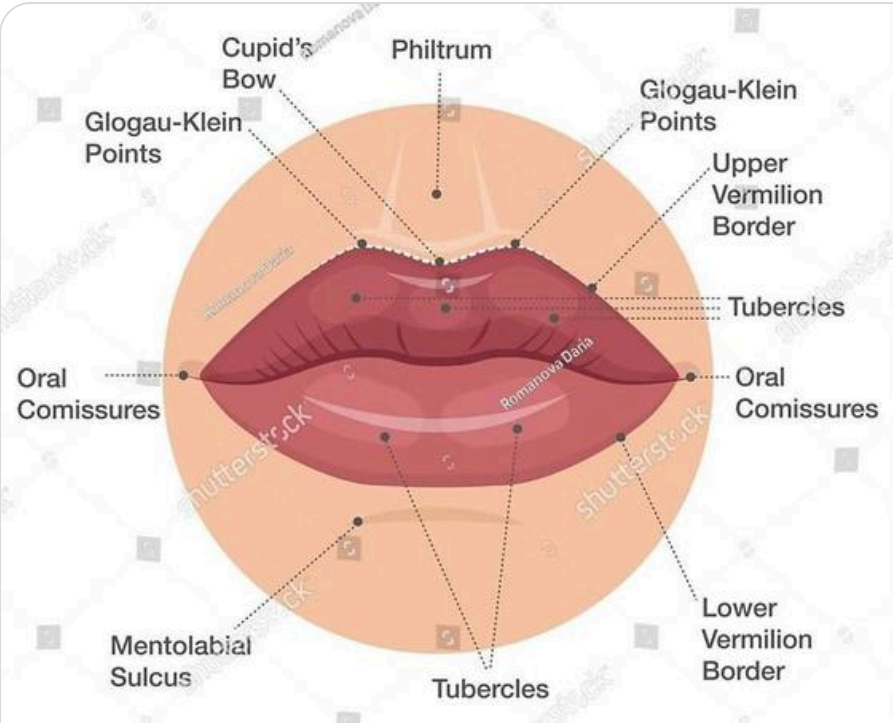


FIGURE 1 Lip landmarks: vermilion, philtrum, Cupid's bow and commissures.
Anatomy atlas · ENT Steps p. 166

1.2 The palate

The *incisive foramen* divides the palate in two. In front of it lies the **primary palate**: the premaxilla with the four incisors, the alveolus and the lip. Behind it lies the **secondary palate**: the hard palate, formed by the palatine processes of the maxillae and the palatine bones, and the soft palate.

In the soft palate, the *levator veli palatini* forms a sling that lifts the velum against the posterior pharyngeal wall, and the *tensor veli palatini* opens the Eustachian tube. The tensor is the only palatal muscle supplied by *V3*; the levator, palatoglossus, palatopharyngeus and musculus uvulae are all supplied by the vagus through the pharyngeal plexus. The hard palate is supplied by the greater palatine artery through the greater palatine foramen, the pedicle of every palatal flap, and the soft palate by the lesser palatine, ascending palatine and ascending pharyngeal arteries.

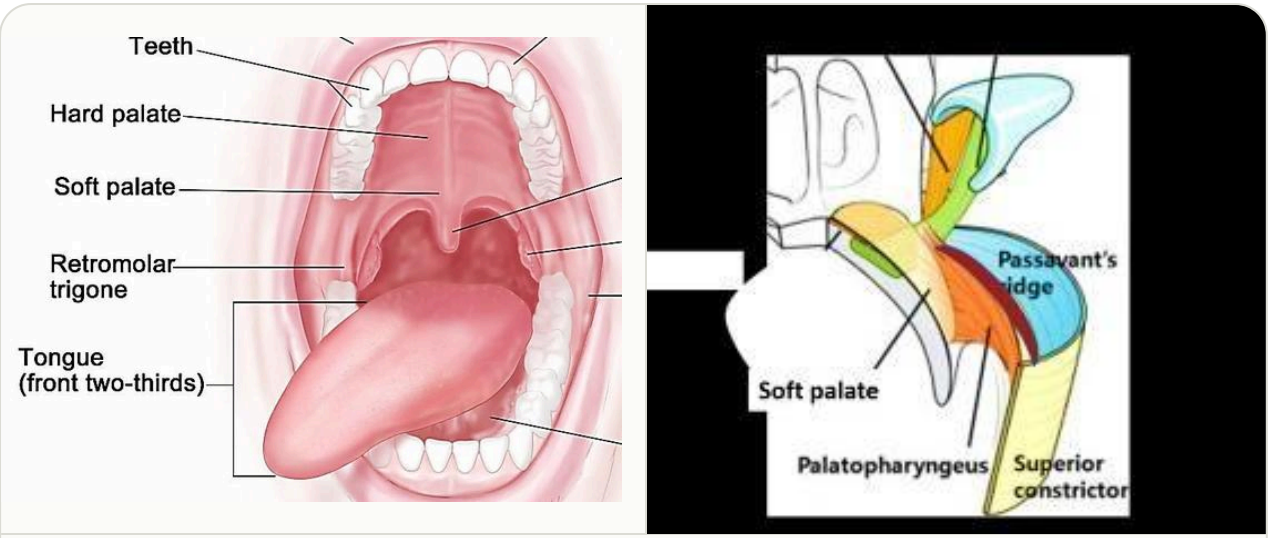


FIGURE 2 The hard and soft palate, and Passavant ridge where the velum closes against the pharyngeal wall.

Anatomy atlas · ENT Steps p. 172, ENT Steps p. 205

1.3 Embryology

The lip and primary palate form between the fourth and seventh weeks. The paired *maxillary prominences* of the first arch grow medially to fuse with the *medial nasal prominences* of the frontonasal process, and the medial nasal prominences join to form the intermaxillary segment, which becomes the philtrum, premaxilla and four incisors. Failure of fusion causes *cleft lip*, with or without a cleft alveolus.

The secondary palate forms between the sixth and twelfth weeks. The *palatine shelves* grow down beside the tongue, rise to the horizontal as the tongue drops, and fuse from front to back, from the incisive foramen to the uvula. Failure causes *cleft palate*. If the tongue cannot drop, as with a small mandible, the shelves cannot meet and the wide U-shaped cleft of the *Pierre Robin sequence* results.

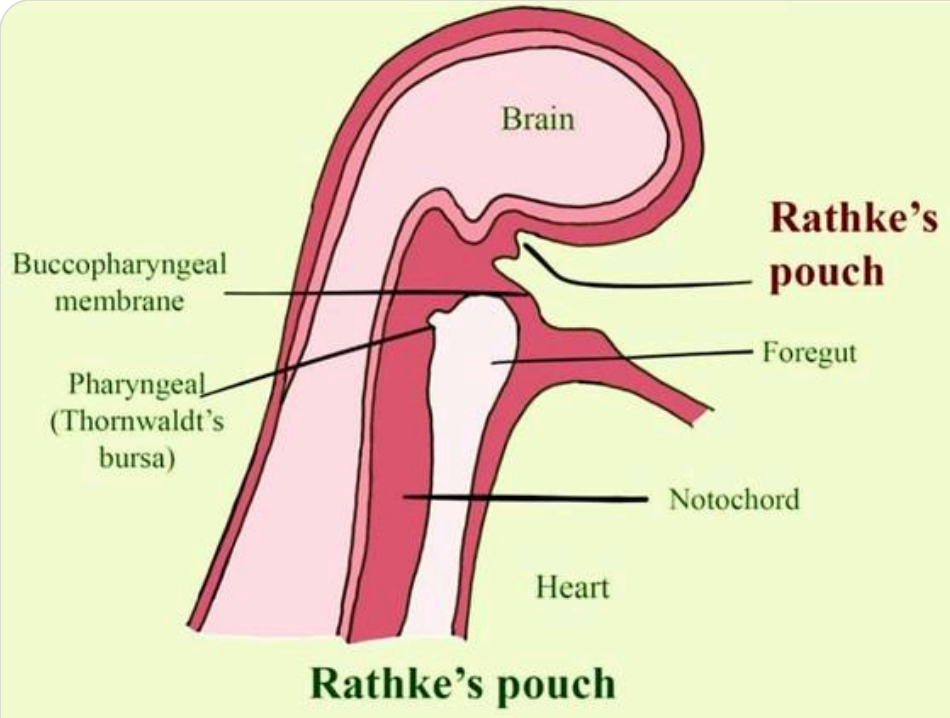


FIGURE 3 Rathke pouch and the buccopharyngeal membrane in the early embryo.
Anatomy atlas · ENT Steps p. 202

1.4 Epidemiology and causes

TWO DIFFERENT CONDITIONS

FEATURE	CLEFT LIP WITH OR WITHOUT PALATE	ISOLATED CLEFT PALATE
Frequency	About 1 in 700 to 1000 births; varies with ethnicity (commoner in Asian and Native American, less in African populations)	About 1 in 1500 to 2000; uniform across ethnicities
Sex	Boys more than girls	Girls more than boys
Associated syndromes	Up to about 30%	Up to about 50%
Side	80% unilateral, two-thirds of these on the left	—

Clefting is multifactorial. The risk to a sibling is about 4% and rises as more relatives are affected. Smoking and alcohol, anticonvulsants such as phenytoin and valproate, retinoids, steroids, folate deficiency and amniotic bands all contribute.

Several syndromes must be recognised. **Van der Woude syndrome**, autosomal dominant through IRF6, combines *lower lip pits* with cleft lip or palate and is the

commonest syndromic cleft. The **Pierre Robin sequence** runs from micrognathia to glossoptosis to a wide U-shaped cleft palate, with airway obstruction and feeding difficulty. **Stickler syndrome** is the commonest syndrome underlying Robin sequence; it is a collagen disorder with high myopia, *retinal detachment* (so eye screening is essential), hearing loss and joint problems. The **22q11.2 deletion (velocardiofacial or DiGeorge) syndrome** brings cleft palate or velopharyngeal insufficiency with heart defects, hypocalcaemia, immune deficiency and learning difficulty, and its carotids may lie *medially* behind the pharynx, a danger at pharyngeal surgery. Treacher Collins syndrome, hemifacial microsomia, the CHARGE association and Apert syndrome complete the list.

2 CHAPTER TWO Classification

Describing a cleft.

2.1 Lip and palate

A cleft lip is unilateral or bilateral. It is *complete* when it runs through the nasal sill, *incomplete* when the sill is intact, sometimes with a *Simonart band* of tissue bridging the gap, and *microform* when there is only a notch and a muscle gap under intact skin.

A cleft palate is *unilateral* when one palatal shelf remains attached to the vomer and *bilateral* when the vomer lies free in the midline. Veau's classification grades the extent: group I involves the soft palate only, group II the soft and hard palate up to the incisive foramen, group III is a complete unilateral cleft through lip, alveolus and palate, and group IV a complete bilateral cleft.

2.2 The submucous cleft

In a submucous cleft the mucosa is intact but the muscles are separated beneath it. The classic signs form *Calnan's triad*: a *bifid uvula*, a *zona pellucida* (a translucent bluish strip in the midline of the soft palate), and a *notch in the posterior edge of the hard palate* that can be felt with a finger. An *occult* submucous cleft has none of these

signs and is seen only on nasendoscopy, as a missing muscle bulge on the nasal surface of the velum.

CHECK THE PALATE BEFORE ADENOIDECTOMY

A bifid uvula, a palpable notch or hypernasal speech means a possible submucous cleft or velopharyngeal weakness. A full adenoidectomy can unmask *velopharyngeal insufficiency*, because the adenoid was helping the palate seal. Avoid adenoidectomy, or do a limited (superior) one, and involve the cleft team.

3 CHAPTER THREE Early care

Feeding, airway, ears.

3.1 The newborn

Many clefts are now diagnosed by antenatal ultrasound, which allows the family to be counselled before birth. After birth, feeding is the first problem: a baby with a cleft palate cannot create suction, so breastfeeding is usually impossible except with a cleft lip alone. Squeezy bottles or special teats such as the Haberman or pigeon, with feeding upright, are used, and weight gain is watched closely.

The airway matters most in Robin sequence. Prone or side positioning and a nasopharyngeal airway manage most babies; severe obstruction needs a tongue–lip adhesion, *mandibular distraction osteogenesis* or a tracheostomy, guided by sleep studies. Every baby is examined for other anomalies, especially cardiac and syndromic, reviewed by a geneticist, and given a *hearing screen*. Presurgical orthopaedics, by lip taping or *nasoalveolar moulding*, narrows a wide cleft and shapes the nose before lip repair.

3.2 The ears

Because the tensor and levator insert abnormally, the Eustachian tube cannot open, and otitis media with effusion affects *almost every child (over 90%)* with a cleft palate. Hearing is tested from birth and throughout childhood, and persistent hearing loss is treated with ventilation tubes or hearing aids, the tubes often being

inserted at the time of lip or palate repair. The ears usually improve after palate repair and with growth, but the long-term risks of chronic otitis media, retraction pockets and cholesteatoma remain.

OPERATIVE GUIDES IN THEATRE 1 operation

Mandibular distraction.

Mandibular Distraction
Mandibular distraction

4 CHAPTER FOUR Surgery

To a timetable.

4.1 The timetable

A TYPICAL CLEFT CARE TIMELINE

AGE	STEP
Antenatal and birth	Diagnosis, counselling, feeding plan, airway, hearing screen, genetics
Weeks 1 to 12	Presurgical orthopaedics (lip taping, nasoalveolar moulding) if needed
About 3 months	Cleft lip repair, with or without primary rhinoplasty (traditional "rule of 10s": 10 weeks, 10 lb, haemoglobin 10 g/dL)
6 to 12 (up to 18) months	Cleft palate repair, with ear tubes as needed
3 to 5 years	Speech assessment; velopharyngeal surgery if needed; fistula repair
8 to 11 years (mixed dentition, before the canine erupts)	Alveolar bone graft (cancellous bone from the iliac crest)
Teens	Orthodontics; orthognathic surgery (maxillary advancement) after growth

AGE	STEP
16 to 18 years	Definitive septorhinoplasty; lip revision

4.2 Cleft lip repair

Lip repair has four aims: to join the *orbicularis oris* into a working sphincter, to align the white roll and vermilion exactly, to create a symmetrical philtrum, nostril sill and alar base, and to correct the nose.

The unilateral cleft lip nose has a characteristic deformity. The cleft-side alar cartilage is displaced down, back and out with a flat, low dome, the alar base is wide, the columella and caudal septum deviate to the *non-cleft* side, and the maxilla on the cleft side is hypoplastic.

In the **Millard rotation–advancement repair**, the *non-cleft (medial) side* is *rotated* down to level Cupid's bow, and the *cleft (lateral) side* is *advanced* into the gap beneath the nostril. The muscle is released from its abnormal insertions and repaired across the cleft, and the nasal floor is closed. The Fisher anatomical subunit repair and triangular-flap repairs are the alternatives.

The **bilateral cleft lip** is harder, because of a protruding premaxilla, a prolabium without muscle and a short columella; it is repaired in one or two stages, building the muscle across behind the prolabium. The complications of lip repair are dehiscence, a vermilion notch (the "whistle deformity"), a mismatched white roll, scar hypertrophy and nasal asymmetry.

4.3 Cleft palate repair

The palate is repaired at about 9 to 12 months, and certainly before 18 months. The timing balances *better speech with early repair* against *better maxillary growth with later repair*. The aims are to separate the mouth from the nose and, above all, to *rebuild the levator sling* by intravelar veloplasty and lengthen the palate for normal speech, because in a cleft the levator runs front to back and inserts into the back of the hard palate instead of meeting its partner in the midline.

The **two-flap palatoplasty (Bardach)** raises mucoperiosteal flaps on the greater palatine pedicles, closes the nasal and oral layers and adds an intravelar veloplasty;

it is used for complete clefts. The **von Langenbeck repair** uses bipedicled flaps with lateral relaxing incisions. The **Furlow double-opposing Z-plasty** places opposite Z-plasties on the oral and nasal layers, which *lengthens the palate and realigns the levator*, and is used for soft palate clefts, submucous clefts and velopharyngeal insufficiency. The older **Veau–Wardill–Kilner V–Y pushback** leaves more raw bone and restricts growth.

After repair, bleeding and airway obstruction, especially in Robin sequence, are the early risks. An *oronasal fistula* develops in up to about 10%, usually at the junction of hard and soft palate, *velopharyngeal insufficiency* persists in up to 20 to 25%, and maxillary hypoplasia appears with growth. A submucous cleft is repaired only if it causes velopharyngeal insufficiency or feeding or ear problems, by Furlow palatoplasty or intravelar veloplasty.

OPERATIVE GUIDES IN THEATRE 2 operations

Cleft palate and velopharyngeal dysfunction, and surgery of the soft palate.

Cleft Palate and Velopharyngeal Dysfunction
Cleft palate and VPI

Surgery of the Soft Palate
Soft palate

5 CHAPTER FIVE Velopharyngeal dysfunction

Speech and the palate.

5.1 Definitions

Three kinds of velopharyngeal dysfunction are distinguished. **Insufficiency** is a structural deficit: the palate is too short or scarred, there is a cleft or submucous cleft, or tissue has been lost after adenoidectomy or tumour surgery. **Incompetence** is a neuromuscular deficit, as in palsies or the 22q11 deletion. **Mislearning** is an articulation pattern that needs speech therapy only.

5.2 Features

The child has *hypernasal* resonance on vowels and voiced oral consonants, while the nasal sounds /m/, /n/ and /ng/ remain normal. *Nasal air emission* and turbulence accompany pressure consonants such as p, b, t, d and s; consonants are weak; *compensatory articulations* such as glottal stops develop; and food and liquid may regurgitate through the nose.

5.3 Assessment

A speech and language therapist makes a perceptual assessment, and the velopharynx is imaged during speech by *nasendoscopy*, which shows the size and site of the gap and the closure pattern, and by *multiview videofluoroscopy*. The closure pattern may be coronal (the commonest), sagittal, circular, or circular with a Passavant ridge, and it determines the operation. Before any pharyngeal surgery the child is checked for the 22q11 deletion with *medialised carotids*, ideally by MRA, and for obstructive sleep apnoea.

5.4 Treatment

Surgery corrects structure and therapy corrects articulation, so speech therapy alone is used for mislearning and small gaps. For a short palate or malpositioned muscle with a small gap, the palate is re-repaired with a Furlow double-opposing Z-plasty or a radical intravelar veloplasty. Small central gaps can be closed by *augmenting the posterior pharyngeal wall* with fat, implants or fillers, at the risk of migration and extrusion.

For a **large central gap with good lateral wall movement**, a *superiorly based pharyngeal flap* is raised from the posterior wall and attached to the velum, leaving two lateral ports. For a **circular or bow-tie gap with poor lateral wall movement**, a *sphincter pharyngoplasty* (Orticochea, Jackson or Hynes) transposes the posterior tonsillar pillars with palatopharyngeus and sutures them to the posterior wall, narrowing the port while leaving a central opening. A speech bulb or palatal lift helps poor surgical candidates.

The main risks of pharyngeal surgery are *obstructive sleep apnoea* and hyponasality, so oxygen saturation is monitored after the operation.

OPERATIVE GUIDES IN THEATRE 1 operation

Advanced palatal surgery: expansion sphincter and transpalatal advancement pharyngoplasty.

Advanced Palatal Surgery: Expansion Sphincter Pharyngoplasty & Transpalatal Advancement Pharyngoplasty
Pharyngoplasty

6 CHAPTER SIX Later care

Bone, jaws, nose.

6.1 Through growth

The alveolar cleft is grafted in the mixed dentition, at about 8 to 11 years and before the permanent canine erupts, with cancellous bone from the iliac crest. The graft closes the alveolar cleft and any oronasal fistula there and supports the teeth and the alar base.

About a quarter of children develop maxillary hypoplasia with a class III bite and need a *Le Fort I advancement* or distraction once growth stops. The cleft lip nose, corrected in part by a primary rhinoplasty at lip repair, is finally treated by a definitive *septorhinoplasty* with cartilage grafts after facial growth. Dental care and orthodontics continue throughout, and psychological support for the child and family matters most at school age and in adolescence.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Primary versus secondary palate;** the incisive foramen; the embryology of each.

- **The nerve supply of the palatal muscles** (tensor by V3).
- **Cleft lip versus isolated cleft palate:** epidemiology and syndromes (Van der Woude, Robin, Stickler, 22q11).
- **The submucous cleft triad,** and why it matters before adenoidectomy.
- **The timetable of care,** and the rule of 10s.
- **Millard repair:** which side rotates.
- **Furlow palatoplasty and intravelar veloplasty.**
- **Otitis media with effusion in cleft palate:** why, and how it is managed.
- **Velopharyngeal insufficiency:** signs, nasendoscopy, pharyngeal flap versus sphincter pharyngoplasty, OSA risk.

7.2 Common pitfalls

COMMON PITFALLS

- Adenoidectomy in a child with an unrecognised submucous cleft.
- Missing Stickler syndrome (retinal detachment) in Robin sequence.
- Pharyngeal surgery in 22q11 deletion without imaging the carotids.
- Neglecting hearing surveillance after the palate is repaired.

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CLINICAL · MOUTH & THROAT

The syndromal child

Children with congenital anomalies make up almost half of paediatric hospital admissions, and many pass through the ENT clinic. There are hundreds of syndromes, and memorising all of them is neither possible nor useful. What the ENT surgeon needs is to recognise when a common problem, such as hearing loss, a blocked nose, an effusion or obstructive sleep apnoea, may be part of a syndrome; to know the handful of syndromes that recur in ENT practice; and to anticipate the airway, anaesthetic and hearing problems they bring.

AT A GLANCE

- **Syndrome, sequence, association:** single cause, a cascade from one anomaly, or a non-random cluster.
- **Down syndrome:** persistent effusion (hearing aids often first), OSA in about 60%, subglottic stenosis, atlantoaxial instability.
- **Treacher Collins** is bilateral with lower lid colobomas; **Goldenhar** is unilateral with upper lid colobomas and epibulbar dermoids.
- **Syndromic sensorineural loss:** ECG for long QT (Jervell and Lange-Nielsen); Pendred, Waardenburg, Usher, BOR, Alport, CHARGE.
- **22q11.2 deletion:** velopharyngeal insufficiency, glottic web, medial carotids — image before pharyngeal surgery.

1 CHAPTER ONE Definitions

Syndrome, sequence, association.

1.1 The terms

A **syndrome** is a pattern of malformations known or assumed to have a single cause, such as Down syndrome or Treacher Collins syndrome. A **sequence** is a cascade of anomalies that follows from one initial malformation: in the *Pierre Robin sequence*, a small mandible lets the tongue fall back, which stops the palatal shelves from fusing and obstructs the airway. An **association** is a non-random clustering of anomalies

without a known single cause, such as VACTERL. CHARGE was long called an association until mutations in *CHD7* were found, and it is now a syndrome.

1.2 The ENT presentations that should raise suspicion

COMMON ENT PROBLEMS AND THE SYNDROMES BEHIND THEM

PROBLEM	SYNDROMES TO CONSIDER
Conductive hearing loss	Down, Goldenhar, Treacher Collins, Turner, primary ciliary dyskinesia (Kartagener), cleft palate syndromes
Sensorineural hearing loss	Usher, Pendred, Jervell and Lange-Nielsen, Waardenburg, branchio-oto-renal, Alport, CHARGE, Alström
Nasal obstruction in the newborn	Choanal atresia (CHARGE), piriform aperture stenosis, solitary median maxillary central incisor
Tongue base and pharyngeal obstruction	Pierre Robin, Stickler, Beckwith–Wiedemann, mucopolysaccharidoses, craniosynostosis syndromes
Laryngeal cleft and tracheo-oesophageal fistula	Opitz G/BBB, Pallister–Hall, VACTERL
Laryngeal web and velopharyngeal insufficiency	22q11.2 deletion (velocardiofacial) syndrome

EVERY SYNDROMAL CHILD IN THEATRE

- **Expect a difficult airway** in mandibular hypoplasia (Treacher Collins, Pierre Robin, Goldenhar), macroglossia (Beckwith–Wiedemann, Down) and mucopolysaccharidoses: experienced anaesthetist and ENT surgeon present.
- **Protect the cervical spine** in Down syndrome and Goldenhar syndrome.
- **Plan postoperative monitoring** after adenotonsillectomy for obstructive sleep apnoea: many need high-dependency care.
- **Image the carotids** before pharyngeal surgery in 22q11.2 deletion, because the internal carotids may lie medially behind the pharyngeal wall.

2 CHAPTER TWO Down syndrome

The commonest syndrome in ENT.

2.1 Genetics and general features

Down syndrome affects about 1 in 700 births and is the commonest chromosomal disorder. About 95% have trisomy 21, 3 to 4% a translocation (the only inheritable form) and a few are mosaic. Risk rises with maternal age. General features include learning disability, hypotonia, congenital heart disease in about 40% (atrioventricular septal defect most typically), hypothyroidism, low IgA with frequent respiratory infections, leukaemia, early Alzheimer's disease and obesity.

2.2 The ears

The ear canals are narrow and fill with wax, and Eustachian tube dysfunction makes **otitis media with effusion** almost universal and longer-lasting than in other children. Ventilation tubes work, but insertion is technically hard, the benefit is shorter, and otorrhoea, early extrusion and recurrence are commoner, so **hearing aids are often offered first**. Chronic otitis media and cholesteatoma develop more often, so follow-up is long term. Sensorineural loss is less common but occurs earlier with age.

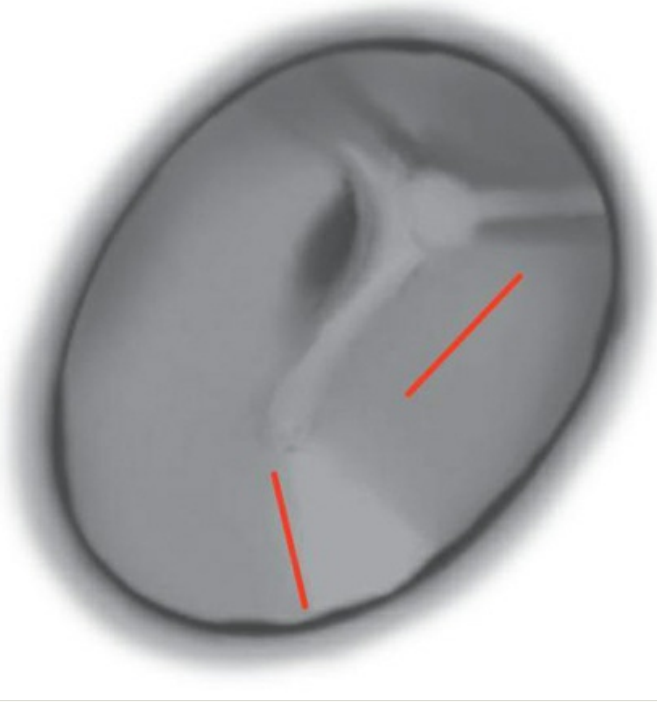


FIGURE 1 Grommets for effusion: harder to place and shorter-lived in the narrow canals of Down syndrome, which is why hearing aids are often preferred.

Dhingra, fig. 10.3

2.3 The airway

Midfacial hypoplasia, relative macroglossia, hypotonia, obesity and adenotonsillar hypertrophy give **obstructive sleep apnoea in about 60%**. Oximetry or, ideally, polysomnography is done before adenotonsillectomy, because up to three-quarters need breathing support afterwards and surgery alone often does not cure it; CPAP or further surgery may follow. A mild congenital **subglottic stenosis** is common, so the endotracheal tube is usually half a size smaller than predicted for age.

2.4 The neck

Atlantoaxial instability is present in about one in five, and neck hyperextension during surgery such as adenotonsillectomy can injure the cord. Plain films do not predict who is at risk, so the neck of every child with Down syndrome is handled with care. Grisel syndrome, non-traumatic atlantoaxial subluxation after inflammation, has been linked to excessive diathermy during adenoidectomy.

3 CHAPTER THREE First and second arch syndromes

The face, the ear and the jaw.

3.1 Treacher Collins syndrome

Treacher Collins syndrome (mandibulofacial dysostosis) affects about 1 in 50,000. It is autosomal dominant, usually from mutations in *TCOF1* (the "treacle" gene), with half arising as new mutations. It is **bilateral and symmetrical**: down-slanting palpebral fissures, **lower lid colobomas**, absent or hypoplastic zygomas, micrognathia, and microtia with canal atresia and a conductive loss, usually managed with **bone-conduction hearing devices**. About a third have a cleft palate. The small mandible, especially with choanal atresia, can cause airway obstruction at birth that needs a tracheostomy; distraction osteogenesis may later lengthen the mandible. Intubation is difficult throughout life.



FIGURE 2 First, anotia. Second, microtia (a "peanut" ear), seen in first and second arch syndromes.

Dhingra, figs 8.2 and 8.3

3.2 Goldenhar syndrome (hemifacial microsomia)

Goldenhar syndrome is the commonest first and second arch anomaly, affecting about 1 in 5,600 to 26,500, usually sporadically. It is **unilateral and asymmetrical**, more often on the right: hypoplasia of the mandible, temporomandibular joint, muscles and orbit, preauricular tags, microtia or anotia with canal atresia, ossicular anomalies, and sometimes facial nerve and inner ear anomalies. Associated features include **epibulbar dermoids, upper lid colobomas**, and cervical vertebral fusion. The OMENS+ system grades the orbit, mandible, ear, nerves and soft tissues.

LOWER LID OR UPPER LID?

A coloboma of the **lower** lid suggests Treacher Collins syndrome (bilateral and symmetrical). A coloboma of the **upper** lid, with an epibulbar dermoid, suggests Goldenhar syndrome (unilateral and asymmetrical).

3.3 Pierre Robin sequence and Stickler syndrome

The Pierre Robin sequence combines micrognathia, glossoptosis and a U-shaped cleft palate. Airway obstruction is managed by prone positioning, a nasopharyngeal airway, tongue–lip adhesion, mandibular distraction or tracheostomy, and feeding often needs support. **Stickler syndrome**, a collagen disorder, is the commonest associated syndrome and adds high myopia with a risk of retinal detachment, joint problems and sensorineural or mixed hearing loss; every child with Robin sequence should have an eye examination.

4 CHAPTER FOUR Syndromes with sensorineural loss

Look beyond the ear.

4.1 The ones to know

SYNDROMIC SENSORINEURAL HEARING LOSS

SYNDROME	INHERITANCE AND GENE	FEATURES THAT MATTER
Waardenburg	Usually dominant; *PAX3* (types 1 and 3), *MITF* (type 2), *SOX10* (type 4)	White forelock, heterochromia, dystopia canthorum in type 1; about 2 to 5% of congenitally deaf children
Pendred	Recessive; *SLC26A4* (pendrin)	Enlarged vestibular aqueduct, goitre later in childhood; fluctuating loss after minor head injury
Jervell and Lange-Nielsen	Recessive; *KCNQ1*, *KCNE1*	Long QT syndrome, syncope and sudden death; ECG for every child with profound congenital loss
Usher	Recessive; several genes	Retinitis pigmentosa; types 1 to 3 by severity of loss and vestibular function
Branchio-oto-renal	Dominant; *EYA1*	Preauricular pits, branchial fistulae, renal anomalies; mixed loss
Alport	Mostly X-linked; *COL4A5*	Haematuria and renal failure; progressive high-frequency loss; lens anomalies
CHARGE	*CHD7*	Coloboma, heart defects, choanal atresia, retardation of growth, genital and ear anomalies, with absent semicircular canals

4.2 CHARGE syndrome in ENT

CHARGE stands for coloboma, heart defects, choanal atresia, retarded growth and development, genital hypoplasia and ear anomalies. The ears are typically short and wide with a small or absent lobe; the inner ear often lacks semicircular canals, so balance and motor milestones are delayed; and cranial nerve palsies, including

facial palsy and swallowing problems, are common. **Bilateral choanal atresia** can be the presenting emergency.

5 CHAPTER FIVE Other syndromes ENT surgeons meet

Airway, palate and skull.

5.1 22q11.2 deletion syndrome

The 22q11.2 deletion (velocardiofacial or DiGeorge syndrome) is one of the commonest microdeletion syndromes. It causes congenital heart disease, hypocalcaemia from absent parathyroids, immune deficiency from thymic hypoplasia, learning and psychiatric problems, and characteristic facies. In ENT it causes **velopharyngeal insufficiency** with hypernasal speech (often with an overt or submucous cleft), **anterior glottic webs**, and **medially placed internal carotid arteries** that must be imaged before pharyngoplasty.

5.2 Craniosynostosis syndromes

Crouzon and **Apert** syndromes, caused mainly by *FGFR2* mutations, combine premature skull suture fusion with midfacial hypoplasia and proptosis; Apert adds fusion of the fingers and toes. Up to half develop obstructive sleep apnoea from the small midface, so overnight studies are done readily, and midface advancement by distraction may be needed. Conductive loss from middle ear anomalies and effusion is common.

5.3 Macroglossia and storage disorders

Beckwith–Wiedemann syndrome (11p15) causes macroglossia, overgrowth, abdominal wall defects, neonatal hypoglycaemia and a risk of embryonal tumours; tongue reduction may be needed for airway or feeding problems. The **mucopolysaccharidoses** (Hurler, Hunter and others) deposit glycosaminoglycans in the airway, adenoids, tonsils and middle ear, causing progressive obstruction,

sleep apnoea and conductive loss; the airway becomes increasingly difficult with age, and enzyme replacement or stem cell transplant changes the course.

5.4 Turner syndrome

Turner syndrome (45,X) affects about 1 in 2,500 girls: short stature, a webbed neck and ovarian failure, with recurrent otitis media, effusion and cholesteatoma in childhood and a characteristic mid-frequency sensorineural loss in adult life.

6 CHAPTER SIX Approach

The syndromal child in clinic.

6.1 Principles

Ask about pregnancy, birth, development, family history and other medical problems; look at the face, eyes, ears, neck, hands and skin; and listen to parents, who often know a great deal about their child's condition and should be asked what they expect from the consultation. When an ENT finding suggests a syndrome, refer to clinical genetics, and arrange the relevant tests: an ECG, renal ultrasound, eye examination, thyroid tests or imaging. Care is multidisciplinary, involving paediatrics, audiology, speech and language therapy, cleft and craniofacial teams, anaesthesia and teachers of the deaf.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - **Syndrome, sequence and association,** with an example of each.

- **Down syndrome:** effusion and hearing aids, obstructive sleep apnoea and postoperative care, subglottic stenosis, atlantoaxial instability.
- **Treacher Collins versus Goldenhar.**
- **Syndromes with sensorineural loss:** Waardenburg, Pendred, Jervell and Lange-Nielsen, Usher, branchio-oto-renal, Alport.
- **CHARGE** and choanal atresia.
- **22q11.2 deletion:** velopharyngeal insufficiency and the medial carotid.

7.2 Common pitfalls

COMMON PITFALLS

- Hyperextending the neck of a child with Down syndrome during adenotonsillectomy.
- Discharging a child with Down syndrome and severe sleep apnoea on the day of adenotonsillectomy.
- Missing long QT syndrome in a profoundly deaf child.
- Operating on the pharynx in 22q11.2 deletion without imaging the carotids.

Written for this app from Oxford Textbook of Otolaryngology ch. 90, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 19. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · MOUTH & THROAT

Oral potentially malignant disorders and oral cancer

Oral cancer is the head and neck cancer that patients and dentists can see, yet it is still often diagnosed late. Early disease is cured in 80 to 90% of patients; advanced disease in only 30 to 40%. The surgeon's job therefore begins before the cancer, with the potentially malignant disorders that precede many of these tumours, and continues through a staging system now built around depth of invasion, a firm rule to treat the neck electively, and adjuvant therapy chosen from the pathology report.

AT A GLANCE

- **Erythroplakia** and **proliferative verrucous leukoplakia** carry the highest risk; dysplasia transforms in about 5 to 18%.
- **Depth of invasion** (not thickness) drives T stage in AJCC 8: T1 ≤2 cm and DOI ≤5 mm; T3 >4 cm or DOI >10 mm.
- **Elective neck dissection** (levels I–III) improves survival in early oral cancer (D'Cruz 2015); sentinel node biopsy is emerging.
- **Margins:** clear ≥5 mm, close 1–5 mm, involved <1 mm. Marginal mandibulectomy for abutment, segmental for medullary invasion.
- **Adjuvant:** radiotherapy for pT3–4, N2+, close margins, PNI, LVI; chemoradiotherapy for ENE or involved margins.

1 CHAPTER ONE Potentially malignant disorders

Before the cancer.

1.1 The term and the lesions

In 2005 the WHO replaced "precancer" with **oral potentially malignant disorders**, because these lesions carry a risk of cancer but do not always precede it. **Leukoplakia** is a white patch that cannot be characterised as any other definable lesion; it is the commonest, and smoking increases its risk. **Erythroplakia**, a red velvety patch, is much rarer but carries the highest risk, and most erythroplakias already show severe dysplasia or carcinoma when biopsied. **Proliferative verrucous leukoplakia** is a multifocal, relentlessly spreading white lesion, typically

in older women who do not smoke, with a very high lifetime cancer risk. **Oral submucous fibrosis**, from areca nut chewing, and **oral lichen planus**, especially the erosive form, complete the main list.

FEATURES THAT RAISE THE RISK OF TRANSFORMATION

- Red or mixed red-and-white lesions (erythroleukoplakia), and proliferative verrucous leukoplakia.
- Sites on the lateral and ventral tongue and floor of mouth.
- Multiple, large or long-standing lesions.
- Lesions in non-smokers and in women.
- Epithelial dysplasia on biopsy, of which about 5 to 18% transform.

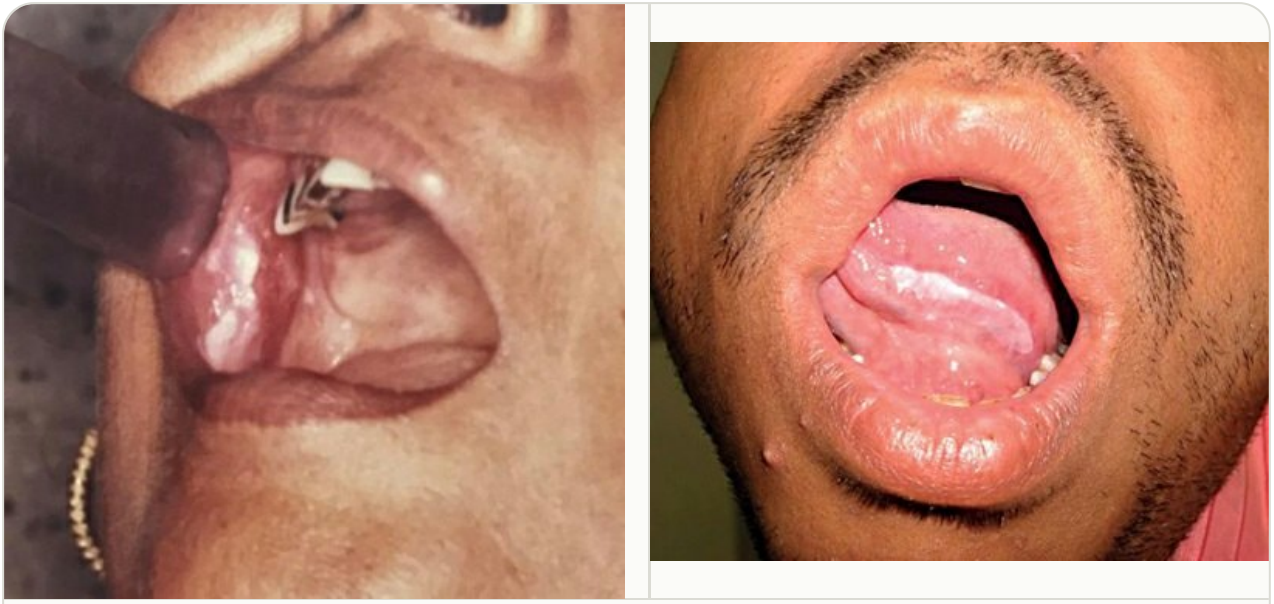


FIGURE 1 First, leukoplakia of the cheek after pan and tobacco use. Second, leukoplakia on the lateral border of the tongue.
Dhingra, figs 44.7 and 44.8

1.2 Assessment of a white or red patch

Inspect and palpate the whole mouth, note size, colour, texture and induration, examine the neck, remove irritants (a sharp tooth, a denture), treat candida, and review in two to three weeks. Any lesion that persists is **biopsied**, taking the reddest, most indurated or most irregular part; histology grades dysplasia as mild, moderate or severe. Adjuncts such as toluidine blue or Lugol's iodine staining,

autofluorescence and narrow-band imaging help choose biopsy sites and map margins, but none replaces a biopsy.

1.3 Management

Stop tobacco, alcohol and areca nut. Moderate and severe dysplasia, erythroplakia and accessible high-risk lesions are excised, usually with a laser or scalpel, and all patients are followed long term, because new lesions and cancers arise elsewhere in the mouth (field change).

2 CHAPTER TWO
Oral squamous cell carcinoma

Who, where and how it presents.

2.1 Epidemiology and risk factors

More than 90% of oral cancers are squamous cell carcinomas. The main risk factors are **tobacco** (smoked and chewed), **alcohol**, which multiplies the risk of tobacco, and **betel quid with areca nut**, which makes oral cancer one of the commonest cancers in South Asia. HPV plays only a small part, unlike in the oropharynx. An increasing number of young non-smokers, often women, develop tongue cancer without the usual risk factors, and patients with Fanconi anaemia have a very high risk.



FIGURE 2 Leukoplakia where a quid of tobacco and lime was held.
Dhingra, fig. 44.9

2.2 Sites and presentation

In Western countries the lateral border of the **tongue** and the **floor of mouth** are the commonest sites; in South Asia the **buccal mucosa** and gingivobuccal sulcus, where the quid is held, predominate. Patients present with a non-healing ulcer, a lump, a red or white patch, pain referred to the ear, bleeding, loose teeth or ill-fitting dentures, trismus, numbness of the lip (mental nerve involvement) or a neck lump. Any single ulcer that has not healed in three weeks must be biopsied.

ORAL CAVITY SUBSITES AND THEIR PARTICULAR PROBLEMS

SUBSITE	PARTICULAR POINTS
Oral tongue	Commonest site in the West; depth of invasion drives nodal spread; may cross the midline
Floor of mouth	Early spread to the mandible and both sides of the neck; can obstruct the submandibular ducts
Buccal mucosa and gingivobuccal sulcus	Commonest in betel users; spread to skin, mandible, maxilla and infratemporal fossa
Lower alveolus and retromolar trigone	Early bone invasion; retromolar disease spreads to the pterygoid region
Hard palate and upper alveolus	Minor salivary tumours are common here; maxillectomy defects need obturators or flaps

SUBSITE	PARTICULAR POINTS
Lip	Usually lower lip, sun-exposed; excellent prognosis; treated by surgery or radiotherapy

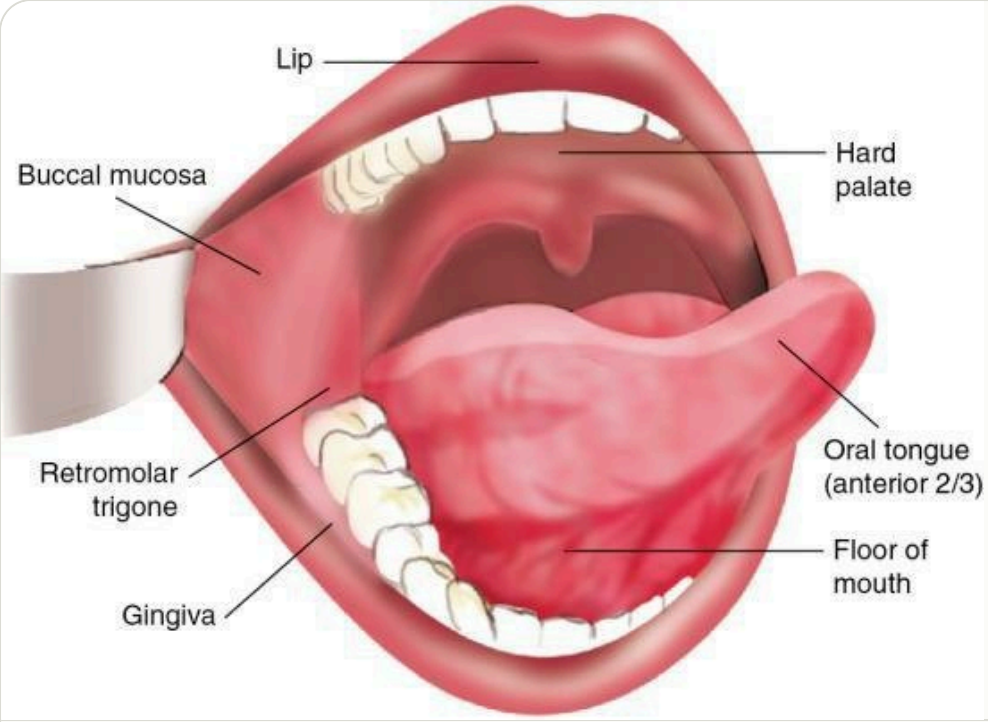


FIGURE 3 The subsites of the oral cavity.
Dhingra, fig. 42.1

3 CHAPTER THREE Staging and investigation

Depth, bone and nodes.

3.1 Investigation

Examination under anaesthesia confirms extent and allows biopsy. **MRI** shows soft tissue extent, depth, marrow invasion and perineural spread better; **CT** shows cortical bone erosion; and an **orthopantomogram** gives dental and mandibular detail. Ultrasound with FNA assesses neck nodes. **FDG PET-CT** is used for advanced or high-grade disease and low neck nodes, to find distant metastases and second

primaries, and for suspected recurrence. Dental assessment and any extractions are done before radiotherapy to reduce osteoradionecrosis.

3.2 The AJCC eighth edition

The eighth edition added **depth of invasion** (DOI), measured from the level of the adjacent normal basement membrane to the deepest point of invasion, because it predicts nodal spread better than surface size; it is not the same as tumour thickness, which includes exophytic growth.

T STAGING OF ORAL CAVITY CANCER (AJCC 8TH)

T	SIZE AND DEPTH OF INVASION
T1	2 cm or less, DOI 5 mm or less
T2	2 cm or less with DOI over 5 mm, or 2 to 4 cm with DOI 10 mm or less
T3	Over 4 cm, or DOI over 10 mm
T4a	Invades through cortical bone of mandible or maxilla, into the maxillary sinus or the skin of the face
T4b	Invades masticator space, pterygoid plates or skull base, or encases the internal carotid artery

Superficial erosion of a tooth socket by a gingival primary is not enough to make a tumour T4. **Extranodal extension** now upstages nodal disease, clinically when obvious and pathologically on the specimen.

4 CHAPTER FOUR Treatment

Surgery first for most.

4.1 The primary

Surgery followed by risk-based adjuvant therapy gives the best control for operable oral cancer at every stage. The aim is complete excision with a **5 mm** margin of normal tissue in three dimensions; on the specimen, margins of 5 mm or more are

clear, 1 to 5 mm *close*, and under 1 mm *involved*, and positive margins reduce survival. Tumours near the mandible need a **marginal mandibulectomy** (removing the alveolar rim and keeping continuity) when the bone is abutted or superficially eroded, and a **segmental mandibulectomy** when tumour invades the medullary bone or the mandible is thin and edentulous. Radiotherapy or brachytherapy may be preferred for some lip and hard palate cancers where surgery would cost too much function.



FIGURE 4 First, carcinoma of the buccal mucosa. Second, carcinoma of the lateral tongue with leukoplakia of the floor of mouth.
Dhingra, figs 44.11 and 44.12

4.2 The neck

The neck is treated electively in early oral cancer. A large Indian randomised trial (D'Cruz, 2015) showed that **elective neck dissection** of levels I to III in clinically node-negative early cancer improved three-year overall survival compared with waiting and operating when nodes appeared (about 80% versus 68%). Elective treatment is therefore standard when DOI is more than about 3 to 4 mm, and is considered at lower depths for high-risk sites. **Sentinel node biopsy** is an emerging alternative in early disease, with less morbidity. Node-positive necks have a comprehensive dissection of levels I to V (sometimes I to IV), and the opposite side is treated for midline tumours or extensive nodal disease.

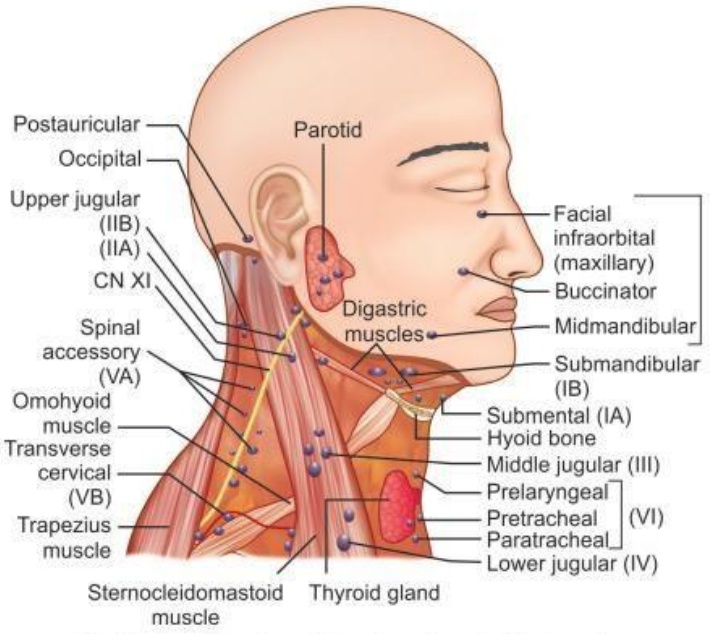


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

FIGURE 5 The lymph node levels of the neck: levels I to III drain the oral cavity first.

Anatomy atlas · ENT Steps p. 285

4.3 Reconstruction

Reconstruction restores swallowing, speech, mastication and appearance, replacing like with like. Small defects heal by secondary intention, are grafted or are closed with local flaps; tongue and floor of mouth defects are commonly reconstructed with a **radial forearm** or **anterolateral thigh** free flap, and segmental mandibular defects with a **fibula** or other vascularised bone flap, which can later carry dental implants.

4.4 Adjuvant therapy

Postoperative **radiotherapy** is given for pT3–T4 tumours, multiple or level IV–V nodes, close margins, perineural invasion and lymphovascular invasion. **Concurrent chemoradiotherapy** with cisplatin is given for **extranodal extension** and **involved margins**, the two features that most predict recurrence. Unresectable T4b disease and patients unfit for surgery receive chemoradiotherapy or palliative treatment; borderline resectable tumours may be downstaged with **induction chemotherapy**.

THE SUSPICIOUS ORAL LESION

1. ## Persistent ulcer, red or white patch after removing irritants

- → biopsy the most suspicious area
- → **dysplasia**
  stop tobacco, alcohol and areca, excise moderate or severe dysplasia, follow up long term
- → **carcinoma**
  step 2

2. ## Stage

- → MRI or CT of primary and neck, orthopantomogram, chest imaging or PET-CT
- → dental assessment

3. ## Treat the primary

- → wide excision with 5 mm margins
- → marginal or segmental mandibulectomy by bone invasion
- → free flap reconstruction where needed

4. ## Treat the neck

- → **N0 with DOI over 3 to 4 mm**
  elective neck dissection levels I to III or sentinel node biopsy
- → **N+**
  comprehensive neck dissection

5. ## Pathology

- → **close margins, PNI, LVI, pT3–4, N2+**
  radiotherapy
- → **ENE or involved margin**
  chemoradiotherapy

5 CHAPTER FIVE Complications and follow-up

After treatment.

5.1 Complications

Haemorrhage, infection (more after reconstruction, in drinkers and with tracheostomy), seroma and chyle leak follow neck surgery; marginal mandibular and accessory nerve weakness are recognised risks. **Carotid blowout** is rare after primary surgery but feared in irradiated and recurrent necks, especially with salivary fistula or flap necrosis. Radiotherapy causes dry mouth, dental decay, trismus and osteoradionecrosis, so lifelong dental care and jaw exercises matter.

5.2 Follow-up

Patients are reviewed frequently for the first two years, when most recurrences occur, and for at least five years, with imaging as needed. Second primary cancers in the mouth, pharynx, larynx, oesophagus and lung are common in smokers, and stopping tobacco and alcohol is the single most valuable intervention after treatment.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Potentially malignant disorders:** leukoplakia, erythroplakia, proliferative verrucous leukoplakia, submucous fibrosis, and the risk features.
- **Depth of invasion** and the AJCC eighth edition T stages.
- **Elective neck dissection in early oral cancer:** the D'Cruz trial; sentinel node biopsy.

- **Margins:** clear, close and involved.
- **Marginal versus segmental mandibulectomy.**
- **Adjuvant therapy:** indications for radiotherapy and for chemoradiotherapy.

6.2 Common pitfalls

COMMON PITFALLS

- Treating a persistent ulcer with repeated topical steroids instead of biopsying it.
- Watching the N0 neck in a tongue cancer with significant depth of invasion.
- Confusing depth of invasion with tumour thickness.
- Forgetting dental assessment before radiotherapy.

Written for this app from Oxford Textbook of Otolaryngology ch. 57, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 49. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

Imaging for this region

The core imaging conditions of this region, with their key facts and a representative scan. For the normal anatomy and a search pattern, read Lesson 12, The oral cavity and oropharynx in the imaging volume.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

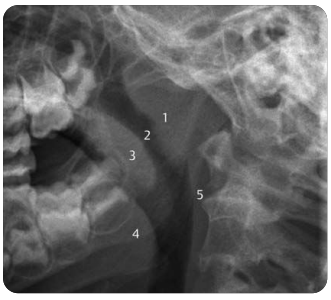
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions



Plain radiograph for adenoid evaluation.
Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Nasopharyngeal Carcinoma

IMAGING FINDINGS

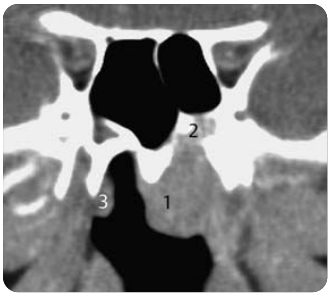
- Asymmetric mass in the fossa of Rosenmüller
- Parapharyngeal and skull base spread; clival marrow replaced on T1
- Retropharyngeal and level II and V nodes

TOP DIFFERENTIALS

- Adenoidal lymphoid hyperplasia (symmetric, striated)
- Lymphoma
- Minor salivary gland tumour

CHECKLIST

- MRI for staging: skull base, cavernous sinus, perineural spread
- Middle ear effusion from Eustachian tube obstruction



CT, coronal with intravenous contrast enhancement. Note the soft-tissue mass in the roof of the nasopharynx of the left side (1), with invasion in the…



Coronal NECT of paranasal sinuses, performed in a 62-year-old Asian man for evaluation of nasal stuffiness and epistaxis, demonstrates an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Tonsillar Abscess

IMAGING FINDINGS

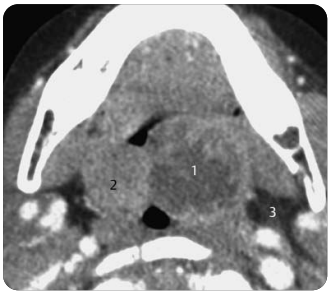
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

TOP DIFFERENTIALS

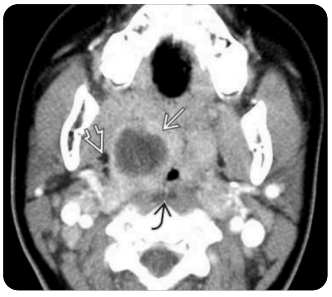
- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

CHECKLIST

- Airway
- Internal jugular vein thrombosis and carotid



CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Tonsillar Carcinoma

IMAGING FINDINGS

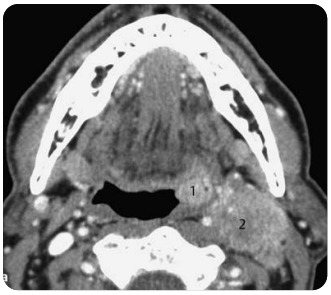
- Asymmetric enhancing tonsillar mass, sometimes small
- Spread to soft palate, tongue base and parapharyngeal space
- Cystic or necrotic level II nodes (HPV)

TOP DIFFERENTIALS

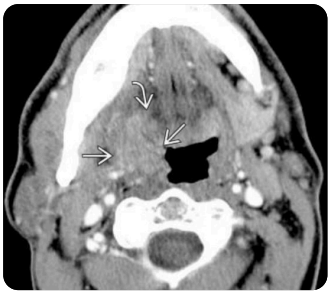
- Tonsillar lymphoid hyperplasia
- Lymphoma
- Peritonsillar abscess

CHECKLIST

- Medial pterygoid, mandible and carotid
- Both sides of the neck



CT with intravenous contrast enhancement, axial. In the left tonsillar fossa, a small and enhancing tumor is visible (1), with enlarged level II lymph…



Axial CECT in alcoholic smoker presenting with multiple right neck masses shows an enlarged right palatine tonsil mass…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 492.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Oropharyngeal Tumor, Base of Tongue

IMAGING FINDINGS

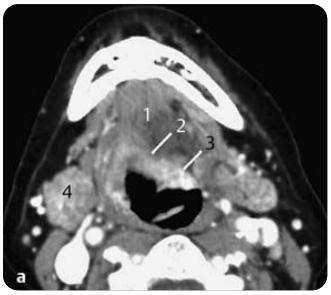
- Asymmetric fullness of the lingual tonsil with enhancement
- Spread to vallecula, pre-epiglottic space and floor of mouth
- Level II nodes, often bilateral

TOP DIFFERENTIALS

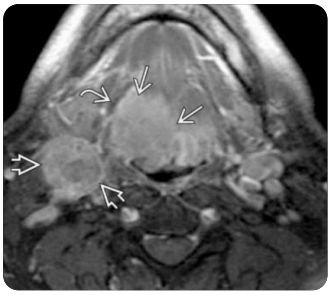
- Lingual tonsil lymphoid hyperplasia
- Lymphoma
- Thyroglossal or vallecular cyst

CHECKLIST

- Crossing the midline and neurovascular bundles
- Pre-epiglottic and laryngeal invasion



CT with intravenous contrast enhancement, axial. Axial slice at the level of the mandible and floor of the mouth (1). At clinical evaluation, an…



Axial T1WI C+ FS MR in a 58-year-old alcoholic presenting with persistent right neck mass after dental procedure demonstrates enlarged,…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 488.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Zenker Diverticulum (Pharyngeal Pouch)

IMAGING FINDINGS

- Outpouching from the posterior hypopharynx above the cricopharyngeus on barium swallow
- Air–fluid level behind the oesophagus on CT
- Residual barium after the swallow

TOP DIFFERENTIALS

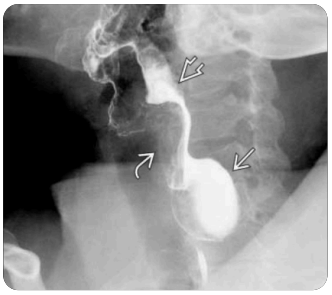
- Lateral pharyngeal or Killian–Jamieson diverticulum
- Paratracheal air cyst

CHECKLIST

- Size and neck of the pouch
- Aspiration on the swallow



Barium swallow, anteroposterior view. After swallowing contrast, some contrast has remained in the hypopharynx (1) due to physiologic…



Barium esophagram demonstrates a large diverticulum with retained layering contrast from the posterior lateral junction of the hypopharynx…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 258.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

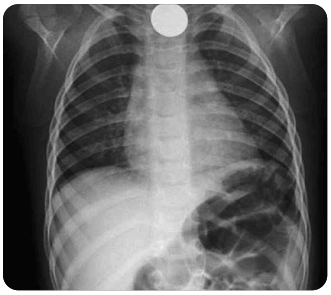
- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

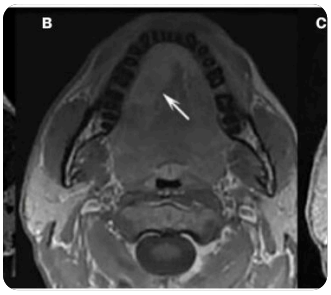
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.